

CHAPTER 15

AUXILIARY POWER PLANT SYSTEM

CHAPTER OVERVIEW

SECTION	TITLE
15-1	Equipment Description and Data
15-2	Troubleshooting
15-3	On Aircraft Inspections
15-4	Maintenance Procedures
15-5	Maintenance Procedures (BENCH)

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SECTION 15-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
15-1-1	Auxiliary Power Unit System Description and Data
15-1-2	Auxiliary Power Unit System With 5 BIT DESU

15-1-1/(15-1-2 Blank)

15-1-1 AUXILIARY POWER UNIT SYSTEM DESCRIPTION AND DATA.

15-1-1.1 Auxiliary Power Unit System Description.

The Auxiliary Power Unit (APU) System (Figure 15-1-1-1) consists of:

- a. APU
- b. Accessories
- c. Controls
- d. Monitoring System
- e. Starting System

The APU System provides pneumatic power for main engine starting and cabin heating, electrical and hydraulic power for ground operations, and in-flight emergency electrical power.

APU System Accessories include:

- a. Prime/Boost Pump
- b. Hydraulic Accumulator
- c. Hydraulic Handpump
- d. Hydraulic Utility Module
- e. Hydraulic Backup Pump
- f. AC Generator
- g. Hydraulic Start Motor

The prime/boost pump, above the No. 1 fuel cell, is used to prime engine and APU fuel lines, and provide fuel under boost pressure, for APU operation at high altitudes.

The hydraulic accumulator, in the rear midsection cabin ceiling, provides the hydraulic pressure used to drive the APU starter.

The APU starter is a hydraulic start motor mounted on the APU.

If the APU fails during light-off, the hydraulic accumulator is recharged manually by pumping the hydraulic handpump.

Once the APU is operating, the AC generator, mounted on the APU, provides electrical power to the helicopter systems.

The hydraulic utility module and backup pump, on the left forward deck within the main rotor pylon, automatically recharges the depleted hydraulic Accumulator for the next APU start.

The APU controls are in the cockpit on the upper console. The controls consist of:

- a. APU Control Switch
- b. APU Fire Detector/Fuel Shut-Off Selector T-Handle

Indicator lights on the caution/advisory panel permit cockpit monitoring of the APU System. Monitored conditions include:

- a. APU OIL TEMP HI
- b. APU FAIL
- c. APU ON
- d. APU GEN ON
- e. PRIME BOOST PUMP ON
- f. APU ACCUM LOW

15-1-1.2 Principles of Operation.

15-1-1.2.1 Power Distribution.

Electrical power of 28 VDC is supplied by BATT BUS through APU CONTR INST circuit breaker, lower console circuit breaker panel, to the APU power transfer relay, in the left relay panel.

Electrical power of 28 VDC is supplied by BATT UTIL BUS through APU CONTR INST circuit breaker, lower console circuit breaker panel, to APU power transfer relay, in the left relay panel.

The APU is started with the APU CONTR ON/OFF switch on the upper console. Placing the

switch ON sends a 28 VDC start signal, through the APU T-handle switch, to the Digital Electronic Sequence Unit (DESU).

Initially, this signal comes from the power transfer relay, through diode CR1, both in the left relay panel, to the APU CONTR switch.

However, during switchover from battery power to either external power or the helicopter's main engine generators or APU generator, the 28 VDC to the APU CONTR INST circuit breaker (BATT BUS) is momentarily de-energized. This condition would cause the APU to shut down due to loss of start signal. To prevent inadvertent drop out of the APU, relay K47, in the left relay panel, has a 0.5 second time delay at dropout. Contacts A1 and A2, previously closed, will remain closed during the BUS switch over allowing 28 VDC from the APU CONTR INST circuit breaker (BATT UTIL BUS) to be applied through diode CR2 to the APU CONTR switch to maintain the start signal connection.

15-1-1.2.2 Accumulator.

A nitrogen (N₂) charged hydraulic accumulator is installed in the cabin overhead next to the start valve. The accumulator provides the hydraulic pressure required by the APU hydraulic starter.

If the APU does not start, the accumulator may be manually recharged with a handpump or the backup hydraulic pump (electric).

When the APU is in operation, the AC generator provides power for the helicopter systems. During this operation, the accumulator may be recharged for the next start.

15-1-1.2.3 APU Controls and Indicators.

Two APU controls are mounted on the upper console: an APU control switch and an APU fire extinguishing T-handle. Setting the APU control switch to ON opens a hydraulic start valve to release the accumulator hydraulic charge to the APU starter. It also opens the APU shutoff valve to allow fuel to be supplied to the APU (refer to Fuel Prime Boost System (PARA 10-2-3)). The T-handle lights when a sensor detects fire in the

APU compartment. Pulling the T-handle arms the fire extinguisher, removes 28 VDC from the APU control switch, and cuts fuel, thus shutting down the APU.

The electrically controlled and hydraulically started APU runs up to its rated speed automatically. Sequencing and shutdown for a APU System failure are controlled by the DESU Caution/Advisory Panel capsules include:

- a. # 1 ENGINE STARTER
- b. # 2 ENGINE STARTER
- c. APU OIL TEMP HI
- d. APU FAIL
- e. APU ON
- f. APU GEN ON
- g. PRIME BOOST PUMP ON
- h. BACK UP PUMP ON
- i. APU ACCUM LOW
- j. BATT LOW CHARGE
- k. EXT PWR CONNECTED

15-1-1.2.4 Digital Electronic Sequence Unit.

The DESU, located in the rear cabin ceiling, is a microprocessor-based electronic component that monitors and controls all APU operations.

The DESU also shuts down the APU when monitored values exceed established operating parameters.

APU System status is indicated by the five Built-In-Test Equipment (BITE) indicators on the DESU. These indicators cycle from black to white in various combinations (codes) to indicate APU status and fault isolation. Two categories of codes are displayed: normal sequence required during APU start sequence, and normal and failure indications associated with operation. Decals on the side of the

DESU decode the BITE indicators and provide information relative to APU operation. The BITE indicators are electrically operated and magnetically latched to retain coded indications after electrical power has been removed. If the APU shuts down automatically, and the BATT or APU control switches are cycled, the BITE indicators will be cleared.

The DESU initiates the starting events programmed in the microprocessor.

Prior to initiating each event, the microprocessor makes an internal check to ensure that the DESU is operating properly. It compares APU input signals to previously stored data to ensure normal operation and scheduled event timing.

APU speed (provided by the speed sensor), EGT (provided by a thermocouple), oil temperature, and pressure are continuously monitored. APU speed is the basis for APU start sequencing events.

The DESU operates in two modes:

- a. Start sequence mode - BITE indicates the speed at which a start sequence event occurs.
- b. Operational mode - BITE latches onto a malfunction.

A start command from the DESU sends 28 VDC to the APU hydraulic start valve. There is a manual override lever on the valve should the valve solenoid fail. Opening the valve releases the accumulator hydraulic charge. The charge spins the APU starter that rotates the APU compressor and fuel pump. The DESU also controls ignition and signals the fuel valves to open. This accelerates the APU to 100% speed.

When speed reaches 90%, and after a time delay of 1.5 seconds, the maximum fuel valve opens. It allows a governor to control the fuel flow. The APU ON capsule on the instrument panel and the legends on the pilot's and copilot's inboard MFDs will then go on.

The bypass valve permits the APU to bleed air during starting in order to prevent the possibility of

compressor stall. When the APU is in full operation, it may be used as an air source for main engine starting. The AIR SOURCE HEAT/START switch is first set to APU. DC power is routed from the AIR SOURCE HEAT/START circuit breaker on the copilot's circuit breaker panel through switch S18. This power is then applied to energize relay K22.

When either No. 1 or No. 2 engine start button is pressed, the start circuits are energized. This provides 28 VDC to turn on the ENG 1 STARTER ON or the ENG 2 STARTER ON legend. The 28 VDC is also routed through a diode to relay K22. The 28 VDC signal goes through the closed contacts of K22 and to the DESU. This commands the bypass valve to close to allow more airflow for engine starting.

The prime boost pump motor is turned on by the FUEL PUMP switch, S37, on the upper console. This switch also opens the APU fuel shutoff valve. This circuit receives its power through the FUEL PRIME BOOST circuit breaker from the BATT BUS. The PRIME BST PUMP ON legend will go on when the FUEL PUMP switch S37 is set to APU BOOST or FUEL PRIME.

During APU start, acceleration, and operation, the DESU controls and monitors the APU. BITE indicators show APU and DESU status during start and operation. The APU starter rotates the turbine up to 20% - 30% rated speed. When the turbine is at 5% speed, a start fuel valve in the APU opens and ignition occurs. When speed reaches 14%, the main fuel valve opens and at 30% the EGT is read by DESU. The hydraulic and fuel start valves are closed at 70% speed.

When the APU reaches 5% speed, the 40 second counter will start and at 90% speed, the 40 second counter in the DESU shuts down. During this 40 second period, the DESU will not shut the APU down for oil low pressure, over or under speed conditions, and oil high temperature. A 1.5 second timer permits final APU checks to be made before reaching 105% unloaded speed. At stable 100% speed, the timer shuts down.

15-1-1.2.5 Backup Hydraulic Pump.

The electrical power generated by the APU generator can operate the backup hydraulic pump to recharge the APU accumulator.

When the APU is started by the accumulator, its hydraulic fluid charge is depleted. This is signaled by an APU ACCUM LOW capsule on caution/advisory panel. The BACKUP HYD PUMP switch on the upper console controls backup hydraulic pump operation.

When the switch is placed to the ON position with the helicopter either on the ground or airborne, the backup hydraulic pump will run continuously. The pump motor contains a thermal switch that interrupts electrical power to the motor when the helicopter is on the ground and motor temperature rises above a safe value.

With the BACKUP HYD PUMP switch in either AUTO or OFF and the helicopter airborne, the backup hydraulic pump will operate for a minimum of 90 seconds to fully charge the APU accumulator and turn off the APU ACCUM LOW capsule on the caution/advisory panel.

When the BACKUP HYD PUMP switch is placed to OFF and the helicopter is on the ground, the weight-on-wheels switch will disable the system and the backup hydraulic pump will not function. For a more detailed discussion of the backup hydraulic pump and its operation, refer to Hydraulic Power Systems (PARA 7-1-1).

15-1-1.2.6 Fuel System.

Fuel is fed to the gear fuel pump inlet through a ten-micron filter. The fuel pump is protected from overpressure by a bypass relief valve. The fuel flow is controlled by a hydromechanical governor and metering valve which varies the size of the fuel flow orifice. The orifice diameter is set by the pressure differential between ambient and Compressor Discharge Pressure (CDP). This ensures a uniform fuel flow at low as well as elevated pressures.

At 5% rated speed, the start fuel valve opens and stays open until speed reaches 70%. The start fuel valve feeds fuel into the start fuel nozzle. When the

APU reaches 14% speed, the main fuel valve is opened by a 28 VDC DESU signal. The main fuel valve now remains open during normal operation. The fuel flows from the accelerator to the main fuel nozzles. The maximum fuel valve opens electrically when the APU reaches 90% speed and after a time delay of 1.5 seconds.

When the maximum fuel valve opens, it permits fuel to bypass the acceleration schedule. The main fuel line contains a fuel manifold which divides fuel flow to the six main fuel nozzles.

After the start fuel valve closes, a purge drain harness (T-fittings bracket assembly) uses combustor pressure through the start fuel nozzle to force any residual fuel out of the nozzle and start fuel circuit to prevent carbon buildup on the nozzle.

15-1-1.3 Mechanical Description.

15-1-1.3.1 APU.

The APU consists of a gas turbine power section, reduction gear drive, and appropriate controls and accessories. It has a self-contained oil system and is controlled by the DESU. The gas turbine power section is a simple centrifugal compressor and a single-stage radial inflow turbine. The centrifugal compressor and turbine are mounted back-to-back on the end of a high-speed shaft. Air passing through an inlet screen and over the bearings to the compressor prolongs bearing and lubricant life. Turbine-furnished power drives the compressor and the high-speed output shaft.

The combustor is an annular type with an internal liner and six air-atomizing fuel injector nozzles. The combustor is mounted onto the turbine housing and has on its periphery a compressor discharge plenum chamber. Ignition is furnished by a separate pressure atomizing fuel nozzle and igniter plug. The igniter plug, fuel nozzle, and main fuel manifold are installed in the combustion chamber. The thermocouple is located in the APU exhaust and measures Exhaust Gas Temperature (EGT).

During operation above 70% of the gas turbine speed, compressor discharge air is bled through the start fuel nozzle. The bleed-air passes through a purge drain harness and into the start fuel line. The

bleed-air keeps the nozzle continuously clean (no coking) for the next start.

A high-speed pinion gear and shaft, connected to the main rotor shaft, provides the input to the reduction gear drive. The pinion is supported by three planetary gears. An internal gear ring, installed around the three planetary gears, reduces turbine shaft speed from 61,565 rpm to 12,000 rpm at the axial pad, which also provides a mounting pad for the APU generator. A set of gears extending from the output shaft are used to drive the fuel pump and accelerator, and the APU starter mounted on the accessory pads at 4,235 and 8,229 rpm respectively. Other reduction gears drive the lube pump.

The accessory gear box is mounted to the top of the reduction drive assembly. The gear box provides the mount for the hydraulic starter, fuel pump, and accelerator. A speed sensor on the accessory drive assembly provides a pulse signal to the DESU. The DESU converts the pulses to an RPM signal which is used to operate the main fuel and maximum fuel valves.

15-1-1.3.2 APU T-Handle.

When the Fire Detector System detects a fire in the APU compartment, the APU T-handle and master warning panel FIRE capsules will go on red. The pilot can then pull out the APU fire extinguishing T-handle. This does several things: (a) provides an APU stop command through S102 to the DESU, (b) shuts off the prime boost pump if it is running, and (c) de-energizes the APU fuel shutoff valve.

15-1-1.3.3 APU AC Generator.

The helicopter's electrical requirements are supplied when the GENERATORS-APU switch is

positioned to ON. When the AC generator is in operation, the APU GEN ON legends on pilot's and copilot's inboard MFDs go on. The APU generator supplies electrical power to the AC PRIM BUSES and the AC ESNTL BUS. For a more detailed discussion of the APU generator and its operation, refer to AC Power System (PARA 9-1-1).

15-1-1.3.4 APU Accumulator.

The accumulator consists of:

- a. Gas Chamber
- b. Reservoir
- c. Hydraulic Pressure Chamber

The gas chamber contains N₂. The N₂, which can be compressed to act like a spring, forces hydraulic fluid from the pressure chamber. The pressure line is connected to the start valve and starter.

N₂ quantity is measured by a pressure gage in the cabin overhead near the handpump.

The handpump enables accumulator recharging if backup hydraulic pump operation is not possible. The handpump is a short-stroke type with a removable handle.

The accumulator piston is connected to a tape position indicator that is marked in percent of hydraulic charge. When the tape position indicator is at 100% (full left), the pressure chamber is fully charged. When the tape position indicator is at 0%, the accumulator is empty and needs recharging. A second accumulator and pressure switch are part of a winterization kit. This supplies double the normal fluid volume charge to motorize the APU in cold weather.

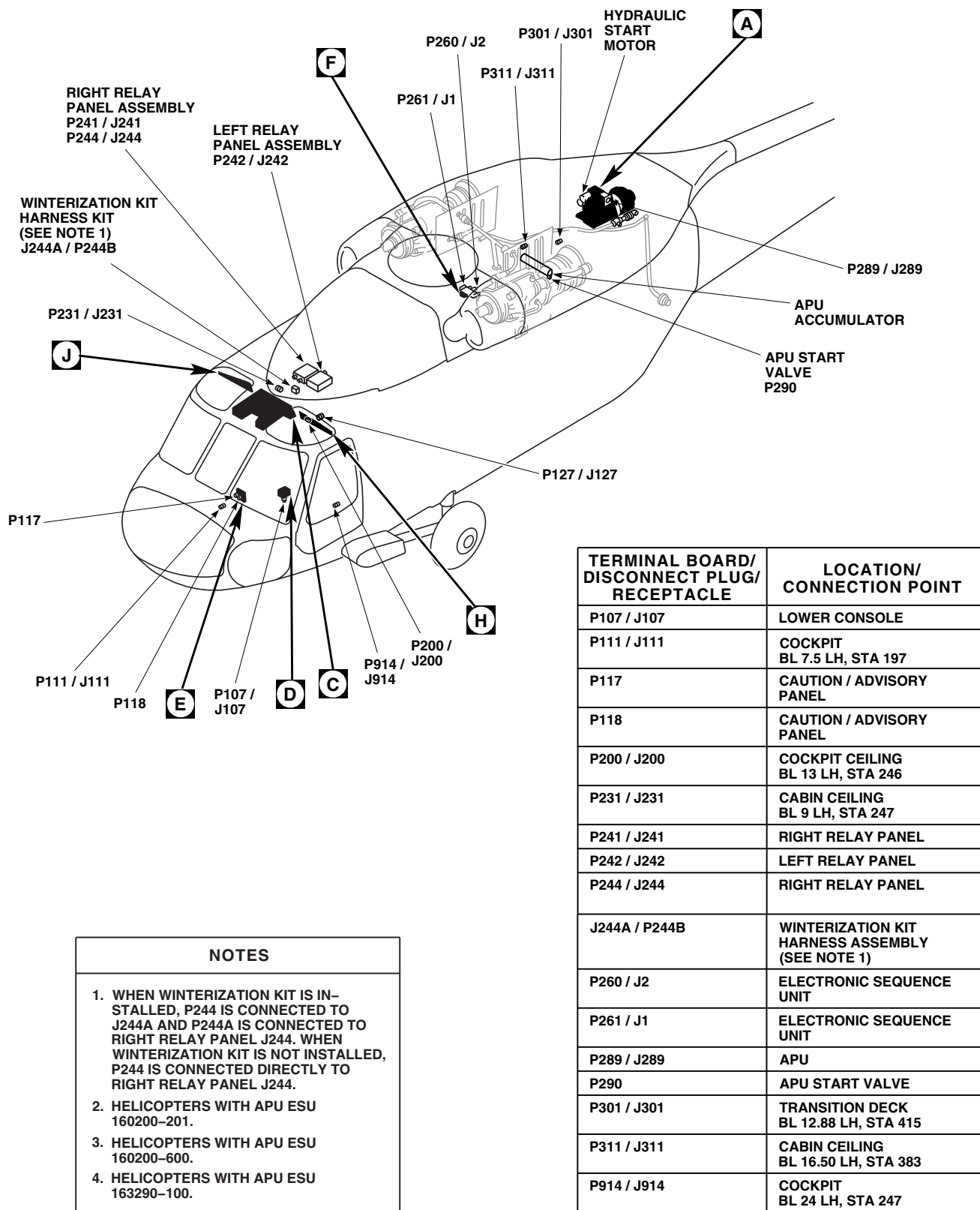
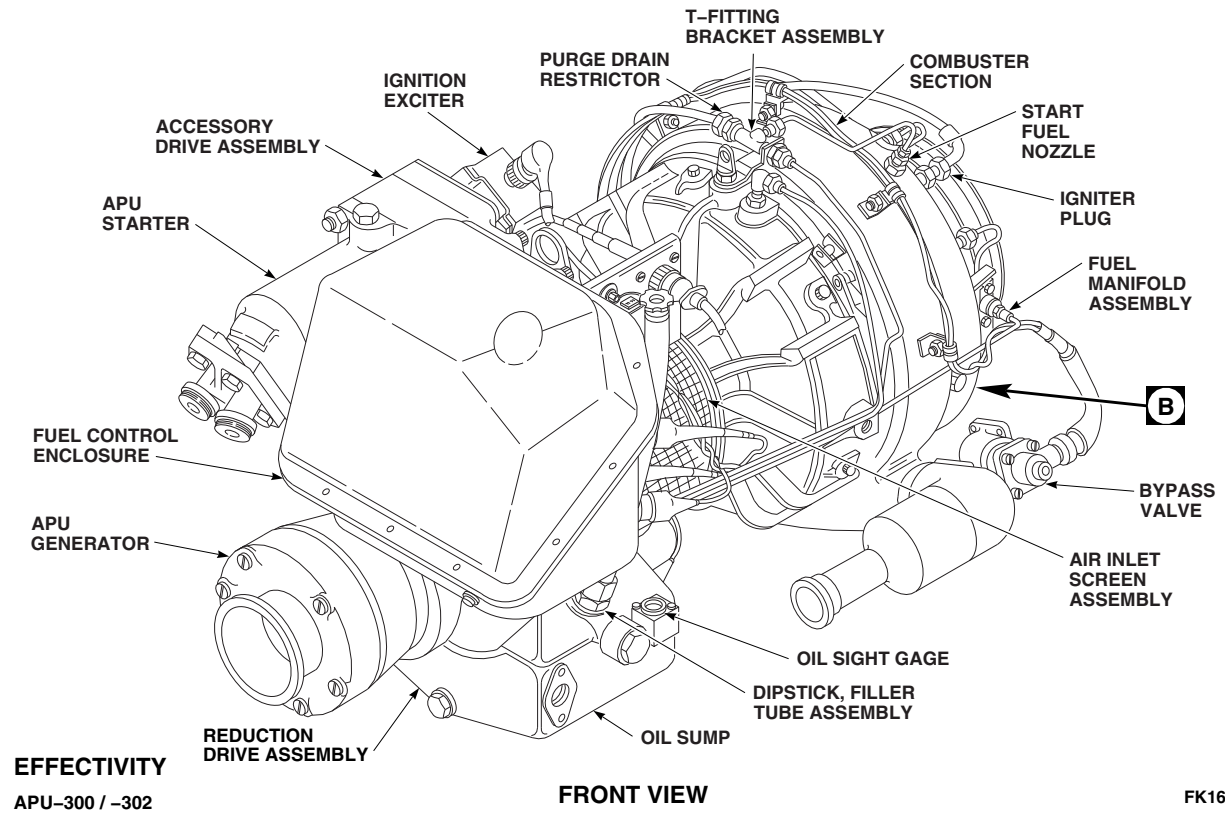
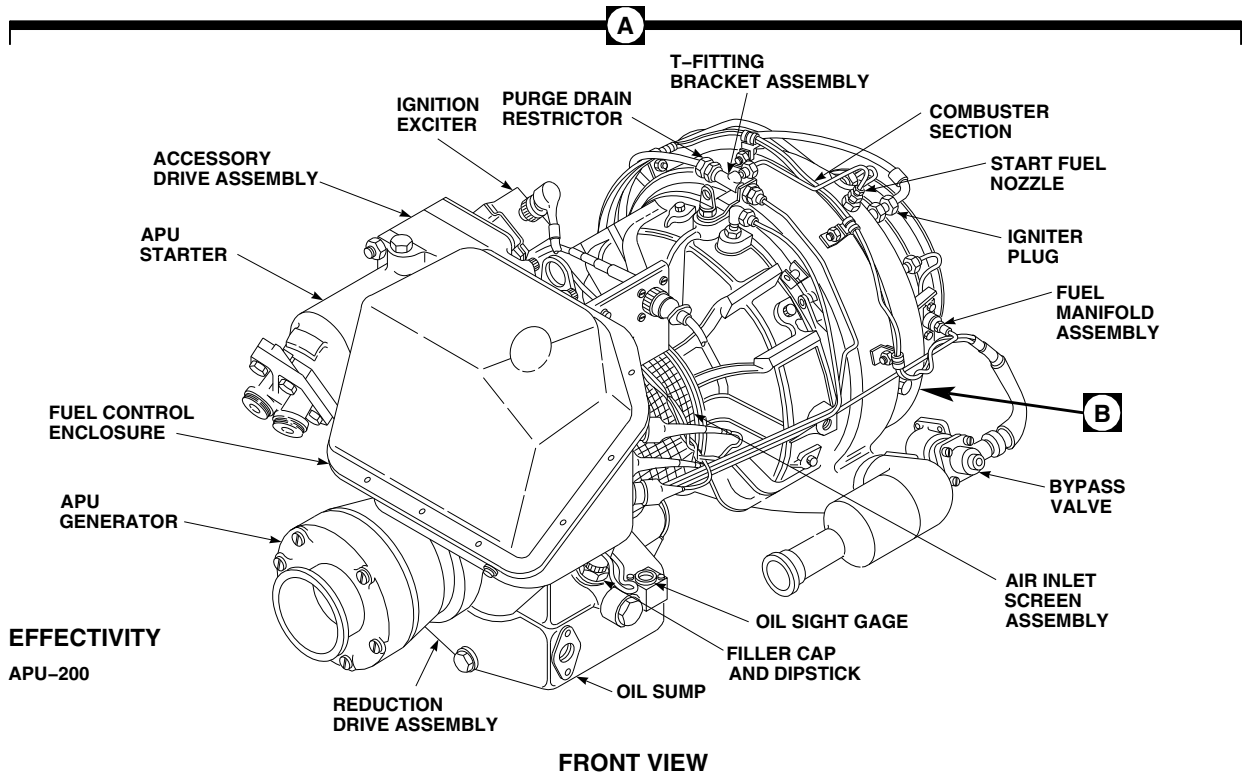
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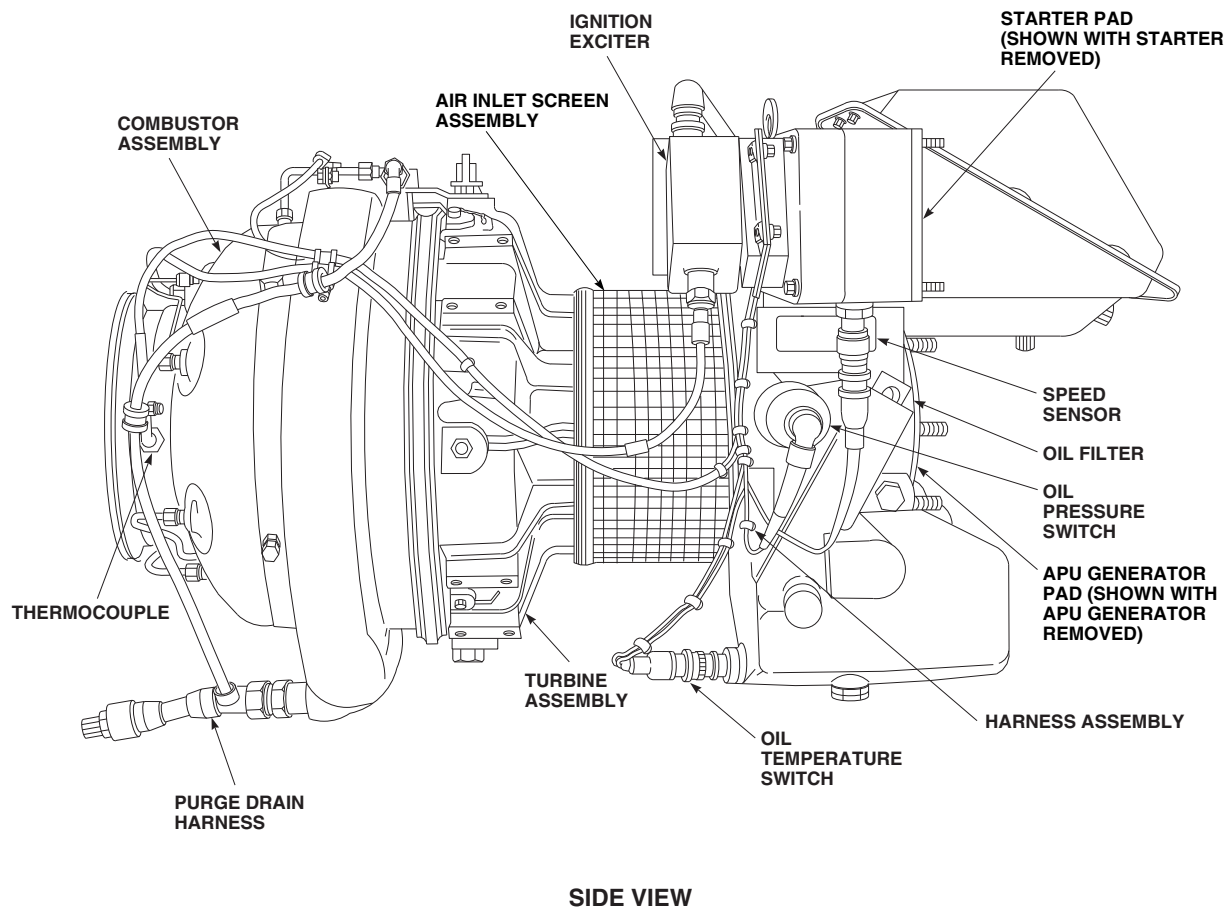
FIGURE 15-1-1-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 1 OF 7)



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FIGURE 15-1-1-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 2 OF 7)

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FIGURE 15-1-1-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 3 OF 7)

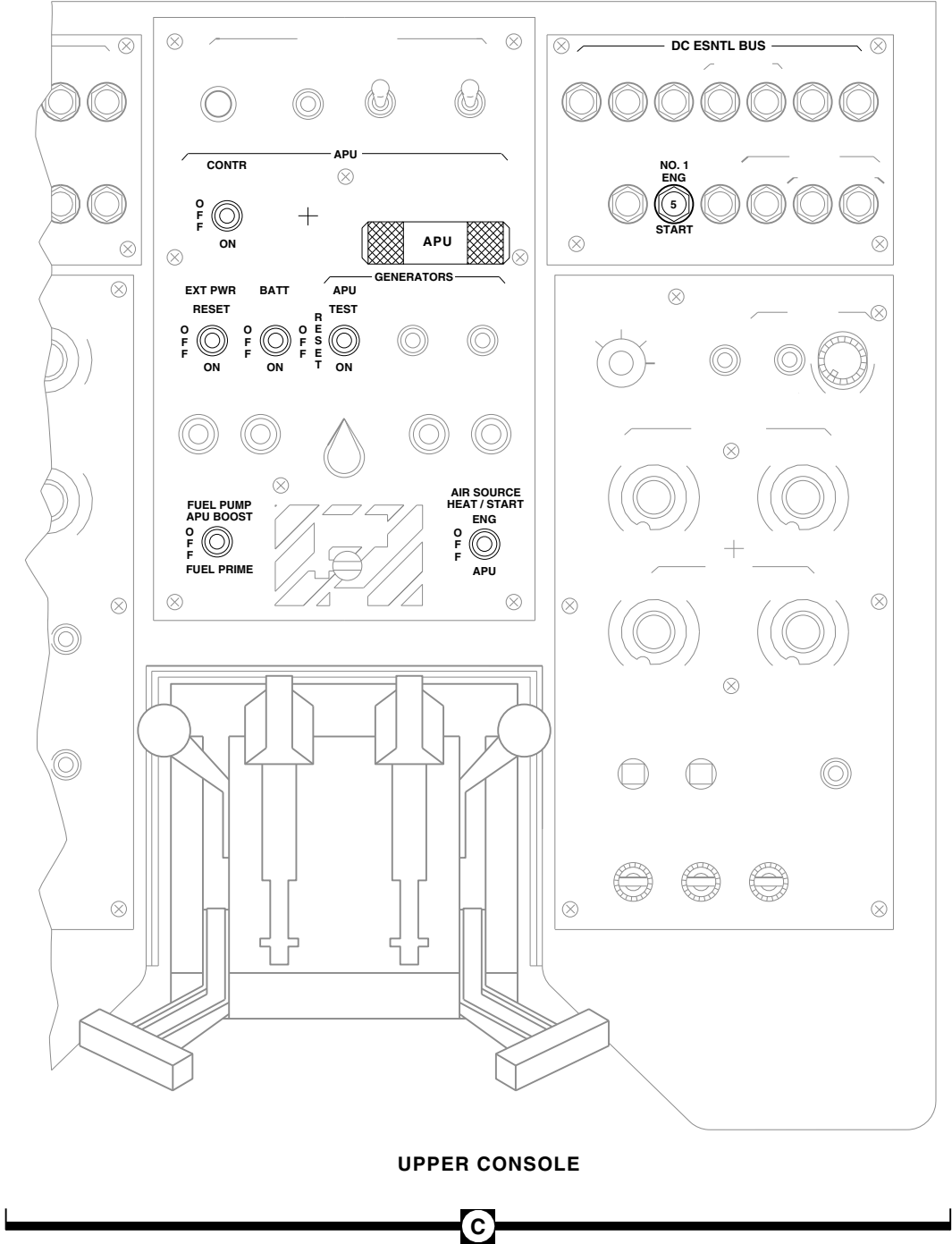
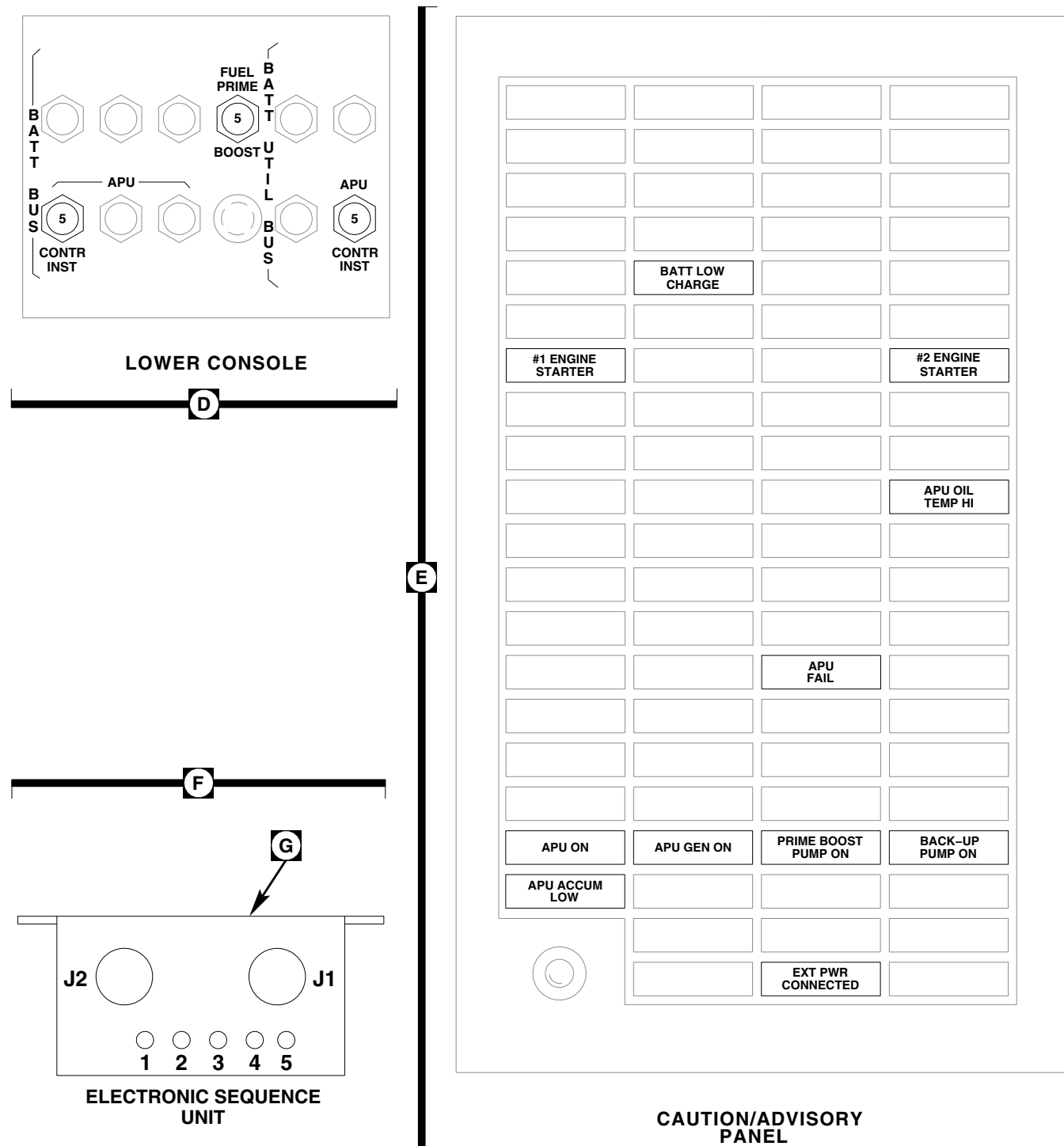


FIGURE 15-1-1-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 4 OF 7) |

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FIGURE 15-1-1-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 5 OF 7)

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T-62T-40-1				DECODED BITE INFORMATION	
BITE # 1 2 3 4					
● ● ● ●	RESET			START SEQUENCE	OPERATION
○ ● ● ●	START FUEL VALVE & EXCITER SIGNAL OUT (5%)				
○ ○ ● ●	MAIN FUEL VALVE SIGNAL OUT (14%)				
○ ○ ○ ●	START FUEL VALVE & EXCITER SIGNAL OFF(70%)				
○ ○ ○ ○	90% RPM SWITCH ON				
● ● ● ●	READY FOR SERVICE (90% + 1.5 SEC)				
○ ● ● ●	PROCESSOR BOARD FAILURE				
● ○ ● ●	SENSOR / DATA BOARD FAILURE				
○ ○ ● ●	OVERTEMPERATURE				
● ● ○ ●	OVERSPEED				
○ ● ○ ●	UNDERSPEED				
● ○ ○ ●	FAIL TO START				
○ ○ ○ ●	LOW OIL PRESSURE				
● ● ● ○	HIGH OIL TEMPERATURE (WARNING)				
○ ● ● ○	SHORTED OIL PRESSURE SWITCH				
● ○ ● ○	SHORTED THERMOCOUPLE PROBE (WARNING)				
● ● ○ ○	OPEN THERMOCOUPLE				
● ○ ○ ○	PROCESSOR SEQUENCE FAIL				
○ ○ ○ ○	NO DATA				
○ ● ○ ○	LOSS OF SPEED DATA				

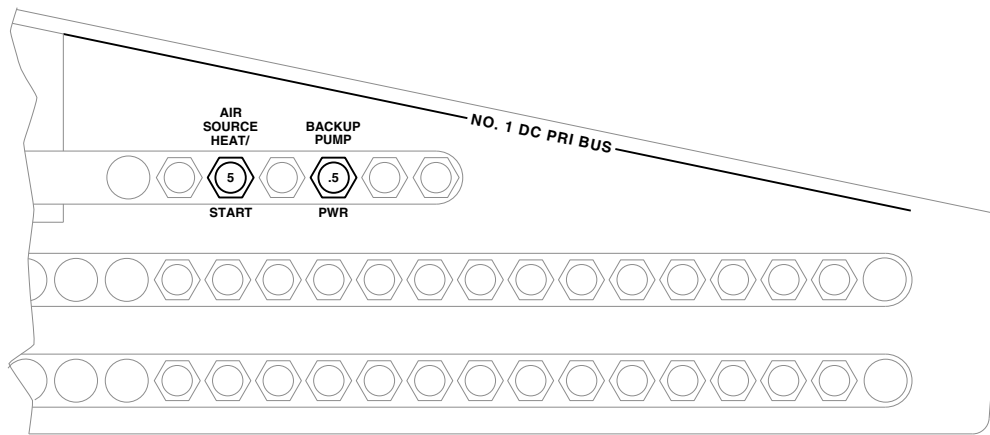
MALFUNCTION BITE INDICATIONS/
 START SEQUENCE DECAL
 (SEE NOTE 1)

T-62T-40-1				DECODED BITE INFORMATION	
BITE # 1 2 3 4					
● ● ● ●	RESET				START SEQUENCE
○ ● ● ●	START FUEL VALVE & EXCITER SIGNAL OUT (5%)				
○ ○ ● ●	MAIN FUEL VALVE SIGNAL OUT (14%)				
○ ○ ○ ●	START FUEL VALVE & EXCITER SIGNAL OFF (70%)				
○ ○ ○ ○	90% RPM SWITCH ON				OPERATION
● ● ● ●	READY FOR SERVICE (90% + 1.5 SEC)				
○ ● ● ●	PROCESSOR BOARD FAILURE				
● ○ ● ●	SENSOR / DATA BOARD FAILURE				
○ ○ ● ●	OVERTEMPERATURE				
● ● ○ ●	OVERSPEED				
○ ● ○ ●	UNDERSPEED				
● ○ ○ ●	FAIL TO START				
○ ○ ○ ●	LOW OIL PRESSURE				
● ● ● ○	HIGH OIL TEMPERATURE (WARNING)				
○ ● ● ○	SHORTED OIL PRESSURE SWITCH				
● ○ ● ○	SHORTED THERMOCOUPLE PROBE (WARNING)				
● ● ○ ○	OPEN THERMOCOUPLE				
● ○ ○ ○	PROCESSOR SEQUENCE FAIL				
○ ○ ○ ○	NO DATA				
○ ○ ● ○	EXTERNAL SHORT				
○ ● ○ ○	LOSS OF SPEED DATA				

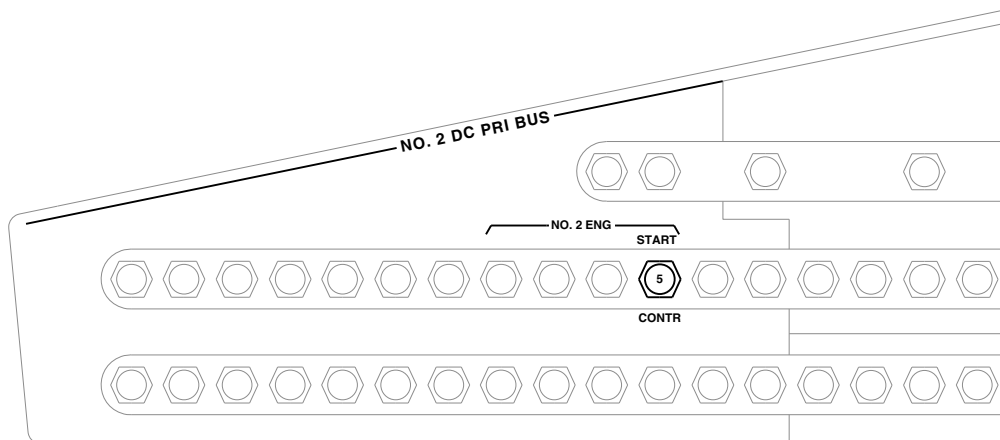
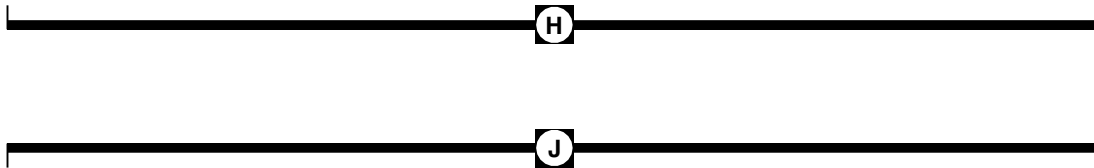
MALFUNCTION BITE INDICATIONS/
 START SEQUENCE DECAL
 (SEE NOTE 2)

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FIGURE 15-1-1-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 6 OF 7)



COPILOT'S CIRCUIT BREAKER PANEL



PILOT'S CIRCUIT BREAKER PANEL

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FIGURE 15-1-1-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 7 OF 7)

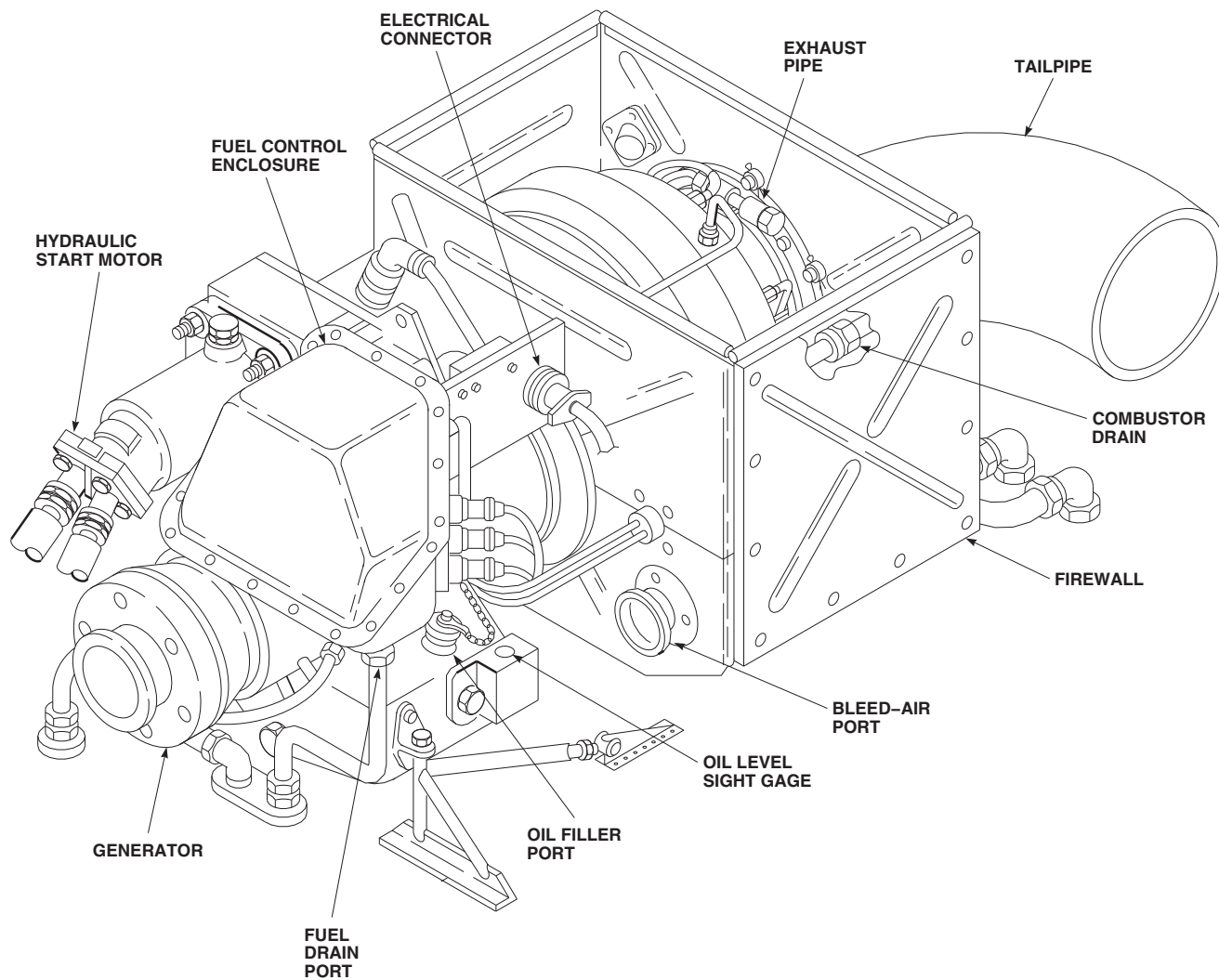
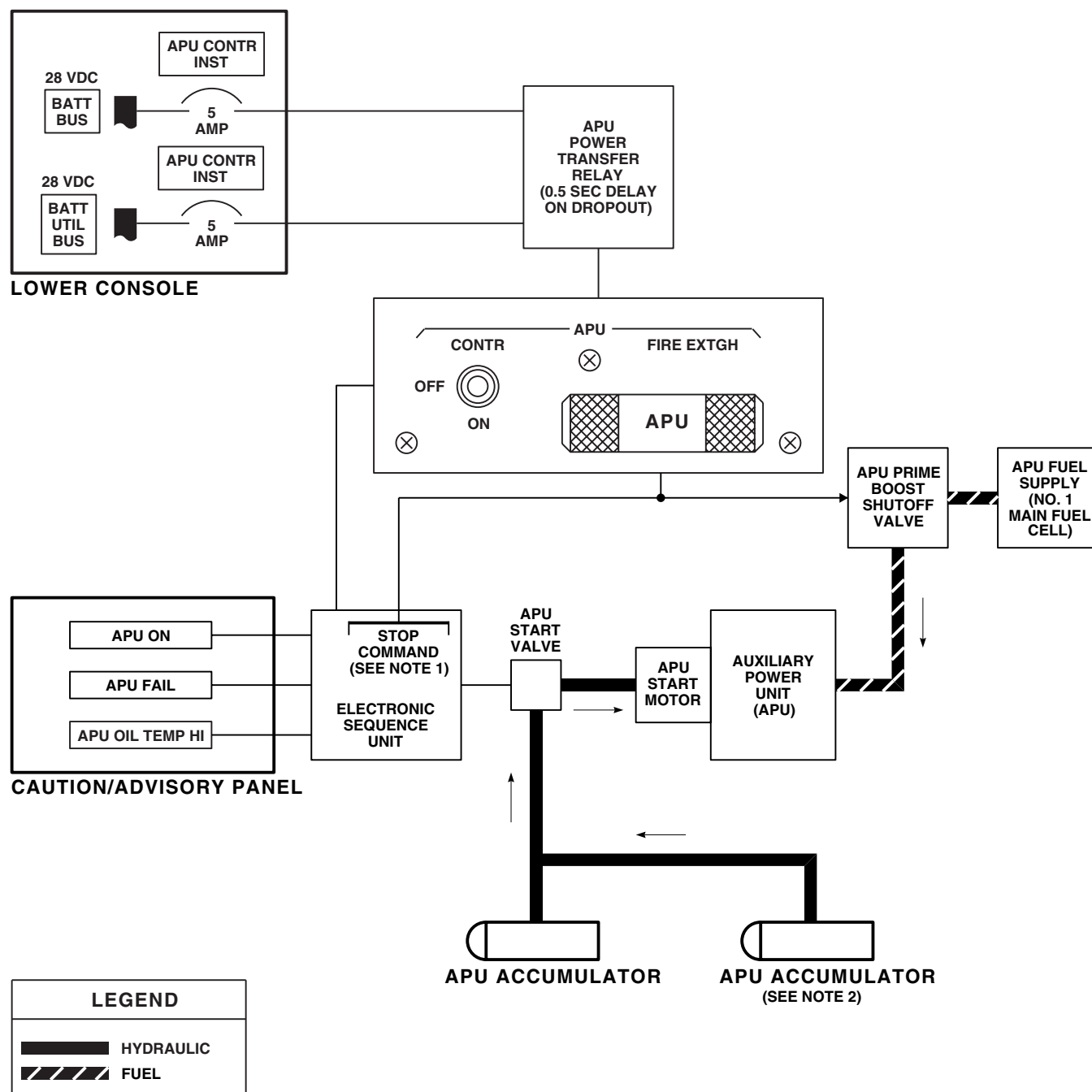


FIGURE 15-1-1-2. AUXILIARY POWER UNIT INSTALLED

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NOTES

1. STOP COMMAND WILL SHUT OFF ANY FUEL VALVES THAT ARE OPEN AND TURN OFF IGNITION, IF ON.
2. A SECOND ACCUMULATOR IS USED ON HELICOPTERS WITH WINTERIZATION KIT INSTALLED.

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FIGURE 15-1-3. AUXILIARY POWER UNIT SIMPLIFIED BLOCK DIAGRAM

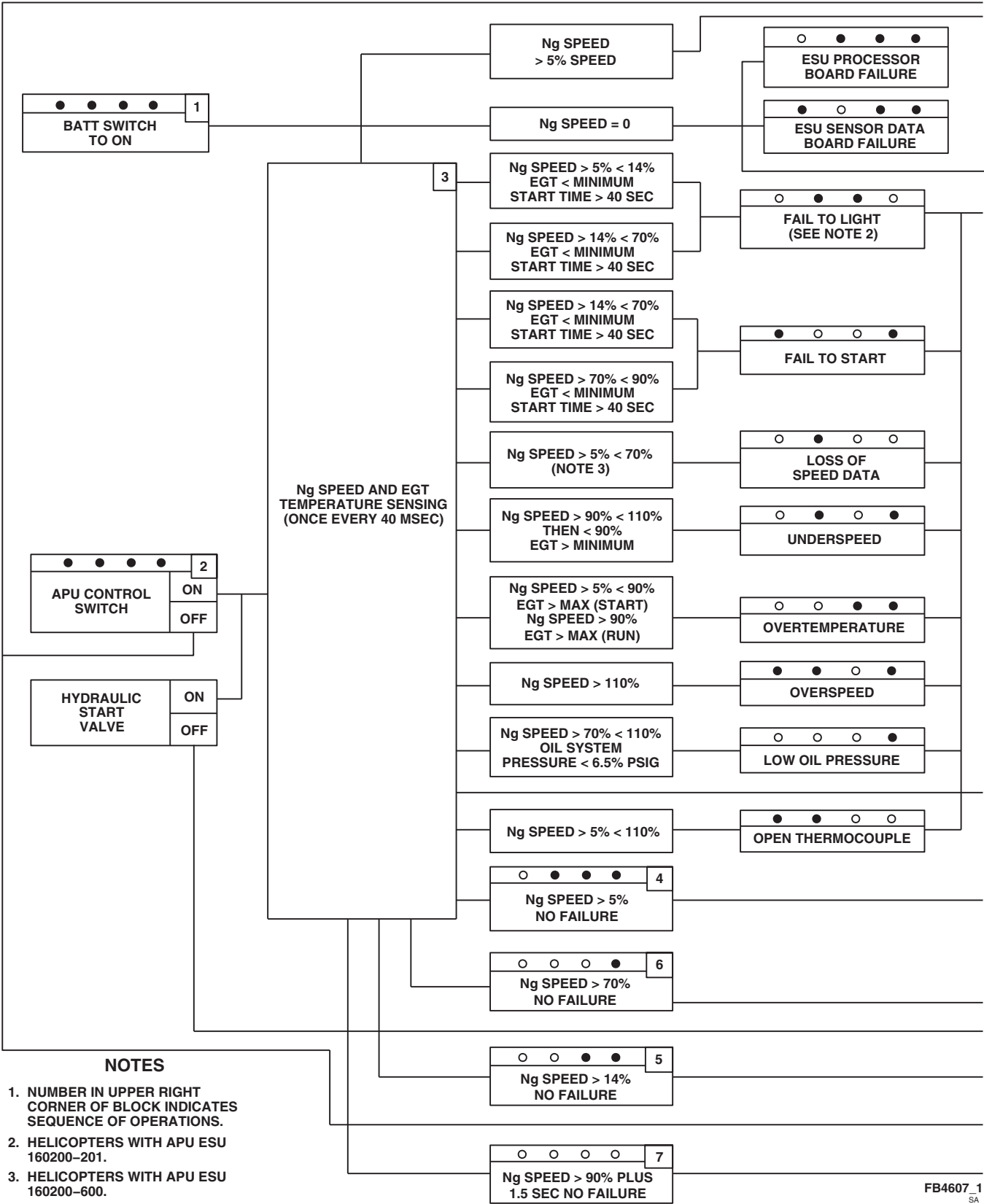
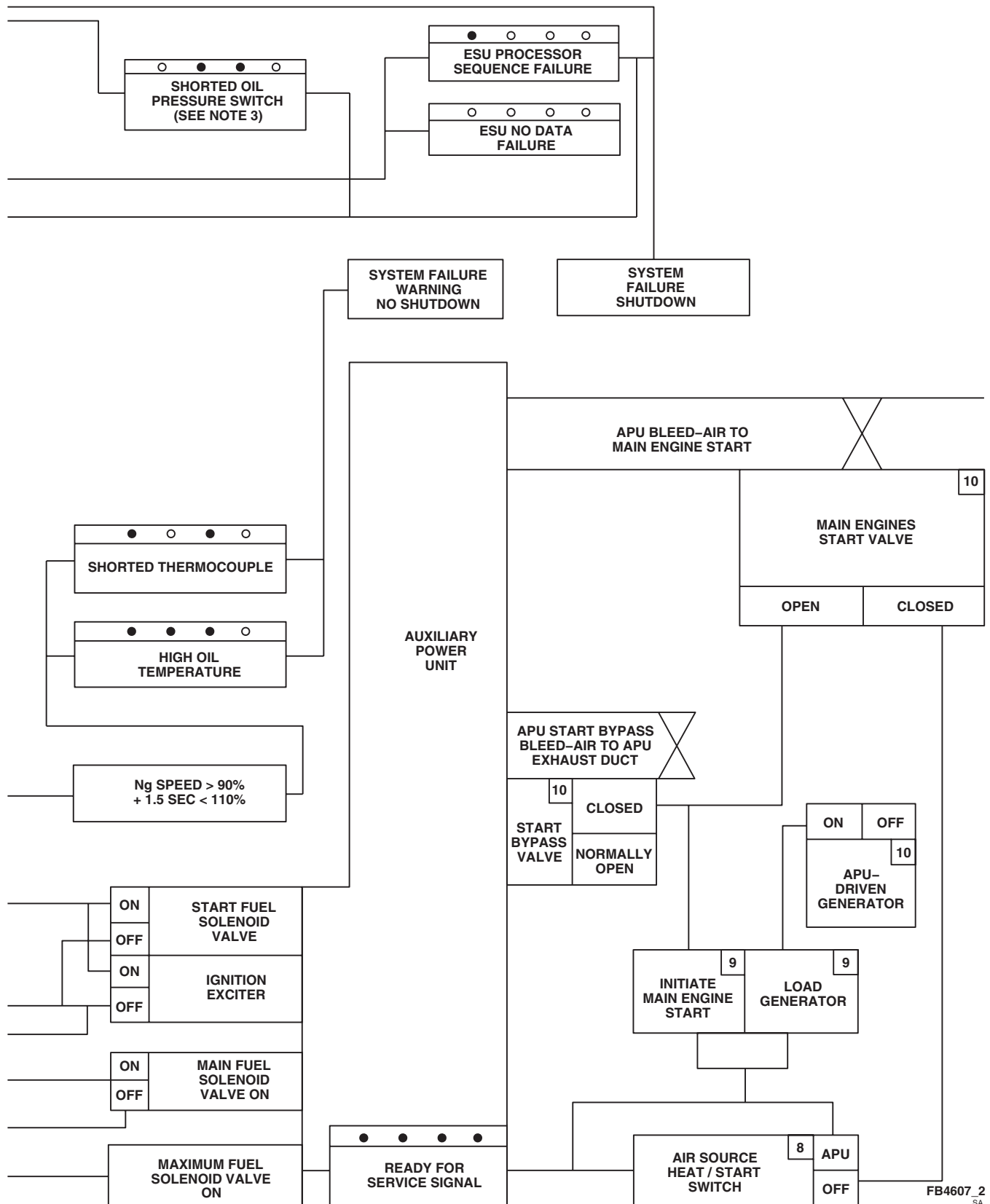


FIGURE 15-1-1-4. AUXILIARY POWER UNIT SYSTEM SEQUENCE CHART (SHEET 1 OF 2)



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FIGURE 15-1-1-4. AUXILIARY POWER UNIT SYSTEM SEQUENCE CHART (SHEET 2 OF 2)

15-1-2 AUXILIARY POWER UNIT SYSTEM WITH 5 BIT DESU.

15-1-2.1 System Description.

The Auxiliary Power Unit System with Digital Electronic Sequence Unit (DESU) (Figure 15-1-2-1 and Figure 15-1-2-2) consists of:

- a. APU
- b. Accessories
- c. Controls
- d. Monitoring System
- e. Starting System

The system provides pneumatic power for main engine starting and cabin heating, electrical and hydraulic power for ground operations, and in-flight emergency electrical power.

APU System accessories include:

- a. Prime/Boost Pump
- b. Hydraulic Accumulator
- c. Hydraulic Handpump
- d. Hydraulic Utility Mode
- e. Hydraulic Backup Pump
- f. AC Generator
- g. Hydraulic Start Motor

The prime/boost pump, above the No. 1 fuel tank, is used to prime engine and APU fuel lines, and to provide fuel under boost pressure for APU operation at high altitudes.

The hydraulic accumulator, in the rear midsection cabin ceiling, provides the hydraulic pressure used to drive the APU starter.

The APU starter is a hydraulic start motor mounted on the APU.

If the APU fails during light-off, the hydraulic accumulator is recharged manually by pumping the hydraulic handpump.

Once the APU is operating, the AC generator, mounted on the APU, provides electrical power to the helicopter systems.

The hydraulic utility module and backup pump, on the left front deck within the main rotor pylon, automatically recharges the depleted hydraulic accumulator for the next APU start.

The APU controls are in the cockpit on the upper console. The controls consist of:

- a. APU Control Switch
- b. APU Fire Detector/Fuel Shut-Off Selector T-Handle

Indicator lights on the caution/advisory panel permit cockpit monitoring of the APU System. Monitored conditions include:

- a. APU OIL TEMP HI
- b. APU FAIL
- c. APU ON
- d. APU GEN ON
- e. PRIME BOOST PUMP ON
- f. APU ACCUM LOW

15-1-2.2 Principles of Operation.

15-1-2.2.1 Accumulator.

A nitrogen (N₂) charged hydraulic accumulator is installed in the cabin overhead next to the start valve. The accumulator provides the hydraulic pressure required by the APU hydraulic starter. If the APU does not start, the accumulator may be manually recharged with a handpump or the backup hydraulic pump (electric). When the APU is in operation, the AC generator provides power for the helicopter systems. During this operation, the accumulator may be recharged for the next start.

15-1-2.2.2 Digital Electronic Sequence Unit.

The DESU, located in the rear cabin ceiling, is a microprocessor-based electronic component that monitors and controls all APU operations. The DESU also shuts down the APU when monitored values exceed established operating parameters. System status is indicated by instrument panel capsules, pilot's and copilot's inboard MFD legends, and by the five Built-In-Test Equipment (BITE) indicators on the DESU. These indicators cycle from black to white in various combinations (codes) to indicate APU status and fault isolation. Two categories of codes are displayed: normal sequence required during APU start sequence and normal and failure indications associated with operation. Decals on the side of the DESU decode the BITE indicators and provide information relative to APU operation. The BITE indicators are electrically operated and magnetically latched to retain coded indications after electrical power has been removed. If the APU shuts down automatically, and the BATT or APU control switches are cycled, the BITE indicators will be cleared.

The DESU initiates the starting events programmed in the microprocessor. Prior to initiating each event, the microprocessor makes an internal check to ensure that the DESU is operating properly. It compares APU input signals to previously stored data to ensure normal operation and scheduled event timing. APU speed (provided by the speed sensor), EGT (provided by a thermocouple), oil temperature, and pressure are continuously monitored. APU speed is the basis for APU start sequencing events. The DESU operates in two modes:

- a. Start sequence mode - BITE indicates the speed at which a start sequence event occurs.
- b. Operational mode - BITE latches onto a malfunction.

Setting the APU control switch S4 to ON sends a 28 VDC start signal from the BATT BUS APU CONTR INST circuit breaker to the DESU. Normally, start command power is from the BATT BUS through diode to the APU control switch S4. During switchover from battery power to external power or the helicopter main generators, the BATT

BUS is momentarily de-energized. This condition would cause the APU to shut down due to the loss of start/run command. To prevent inadvertent dropout of the APU, a 0.5 second time-delay relay, K47, will maintain its contacts closed during the BUS-switchover period. This allows battery power from the BATT UTIL BUS to be applied through diode to the APU control switch to maintain the start/run command. The BATT UTIL BUS is hard-wired to the battery and does not switch power sources.

A start command from the DESU sends 28 VDC to the APU hydraulic start valve. There is a manual override lever on the valve should the valve solenoid fail. Opening the valve releases the accumulator hydraulic charge. The charge spins the APU starter that rotates the APU compressor and fuel pump. The DESU also controls ignition and signals the fuel valves to open. This accelerates the APU to 100% speed.

When speed reaches 90% and after a time delay of 1.5 seconds, the maximum fuel valve opens. It allows a governor to control the fuel flow. The APU ON capsule on the instrument panel and the legends on the pilot's and copilot's inboard MFDs will then go on.

The bypass valve permits the APU to bleed air during starting in order to prevent the possibility of compressor stall. When the APU is in full operation, it may be used as an air source for main engine starting. The AIR SOURCE HEAT/START switch is first set to APU. DC power is routed from the AIR SOURCE HEAT/START circuit breaker on the copilot's circuit breaker panel through switch S18. This power is then applied to energize relay K22.

When either No. 1 or No. 2 engine start button is pressed, the start circuits are energized. This provides 28 VDC to turn on the ENG 1 STARTER ON or the ENG 2 STARTER ON legend. The 28 VDC is also routed through a diode to relay K22. The 28 VDC signal goes through the closed contacts of K22 and to the DESU. This commands the bypass valve to close to allow more airflow for engine starting.

The prime boost pump motor is turned on by the FUEL PUMP switch, S37, on the upper console.

This switch also opens the APU fuel shutoff valve. This circuit receives its power through the FUEL PRIME BOOST circuit breaker from the BATT BUS. The PRIME BST PUMP ON legend will go on when the FUEL PUMP switch S37 is set to APU BOOST or FUEL PRIME.

15-1-2.2.3 APU Controls and Indicators.

Two APU controls are mounted on the upper console: an APU control switch and an APU fire extinguishing T-handle. Setting the APU control switch to ON opens a hydraulic start valve to release the accumulator hydraulic charge to the APU starter. It also opens the APU shutoff valve to allow fuel to be supplied to the APU (refer to Fuel Prime/Boost System (PARA 10-2-3)). The T-handle lights whenever a sensor detects fire in the APU compartment. Pulling the T-handle arms the fire extinguisher, removes 28 VDC from the APU control switch and cuts fuel, thus shutting down the APU.

The electrically controlled and hydraulically started APU runs up to its rated speed automatically. Sequencing and shutdown for a system failure are controlled by the DESU. Caution/Advisory Panel capsules include:

- a. # 1 ENGINE STARTER
- b. # 2 ENGINE STARTER
- c. APU OIL TEMP HI
- d. APU FAIL
- e. APU ON
- f. APU GEN ON
- g. PRIME BOOST PUMP ON
- h. BACK UP PUMP ON
- i. APU ACCUM LOW
- j. BATT LOW CHARGE
- k. EXT PWR CONNECTED

15-1-2.2.4 Backup Hydraulic Pump.

The electrical power generated by the APU generator can operate the backup hydraulic pump to recharge the APU accumulator. When the APU is started by the accumulator, its hydraulic fluid charge is depleted. This is signaled by an APU ACCUM LOW capsule on caution/advisory panel. The BACKUP HYD PUMP switch on the upper console controls backup hydraulic pump operation. When the switch is placed to the ON position with the helicopter either on the ground or airborne, the backup hydraulic pump will run continuously. The pump motor contains a thermal switch that interrupts electrical power to the motor when the helicopter is on the ground and motor temperature rises above a safe value. With the BACKUP HYD PUMP switch in either AUTO or OFF and the helicopter airborne, the backup hydraulic pump will operate for a minimum of 90 seconds to fully charge the APU accumulator and turn off the APU ACCUM LOW capsule on the caution/advisory panel. When the BACKUP HYD PUMP switch is placed to OFF and the helicopter is on the ground, the weight-on-wheels switch will disable the system and the backup hydraulic pump will not function. For a more detailed discussion of the backup hydraulic pump and its operation, refer to Hydraulic Power Systems (PARA 7-1-1).

15-1-2.2.5 DESU Operation.

During APU start, acceleration, and operation, the DESU controls and monitors the APU. BITE indicators show APU and DESU status during start and operation. The APU starter rotates the turbine up to 20% - 30% rated speed. When the turbine is at 5% speed, a start fuel valve in the APU opens and ignition occurs. When speed reaches 14%, the main fuel valve opens and at 30% the EGT is read by DESU. The hydraulic and fuel start valves are closed at 70% speed.

When the APU reaches 5% speed, the 40 second counter will start and at 90% speed, the 40 second counter in the DESU shuts down. During this 40 second period, the DESU will not shut the APU down for oil low pressure, over or under speed conditions, and oil high temperature. A 1.5 second timer permits final APU checks to be made before

reaching 105% unloaded speed. At stable 100% speed, the timer shuts down.

15-1-2.2.6 Fuel System.

The major components of the Fuel System internal to the APU are shown in Figure 15-1-2-2.

15-1-2.2.7 Fuel System Operation.

Fuel is fed to the gear fuel pump inlet through a ten-micron filter. The fuel pump is protected from overpressure by a bypass relief valve. The fuel flow is controlled by a hydromechanical governor and metering valve which varies the size of the fuel flow orifice. The orifice diameter is set by the pressure differential between ambient and Compressor Discharge Pressure (CDP). This ensures a uniform fuel flow at low as well as elevated pressures.

At 5% rated speed, the start fuel valve opens and stays open until speed reaches 70%. The start fuel valve feeds fuel into the start fuel nozzle. When the APU reaches 14% speed, the main fuel valve is opened by a 28 VDC DESU signal. The main fuel valve now remains open during normal operation. The fuel flows from the accelerator to the main fuel nozzles. The maximum fuel valve opens electrically when the APU reaches 90% speed and after a time delay of 1.5 seconds.

When the maximum fuel valve opens, it permits fuel to bypass the acceleration schedule. The main fuel line contains a fuel manifold which divides fuel flow to the six main fuel nozzles.

After the start fuel valve closes, a purge drain harness (tee fittings bracket assembly) uses combustor pressure through the start fuel nozzle to force any residual fuel out of the nozzle and start fuel circuit to prevent carbon buildup on the nozzle.

15-1-2.3 Mechanical Description.

15-1-2.3.1 APU.

The APU consists of a gas turbine power section, reduction gear drive, and appropriate controls and accessories. It has a self-contained oil system and is controlled by the DESU. The gas turbine power

section is a simple centrifugal compressor and a single-stage radial inflow turbine. The centrifugal compressor and turbine are mounted back-to-back on the end of a high-speed shaft. Air passing through an inlet screen and over the bearings to the compressor prolongs bearing and lubricant life. Turbine-furnished power drives the compressor and the high-speed output shaft.

The combustor is an annular type with an internal liner, and six air-atomizing fuel injector nozzles. The combustor is mounted onto the turbine housing and has on its periphery a compressor discharge plenum chamber. Ignition is furnished by a separate pressure atomizing fuel nozzle and igniter plug. The igniter plug, fuel nozzle, and main fuel manifold are installed in the combustion chamber. The thermocouple is located in the APU exhaust and measures Exhaust Gas Temperature (EGT).

During operation above 70% of the gas turbine speed, compressor discharge air is bled through the start fuel nozzle. The bleed-air passes through a purge drain harness and into the start fuel line. The bleed-air keeps the nozzle continuously clean (no coking) for the next start.

A high-speed pinion gear and shaft, connected to the main rotor shaft, provides input to the reduction gear drive. The pinion is supported by three planetary gears. An internal gear ring, installed around the three planetary gears, reduces turbine shaft speed from 61,565 rpm to 12,000 rpm at the axial pad, which also provides a mounting pad for the APU generator. A set of gears extending from the output shaft are used to drive the fuel pump and accelerator, and the APU starter mounted on the accessory pads at 4,235 and 8,229 rpm respectively. Other reduction gears drive the lube pump.

The accessory gear box is mounted to the top of the reduction drive assembly. The gear box provides the mount for the hydraulic starter, fuel pump, and accelerator. A speed sensor on the accessory drive assembly provides a pulse signal to the DESU. The DESU converts the pulses to an RPM signal which is used to operate the main fuel and maximum fuel valves.

15-1-2.3.2 APU T-Handle.

When the Fire Detector System detects a fire in the APU compartment, the APU T-handle and master warning panel FIRE capsules will go on red. The pilot can then pull out the APU fire extinguishing T-handle. This does several things: (a) provides an APU stop command through S102 to the DESU, (b) shuts off the prime boost pump if it is running, and (c) de-energizes the APU fuel shutoff valve.

15-1-2.3.3 APU AC Generator.

The helicopter's electrical requirements are supplied when the GENERATORS-APU switch is positioned to ON. When the AC generator is in operation, the APU GEN ON legends on pilot's and copilot's inboard MFDs go on. The APU generator supplies electrical power to the AC PRI BUSES and the AC ESNTL BUS. For a more detailed discussion of the APU generator and its operation, refer to AC Power System (PARA 9-1-1).

15-1-2.3.4 APU Accumulator.

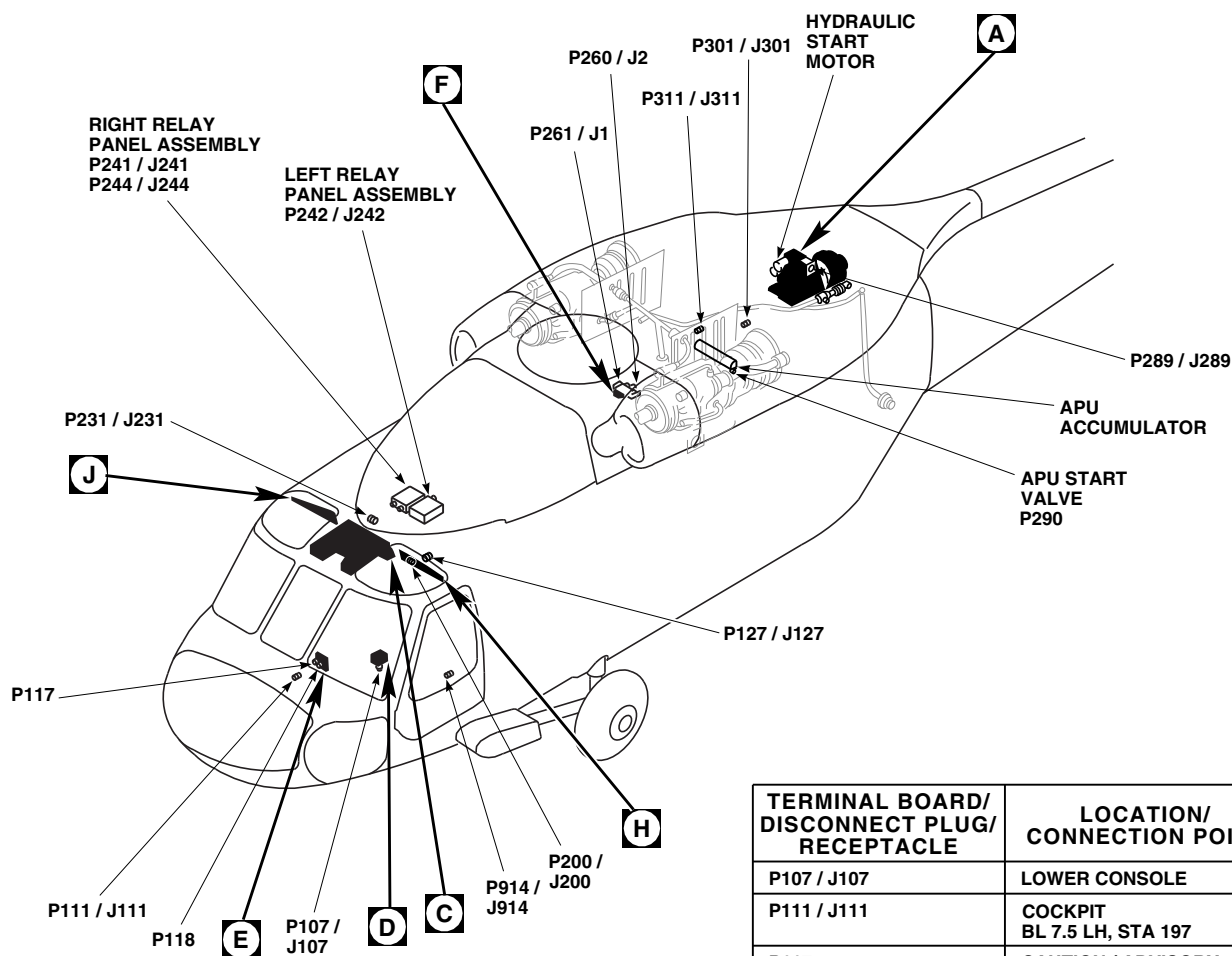
The accumulator consists of:

- a. Gas Chamber

- b. Reservoir

- c. Hydraulic Pressure Chamber

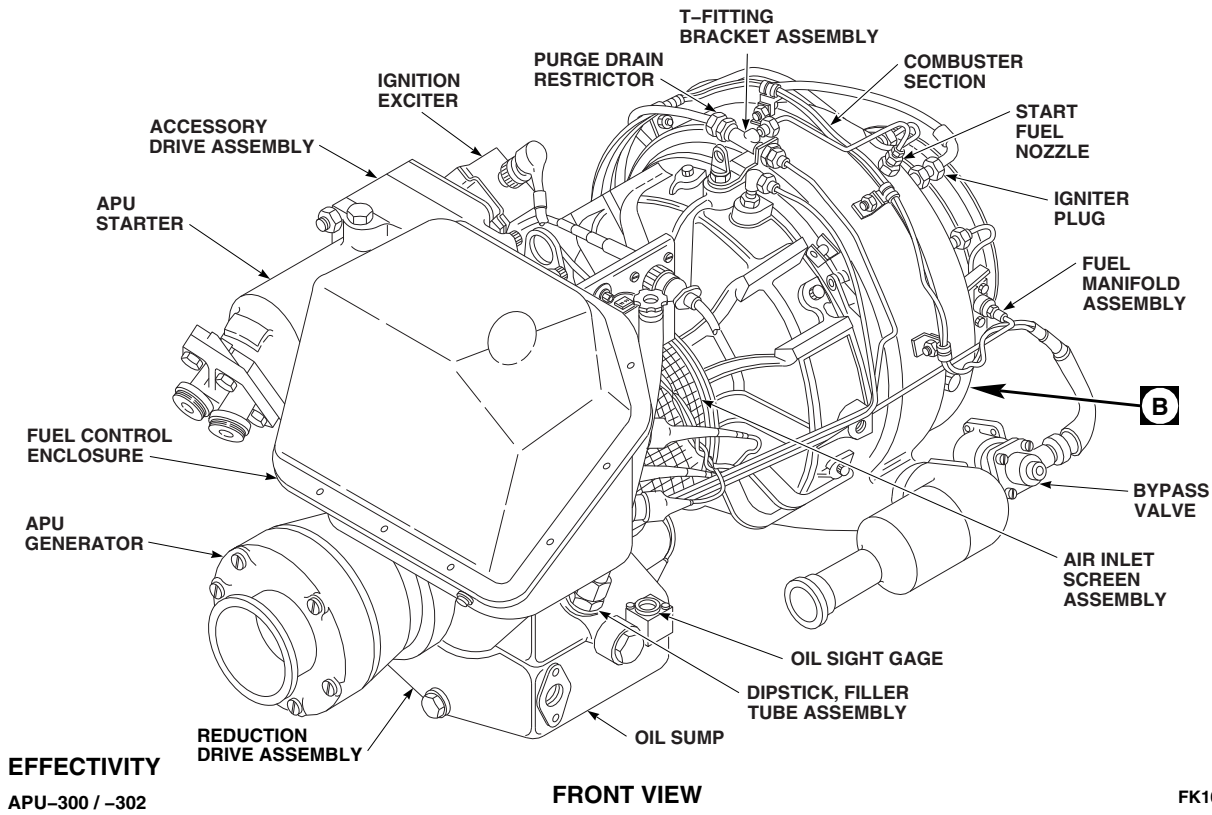
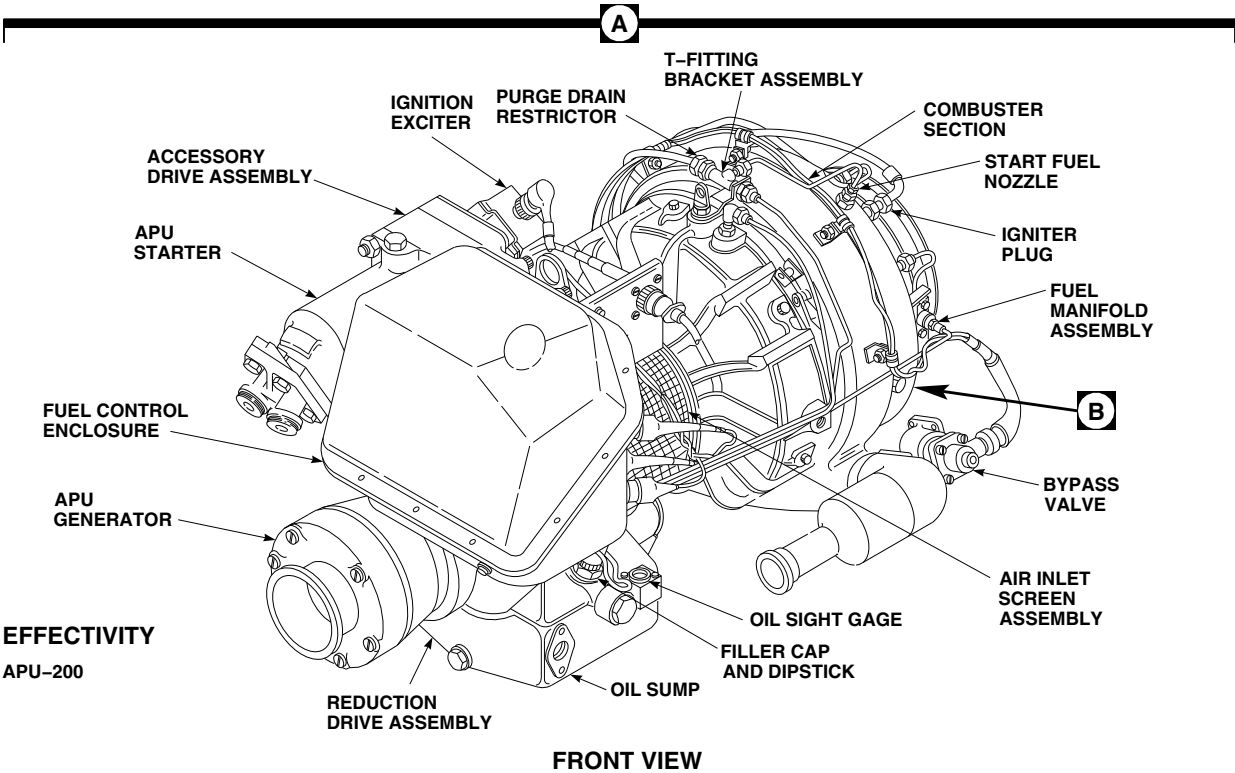
The gas chamber contains N₂. The N₂, which can be compressed to act like a spring, forces hydraulic fluid from the pressure chamber. The pressure line is connected to the start valve and starter. N₂ quantity is measured by a pressure gauge in the cabin overhead near the handpump. The handpump enables accumulator recharging if backup hydraulic pump operation is not possible. The handpump is a short-stroke type with a removable handle. The accumulator piston is connected to a tape position indicator that is marked in percent of hydraulic charge. When the tape position indicator is at 100% (full left), the pressure chamber is fully charged. When the tape position indicator is at 0%, the accumulator is empty and needs recharging. A second accumulator and pressure switch are part of a winterization kit. This supplied double the normal fluid volume charge to motorize the APU in cold weather.



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P107 / J107	LOWER CONSOLE
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P117	CAUTION / ADVISORY PANEL
P118	CAUTION / ADVISORY PANEL
P200 / J200	COCKPIT CEILING BL 13 LH, STA 246
P231 / J231	CABIN CEILING BL 9 LH, STA 247
P241 / J241	RIGHT RELAY PANEL
P242 / J242	LEFT RELAY PANEL
P244 / J244	RIGHT RELAY PANEL
P260 / J2	DIGITAL ELECTRONIC SEQUENCE UNIT (DESU)
P261 / J1	DIGITAL ELECTRONIC SEQUENCE UNIT (DESU)
P289 / J289	APU
P290	APU START VALVE
P301 / J301	TRANSITION DECK BL 12.88 LH, STA 415
P311 / J311	CABIN CEILING BL 16.50 LH, STA 383
P914 / J914	COCKPIT BL 24 LH, STA 247

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SA

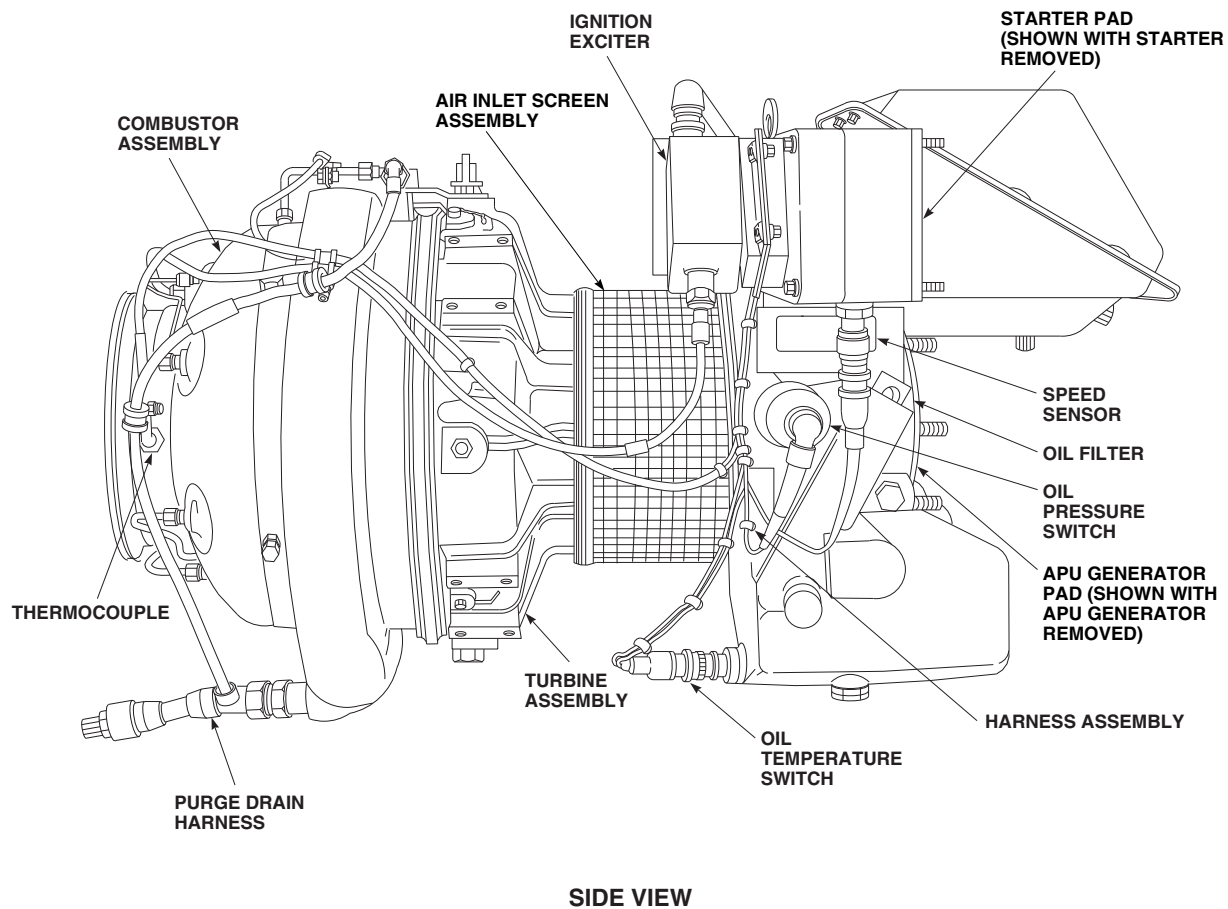
FIGURE 15-1-2-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 1 OF 7)



FK1677_2A
SA

FIGURE 15-1-2-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 2 OF 7)

B

**EFFECTIVITY**

APU-200 / -300 / -302

FK1677_3A
SA

FIGURE 15-1-2-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 3 OF 7)

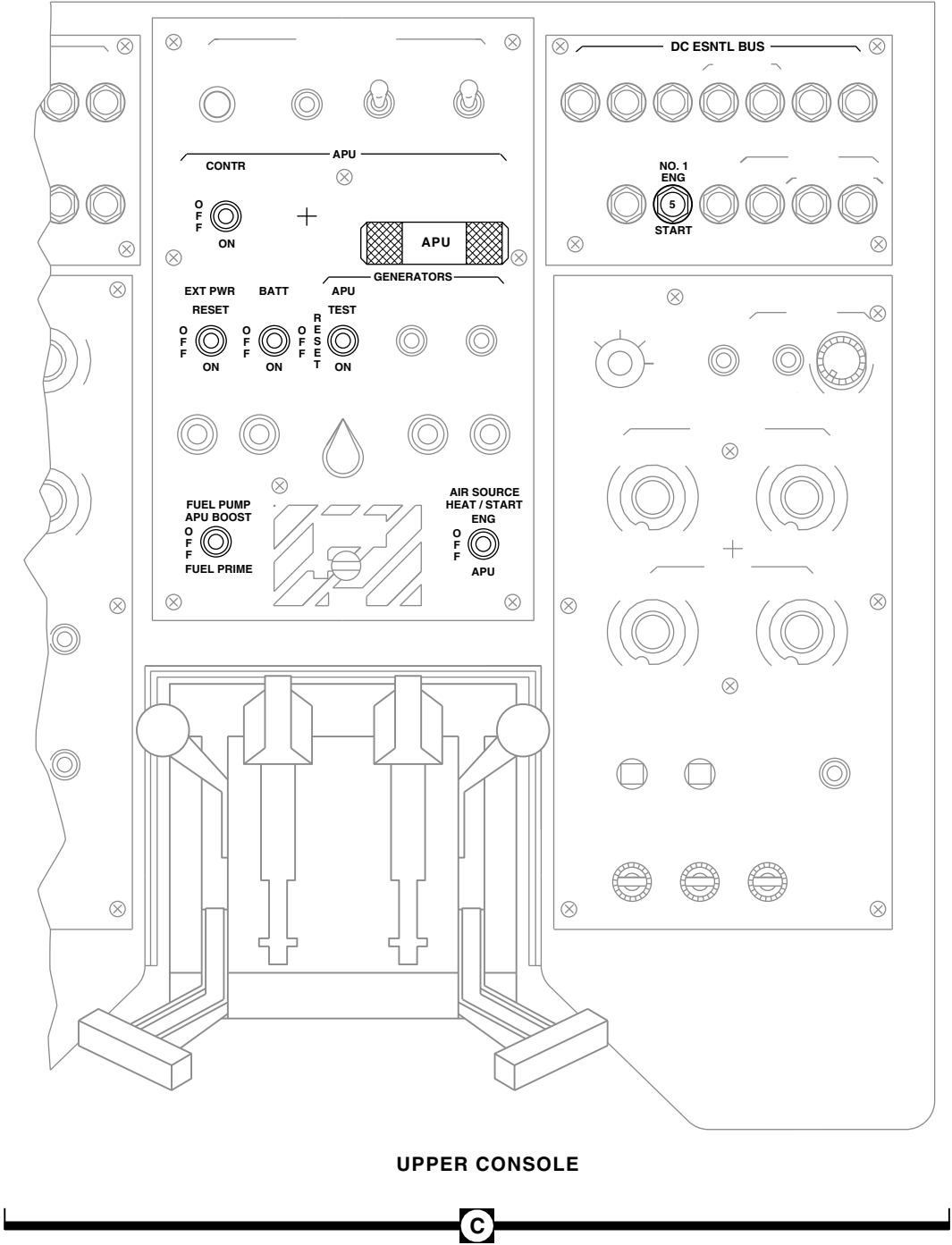
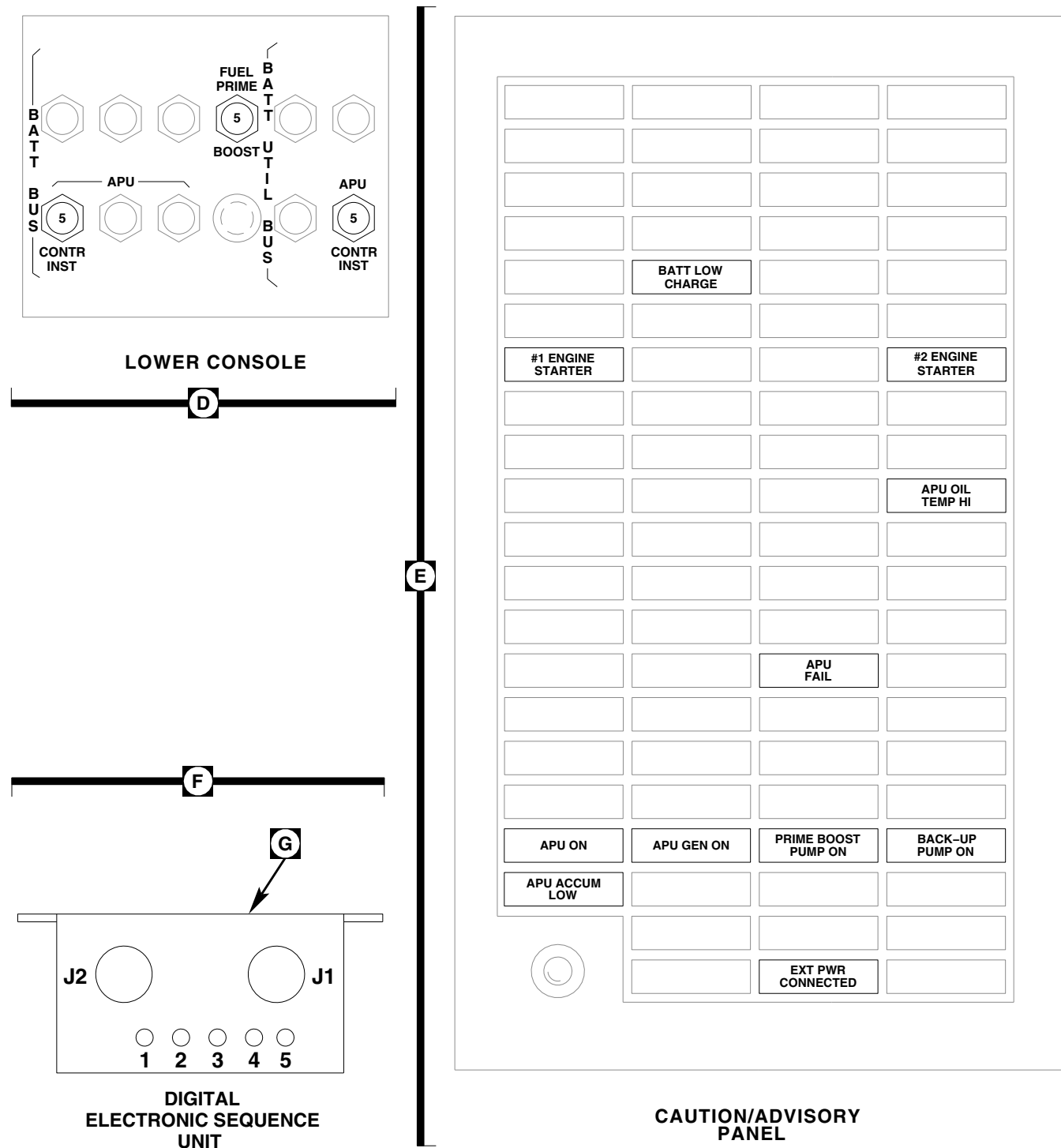


FIGURE 15-1-2-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 4 OF 7)

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FE0718_4B
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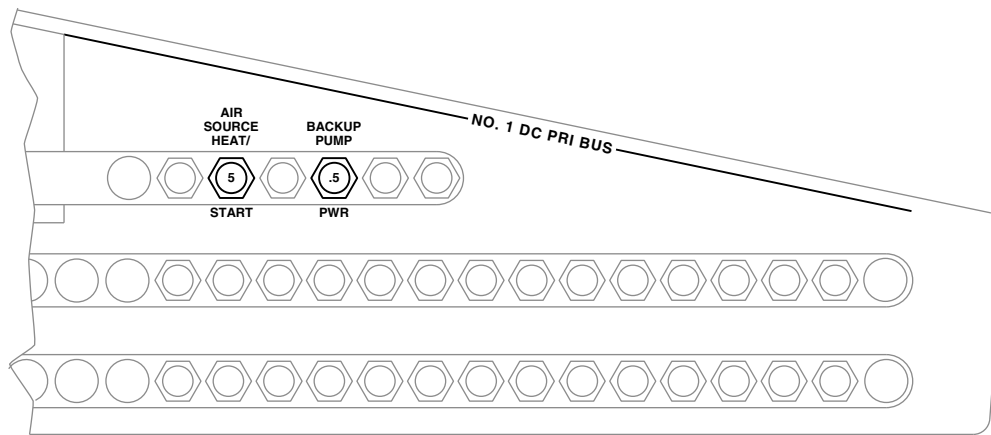
FIGURE 15-1-2-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 5 OF 7)

BITE #					DECODED BITE INFORMATION	
1	2	3	4	5		
●	●	●	●	●	POWER ON RESET	START SEQUENCE
●	●	●	●	○	START COMMAND RECEIVED	
○	●	●	●	○	START FUEL VALVE & EXCITER SIGNAL OUT (5%)	
○	○	●	●	○	MAIN FUEL VALVE SIGNAL OUT (14%)	
○	○	○	●	○	START FUEL VALVE & EXCITER SIGNAL OFF (70%)	
○	○	○	○	○	90% RPM SWITCH ON	OPERATION
●	●	●	●	●	READY TO LOAD (90% + 1.5 SEC)	
●	●	●	●	●	NORMAL OPERATION	
○	●	●	●	●	DESU FAILURE	
●	○	●	●	●	SPARE	
○	○	●	●	●	OVERTEMPERATURE (RUN)	
●	●	○	●	●	OVERSPEED	
○	●	○	●	●	UNDERSPEED	
●	○	○	●	●	FAIL TO ROTATE	
○	○	○	●	●	LOW OIL PRESSURE	
●	●	●	○	●	HIGH OIL TEMPERATURE	
○	●	●	○	●	SHORTED OIL PRESSURE SWITCH	
●	○	●	○	●	SHORTED THERMOCOUPLE	
○	○	●	○	●	EINCAS LINES FAULT	
●	●	○	○	●	OPEN CIRCUIT FAULT THERMOCOUPLE	
○	●	○	○	●	LOSS OF SPEED DATA	
●	○	○	○	●	SPARE	
○	○	○	○	●	POWER INTERRUPT	
●	●	●	○	○	RTL INDICATOR FAILURE	
○	●	●	○	○	MALFUNCTION INDICATOR FAILURE	
●	○	●	○	○	MAXIMUM FUEL VALVE FAILURE	
○	○	●	○	○	HOT INDICATOR FAILURE	
●	●	○	○	○	MAIN FUEL VALVE FAILURE	
○	○	○	○	○	ANTI - SURGE VALVE FAILURE	
●	○	○	○	○	START CIRCUIT FAILURE	
○	○	○	○	○	START FUEL EXCITER FAILURE	
●	●	●	○	○	ILLOGICAL SWITCH INPUTS	
○	●	○	○	○	FAIL TO LIGHT	
●	○	○	○	○	FAIL TO ACCELERATE (START FUEL)	
○	○	●	○	○	FAIL TO ACCELERATE (START & MAIN)	
●	●	○	○	○	FAIL TO ACCELERATE (MAIN FUEL)	
○	○	○	○	○	SLOW ACCELERATE (TIME OUT)	
●	○	○	○	○	OVERTEMPERATURE (START)	
○	○	○	○	○	APU START SWITCH FAILURE	

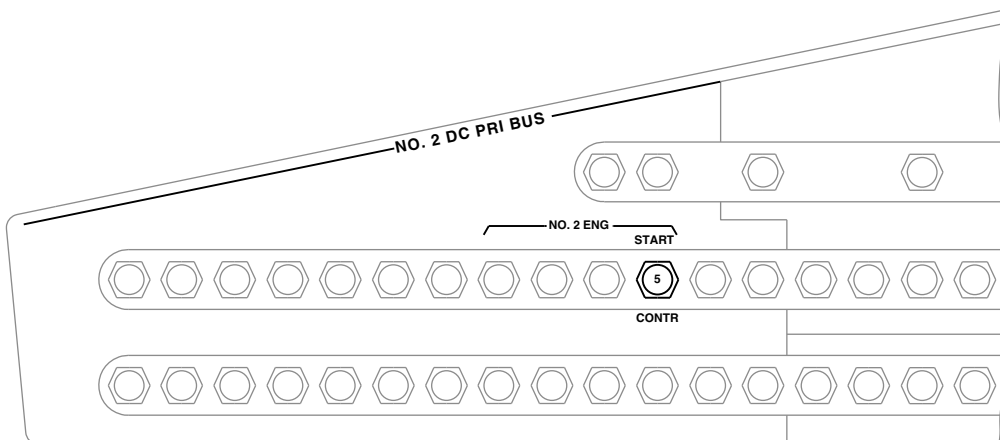
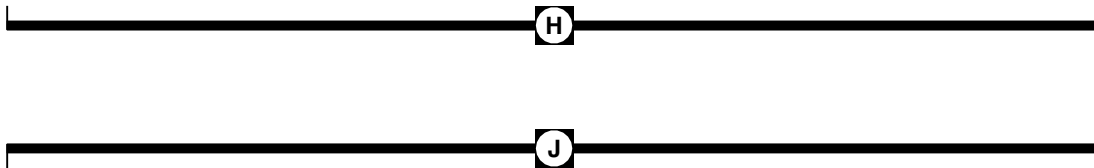
MALFUNCTION BITE INDICATIONS/START SEQUENCE DECAL

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SA

FIGURE 15-1-2-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 6 OF 7)



COPILLOT'S CIRCUIT BREAKER PANEL

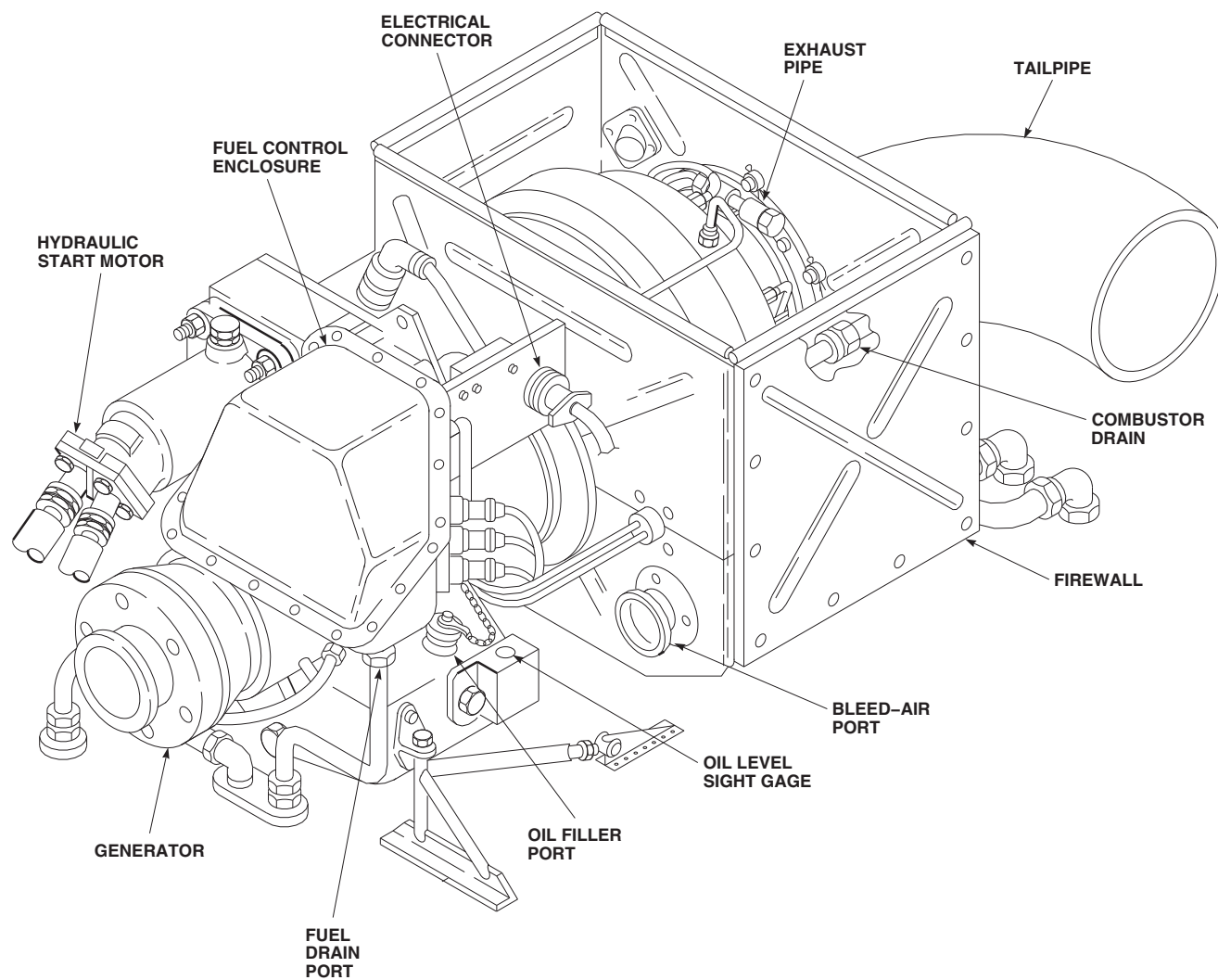


PILOT'S CIRCUIT BREAKER PANEL

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SA

FIGURE 15-1-2-1. AUXILIARY POWER UNIT SYSTEM LOCATION DIAGRAM (SHEET 7 OF 7)

15-1-2-12 Change 11



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FIGURE 15-1-2-2. AUXILIARY POWER UNIT INSTALLED

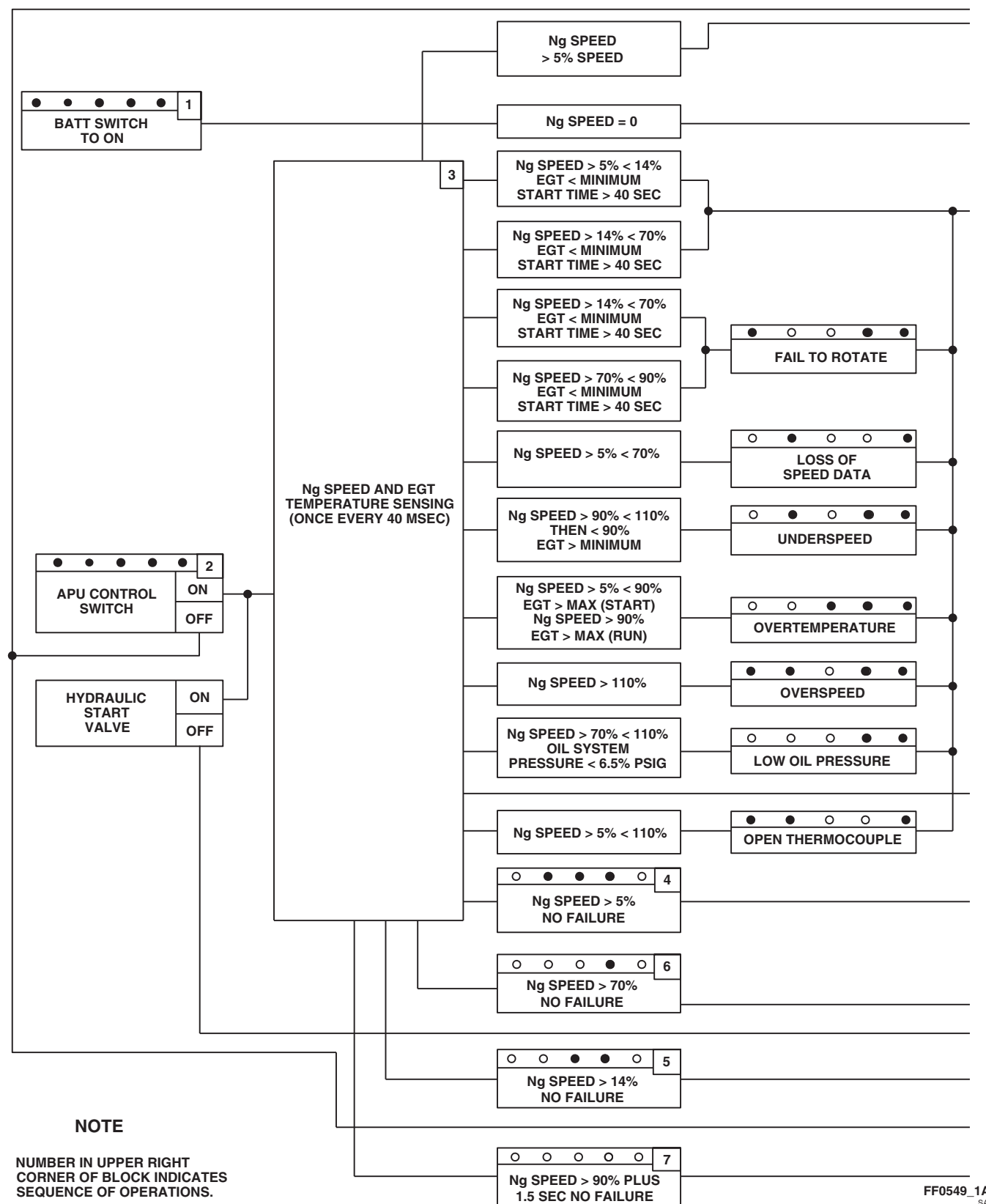


FIGURE 15-1-2-3. AUXILIARY POWER UNIT SYSTEM SEQUENCE CHART (SHEET 1 OF 2)

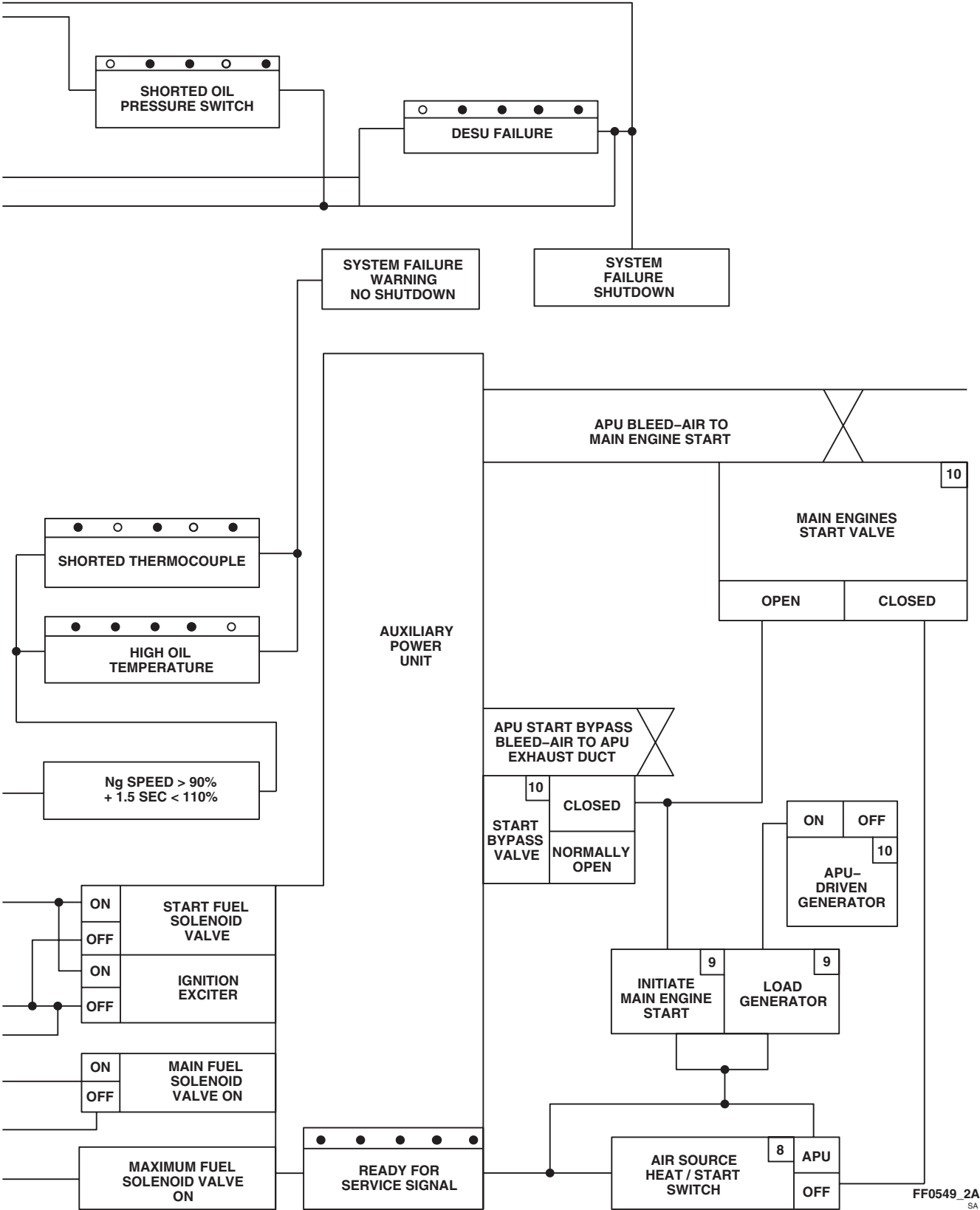


FIGURE 15-1-2-3. AUXILIARY POWER UNIT SYSTEM SEQUENCE CHART (SHEET 2 OF 2)

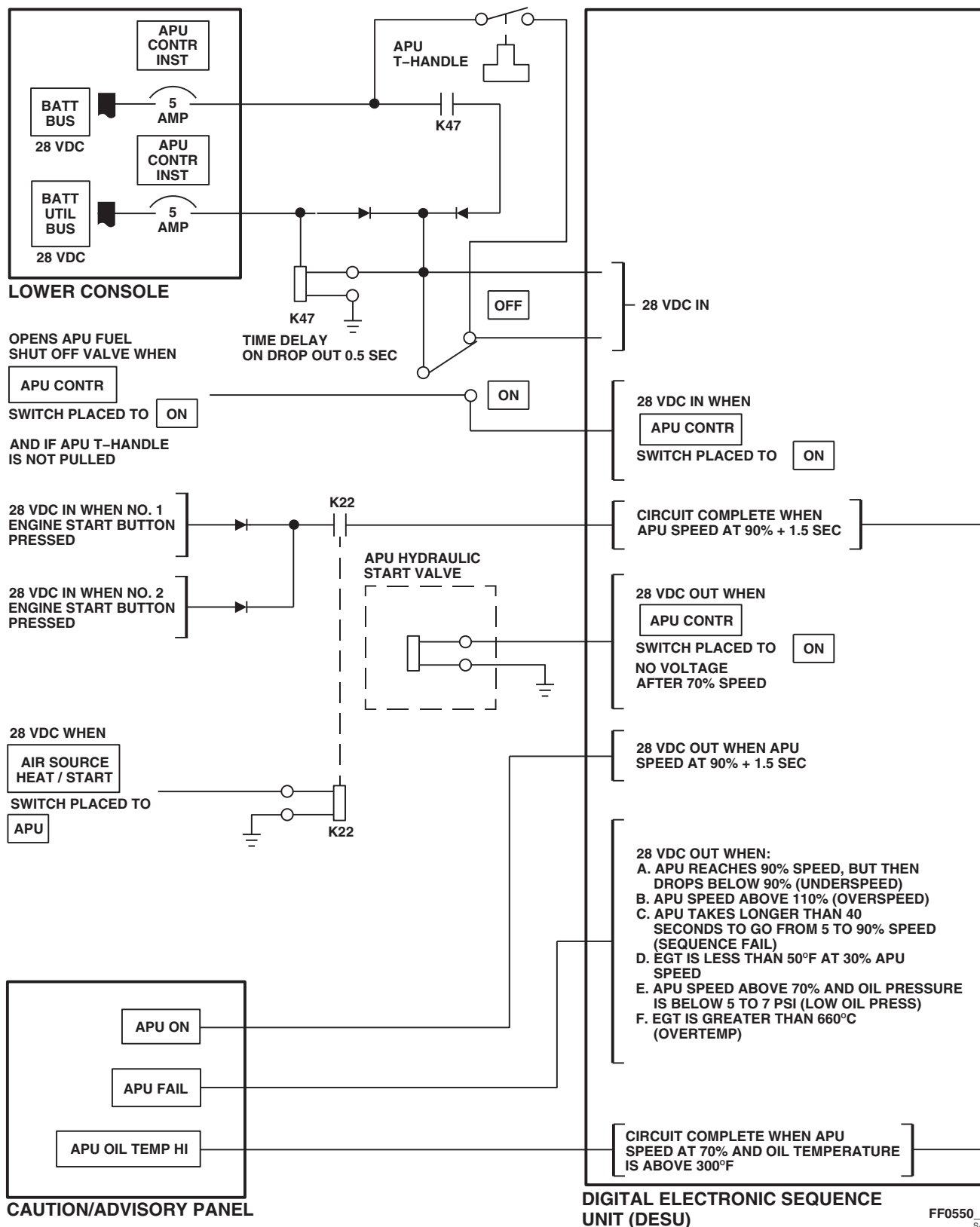


FIGURE 15-1-2-4. AUXILIARY POWER UNIT SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

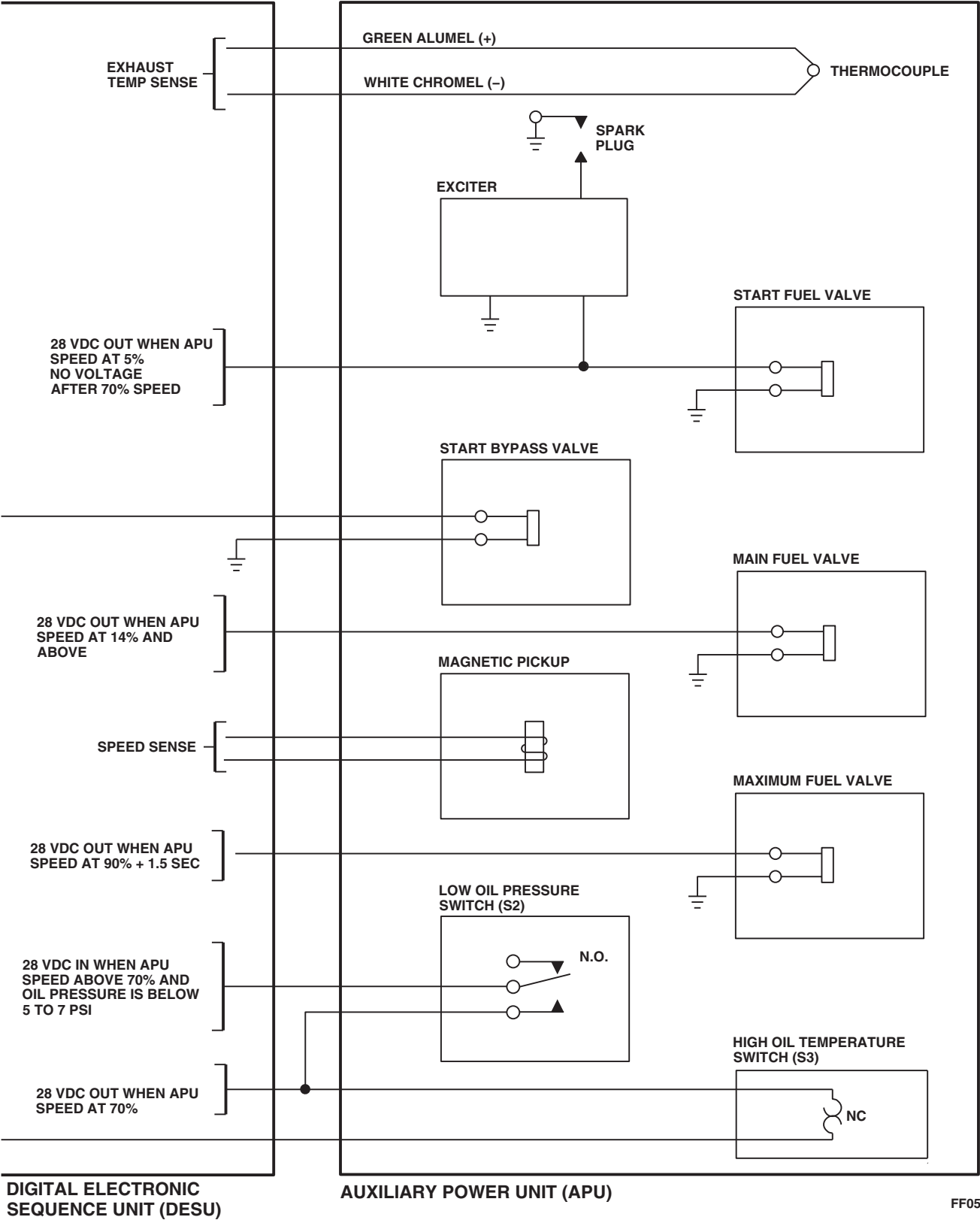


FIGURE 15-1-2-4. AUXILIARY POWER UNIT SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

SECTION 15-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH	TITLE
15-2-1	Auxiliary Power Unit System Using Test Set H296A
15-2-2	Auxiliary Power Unit System Using Test Set H296B-1
15-2-3	Auxiliary Power Unit System With 5 BIT DESU

15-2-1 AUXILIARY POWER UNIT SYSTEM USING TEST SET H296A.

INITIAL SETUP

Applicable Configuration

- Auxiliary Power Unit (APU) 116305-100

Test Equipment

- Multimeter, Fluke 27/AN
- Multimeter, Fluke 45
- Simulator, Temp./Speed, H296A

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16
- PARA 2-4-69
- PARA 7-4-8
- Appropriate Flight Manual

Equipment Conditions

- APU Accumulator Properly Serviced
- Battery Fully Charged
- External Electrical Power Available
- No. 1 Fuel Tank Serviced

WARNING

To prevent injury to personnel and damage to helicopter, have fire guard

check APU fuel drain. DO NOT RESTART APU until fuel drainage has stopped.

CAUTION

- To prevent damage to helicopter, wait at least two minutes after APU shutdown before attempting a restart.
- To prevent possible damage to APU, do not operate APU more than 30 minutes when main engines and rotor are running and outside temperature is greater than 109°F (43°C). Continuous operation of APU is permitted at temperatures up to 124°F (51°C) only if main engines and rotors are not running.
- On helicopters with ESU 160200-201, cycling of the BATT switch during APU coast-down may cause the APU to restart with a possible overtemperature.
- If onboard fire extinguishing is required before a generator becomes operational, RESERVE fire extinguisher must be used to fight fire.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper results, switch positions and order of operation must be strictly followed.

Special Instructions

The APU System operational/troubleshooting procedure is subdivided as follows:

- | | |
|------------------------------------|-----------------------------|
| (1) APU System | (f) Low Oil Pressure |
| (2) Electronic Sequence Unit (ESU) | (g) High Oil Temperature |
| (a) Simulator Setup | (h) Start Bypass Valve |
| (b) Hydraulic Start Valve | (i) Overtemperature |
| (c) Speed Sequence | (j) Open Thermocouple |
| (d) Underspeed | (k) Shorted Thermocouple |
| (e) Overspeed | (l) Disconnecting Simulator |
-

15-2-1.1 APU System.**NOTE****NOTE**

Do not connect external electrical power to helicopter before starting or during APU System operational/troubleshooting procedure.

- a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- b. Make sure upper console BATT switch and GENERATORS APU switch are placed to OFF.
- c. Check bleed air pneumatic tube couplings for loose clamps, cuts, and/or incorrect installation at all connecting points (PARA 7-4-8).

RESULT: All couplings shall be securely installed with clamps and there shall be no cuts in pneumatic tube lines.

- If result is not as specified, repair/replace couplings and lines (PARA 7-4-8).
- d. Remove soundproofing material and panel from cabin ceiling to visually check ESU BITE indicators (PARA 2-4-69).

- An observer must be in the cabin area to watch the ESU BITE indicators during the operational/troubleshooting procedure.

- The observer shall call out the procedural steps to the operator in the cockpit. This will allow the observer to watch for a change in ESU BITE indication as the operator does the next procedural step.

- e. Visually check ESU BITE indicators.

RESULT: All indicators shall be black.

- If result is not as specified, troubleshoot System for indicated malfunction.

NOTE

- Do not attempt to start the APU if BATT LOW CHARGE capsule is on.
- TABLE NO. 15-2-1-1 lists the sequence of events for a normal start.
- The APU will not start if any of the following ESU internal failures appear as bite indications:

15-2-1-2 Change 11

BITE
INDICATION

o ●●●
● o ●●
● o o o
o o o o

FAILURE

Processor Board Failure
Sensor/Data Board Failure
Processor Sequence Fail
No Data

NOTE

- On helicopters with ESU 160200-201, if the APU should start to motor and then shut down during an attempted start, do not place BATT switch to OFF. Any cycling of BATT switch will reset the ESU and malfunction BITE indications (indicating the cause of shutdown) will be lost.
- If the APU shuts down, the observer shall take note of the last APU speed BITE indication before the malfunction BITE indication appears.
- Make sure upper console APU T-handle is pushed in or APU will not start.

f. Place upper console BATT switch to ON. Watch ESU BITE indicators.

RESULT: No. 1 BITE indicator shall cycle from black to white to black. If BITE indicator is white, due to malfunction, it shall only cycle from white to black.

- If result is not as specified, go to TABLE NO. 15-2-1-2.

g. Place upper console FUEL PUMP switch to APU BOOST.

NOTE

- Prime boost pump must be running for all APU motoring and start operations.
- A minimal amount of fuel may be evident during and shortly after start.

RESULT:

- PRIME BOOST PUMP ON capsule on caution/advisory panel shall go on.
- No fuel leakage shall occur from APU combustor drain line (below helicopter).

- If result (1) is not as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3).
- If result (2) is not as specified, go to TABLE NO. 15-2-1-3.

NOTE

If a restart is being attempted, the original malfunction BITE indication will be shown.

RESULT:

- APU shall start and BITE indicators shall cycle as follows:

BITE
INDICATION

o ●●●	5%
o o ●●	14%
o o o ●	70%
o o o o	90%
●●●●	90% + 1.5 seconds (ready for service)

- APU ON capsule on caution/advisory panel shall go on.

- (3) APU FAIL capsule on caution/advisory panel shall be off.



If APU OIL TEMP HI capsule is on, shut down APU immediately.

- (4) APU OIL TEMP HI capsule on caution/advisory panel shall be off.
- (5) No fuel leakage shall occur from APU combustor drain line (below helicopter).
- If shorted thermocouple (● o ● o) is present and a restart attempt fails, go to TABLE NO. 15-2-1-4.
 - On helicopters with ESU 160200-201, if fail to light (o ●● o) is present after a restart attempt fails, go to TABLE NO. 15-2-1-5.
 - On helicopters with ESU 160200-600, if shorted oil pressure switch indication (o ●● o) is shown, go to TABLE NO. 15-2-1-6.
 - If APU does not motor, go to TABLE NO. 15-2-1-7.
 - If result (1) is not as specified, go to TABLE NO. 15-2-1-2.
 - If APU motors, but shuts down before result (2) occurs, and APU FAIL capsule is on, visually check ESU BITE indications and perform the following:

IF BITE INDICATION IS:

● o o ● (fail to start)	TABLE NO. 15-2-1-8
o ● o o (loss of speed data)	TABLE NO. 15-2-1-9
o o ●● (overtemperature)	TABLE NO. 15-2-1-10
o o o ● (low oil pressure)	TABLE NO. 15-2-1-11

IF BITE INDICATION IS:

●● o o (open thermocouple)	TABLE NO. 15-2-1-12
o o ● o (external short after, o ●●● start fuel valve and exciter signal out)	TABLE NO. 15-2-1-13
o o ● o (external short after, o o ●● signal out; 14%)	TABLE NO. 15-2-1-14
o o ● o (external short after, o o o o 90% RPM switch on or ●●●● ready for service)	TABLE NO. 15-2-1-15

- If APU motors (attempts to start), but shuts down after result (2) occurs (APU ON capsule goes off and APU FAIL capsule goes on), go to TABLE NO. 15-2-1-16.
- If APU motors, but shuts down after APU ON capsule goes on, then off, and APU FAIL capsule is off, visually check ESU BITE indications and perform the following:

IF BITE INDICATION IS:

o ● o ● (underspeed)	TABLE NO. 15-2-1-17
●● o ● (overspeed)	TABLE NO. 15-2-1-18
o o ●● (overtemperature)	TABLE NO. 15-2-1-19
o o o ● (low oil pressure)	TABLE NO. 15-2-1-11
●● o o (open thermocouple)	TABLE NO. 15-2-1-12

- If APU operates, but result (2) is not as specified, go to TABLE NO. 15-2-1-20.
- If result (4) is not as specified, go to TABLE NO. 15-2-1-21.

- If result (5) is not as specified, go to TABLE NO. 15-2-1-22.
- If ESU BITE indicators do not cycle as specified, go to TABLE NO. 15-2-1-23.
- i. Recharge APU accumulator by placing GENERATORS APU switch to ON. Visually check accumulator for proper charge (pressure gauge and tape indicator).

RESULT: APU ACCUM LOW capsule on caution/advisory panel shall go off in 90 seconds (180 seconds if winterization kit is installed) and APU accumulator shall be charged.

- If result is not as specified, troubleshoot Hydraulic Systems (PARA 7-2-1).
- If APU shuts down and underspeed BITE indication (o ● o ●) appears on ESU, replace left relay panel (PARA 9-4-2).
- j. Place upper console GENERATORS APU switch to OFF.
- k. Shut down APU (Appropriate Flight Manual).

RESULT:

- (1) APU shall shut down.
 - (2) APU ON capsule shall go off.
- If result (1) is not as specified, place FUEL PUMP switch to OFF and go to TABLE NO. 15-2-1-25.
 - If result (2) is not as specified, disconnect P260 on ESU.
 - (a) If APU ON capsule goes off, replace ESU (PARA 9-4-39).
 - (b) If APU ON capsule does not go off, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

1. Check ESU BITE indicators after APU is shut down.

RESULT: All BITE indicators shall be black.

- If result is not as specified, troubleshoot System for indicated malfunction.
- m. Place upper console FUEL PUMP switch to OFF.

RESULT: PRIME BOOST PUMP ON capsule on caution/advisory panel shall go off.

- If result is not as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3).
- n. Visually check that there is no fuel or fuel vapors coming from combustor drain line (below helicopter).
- If result is not as specified, check APU combustor drain check valve. Replace if necessary (49-20-05).
- o. Place upper console BATT switch to OFF.
- p. If no other check is to be done, install panel and soundproofing material over ESU on cabin ceiling (PARA 2-4-69).

15-2-1.2 Electronic Sequence Unit (ESU).

Simulator Setup.



To prevent possible damage to APU when attempting restart, make sure all items on APU are properly installed and that any airframe or APU electrical connectors that may have been disconnected during troubleshooting are connected.

- a. Connect cables (supplied with simulator) to simulator and helicopter as follows:

- (1) Power cable receptacle labeled P1 to J1 on simulator.
- (2) Test cable receptacle labeled P2 to J2 on simulator.
- (3) Other end of power cable to utility receptacle J257 in cabin ceiling.
- (4) Disconnect P289 from APU. Connect other end of test cable to P289.

- b. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- c. If not already performed, remove soundproofing material and panel from cabin ceiling to visually check ESU BITE indicators (PARA 2-4-69).
- d. Turn on electrical power (PARA 1-6-16).
- e. Make sure upper console APU CONTR switch is placed OFF.
- f. Turn simulator TEMP OUTPUT knob until temperature indicates 752°F (400°C).

Hydraulic Start Valve.

- a. Perform Simulator Setup, if not already performed.
- b. Connect multimeter between P290-1 and ground.
- c. Make sure simulator Test Set (simulator) power is off.
- d. Place upper console APU CONTR switch to ON.

RESULT: Multimeter shall read between 18 and 28 VDC.

- If result is not as specified, go to TABLE NO. 15-2-1-26.
- e. Turn on simulator power.

NOTE

The next step must be done within 29 seconds.

- f. Turn simulator S2 switch to 70%. Turn F1 switch to +1.

RESULT:

- (1) Lamp L1 shall go on, then go off.
 - (2) Lamp L2 shall go on, and stay on.
 - (3) ESU BITE indicators shall cycle to 70% speed (o o o ●).
 - (4) Multimeter shall indicate less than 18 VDC.
- If results are not as specified, replace ESU (PARA 9-4-39).
- g. If no other check is to be performed, go to Disconnecting Simulator.

Speed Sequence.

- a. Perform Simulator Setup, if not already performed.
- b. Turn simulator power on, if not already performed.
- c. Turn simulator F1 switch to -1 and S2 switch to 5%. If necessary, press START/STOP RESET button as required until SECONDS display shows 00.
- d. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

RESULT:

- (1) ESU BITE indications shall remain the same.
- (2) All lamps on simulator shall be off.

- (3) Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● o o ●).
 - (4) APU FAIL capsule on caution/advisory panel shall go on.
 - If result (3) occurs, but result (4) is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
 - e. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.
 - f. Place upper console APU CONTR switch to OFF.
- RESULT: APU FAIL capsule on caution/advisory panel shall go off.
- If result is not as specified, replace ESU (PARA 9-4-39).
 - g. Turn simulator F1 switch to +1 and S2 switch to 5%.
 - h. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

RESULT:

- (1) ESU BITE indicators shall cycle to 5% speed (o ● ● ●).
- (2) On simulator, lamp L1 shall be on.
- (3) Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● o o ●).
- (4) APU FAIL capsule on caution/advisory panel shall go on.
- (5) On simulator, lamp L1 shall go off.
- If result (2) is not as specified, check continuity between P261-17 and P289-17.
 - (a) If continuity is not present, repair/replace wiring (Appendix F).

- (b) If continuity is present, replace ESU (PARA 9-4-39).
 - If result (4) is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
 - i. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.
 - j. Place upper console APU CONTR switch to OFF.
- RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ●).
- If result is not as specified, replace ESU (PARA 9-4-39).
 - k. Turn simulator F1 switch to -2 and S2 switch to 15%.
 - l. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

RESULT:

- (1) ESU BITE indicators shall cycle to 5% speed (o ● ● ●).
- (2) On simulator, lamp L1 shall be on.
- (3) Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● o o ●).
- (4) APU FAIL capsule on caution/advisory panel shall go on.
- (5) On simulator, lamp L1 shall go off.
- If results are not as specified, replace ESU (PARA 9-4-39).
- m. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.

- n. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/ advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- o. Turn simulator S2 switch to 14%.

- p. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

RESULT:

- (1) ESU BITE indicators shall cycle to 14% speed (o o ● ●).
- (2) On simulator, lamps L1 and L2 shall be on.
- (3) Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● o o ●).
- (4) APU FAIL capsule on caution/advisory panel shall go on.
- (5) On simulator, lamps L1 and L2 shall go off.

- If results are not as specified, check continuity between P261-9 and P289-9.

(a) If continuity is not present, repair/replace wiring (Appendix F).

(b) If continuity is present, replace ESU (PARA 9-4-39).

- q. Turn simulator S2 switch to 0. Reset SECONDS display to 00.

- r. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/ advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- s. Turn simulator F1 switch to -1 and S2 switch to 70%.

- t. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

RESULT:

- (1) ESU BITE indicators shall cycle to 15% speed.
- (2) On simulator, lamps L1 and L2 shall be on.
- (3) Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● o o ●).
- (4) APU FAIL capsule on caution/advisory panel shall go on.
- (5) On simulator, lamps L1 and L2 shall go off.

- If results are not as specified, replace ESU (PARA 9-4-39).

- u. Turn simulator F1 switch to 0 and S2 switch to 0. Reset SECONDS display to 00.

- v. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/ advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- w. Turn simulator F1 switch to +1 and S2 switch to 70%.
- x. Place upper console APU CONTR switch to ON and at same time press START/STOP RESET button.

RESULT:

- (1) ESU BITE indicators shall cycle to 70% speed (o o o ●).
- (2) On simulator, lamp L1 shall go on and then go off, and lamp L2 shall go on and stay on.
- (3) Within 29 to 40 seconds, ESU BITE indicators shall cycle to fail to start (● o o ●).
- (4) APU FAIL capsule on caution/advisory panel shall go on.
- (5) On simulator, lamp L2 shall go off.

- If results are not as specified, replace ESU (PARA 9-4-39).

- y. Turn simulator S2 switch to 0. Reset SECONDS display to 00.

- z. Place upper console APU CONTR switch from ON to OFF.

RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- aa. If no other check is to be performed, go to Disconnecting Simulator.

Underspeed.

- a. Perform Simulator Setup, if not already performed.

- b. Turn simulator power on, if not already performed.

- c. Turn simulator S2 switch to 90% and make sure F1 switch is at 0.

- d. Place upper console APU CONTR switch to ON.

RESULT: In approximately 2 seconds, ESU BITE indicators shall cycle to under-speed (o ● o ●) and APU FAIL capsule on caution/advisory panel shall go on.

- If result is not as specified, replace ESU (PARA 9-4-39).

- e. Turn simulator S2 switch to 0 and F1 switch to 0.

- f. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- g. Turn simulator S2 switch to 90% and F1 switch to +5%.

RESULT:

- (1) ESU BITE indicators shall cycle through a normal start and, in approximately 2 seconds, shall cycle to ready for service BITE indication (● ● ● ●).

- (2) APU ON capsule on caution/advisory panel shall go on.

- If results are not as specified, replace ESU (PARA 9-4-39).

- h. Slowly turn simulator F1 switch from +5 to 0.

RESULT:

- (1) ESU BITE indicators shall cycle to under-speed (o ● o ●) when knob F1 is at +1.
 - (2) APU ON capsule on caution/advisory shall go off.
 - (3) APU FAIL capsule on caution/advisory panel shall go on.
- If results are not as specified, replace ESU (PARA 9-4-39).
- i. Turn simulator S2 switch to 0 and F1 switch to 0.
 - j. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- k. If no other check is to be performed, go to Disconnecting Simulator.

Overspeed.

- a. Perform Simulator Setup, if not already performed.
- b. Turn simulator power on, if not already performed.
- c. Turn simulator F1 switch to -5, then turn S2 switch to 110% speed.
- d. Place upper console APU CONTR switch to ON.

RESULT:

- (1) ESU BITE indicators shall cycle from 90% speed to ready for service (● ● ● ●) in approximately 2 seconds.

- (2) APU ON capsule on caution/advisory panel shall go on.

- (3) On simulator, lamps L2 and L3 shall go on.

- If results are not as specified, replace ESU (PARA 9-4-39).

- e. Turn simulator F1 switch to 0.

RESULT:

- (1) ESU BITE indicators shall cycle to over-speed (● ● o ●).

- (2) On simulator, lamps L2 and L3 shall go off.

- (3) APU ON capsule on caution/advisory panel shall go off.

- (4) APU FAIL capsule on caution/advisory panel shall go on.

- If results are not as specified, replace ESU (PARA 9-4-39).

- f. Turn S2 switch to 0%.

- g. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- h. If no other check is to be performed, go to Disconnecting Simulator.

Low Oil Pressure.

- a. Perform Simulator Setup, if not already performed.

- b. Turn simulator power on, if not already performed.

NOTE

Observer must watch ESU BITE indication during the test. The next two steps must be performed within 29 seconds.

- c. Turn simulator F1 switch to -2. Turn S2 switch to 70%.
- d. Place upper console APU CONTR switch to ON.

RESULT: ESU BITE indicators shall cycle up to 15% speed and on simulator, lamps L1 and L2 shall go on.

- If result is not as specified, check continuity between P289-7 and P261-7.
 - (a) If continuity is not present, repair/replace wiring (Appendix F).
 - (b) If continuity is present, replace ESU (PARA 9-4-39).

- e. Turn simulator F1 switch to 0 and S2 switch to 0.
- f. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).
- g. Turn simulator F1 switch to +1 and S2 switch to 70%.
 - h. Place upper console APU CONTR switch to ON.

RESULT:

- (1) ESU BITE indicators shall cycle up to 70% speed (o o o ●).

- (2) On simulator, lamp L1 shall go on, then go off. Lamp L2 shall go on and stay on.

- If results are not as specified, replace ESU (PARA 9-4-39).

- i. Place simulator S3 switch to LOP.

RESULT:

- (1) ESU BITE indicators shall cycle from 70% speed to low oil pressure indication (o o o ●) (BITE indicators will momentarily move, but shall remain the same).

- (2) On simulator, lamp L6 shall go on when switch is pushed to LOP.

- (3) APU FAIL capsule on caution/advisory panel shall go on.

- If results are not as specified, replace ESU (PARA 9-4-39).

- j. Turn simulator F1 switch to 0 and S2 switch to 0.

- k. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- l. Turn simulator F1 switch to +1 and S2 switch to 90%.

- m. Place upper console APU CONTR switch to ON.

RESULT:

- (1) ESU BITE indicators shall cycle to 90% speed, and in approximately 2 seconds, cycle to ready for service (● ● ● ●).

- (2) APU ON capsule on caution/advisory panel shall go on.

- If results are not as specified, replace ESU (PARA 9-4-39).

n. Place simulator S3 switch to LOP.

RESULT: ESU BITE indicators shall cycle to low oil pressure (o o o ●) and APU FAIL capsule on caution/advisory panel shall go on.

- If result is not as specified, replace ESU (PARA 9-4-39).

o. Turn simulator F1 switch to 0 and S2 switch to 0.

p. Place upper console APU CONTR switch to OFF.

RESULT: APU FAIL capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

q. If no other check is to be performed, go to Disconnecting Simulator.

High Oil Temperature.

a. Perform Simulator Setup, if not already performed.

b. Turn simulator power on, if not already performed.

c. Turn simulator F1 switch to +1 and S2 switch to 90%.

d. Place upper console APU CONTR switch to ON.

RESULT:

- (1) ESU BITE indicators shall cycle from 90% speed to ready for service (● ● ● ●) in approximately 2 seconds.

- (2) On simulator, lamp L1 shall be off, lamps L2 and L3 shall be on.

- (3) APU ON capsule on caution/advisory panel shall be on.

- If results are not as specified, replace ESU (PARA 9-4-39).

NOTE

If ESU BITE indicators indicate high oil temperature (● ● ● o) before S3 switch is used, replace ESU (PARA 9-4-39).

e. Place simulator S3 switch to HI OIL TEMP.

RESULT:

- (1) ESU BITE indicators shall cycle from ready for service to high oil temperature (● ● ● o).

- (2) On simulator, lamp L5 shall go on.

- (3) APU OIL TEMP HI capsule on caution/advisory panel shall go on.

- If result (1) is not as specified, replace ESU (PARA 9-4-39).

- If result (1) occurs, but result (3) is not as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

f. Place upper console APU CONTR switch to OFF.

RESULT: APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

g. If no other check is to be performed, go to Disconnecting Simulator.

Start Bypass Valve.

- a. Perform Simulator Setup, if not already performed.
- b. Turn simulator power on, if not already performed.
- c. Place upper console AIR SOURCE HEAT/START switch to APU.
- d. Turn simulator S2 switch to 90%. Turn F1 switch to +2.
- e. Place upper console APU CONTR switch to ON.

RESULT:

- (1) ESU BITE indicators shall cycle from 90% speed to ready for service indication (● ● ● ●) in approximately 2 seconds.
 - (2) APU ON capsule on caution/advisory panel shall be on.
- If results are not as specified, check continuity between ESU and P289.
 - (a) If continuity is not present, repair/replace wiring (Appendix F).
 - (b) If continuity is present, replace ESU (PARA 9-4-39).
- f. Press, then release No. 1 engine start button.

RESULT: Whenever No. 1 engine start button is pressed, lamp L4 on simulator shall go on and #1 ENGINE STARTER capsule, on caution/advisory panel, shall go on.

- If result is not as specified, go to TABLE NO. 15-2-1-26.
- g. Press, then release No. 2 engine start button.

RESULT: Whenever No. 2 engine start button is pressed, lamp L4 on simulator shall go on and #2 ENGINE STARTER capsule, on caution/advisory panel, shall go on.

- If result is not as specified, go to TABLE NO. 15-2-1-26.
- h. Place upper console APU CONTR switch to OFF.
 - i. Place upper console AIR SOURCE HEAT/START switch to OFF.
 - j. If no other check is to be performed, go to Disconnecting Simulator.

Overtemperature.

- a. Perform Simulator Setup, if not already performed.
- b. Turn simulator power on, if not already performed.

NOTE

The next three steps must be done in less than 29 seconds.

- c. Turn simulator F1 switch to -5. Turn S2 switch to 90%.
- d. Place upper console APU CONTR switch to ON.
- e. Slowly turn simulator TEMP OUTPUT - R1 knob until ESU BITE indicators cycle to overtemperature (o o ● ●).

RESULT:

- (1) ESU BITE indicators shall cycle up to 70% BITE indication.
- (2) As knob is turned, ESU BITE indicators shall cycle to overtemperature between 1364°F and 1429°F (740°C and 776°C).

- (3) APU FAIL capsule on caution/advisory panel shall go on.

- If results are not as specified, replace ESU (PARA 9-4-39).

- f. Turn simulator F1 switch to 0 and S2 to 0. Turn TEMP OUTPUT knob R1 until temperature indicates 752°F (400°C).

- g. Place upper console APU CONTR switch to OFF.

RESULT: APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- h. Turn simulator F1 switch to +5. Turn S2 switch to 90%.

- i. Place upper console APU CONTR switch to ON.

RESULT: ESU BITE indicators shall cycle up to ready for service indication (● ● ● ●) and APU ON capsule on caution/advisory panel shall be on.

- If result is not as specified, replace ESU (PARA 9-4-39).

- j. Slowly turn simulator TEMP OUTPUT knob until ESU BITE indicators cycle to overtemperature.

RESULT: As knob is turned, ESU BITE indicators shall cycle to overtemperature between 1242°F and 1282°F (672°C and 694°C), and APU FAIL capsule on caution/advisory panel shall be on.

- If result is not as specified, replace ESU (PARA 9-4-39).

- k. Turn simulator F1 switch to 0 and S2 switch to 0.

- l. Place upper console APU CONTR switch to OFF.

RESULT: APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- m. Turn TEMP OUTPUT knob until temperature indicates approximately 752°F (400°C).

- n. If no other check is to be performed, go to Disconnecting Simulator.

Open Thermocouple.

- a. Perform Simulator Setup, if not already performed.

- b. Turn simulator power on, if not already performed.

- c. Place upper console APU CONTR switch to ON.

RESULT:

- (1) ESU BITE indicators shall cycle up to ready for service indication (● ● ● ●).

- (2) APU ON capsule on caution/advisory panel shall go on.

- (3) On simulator, lamps L2 and L3 shall go on.

- If results are not as specified, replace ESU (PARA 9-4-39).

NOTE

Disregard initial ESU BITE indications when connector is disconnected.

- d. Disconnect P2 from test receptacle J2.

RESULT:

- (1) Simulator lamps L2 and L3 shall go off.
 - (2) APU ON capsule on caution/advisory panel shall go off.
 - (3) Approximately 12 to 24 seconds after connector was pulled, BITE indicators shall cycle to open thermocouple indication (● ● o o).
 - (4) APU FAIL capsule on caution/advisory panel shall be on.
- If results are not as specified, check continuity between the following points:

P289-1	and	P289-2
P261-1	and	P261-2

- (a) If continuity is not present, repair/replace wiring (Appendix F), as required.
 - (b) If continuity is present, replace ESU (PARA 9-4-39).
- e. Reconnect P2 to J2 receptacle on simulator.
 - f. Turn simulator F1 switch to 0 and S2 switch to 0.
 - g. Place upper console APU CONTR switch to OFF.

RESULT: APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).

- h. If no other check is to be performed, go to Disconnecting Simulator.

Shorted Thermocouple.

- a. Perform Simulator Setup, if not already performed.
- b. Turn simulator power on, if not already performed.
- c. Turn simulator F1 switch to +5 and S2 switch to 90%.
- d. Place upper console APU CONTR switch to ON.
- e. On helicopters with APU ESU 160200-201, on simulator, rotate R1 switch to 50 - 55°F (10 - 13°C).

RESULT: ESU BITE indicator shall not be ● o
● o shorted thermocouple.

- If result is not as specified, go to TABLE NO. 15-2-1-4.
- f. On helicopters with ESU 160200-600, on simulator, rotate R1 switch to 244 - 266°F (118 - 130°C).

RESULT: ESU BITE indicator shall not be ● o
● o shorted thermocouple.

- If result is not as specified, go to TABLE NO. 15-2-1-4.
- g. Place upper console APU CONTR switch to OFF.

RESULT: APU OIL TEMP HI capsule on caution/advisory panel shall go off and ESU BITE indicators shall cycle to reset (● ● ● ●).

- If result is not as specified, replace ESU (PARA 9-4-39).
- h. If no other check is to be performed, go to Disconnecting Simulator.

Disconnecting Simulator.

- a. Place upper console APU CONTR switch to OFF.

RESULT: ESU BITE indicators shall cycle to reset (● ● ● ●).

S1 - OFF

- If result is not as specified, replace ESU (PARA 9-4-39).

b. Turn off electrical power (PARA 1-6-16).

c. Make sure simulator switches are placed as follows:

F1	-	0
S2	-	0%
R1	-	752 ° F (400°C)
S3	-	NORM

d. Make sure upper console AIR SOURCE/ HEAT START switch is placed OFF.

e. Disconnect power cable from receptacle in cabin ceiling and simulator. Disconnect test cable from P289 and simulator. Connect P289 to APU.

f. Install test cables in simulator.

g. Make sure P290 is connected to APU hydraulic start valve.

h. On cabin ceiling, install panel and soundproofing over ESU (PARA 2-4-69).

CHART NO. 15-2-1-1. NORMAL START SEQUENCE.

APU Speed Sequence (%RPM)

EVENT

A.5%

1. Start fuel solenoid opens

2. Ignition exciter goes on

3. 20 - 30 second counter (inside ESU) starts

*4. EGT overtemperature shutdown circuit (inside ESU) limit extended to 1,376°F (740°C)

5. Fuel pressure built up due to motoring speed

*6. Thermocouple checked for open condition (inside ESU)

7. ESU internal checks for processor sequence, sensor/data board, and data failures

8. Ignition starts

*9. Overspeed circuit activates

***BITE INDICATION: o ● ● ●

B.14%

1. Main fuel solenoid opens

2. Ignition continues

CHART NO. 15-2-1-1. NORMAL START SEQUENCE. (Cont)

APU Speed Sequence (%RPM)

EVENT

	3. ESU internal self-checks continue
	*4. Thermocouple checked for open condition (inside ESU)
	***BITE INDICATION: o o ● ●
C.30%	*1. Inside ESU - APU exhaust gas temperature is checked for being above 50°F (10°C)
	2. ESU internal self-checks continue
D.70%	1. APU start valve (hydraulic) - closes
	2. Start fuel valve - closes
	3. Purge valve - opens (due to air pressure)
	4. Ignition - off
	*5. Oil pressure circuit - on
	6. ESU internal self-checks are made
	*7. Thermocouple checked for open condition (inside ESU)
	***BITE INDICATION: o o o ●
E.90%	1. This occurs in ESU:
	*a. 20 - 30 second counter - disabled
	*b. APU fail circuit - deactivated
	*c. Underspeed circuit - activated
	d. 1.5 second timer - activated
	e. ESU internal self-checks continue
	*f. Thermocouple checked for open condition (inside ESU)
	***BITE INDICATION: o o o o

CHART NO. 15-2-1-1. NORMAL START SEQUENCE. (Cont)**APU Speed Sequence (%RPM)****EVENT**

F.90% -1.5 Seconds

1. Maximum fuel valve - opens

2. APU ON capsule (caution/advisory panel) - on

3. ESU internal self-checks continue

**4. Thermocouple checked for short condition
(inside ESU)

**5. High oil temperature check

*6. EGT shutdown limit - reduced to 1,240 - 1,280°F
(656 - 679°C)

***BITE INDICATION: ● ● ● ●

G. APU Operating - 102% (Unloaded) APU Operating - 100% (Loaded)

*ESU checks on System operation. If check is not satisfactory, APU will shut down.

**Warning indication that will appear as a BITE indication, but will not cause APU to shut down.

***BITE indication will appear on ESU when APU has reached corresponding speed during start sequence.

TABLE NO. 15-2-1-2. ESU BITE Indicators Do Not Cycle When BATT Switch Is Placed ON.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P260-5 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3.
2. Check continuity between P260-6 and ground.	Step 1. If continuity is present, replace ESU (PARA 9-4-39). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10.
3. Install an insulated jumper wire between P242-n and P242-q. Go to 4.	
4. Check for 28 VDC between P260-5 and ground.	

TABLE NO. 15-2-1-2. ESU BITE Indicators Do Not Cycle When BATT Switch Is Placed ON. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, go to 5 .
	Step 2. If voltage is not as specified, go to 7 .
5. Remove jumper wire from P242. Go to 6.	
6. Replace left relay panel (PARA 9-4-2). Go to 10.	
7. Remove jumper wire from P242. Go to 8.	
8. Check for 28 VDC between P242-q and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between P242- <u>n</u> and P260-5 (Appendix F). Go to 10 .
	Step 2. If voltage is not as specified, go to 9 .
9. Check for 28 VDC between APU CONTR INST circuit breaker terminal 1 and ground.	
	Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P242-q (Appendix F). Go to 10 .
	Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 10 .
10. Procedure completed.	

TABLE NO. 15-2-1-3. Fuel Leaks From APU Combustor Drain Line With Prime Boost Pump Running.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect P305 from start fuel valve, P306 from main fuel valve, and P303 from ignition exciter. Go to 2.	
2. Check for fuel leaks.	
	Step 1. If fuel leaks are present, go to 3 .
	Step 2. If fuel leaks are not present, replace ESU (PARA 9-4-39). Go to 5 .

**TABLE NO. 15-2-1-3. Fuel Leaks From APU Combustor Drain Line With Prime Boost Pump Running.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
3.	Disconnect line to main fuel manifold and purge fuel into container. Go to 4.
4.	Using APU CONTR switch, motor APU and check for fuel flow. Step 1. If fuel flows, replace main fuel valve (49-20-05). Go to 5. Step 2. If fuel does not flow, replace start fuel valve (49-20-05). Go to 5.
5.	Procedure completed.

TABLE NO. 15-2-1-4. ESU BITE Indicator Shows o ● o (Shorted Thermocouple).

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Disconnect P289 from APU. Go to 2.
2.	Check continuity between P261-1 and P261-2. Step 1. If continuity is present, repair/replace wiring (Appendix F), as required between, then go to 6: P261-1 and P289-1 P261-2 and P289-2 Step 2. If continuity is not present, go to 3.
3.	Replace APU wiring harness (49-20-05). Go to 4.
4.	Start and operate APU (Appropriate Flight Manual). Go to 5.
5.	Does APU operate normally? Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 6. Step 2. If APU does not operate normally and ESU BITE indicators show o ● ● o (fail to light), shut down APU (Appropriate Flight Manual). Replace ESU for helicopters with ESU 160200-201 (PARA 9-4-39), then go to 6.
6.	Procedure completed.

TABLE NO. 15-2-1-5. ESU BITE Indicator Shows o ● ● o (Fail To Light) For Helicopters With ESU 160200-201.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check for fuel vapor at exhaust duct or overboard drain.	Step 1. If vapor is present, go to 2. Step 2. If vapor is not present, go to 13.
2. Perform ESU Speed Sequence procedure to verify signal M to ignition exciter.	Step 1. If speed sequence is good, replace igniter plug (49-20-05). Go to 3.
3. Start and operate APU (Appropriate Flight Manual). Go to 4.	
4. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual). Go to 5.
5. Install an insulated jumper wire between P303-A and P303-B. Go to 6.	
6. Check continuity between J289-17 and ground.	Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, go to 11.
7. Remove jumper wire. Go to 8.	
8. Replace exciter cable (49-20-05). Go to 9.	
9. Start and operate APU (Appropriate Flight Manual). Go to 10.	
10. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual). Replace ignition exciter (49-20-05), then go to 24.

TABLE NO. 15-2-1-5. ESU BITE Indicator Shows o ● ● o (Fail To Light) For Helicopters With ESU 160200-201. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
11. Remove jumper wire. Go to 12.	
12. Replace APU wiring harness (49-20-05). Go to 13.	
13. Perform ESU Speed Sequence procedure to verify signal M to start fuel valve.	
Step 1. If speed sequence is good, replace start fuel nozzle (49-20-05). Go to 14.	
14. Start and operate APU (Appropriate Flight Manual). Go to 15.	
15. Does APU operate normally?	
Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24.	
Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual). Go to 16.	
16. Install an insulated jumper wire between P305-A and P305-B. Go to 17.	
17. Check continuity between J289-17 and ground.	
Step 1. If continuity is present, go to 18.	
Step 2. If continuity is not present, go to 22.	
18. Remove jumper wire. Go to 19.	
19. Replace fuel start valve (49-20-05). Go to 20.	
20. Start and operate APU (Appropriate Flight Manual). Go to 21.	
21. Does APU operate normally?	
Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24.	
Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual). Replace start nozzle purge valve (49-20-05), then go to 24.	
22. Remove jumper wire. Go to 23.	

TABLE NO. 15-2-1-5. ESU BITE Indicator Shows o ● ● o (Fail To Light) For Helicopters With ESU 160200-201. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
23. Replace APU wiring harness (49-20-05). Go to 24.	
24. Procedure completed.	

TABLE NO. 15-2-1-6. ESU BITE Indicator Shows o ● ● o (Shorted Oil Pressure Switch) For Helicopters With ESU 160200-600.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between: J289-7 and J289-16 J289-7 and J289-23	
Step 1. If continuity is present, go to 2.	
Step 2. If continuity is not present, replace APU wiring harness (49-20-05). Go to 7.	
Step 3. If trouble remains, replace low oil pressure switch (49-20-05). Go to 7.	
2. Check for 28 VDC between P304-C and ground.	
Step 1. If voltage is as specified, replace low oil pressure switch (49-20-05). Go to 7.	
Step 2. If voltage is not as specified, go to 3.	
3. Check for 28 VDC between P289-16 and ground.	
Step 1. If voltage is as specified, replace APU wiring harness (49-20-05). Go to 7.	
Step 2. If voltage is not as specified, go to 4.	
4. Check for 28 VDC between ESU J1-16 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P261-16 and P289-16 (Appendix F). Go to 7.	
Step 2. If voltage is not as specified, go to 5.	
5. Check for 28 VDC between P260-5 and ground.	

TABLE NO. 15-2-1-6. ESU BITE Indicator Shows o ● ● o (Shorted Oil Pressure Switch) For Helicopters With ESU 160200-600. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, replace ESU (PARA 9-4-39). Go to 7. Step 2. If voltage is not as specified, go to 6.
6. Check continuity between J242-q and J242-n.	Step 1. If continuity is present, repair/replace wiring between P242-n and P260-5 (Appendix F). Go to 7. Step 2. If continuity is not present, replace left relay panel (PARA 9-4-2). Go to 7.
7. Procedure completed.	

TABLE NO. 15-2-1-7. APU Motor Does Not Motor (No Rotation).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check for any of the following BITE indications:	
o ● ● ● ● o ● ● ● o o o o o o o	
	Step 1. If any indication listed is present, replace ESU (PARA 9-4-39). Go to 9. Step 2. If any indication listed is not present, go to 2.
2. Make sure APU CONTR switch is placed OFF. Go to 3.	
3. Press APU start valve manual override lever. Go to 4.	
4. Check that APU start motor operates.	Step 1. If motor operates, go to 5. Step 2. If motor does not operate, go to 6.
5. Perform APU Start Valve procedure.	Step 1. If APU start valve is good, replace APU start valve (PARA 7-4-53). Go to 9.

TABLE NO. 15-2-1-7. APU Motor Does Not Motor (No Rotation). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
6. Disconnect start motor from APU accessory pad. Go to 7.	
7. Using a ½ inch Allen wrench, turn splined shaft on accessory pad. Check that shaft turns.	Step 1. If shaft turns, go to 8. Step 2. If shaft does not turn, replace seized APU (PARA 15-4-1). Go to 9.
8. Attempt to turn splined shaft of APU start motor.	Step 1. If shaft turns, replace APU hydraulic start valve (PARA 7-4-53). Go to 9. Step 2. If shaft does not turn, replace APU start motor (PARA 15-4-7). Go to 9.
9. Procedure completed.	

TABLE NO. 15-2-1-8. ESU BITE Indicator Shows ● o o ● (Fail To Start).

TEST OR INSPECTION	CORRECTIVE ACTION
1. What was APU speed BITE indication before it changed to fail to start indication?	Step 1. If APU speed was over 14% and less than 70% (o o ●●), go to 2. Step 2. If APU speed was over 70% and less than 90% (o o o ●), go to 15.
2. Check that main fuel valve actuates (determined by feeling the valve for a thump) during start sequence.	Step 1. If valve actuates, go to 3. Step 2. If valve does not actuate, go to 13.
3. Loosen nut on inlet fuel filter inlet. Place rag around nut. Go to 4.	
4. On upper console panel, place FUEL PUMP switch to APU BOOST. Go to 5.	
5. Check that fuel is going into rag.	

TABLE NO. 15-2-1-8. ESU BITE Indicator Shows ● o o ● (Fail To Start). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If fuel is present, go to 6. Step 2. If fuel is not present, go to 11.
6. Tighten nut. Go to 7.	
7. Loosen nut on inlet fuel filter outlet. Place rag around nut. Go to 8.	
8. Place upper console FUEL PUMP switch to APU BOOST. Go to 9.	
9. Check that fuel is going into rag.	Step 1. If fuel is present, go to 10. Step 2. If fuel is not present, replace filter (49-20-05). Go to 24.
10. Check for restricted outlet filter.	Step 1. If filter is restricted, replace fuel pump (49-20-05). Go to 24. Step 2. If filter is not restricted, replace main fuel nozzles/manifold assembly (49-20-05). Go to 24.
11. Tighten nut. Go to 12.	
12. Troubleshoot Fuel Prime/Boost System (PARA 10-2-3). Go to 24.	
13. Check for 28 VDC between P306-A and ground.	Step 1. If voltage is as specified, go to 14. Step 2. If voltage is not as specified, perform ESU Speed Sequence procedure steps a. through m. and step aa. to verify power to main fuel valve. Go to 24.
14. Check continuity between P306-B and ground.	Step 1. If continuity is present, replace main fuel valve (49-20-05). Go to 24. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 24. Step 3. If trouble remains, replace APU wiring harness (49-20-05). Go to 24.
15. Make sure all bleed air ducts are closed and driven equipment is unloaded. Check air inlet for Foreign Object Debris (FOD). Go to 16.	
16. Restart APU (Appropriate Flight Manual). Go to 17.	

TABLE NO. 15-2-1-8. ESU BITE Indicator Shows ● o o ● (Fail To Start). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
17. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Go to 18.
18. Replace fuel manifold (49-20-05). Go to 19.	
19. Restart APU (Appropriate Flight Manual). Go to 20.	
20. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Go to 21.
21. Replace fuel pump/acceleration control (49-20-05). Go to 22.	
22. Restart APU (Appropriate Flight Manual). Go to 23.	
23. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Replace APU (PARA 15-4-1), then go to 24.
24. Procedure completed.	

TABLE NO. 15-2-1-9. ESU BITE Indicator Shows o ● o o (Loss Of Speed Data) For Helicopters With ESU 160200-600.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 1,190 to 1,450 ohms between P261-12 and P261-13.	

TABLE NO. 15-2-1-9. ESU BITE Indicator Shows o ● o o (Loss Of Speed Data) For Helicopters With ESU 160200-600. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 1. If resistance is as specified, replace ESU (PARA 9-4-39). Go to 8. Step 2. If resistance is not as specified, go to 2. 2. Install an insulated jumper wire between P302-A and P302-B. Go to 3. 3. Check continuity between P261-12 and P261-13. Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, go to 6. 4. Remove jumper wire from P302. Go to 5. 5. Replace magnetic pickup (speed sensor) (49-20-05). Go to 8. 6. Remove jumper wire from P302. Go to 7. 7. Replace APU wiring harness (49-20-05). Go to 8. Step 1. If trouble remains, repair/replace wiring (Appendix F), as required between, then go to 8: P261-12 and P289-12 P261-13 and P289-13 8. Procedure completed.

TABLE NO. 15-2-1-10. ESU BITE Indicator Shows o o ● ● (Overtemperature).

TEST OR INSPECTION CORRECTIVE ACTION
1. Perform ESU Overtemperature procedure. Step 1. If APU overtemperature is good, go to 2. 2. Replace combustor drain valve (49-20-05). Go to 3. 3. Start and operate APU (Appropriate Flight Manual). Go to 4.

TABLE NO. 15-2-1-10. ESU BITE Indicator Shows o o ● ● (Overtemperature). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
4. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 8. Step 2. If APU does not operate normally and ESU BITE indicator shows overtemperature, shut down APU (Appropriate Flight Manual). Go to 5.
5. Replace maximum fuel valve (49-20-05). Go to 6.	
6. Start and operate APU (Appropriate Flight Manual). Go to 7.	
7. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 8. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Replace fuel pump/acceleration control (49-20-05), then go to 8.
8. Procedure completed.	

TABLE NO. 15-2-1-11. ESU BITE Indicator Shows o o o ● (Low Oil Pressure).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check APU oil sump for proper level.	Step 1. If oil level is good, go to 2. Step 2. If oil level is not good, service APU oil sump, as required (PARA 1-3-9). Go to 5.
2. Perform ESU Low Oil Pressure procedure.	
	Step 1. If APU oil pressure is good, replace APU low oil pressure switch (49-20-05). Go to 3.
3. Start and operate APU (Appropriate Flight Manual). Go to 4.	
4. Does APU operate normally?	

TABLE NO. 15-2-1-11. ESU BITE Indicator Shows o o o • (Low Oil Pressure). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 5. Step 2. If APU does not operate normally and ESU BITE indicator shows low oil pressure, shut down APU (Appropriate Flight Manual). Replace APU wiring harness (49-20-05), then go to 5.
5. Procedure completed.	

TABLE NO. 15-2-1-12. ESU BITE Indicator Shows • • o o (Open Thermocouple).

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between P261-1 and P261-2. Step 1. If continuity is present, replace ESU (PARA 9-4-39). Go to 3. Step 2. If continuity is not present, go to 2. 2. Check continuity between J289-1 and J289-2. Step 1. If continuity is present, repair/replace wiring (Appendix F), as required between, then go to 3. P261-1 and P289-1 P261-2 and P289-2 Step 2. If continuity is not present, replace APU thermocouple and harness (49-20-05). Go to 3. 3. Procedure completed.

TABLE NO. 15-2-1-13. ESU BITE Indicator Shows o o • o (External Short) After o • • • (Start Fuel Valve And Exciter Signal Out) For Helicopters With ESU 163290-100.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between J303-B and ground.

TABLE NO. 15-2-1-13. ESU BITE Indicator Shows o o ● o (External Short) After o ● ● ● (Start Fuel Valve And Exciter Signal Out) For Helicopters With ESU 163290-100. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
Step 1. If continuity is present, replace ignition exciter (49-20-05). Go to 5 . Step 2. If continuity is not present, go to 2 .	
2. Check continuity between J305-A and ground.	
Step 1. If continuity is present, replace start fuel valve (49-20-05). Go to 5 . Step 2. If continuity is not present, go to 3 .	
3. With P303 and P305 disconnected, check continuity between J289-17 and ground.	
Step 1. If continuity is present, replace APU wiring harness (49-20-05). Go to 5 . Step 2. If continuity is not present, go to 4 .	
4. Check continuity between P261-17 and ground.	
Step 1. If continuity is present, repair/replace wiring between P261-17 and P289-17 (Appendix F). Go to 5 . Step 2. If continuity is not present, replace ESU (PARA 9-4-39). Go to 5 .	
5. Procedure completed.	

TABLE NO. 15-2-1-14. ESU BITE Indicator Shows o o ● o (External Short) After o o ● ● (Main Fuel Valve Signal Out; 14%) For Helicopters With ESU 163290-100.

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check continuity between J306-A and ground.	
Step 1. If continuity is present, replace main fuel valve (49-20-05). Go to 4 . Step 2. If continuity is not present, go to 2 .	
2. With P306 disconnected, check continuity between J289-9 and ground.	
Step 1. If continuity is present, replace APU wiring harness (49-20-05). Go to 4 . Step 2. If continuity is not present, go to 3 .	

TABLE NO. 15-2-1-14. ESU BITE Indicator Shows o o ● o (External Short) After o o ● ● (Main Fuel Valve Signal Out; 14%) For Helicopters With ESU 163290-100. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. With J306 disconnected and J289 connected, check continuity between P261-9 and ground.	Step 1. If continuity is present, repair/replace wiring between P261-9 and P289-9 (Appendix F). Go to 4. Step 2. If continuity is not present, replace ESU (PARA 9-4-39). Go to 4.
4. Procedure completed.	

TABLE NO. 15-2-1-15. ESU BITE Indicator Shows o o ● o (External Short) After o o o o (90% RPM Switch On) Or ● ● ● ● (Ready For Service) For Helicopters With ESU 163290-100.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between J307-A and ground.	Step 1. If continuity is present, replace maximum fuel valve (49-20-05). Go to 4. Step 2. If continuity is not present, go to 2.
2. With P307 disconnected, check continuity between J289-15 and ground.	Step 1. If continuity is present, replace APU wiring harness (49-20-05). Go to 4. Step 2. If continuity is not present, go to 3.
3. Check continuity between P261-15 and ground.	Step 1. If continuity is present, repair/replace wiring between P261-15 and P289-15 (Appendix F). Go to 4. Step 2. If continuity is not present, replace ESU (PARA 9-4-39). Go to 4.
4. Procedure completed.	

TABLE NO. 15-2-1-16. APU Motors But Shuts Down (APU ON Capsule Goes Off And APU FAIL Capsule Goes On).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check for any of the following BITE indications:	o • • • • o • • • o o o o o o o Step 1. If any indication listed is present, replace ESU (PARA 9-4-39). Go to 10. Step 2. If any indication listed is not present, go to 2.
2. Reset ESU. Go to 3.	
3. Check for 1,190 to 1,450 ohms between P261-12 and P261-13.	Step 1. If resistance is as specified, replace ESU (PARA 9-4-39). Go to 10. Step 2. If resistance is not as specified, go to 4.
4. Install an insulated jumper wire between P302-A and P302-B. Go to 5.	
5. Check continuity between P261-12 and P261-13.	Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, go to 8.
6. Remove jumper wire from P302. Go to 7.	
7. Remove and reinstall magnetic pickup (speed sensor) to be sure of correct adjustment (49-20-05).	
	Step 1. If trouble remains, replace magnetic pickup (speed sensor) (49-20-05). Go to 10.
8. Remove jumper wire from P302. Go to 9.	
9. Repair/replace wiring (Appendix F), as required between, then go to 10. P261-12 and P289-12 P261-13 and P289-13	
	Step 1. If trouble remains, replace APU wiring harness (49-20-05). Go to 10.

TABLE NO. 15-2-1-16. APU Motors But Shuts Down (APU ON Capsule Goes Off And APU FAIL Capsule Goes On). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
10. Procedure completed.	

TABLE NO. 15-2-1-17. ESU BITE Indicator Shows o ● o ● (Underspeed).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check helicopter configuration.	Step 1. ESU 160200-600, go to 2. Step 2. All other ESUs, go to 3.
2. Troubleshoot System for loss of speed data o ● o o (TABLE NO. 15-2-1-9).	Step 1. If trouble remains, go to 3.
3. Make sure all bleed air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 4.	
4. Perform ESU Speed Sequence procedure to verify signal to maximum fuel valve.	Step 1. If speed sequence is good, go to 5.
5. Loosen nut on inlet fuel filter. Place rag around nut. Go to 6.	
6. On upper console, place FUEL PUMP switch to APU BOOST. Go to 7.	
7. Check that fuel is going into rag.	Step 1. If fuel is present, go to 8. Step 2. If fuel is not present, go to 19.
8. Install an insulated jumper wire between P307-A and P307-D. Go to 9.	
9. Check continuity between J289-15 and ground.	Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, go to 17.

TABLE NO. 15-2-1-17. ESU BITE Indicator Shows o • o • (Underspeed). (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
10. Remove jumper wire. Go to 11.	
11. Replace maximum fuel valve (49-20-05). Go to 12.	
12. Start and operate APU (Appropriate Flight Manual). Go to 13.	
13. Does APU operate normally?	
Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 21.	
Step 2. If APU does not operate normally and ESU BITE indications shows underspeed, shut down APU (Appropriate Flight Manual). Go to 14.	
14. Replace APU fuel manifold (49-20-05). Go to 15.	
15. Start and operate APU (Appropriate Flight Manual). Go to 16.	
16. Does APU operate normally?	
Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 21.	
Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Replace APU fuel pump/acceleration control (49-20-05), then go to 21.	
17. Remove jumper wire. Go to 18.	
18. Replace APU wiring harness (49-20-05). Go to 21.	
19. Tighten nut. Go to 20.	
20. Troubleshoot Fuel Prime/Boost System (PARA 10-2-3). Go to 21.	
21. Procedure completed.	

TABLE NO. 15-2-1-18. ESU BITE Indicator Shows ● ● o ● (Overspeed).

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Perform ESU Speed Sequence procedure to verify signal to start fuel valve. Step 1. If speed sequence is good, go to 2. 2. Install an insulated jumper wire between P305-A and P305-B. Go to 3. 3. Check continuity between J289-17 and ground. Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, go to 10. 4. Remove jumper wire. Go to 5. 5. Start and operate APU (Appropriate Flight Manual). Go to 6. 6. Does APU operate normally? Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 12. Step 2. If APU does not operate normally and ESU BITE indicator shows overspeed, shut down APU (Appropriate Flight Manual). Go to 7. 7. Replace APU start fuel valve (49-20-05). Go to 8. 8. Start and operate APU (Appropriate Flight Manual). Go to 9. 9. Does APU operate normally? Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 12. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual), then replace APU fuel pump/acceleration control (49-20-05). Go to 12. 10. Remove jumper wire. Go to 11. 11. Repair/replace wiring, as required, between P261-17 and P289-17 (Appendix F), then go to 12. 12. Procedure completed.

WARNING

Failure to disconnect electrical connectors will permit fuel to enter combustion chamber and create a fire hazard.

TABLE NO. 15-2-1-19. ESU BITE Indicator Shows o o ● ● (Overtemperature).

TEST OR INSPECTION	CORRECTIVE ACTION
1.	Make sure all bleed air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 2.
2.	Perform ESU Overtemperature procedure. Step 1. If APU overtemperature and overspeed are good, go to 3.
3.	Replace combustor drain valve (49-20-05). Go to 4.
4.	Start and operate APU (Appropriate Flight Manual). Go to 5.
5.	Does APU operate normally? Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 15. Step 2. If APU does not operate normally and ESU BITE indicator shows overtemperature, shut down APU (Appropriate Flight Manual). Go to 6.
6.	Disconnect P305, P306, and P307 from fuel valves. Disconnect P303 from ignition exciter. Disconnect fuel line from main fuel valve to the fuel manifold. Go to 7.
7.	Place suitable containers under open fuel lines. Go to 8.
8.	Motor APU and check for fuel flow from main fuel valve. Step 1. If fuel is present, replace main fuel valve (49-20-05). Go to 15. Step 2. If fuel is not present, go to 9.
9.	Leave electrical connectors disconnected. Reconnect fuel line to fuel manifold. Disconnect fuel line at outlet of maximum fuel valve. Go to 10.
10.	Place suitable containers under open fuel lines. Go to 11.

TABLE NO. 15-2-1-19. ESU BITE Indicator Shows o o ● ● (Overtemperature). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
11. Motor APU and check for fuel at maximum fuel valve outlet.	Step 1. If fuel is present, replace maximum fuel valve (49-20-05). Go to 15. Step 2. If fuel is not present, go to 12.
12. Reconnect all electrical connectors and fuel lines. Go to 13.	
13. Start and operate APU (Appropriate Flight Manual). Go to 14.	
14. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual). Go to 15. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Replace fuel pump/acceleration control (49-20-05), then go to 15.
15. Procedure completed.	

TABLE NO. 15-2-1-20. APU Operates, But APU ON Capsule Is Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.	
2. Check that APU ON capsule goes on.	Step 1. If capsule goes on, go to 3. Step 2. If capsule does not go on, go to 10.
3. Install an insulated jumper wire between P260-5 and P260-11. Go to 4.	
4. Place BATT switch to ON. Go to 5.	
5. Check that APU ON capsule goes on.	Step 1. If capsule goes on, replace ESU (PARA 9-4-39). Go to 11. Step 2. If capsule does not go on, go to 6.

TABLE NO. 15-2-1-20. APU Operates, But APU ON Capsule Is Off. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION	
6. Check helicopter configuration.	
Step 1. EMEP , go to 7.	
Step 2. W/O EMEP , go to 9.	
EMEP	7. Remove and check Pin Filter Adapter (PFA) P118 (PARA 9-4-98).
Step 1. If PFA is good, go to 8.	
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 11.	
EMEP	8. Install PFA (PARA 9-4-98). Go to 9.
9. Check for 28 VDC between P118-B and ground.	
Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 11.	
Step 2. If voltage is not as specified, repair/replace wiring between P118-B and P260-11 (Appendix F). Go to 11.	
10. Check APU ON capsule lamps.	
Step 1. If lamps are good, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 11.	
Step 2. If lamps are not good, replace lamps (PARA 8-4-20). Go to 11.	
11. Procedure completed.	

TABLE NO. 15-2-1-21. APU OIL TEMP HI Capsule Goes On, ESU BITE Indicator Shows ● ● ● o (High Oil Temperature).

TEST OR INSPECTION CORRECTIVE ACTION	
1. Check APU oil sump for proper oil level.	
Step 1. If oil level is good, go to 2.	
Step 2. If oil level is not good, service APU oil sump (PARA 1-3-9). Go to 3.	
2. Perform ESU High Oil Temperature procedure.	

TABLE NO. 15-2-1-21. APU OIL TEMP HI Capsule Goes On, ESU BITE Indicator Shows ● ● ● o (High Oil Temperature). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If APU high oil temperature is good, replace APU high oil temperature switch (49-20-05). Go to 3. 3. Procedure completed.

TABLE NO. 15-2-1-22. Fuel Leaks From APU Combustor Drain Line With APU Running.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Disconnect P305 from start fuel valve. Go to 2. 2. Check for fuel leaks. Step 1. If fuel leaks are present, go to 3. Step 2. If fuel leaks are not present, replace ESU (PARA 9-4-39). Go to 5. 3. Cycle valve five times and check that leakage stops. Step 1. If leakage stops, go to 4. Step 2. If leakage does not stop, replace start fuel valve (49-20-05). Go to 5. 4. Check main fuel supply for contamination. Step 1. If contamination is found, purge Fuel System (PARA 10-4-40). Go to 5. Step 2. If contamination is not found, replace start fuel valve (49-20-05). Go to 5. 5. Procedure completed.

TABLE NO. 15-2-1-23. ESU BITE Indicators Do Not Cycle When APU CONTR Switch Is Placed ON.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P260-1 and ground.	Step 1. If voltage is as specified, replace ESU (PARA 9-4-39). Go to 12. Step 2. If voltage is not as specified, go to 2.
2. Check for 28 VDC between J200-R and ground.	Step 1. If voltage is as specified, repair/replace wiring between P200-R and P260-1 (Appendix F). Go to 12. Step 2. If voltage is not as specified, go to 3.
3. Check for 28 VDC between P231-V and ground.	Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 8.
4. Check for 28 VDC between APU T-handle terminal C and ground.	Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, repair/replace wiring between APU T-handle terminal C and J231-V (Appendix F). Go to 12.
5. Check for 28 VDC between APU T-handle switch terminal NC and ground.	Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, replace switch (PARA 12-4-9). Go to 12.
6. Check for 28 VDC between APU CONTR switch terminal 2 and ground.	Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, repair/replace wiring between APU T-handle switch terminal NC and APU CONTR switch terminal 2 (Appendix F). Go to 12.
7. Check for 28 VDC between APU CONTR switch terminal 3 and ground.	Step 1. If voltage is as specified, repair/replace wiring between switch terminal 3 and J200-R (Appendix F). Go to 12. Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 12.
8. Check continuity between P231-V and P242-<u>n</u>.	

**TABLE NO. 15-2-1-23. ESU BITE Indicators Do Not Cycle When APU CONTR Switch Is Placed ON.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.
9. Check continuity between P242-k and ground.	Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.
10. Check for 28 VDC between APU CONTR INST circuit breaker terminal 1 and ground.	Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 12.
11. Check continuity between APU CONTR INST circuit breaker terminal 1 and P242-p.	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 12. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.
12. Procedure completed.	

TABLE NO. 15-2-1-24. APU Does Not Shut Down When APU CONTR Switch Is Placed OFF.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P260-2 and P260-6.	Step 1. If voltage is as specified, replace ESU (PARA 9-4-39). Go to 5. Step 2. If trouble remains, replace main fuel valve or maximum fuel valve (49-20-05). Go to 5. Step 3. If voltage is not as specified, go to 2.
2. Check for 28 VDC between P260-2 and ground.	

TABLE NO. 15-2-1-24. APU Does Not Shut Down When APU CONTR Switch Is Placed OFF. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If voltage is as specified, repair/replace wiring between P260-6 and ground (Appendix F). Go to 5 .	
Step 2. If voltage is not as specified, go to 3 .	
3. Check for 28 VDC between J200-P and ground.	
Step 1. If voltage is as specified, repair/replace wiring between P200-P and P260-2 (Appendix F). Go to 5 .	
Step 2. If voltage is not as specified, go to 4 .	
4. Check for 28 VDC between APU CONTR switch terminal 1 and ground.	
Step 1. If voltage is as specified, repair/replace wiring between switch terminal 1 and J200-P (Appendix F). Go to 5 .	
Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 5 .	
5. Procedure completed.	

TABLE NO. 15-2-1-25. No Power At APU Start Valve.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between P260-9 and P290-1.	Step 1. If continuity is present, go to 2 . Step 2. If continuity is not present, repair/replace wiring between P260-9 and P290-1 (Appendix F), then go to 3 .
2. Check for 28 VDC between P260-1 and ground.	
Step 1. If voltage is as specified, replace ESU (PARA 9-4-39). Go to 3 .	
Step 2. If voltage is not as specified, go to TABLE NO. 15-2-1-24 .	
3. Procedure completed.	

TABLE NO. 15-2-1-26. APU Stall When Either Engine Start Button Is Pressed (A Popping Sound Is Heard).

TEST OR INSPECTION	CORRECTIVE ACTION
1. With APU SOURCE/HEAT START switch placed to APU and with either No. 1 or No. 2 engine start button pressed, check for 28 VDC between P260-10 and ground.	Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 14.
2. Install an insulated jumper wire between P261-18 and P261-19. Go to 3.	
3. Check continuity between P289-18 and P289-19.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, go to 12.
4. Remove jumper wire from P261. Go to 5.	
5. Install an insulated jumper wire between J289-18 and J289-19. Go to 6.	
6. Check continuity between P308-A and P308-B.	Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, go to 10.
7. Remove jumper wire from J289. Go to 8.	
8. Do ESU Start Bypass Valve procedure. Go to 9.	
9. Check that lamp L4 on Test Set comes on.	Step 1. If lamp comes on, replace APU start bypass valve (49-20-05). Go to 27. Step 2. If lamp does not come on, replace ESU (PARA 9-4-39). Go to 27.
10. Remove jumper wire from J289. Go to 11.	
11. Replace APU wiring harness (49-20-05). Go to 27.	
12. Remove jumper wire from P261. Go to 13.	
13. Replace APU wiring harness (49-20-05). Go to 27.	

1. With APU SOURCE/HEAT START switch placed to APU and with either No. 1 or No. 2 engine start button pressed, check for 28 VDC between P260-10 and ground.

Step 1. If voltage is as specified, go to 2.
 Step 2. If voltage is not as specified, go to 14.

2. Install an insulated jumper wire between P261-18 and P261-19. Go to 3.

3. Check continuity between P289-18 and P289-19.

Step 1. If continuity is present, go to 4.
 Step 2. If continuity is not present, go to 12.

4. Remove jumper wire from P261. Go to 5.

5. Install an insulated jumper wire between J289-18 and J289-19. Go to 6.

6. Check continuity between P308-A and P308-B.

Step 1. If continuity is present, go to 7.
 Step 2. If continuity is not present, go to 10.

7. Remove jumper wire from J289. Go to 8.

8. Do ESU Start Bypass Valve procedure. Go to 9.

9. Check that lamp L4 on Test Set comes on.

Step 1. If lamp comes on, replace APU start bypass valve (49-20-05). Go to 27.
 Step 2. If lamp does not come on, replace ESU (PARA 9-4-39). Go to 27.

10. Remove jumper wire from J289. Go to 11.

11. Replace APU wiring harness (49-20-05). Go to 27.

12. Remove jumper wire from P261. Go to 13.

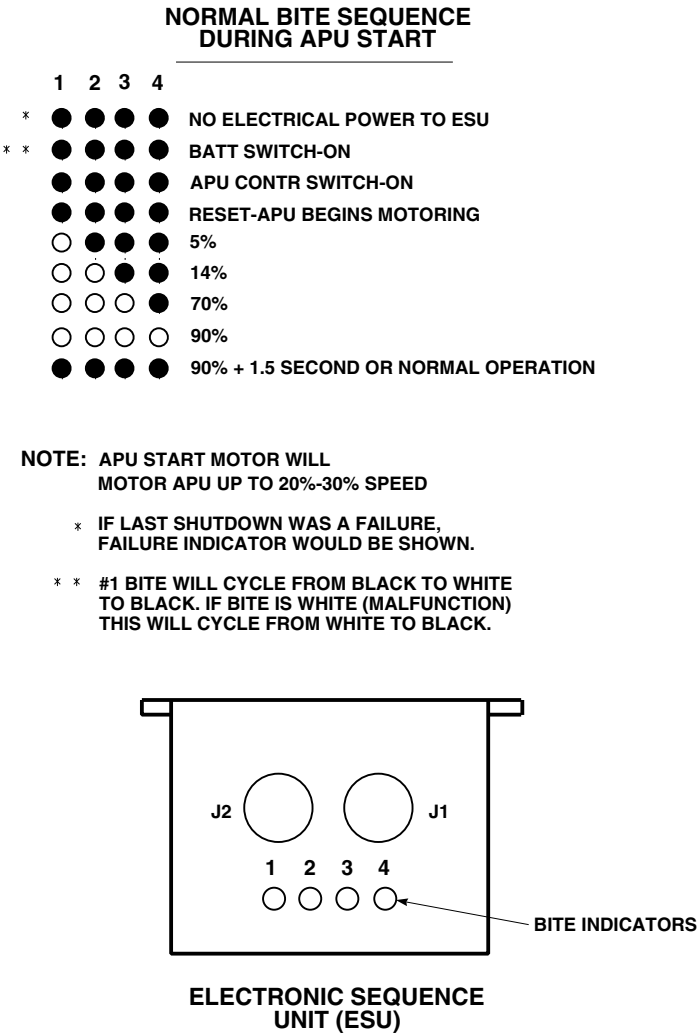
13. Replace APU wiring harness (49-20-05). Go to 27.

TABLE NO. 15-2-1-26. APU Stall When Either Engine Start Button Is Pressed (A Popping Sound Is Heard). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
14. Check helicopter configuration.	Step 1. WINTER , go to 15. Step 2. W/O WINTER , go to 16.
WINTER	15. Check continuity between: J244A-M and P244B-M J244A-S and P244B-S J244A-T and P244B-T J244A-U and P244B-U Step 1. If continuity is present, go to 16. Step 2. If continuity is not present, replace winterization kit harness assembly (PARA 16-6-2). Go to 27. 16. Check for 28 VDC between P244-T and P244-U. Step 1. If voltage is as specified, go to 17. Step 2. If voltage is not as specified, go to 23. 17. Install an insulated jumper wire between P244-T and P244-S. Go to 18. 18. Check for 28 VDC between P260-10 and ground. Step 1. If voltage is as specified, go to 19. Step 2. If voltage is not as specified, go to 21. 19. Remove jumper wire from P244. Go to 20. 20. Replace right relay panel (PARA 9-4-2). Go to 27. 21. Remove jumper wire from P244. Go to 22. 22. Repair/replace wiring between P244-S and P260-10 (Appendix F). Go to 27. 23. Check continuity between P244-U and ground. Step 1. If continuity is present, go to 24. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 27.

TABLE NO. 15-2-1-26. APU Stall When Either Engine Start Button Is Pressed (A Popping Sound Is Heard). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
24. Check for 28 VDC between APU SOURCE HEAT/START circuit breaker terminal 1 and ground.	Step 1. If voltage is as specified, go to 25. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 27.
25. Check continuity between P244-T and AIR SOURCE HEAT/START switch S18 terminal 1.	Step 1. If continuity is present, go to 26. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 27.
26. Check continuity between APU SOURCE HEAT/START circuit breaker terminal 1 and AIR SOURCE HEAT/START switch S18 terminal 2.	Step 1. If continuity is present, replace switch S18 (PARA 9-4-5). Go to 27. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 27.
27. Procedure completed.	



NOTES

- 1. HELICOPTERS WITH APU ESU
160200-201.
- 2. HELICOPTERS WITH APU ESU
160200-600.
- 3. HELICOPTERS WITH APU ESU
163290-100.

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FIGURE 15-2-1-1. ELECTRONIC SEQUENCE UNIT (ESU) BITE INDICATOR CONDITIONS
(SHEET 1 OF 2)

T-62T-40-1					DECODED BITE INFORMATION	
BITE # 1 2 3 4						
● ● ● ●	RESET					START SEQUENCE
○ ● ● ●	START FUEL VALVE & EXCITER SIGNAL OUT (5%)					
○ ○ ● ●	MAIN FUEL VALVE SIGNAL OUT (14%)					
○ ○ ○ ●	START FUEL VALVE & EXCITER SIGNAL OFF(70%)					
○ ○ ○ ○	90% RPM SWITCH ON					
● ● ● ●	READY FOR SERVICE (90% + 1.5 SEC)					OPERATION
○ ● ● ●	PROCESSOR BOARD FAILURE					
● ○ ● ●	SENSOR / DATA BOARD FAILURE					
○ ○ ● ●	OVERTEMPERATURE					
● ● ○ ●	OVERSPEED					
○ ● ○ ●	UNDERSPEED					
● ○ ○ ●	FAIL TO START					
○ ○ ○ ●	LOW OIL PRESSURE					
● ● ● ○	HIGH OIL TEMPERATURE (WARNING)					
○ ● ● ○	SHORTED OIL PRESSURE SWITCH					
● ○ ● ○	SHORTED THERMOCOUPLE PROBE (WARNING)					
● ● ○ ○	OPEN THERMOCOUPLE					
● ○ ○ ○	PROCESSOR SEQUENCE FAIL					
○ ○ ○ ○	NO DATA					
○ ● ○ ○	LOSS OF SPEED DATA					

ESU PART NO. 160200-600

T-62T-40-1				DECODED BITE INFORMATION	START SEQUENCE
BITE #					
1	2	3	4		
●	●	●	●	RESET	OPERATION
○	●	●	●	START FUEL VALVE & EXCITER SIGNAL OUT (5%)	
○	○	●	●	MAIN FUEL VALVE SIGNAL OUT (14%)	
○	○	○	●	START FUEL VALVE & EXCITER SIGNAL OFF (70%)	
○	○	○	○	90% RPM SWITCH ON	
●	●	●	●	READY FOR SERVICE (90% + 1.5 SEC)	
○	●	●	●	PROCESSOR BOARD FAILURE	
●	○	●	●	SENSOR / DATA BOARD FAILURE	
○	○	●	●	OVERTEMPERATURE	
●	●	○	●	OVERSPEED	
○	●	○	●	UNDERSPEED	
●	○	○	●	FAIL TO START	
○	○	○	●	LOW OIL PRESSURE	
●	●	●	○	HIGH OIL TEMPERATURE (WARNING)	
○	●	●	○	SHORTED OIL PRESSURE SWITCH	
●	○	●	○	SHORTED THERMOCOUPLE PROBE (WARNING)	
●	●	○	○	OPEN THERMOCOUPLE	
●	○	○	○	PROCESSOR SEQUENCE FAIL	
○	○	○	○	NO DATA	
○	○	●	○	EXTERNAL SHORT	
○	●	○	○	LOSS OF SPEED DATA	

ESU PART NO. 163290-100

T-62T-40-1				DECODED BITE INFORMATION	
BITE #					
1	2	3	4		START SEQUENCE
●	●	●	●	RESET	OPERATION
○	●	●	●	START FUEL VALVE & EXCITER SIGNAL OUT (5%)	
○	○	●	●	MAIN FUEL VALVE SIGNAL OUT (14%)	
○	○	○	●	START FUEL VALVE & EXCITER SIGNAL OFF (70%)	
○	○	○	○	90% RPM SWITCH ON	
●	●	●	●	READY FOR SERVICE (90% + 1.5 SEC)	
○	●	●	●	PROCESSOR BOARD FAILURE	
●	○	●	●	SENSOR / DATA BOARD FAILURE	
○	○	●	●	OVERTEMPERATURE	
●	●	○	●	OVERSPEED	
○	●	○	●	UNDERSPEED	
●	○	○	●	FAIL TO START	
○	○	○	●	LOW OIL PRESSURE	
●	●	●	○	HIGH OIL TEMPERATURE (WARNING)	
○	●	●	○	FAIL TO LIGHT	
●	○	○	○	SHORTED THERMOCOUPLE PROBE (WARNING)	
●	●	○	○	OPEN THERMOCOUPLE	
●	○	○	○	PROCESSOR SEQUENCE FAIL	
○	○	○	○	NO DATA	

ESU PART NO. 160200-201

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SAFIGURE 15-2-1-1. ELECTRONIC SEQUENCE UNIT (ESU) BITE INDICATOR CONDITIONS
(SHEET 2 OF 2)

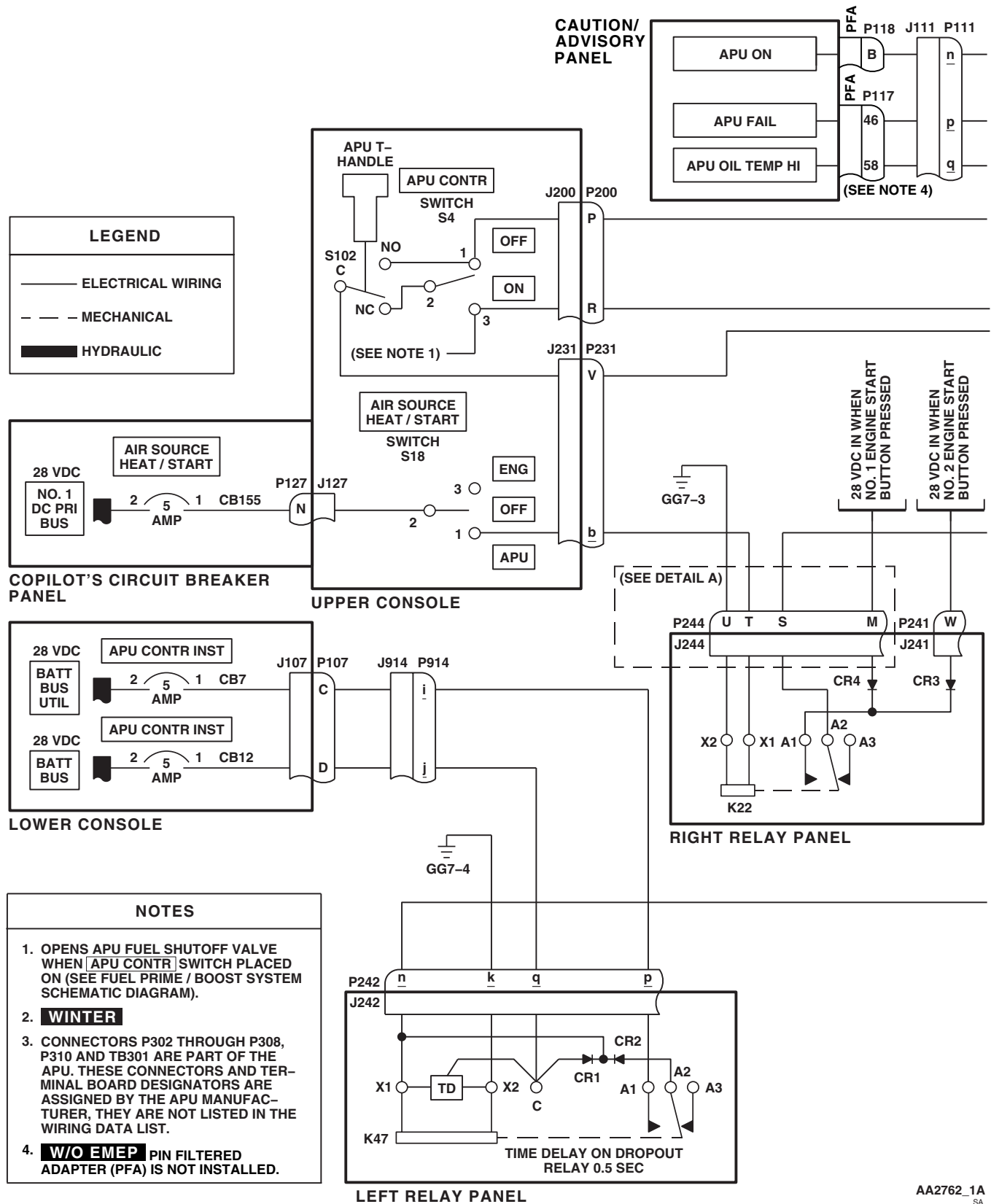


FIGURE 15-2-1-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 4)

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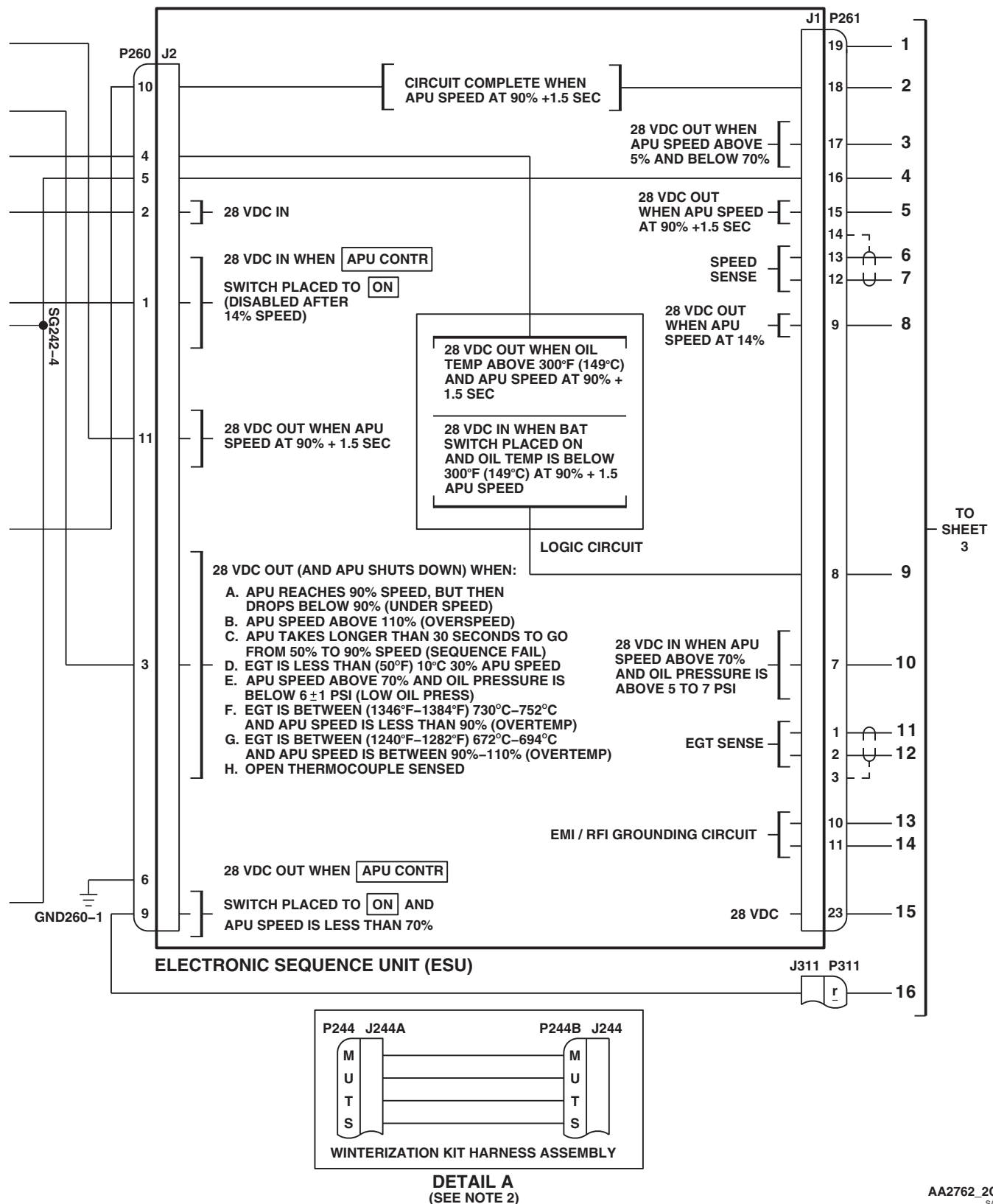


FIGURE 15-2-1-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 4)

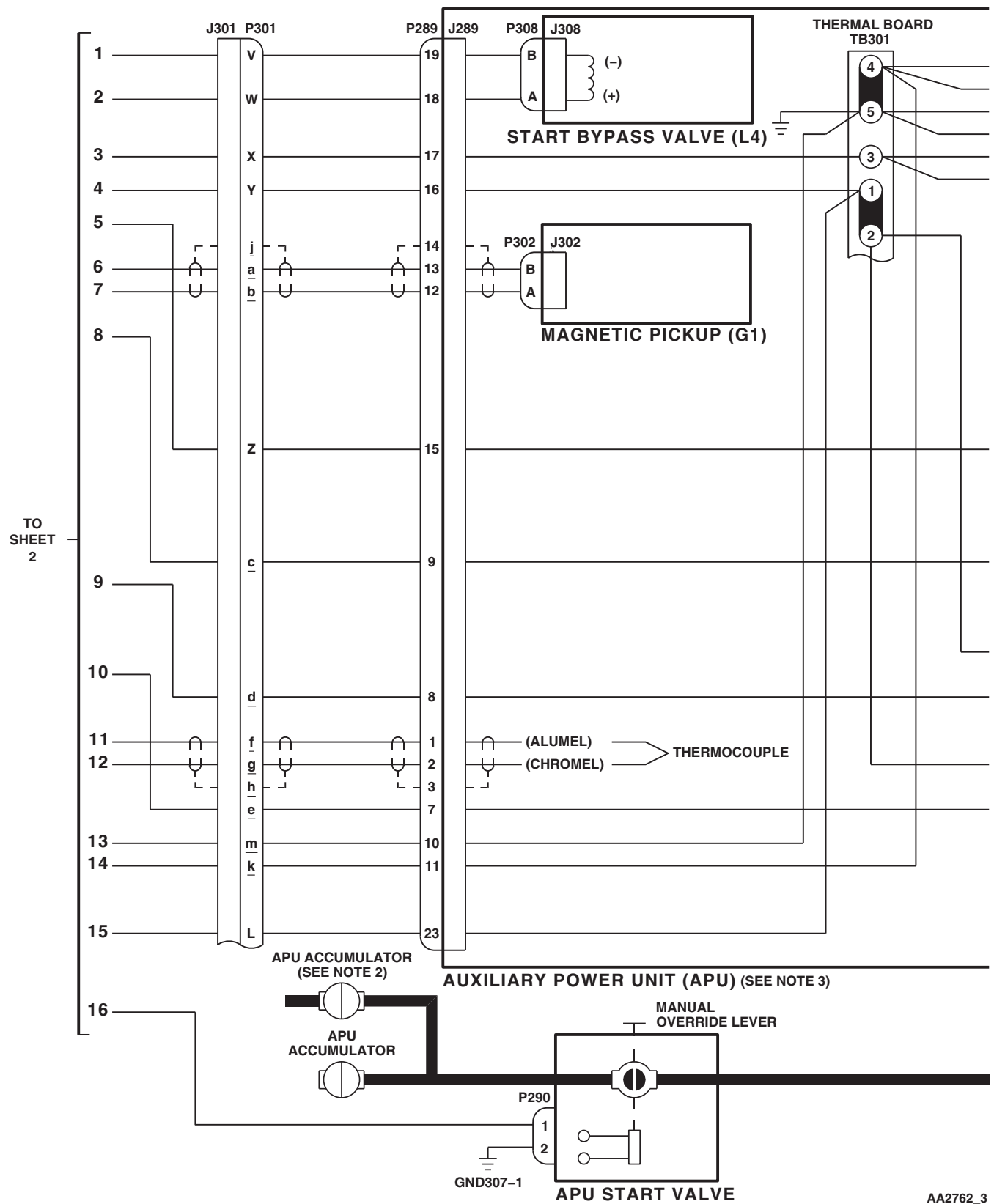


FIGURE 15-2-1-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 4)

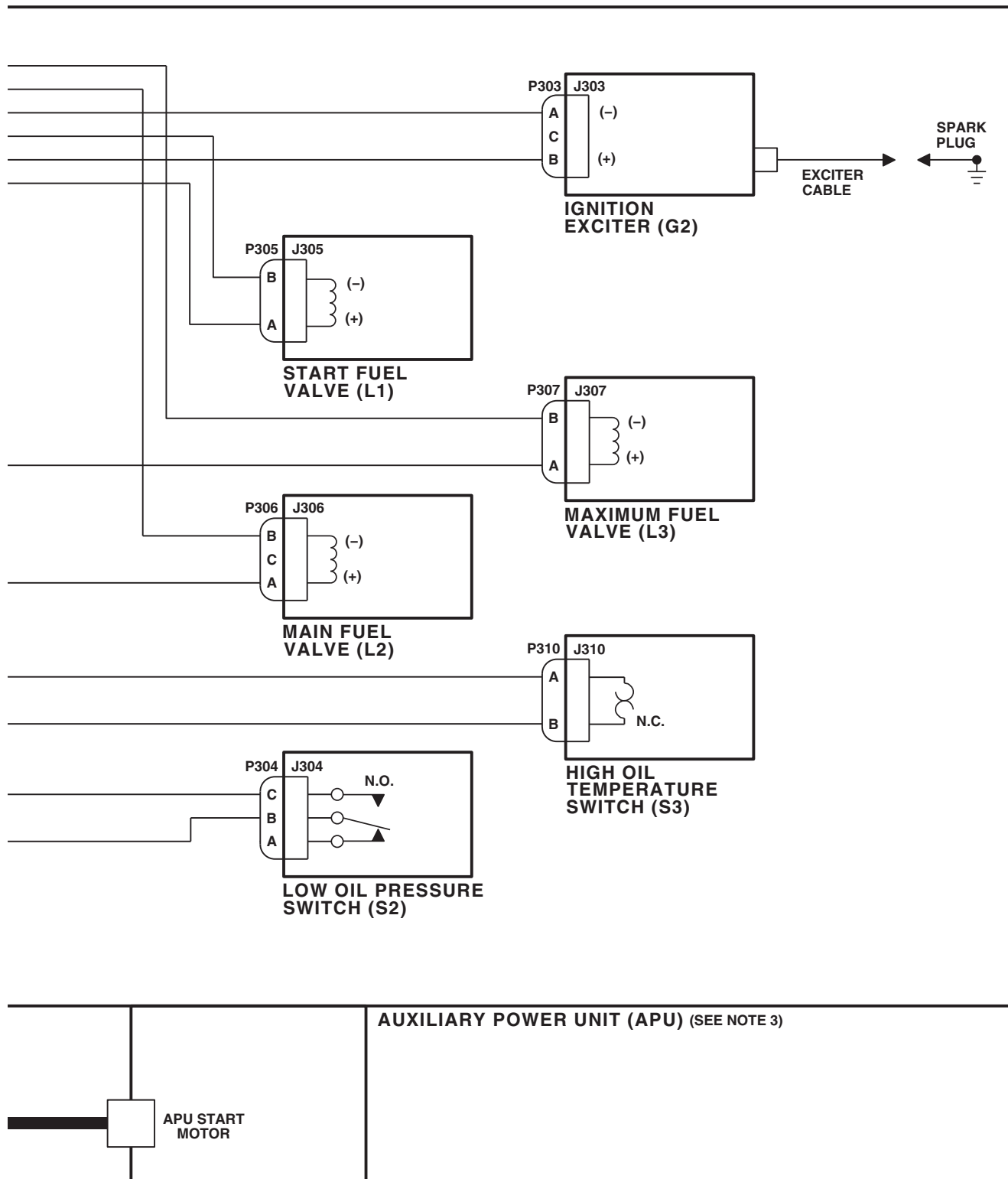
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FIGURE 15-2-1-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 4)

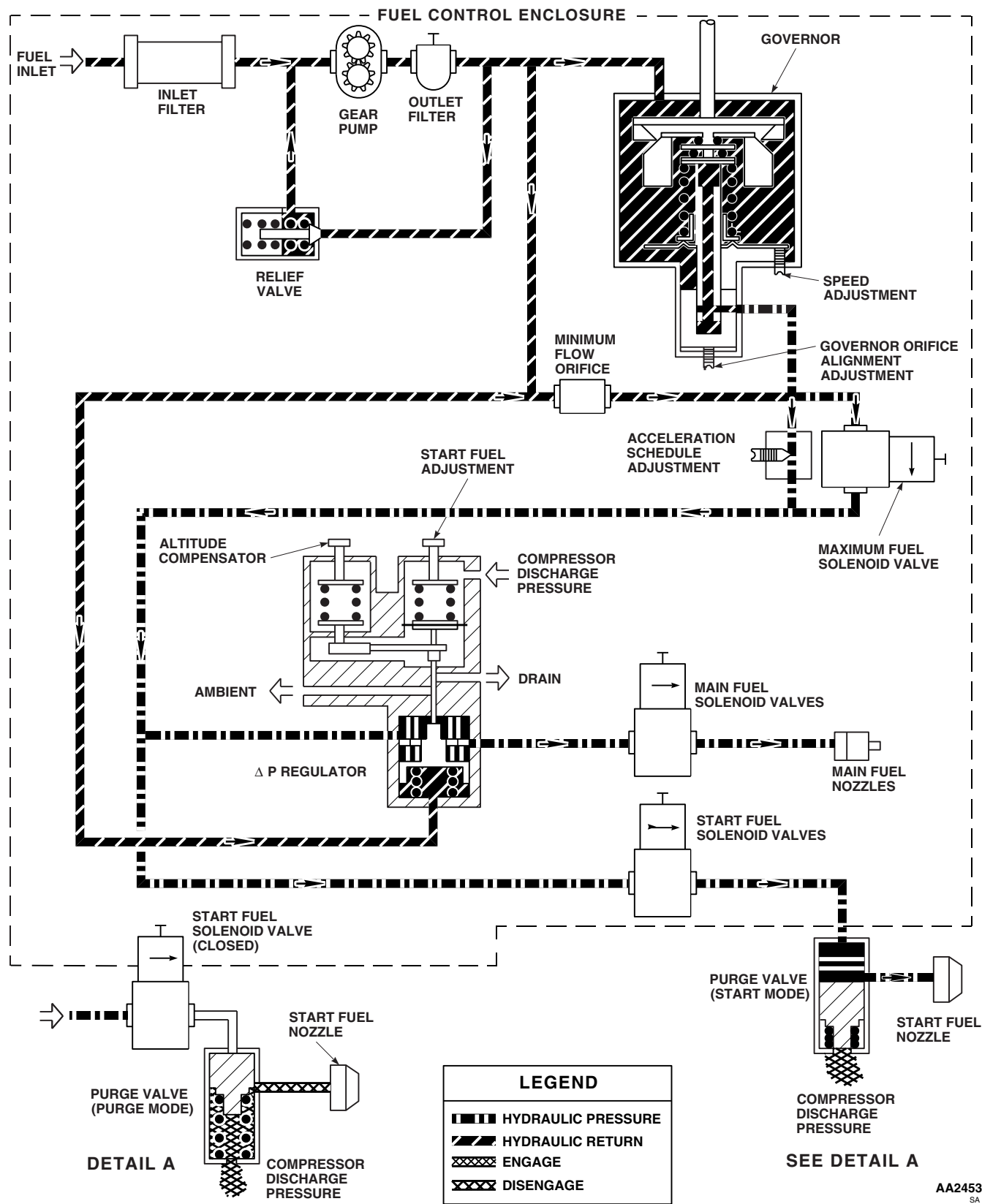


FIGURE 15-2-1-3. AUXILIARY POWER UNIT FUEL SYSTEM SCHEMATIC

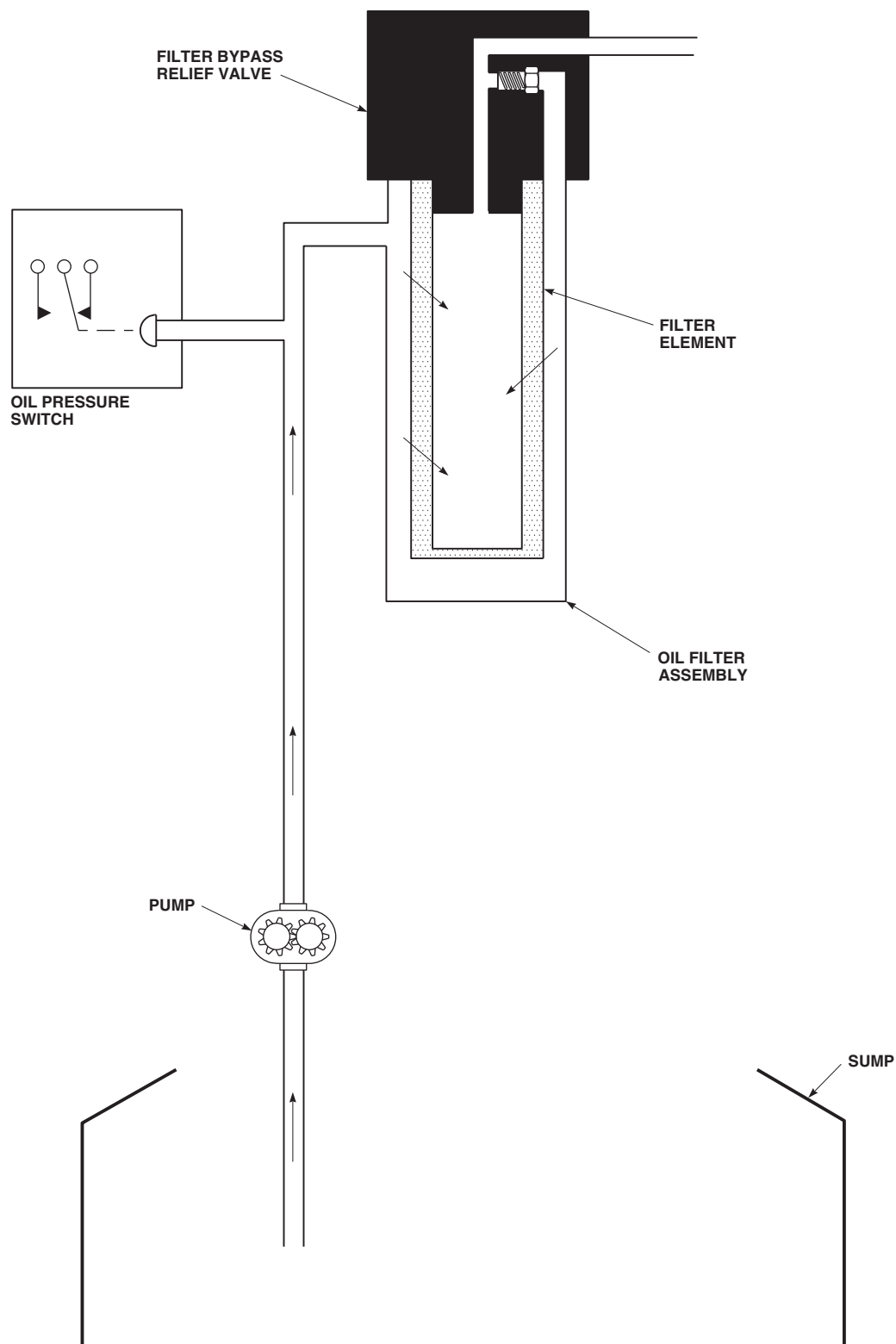
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FIGURE 15-2-1-4. AUXILIARY POWER UNIT LUBRICATION SYSTEM SCHEMATIC

15-2-2 AUXILIARY POWER UNIT SYSTEM USING TEST SET H296B-1.

INITIAL SETUP

Applicable Configuration

- Auxiliary Power Unit (APU) 116305-200, 116305-300, and 116305-302

Test Equipment

- Multimeter, Fluke 27/AN
- Multimeter, Fluke 45
- APU Test Set (Test Set), Simulator, Temp./Speed, H296B-1

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- Appropriate Flight Manual
- PARA 1-6-16
- PARA 7-4-8
- PARA 2-4-69

Equipment Conditions

- APU Accumulator Properly Serviced
- Battery Fully Charged
- External Electrical Power Available (for Dynamic and Static checks)
- No. 1 Fuel Tank Serviced

WARNING

To prevent injury to personnel and damage to helicopter, have fire guard

check APU fuel drain. Wait at least two minutes after APU shutdown before attempting a restart. DO NOT RESTART APU until fuel drainage has stopped.

CAUTION

- To prevent damage to helicopter, wait at least two minutes after APU shutdown before attempting a restart.
- To prevent possible damage to APU, do not operate APU more than 30 minutes when main engines and rotor are running and outside temperature is greater than 109°F (43°C). Continuous operation of APU is permitted at temperatures up to 124°F (51°C) only if main engines and rotors are not running.
- On helicopters with Electronic Sequence Unit (ESU) 160200-201, cycling of the BATT switch during APU coast-down may cause the APU to restart with a possible overtemperature.
- If onboard fire extinguishing is required before a generator becomes operational, RESERVE fire extinguisher must be used to fight fire.

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).
- To prevent discharge of APU accumulator, disconnect P290 from APU start valve.
- For proper results, switch positions and order of operation must be strictly followed.

Special Instructions

The Auxiliary Power Unit System operational/troubleshooting procedure is subdivided as follows:

(1) APU

(2) Dynamic

(3) Static

(4) Shutdown

15-2-2.1 APU.

NOTE

Do not connect external electrical power to helicopter before starting or during APU System operational/troubleshooting procedure.

- a. Make sure all circuit breakers are pushed in except those noted being pulled for maintenance or safety.
- b. Make sure upper console BATT switch and GENERATORS APU switch is placed to RESET/OFF (center position).
- c. Check bleed air pneumatic tube couplings for loose clamps, cuts, and/or incorrect installation at all connecting points (PARA 7-4-8).

RESULT: All couplings shall be securely installed with clamps and there shall be no cuts in pneumatic tube lines.

- If result is not as specified, repair/replace couplings and lines (PARA 7-4-8).
- d. Remove soundproofing material and panel from cabin ceiling to visually check Electronic Sequence Unit (ESU) BITE indicators (PARA 2-4-69).

NOTE

- An observer must be in the cabin area to watch the ESU BITE indicators during the operational/troubleshooting procedure.
- The observer shall call out the procedural steps to the operator in the

cockpit. This will allow the observer to watch for a change in ESU BITE indication as the operator does the next procedural step.

e. Visually check ESU BITE indicators.

RESULT: All indicators shall be black.

- If result is not as specified, troubleshoot System for indicated malfunction.

NOTE

- Do not attempt to start the APU if BATT LOW CHARGE capsule is on.
- TABLE NO. 15-2-2-1 lists the sequence of events for a normal start.
- The APU will not start if any of the following ESU internal failures appear as BITE indications:

BITE INDICATION	FAILURE
o ● ● ●	Processor Board Failure
● o ● ●	Sensor/Data Board Failure
● o o o	Processor Sequence Fail
o o o o	No Data

f. Place upper console BATT switch to ON. Watch ESU BITE indicators.

RESULT: No. 1 BITE indicator shall cycle from black to white to black. If BITE indicator is white, due to malfunction, it shall only cycle from white to black.

- If result is not as specified, go to TABLE NO. 15-2-2-3.

- g. Place upper console FUEL PUMP switch to APU BOOST.

RESULT:

- (1) PRIME BOOST PUMP ON capsule on caution/advisory panel shall go on.
- (2) No fuel leakage shall occur from APU combustor drain line (below helicopter).
- If result (1) is not as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3).
- If result (2) is not as specified, go to TABLE NO. 15-2-2-4.

NOTE

- Prime boost pump must be running for all APU motoring and start operations.
 - A minimal amount of fuel may be evident during and shortly after start.
 - On helicopters with ESU 160200-201, if the APU should start to motor and then shut down during an attempted start, do not place BATT switch to OFF. Any cycling of BATT switch will reset the ESU and malfunction BITE indications (indicating the cause of shutdown) will be lost.
 - If the APU shuts down, the observer shall take note of the last APU speed BITE indication before the malfunction BITE indication appears.
 - Make sure upper console APU T-handle is pushed in or APU will not start.
 - If a restart is being attempted, the original malfunction BITE indication will be shown.
- h. Start APU (Appropriate Flight Manual).

RESULT:

- (1) APU shall start and BITE indicators shall cycle as follows:

**BITE
INDICATION**

- | | |
|---------|---------------------------------------|
| o ● ● ● | 5% |
| o o ● ● | 14% |
| o o o ● | 70% |
| o o o o | 90% |
| ● ● ● ● | 90% + 1.5 seconds (ready for service) |
- (2) APU ON capsule on caution/advisory panel shall go on.
 - (3) APU FAIL capsule on caution/advisory panel shall be off.



If APU OIL TEMP HI capsule is on, shut down APU immediately.

- (4) APU OIL TEMP HI capsule on caution/advisory panel shall be off.
- (5) No fuel leakage shall occur from APU combustor drain line (below helicopter).
- If shorted thermocouple (● o ● o) is present and a restart attempt fails, go to TABLE NO. 15-2-2-5.
- On helicopters with ESU 160200-201, if fail to light (o ● ● o) is present after a restart attempt fails, go to TABLE NO. 15-2-2-6.
- On helicopters with ESU 160200-600, if shorted oil pressure switch indication (o ● ● o) is shown, go to TABLE NO. 15-2-2-7.
- If APU does not motor, go to TABLE NO. 15-2-2-8.
- If result (1) is not as specified, go to TABLE NO. 15-2-2-3.
- If APU motors, but shuts down before result (2) occurs and APU FAIL capsule is on, visually check ESU BITE indications and perform the following:

IF BITE INDICATION IS: ACTION: Go To

● ● ○ ● (fail to start)	TABLE NO. 15-2-2-9
○ ● ○ ○ (loss of speed data)	TABLE NO. 15-2-2-10
○ ○ ● ● (overtemperature)	TABLE NO. 15-2-2-11
○ ○ ○ ● (low oil pressure)	TABLE NO. 15-2-2-12
● ● ○ ○ (open thermocouple)	TABLE NO. 15-2-2-13
○ ○ ● ○ (external short after, ○ ● ● ● start fuel valve and exciter signal out)	TABLE NO. 15-2-2-14
○ ○ ● ○ (external short after, ○ ○ ● ● signal out; 14%)	TABLE NO. 15-2-2-15
○ ○ ● ○ (external short after, ○ ○ ○ ○ 90% rpm switch on or ● ● ● ● ready for service)	TABLE NO. 15-2-2-16

- If APU motors (attempts to start), but shuts down after result (2) occurs (APU ON capsule goes off and APU FAIL capsule goes on), go to TABLE NO. 15-2-2-17.
- If APU motors, but shuts down after APU ON capsule goes on, then off, and APU FAIL capsule is off, visually check ESU BITE indications and perform the following:

IF BITE INDICATION IS: ACTION: Go To

○ ● ○ ● (underspeed)	TABLE NO. 15-2-2-18
● ● ○ ● (overspeed)	TABLE NO. 15-2-2-19
○ ○ ● ● (overtemperature)	TABLE NO. 15-2-2-20
○ ○ ○ ● (low oil pressure)	TABLE NO. 15-2-2-12

IF BITE INDICATION IS: ACTION: Go To

● ● ○ ○ (open thermocouple)	TABLE NO. 15-2-2-13
--------------------------------	---------------------

- If APU operates, but result (2) is not as specified, go to TABLE NO. 15-2-2-21.
- If result (4) is not as specified, go to TABLE NO. 15-2-2-22.
- If result (5) is not as specified, go to TABLE NO. 15-2-2-23.
- If ESU BITE indicators do not cycle as specified, go to TABLE NO. 15-2-2-24.

- Recharge APU accumulator by placing GENERATORS APU switch to ON. Visually check accumulator for proper charge (pressure gauge and tape indicator).

RESULT: APU ACCUM LOW capsule on caution/advisory panel shall go off in 90 seconds and APU accumulator shall be charged.

- If result is not as specified, troubleshoot Hydraulic Systems (PARA 7-2-1).
 - If APU shuts down and underspeed BITE indication (○ ● ○ ●) appears on ESU, replace left relay panel (PARA 9-4-2).
- Place upper console GENERATORS APU switch to OFF.

- With qualified personnel at the controls, start No. 1 engine.

RESULT: No. 1 engine shall start.

- If result is not as specified, perform Dynamic and Static checks.

- With qualified personnel at the controls, start No. 2 engine (Appropriate Flight Manual).

RESULT: No. 2 engine shall start.

- If result is not as specified, perform Dynamic and Static checks.
- m. Shut down No. 1 and No. 2 engines (Appropriate Flight Manual).
- n. Shut down APU (Appropriate Flight Manual).

RESULT:

- (1) APU shall shut down.
- (2) APU ON capsule shall go off.
- If result (1) is not as specified, place FUEL PUMP switch to OFF and go to TABLE NO. 15-2-2-25.
- If result (2) is not as specified, disconnect P260 on ESU.

(a) If APU ON capsule goes off, replace ESU (PARA 9-4-39).

(b) If APU ON capsule does not go off, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- o. Check ESU BITE indicators after APU is shut down.

RESULT: All BITE indicators shall be black.

- If result is not as specified, troubleshoot System for indicated malfunction.
- p. Place upper console FUEL PUMP switch to OFF.

RESULT: PRIME BOOST PUMP ON capsule on caution/advisory panel shall go off.

- If result is not as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3).
- q. Visually check that there is no fuel or fuel vapors coming from combustor drain line (below helicopter).
- If result is not as specified, check APU combustor drain check valve. Replace if necessary (49-20-05).
- r. Place upper console BATT switch to OFF.
- s. If no other check is to be performed, go to Shutdown.

TABLE NO. 15-2-2-1. Normal Start Sequence.

APU SPEED SEQUENCE (%RPM)	EVENT
A.5%	1. Start fuel solenoid opens
	2. Ignition exciter goes on
	3. 20 - 30 second counter (inside ESU) starts
	*4. EGT overtemperature shutdown circuit (inside ESU) limit extended to 1,376°F (740°C)
	5. Fuel pressure built up due to motoring speed
	*6. Thermocouple checked for open condition (inside ESU)

TABLE NO. 15-2-2-1. Normal Start Sequence. (Cont)

APU SPEED SEQUENCE (%RPM)	EVENT
	7. ESU internal checks for processor sequence failure, sensor/data board failure, and data failure
	8. Ignition starts
	*9. Overspeed circuit activates
	***BITE INDICATION: o ● ● ●
B.14%	1. Main fuel solenoid opens
	2. Ignition continues
	3. ESU internal self-checks continue
	*4. Thermocouple checked for open condition (inside ESU)
	***BITE INDICATION: o o ● ●
C.30%	*1. Inside ESU -APU exhaust gas temperature is checked for being above 50°F (10°C)
	2. ESU internal self-checks continue
D.70%	1. APU start valve (hydraulic) - closes
	2. Start fuel valve - closes
	3. Purge valve - opens (due to air pressure)
	4. Ignition - off
	*5. Oil pressure circuit - on
	6. ESU internal self-checks are made
	*7. Thermocouple checked for open condition (inside ESU)
	***BITE INDICATION: o o o ●
E.90%	1. This occurs in ESU:
	*a. 20 - 30 second counter - disabled

TABLE NO. 15-2-2-1. Normal Start Sequence. (Cont)

APU SPEED SEQUENCE (%RPM)	EVENT
	*b. APU fail circuit - deactivated
	*c. Underspeed circuit - activated
	d. 1.5-second timer - activated
	e. ESU internal self-checks continue
	*f. Thermocouple checked for open condition (inside ESU)
	***BITE INDICATION: o o o o
F.90% - 1.5 Seconds	1. Maximum fuel valve - opens
	2. APU ON capsule (caution/advisory panel) - on
	3. ESU internal self-checks continue
	**4. Thermocouple checked for short condition (inside ESU)
	**5. High oil temperature check
	*6. EGT shutdown limit - reduced to 1,240 - 1,280°F (656 - 679°C)
	***BITE INDICATION: ● ● ● ●
G. APU Operating - 102% (Unloaded) APU Operating - 100% (Loaded)	
*ESU checks on System operation. If check is not satisfactory, APU will shut down. **Warning indication that will appear as a BITE indication, but will not cause APU to shut down. ***BITE indication will appear on ESU when APU has reached corresponding speed during start sequence.	

15-2-2.2 Dynamic.

NOTE

An observer must be in the cabin area to watch the ESU BITE indicators, one operator in the cockpit to work the APU

- controls, and one operator to operate the test set during the operational/troubleshooting procedure.
- Perform APU check, if not already performed.
 - Turn off electrical power, if not already performed.

- c. Connect test set (test set cable BH25258), see Figure 15-2-2-6 (Sheet 1), APU Dynamic Test Conditions.
- d. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- e. Turn on external electrical power (PARA 1-6-16).

RESULT:

- (1) Caution/advisory panel EXT PWR CONNECTED capsule shall go on.
- (2) ESU No. 1 BITE indicator shall cycle from black to white back to black.
- If result (1) is not as specified, troubleshoot AC System (PARA 9-2-1).
- If result (2) is not as specified, go to TABLE NO. 15-2-2-3.

- f. Place APU Test Set switches as indicated:

SWITCH	POSITION
POWER (S1)	ON
OPEN T/C/NORM (S5)	NORM
TEMP OUTPUT 752 - 1,472°F (400 - 800°C) (R1)	32°F (0°C)
LOP/NORM/HI OIL TEMP (S3)	NORM
RPM FINE (F1)	0
RPM COARSE (S2)	0%

NOTE

L6 LOW OIL PRESSURE lamp shall be on when APU is below 20%.

- g. Press test set LAMP TEST switch (S4).

RESULT: Test set display lamps L1 through L5 shall go on.

- If result is not as specified, replace test set.
- h. On upper console, place FUEL PUMP switch to APU BOOST.
- i. Make sure that test % RPM, TEMP °CX100, and SECONDS displays indicate 0.

NOTE

Read steps j. and k. completely before proceeding.

- j. Simultaneously place upper console APU CONTR to ON and press test set START/STOP RESET (S6) switch to start timer.

NOTE

- Test set HIGH OIL TEMP lamp (L5) will not go on unless oil temperature becomes greater than 275 - 300°F (135° - 149°C).
- Overtemperature is 1,364 - 1,429°F (740 - 776°C) for less than 90% RPM and 1,242 - 1,281°F (672 - 694°C) if RPM is between 90% and 100%.
- It should not take over 10 seconds in normal temperature, or 40 seconds in cold temperatures for APU to reach 90% RPM.
- k. Observe test set % RPM gauge and timer for proper sequence of test set lamps as indicated in TABLE NO. 15-2-2-2.
- l. Observe ESU BITE indications during APU start.

RESULT: ESU BITE indicators shall cycle to (● ● ●) READY FOR SERVICE.

- If result is not as specified, go to APU System check step h.

- m. On upper console, place GENERATORS APU switch to ON to recharge APU accumulator.
- n. After APU accumulator has been charged within 90 seconds, place GENERATORS APU switch to RESET (center position).

NOTE

Test set LOW OIL PRESSURE lamp (L6) shall go on when APU decelerates to 20%.

- o. Shut down APU (Appropriate Flight Manual).

RESULT: APU shall shutdown.

- If result is not as specified, go to TABLE NO. 15-2-2-25.

- p. If no other check is to be performed, go to Shutdown.

TABLE NO. 15-2-2-2. Test Set Lamp Sequence Chart.

RPM-PERCENT	5%	14%	70%	90%	90% + 1.5 TO 3.5 SEC	
L1 EXCITER/ START FUEL VALVE	ON	ON				
L2 MAIN FUEL VALVE		ON	ON	ON	ON	
L3 MAX FUEL VALVE					ON	
L4 BYPASS VALVE						
L5 HIGH OIL TEMP					ON	
L6 LOW OIL PRESSURE	ON	ON/ OFF				

15-2-2.3 Static.

NOTE

An observer must be in the cabin area to watch the ESU BITE indicators, one operator in the cockpit to work the APU controls, and one operator to operate the test set during the operational/troubleshooting procedure.

- a. Perform APU check, if not already performed.
- b. Turn off electrical power, if not already performed.
- c. Disconnect P290 from APU start valve.
- d. Connect test set (test set cable BH25257), see Figure 15-2-2-6 (Sheet 1), APU Static Test Conditions.
- e. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- f. Place APU Test Set switches as indicated:

SWITCH	POSITION
POWER (S1)	ON
TEMP OUTPUT 752 - 1,472°F (400 - 800°C) (R1)	1,076°F (580°C)
LOP/NORM/HI OIL TEMP (S3)	LOP
RPM FINE (F1)	0
RPM COARSE (S2)	0%
OPEN T/C/NORM (S5)	NORM

- g. Turn on external electrical power (PARA 1-6-16).

RESULT:

- (1) Caution/advisory panel EXT PWR CONNECTED capsule shall go on.
 - (2) ESU No. 1 BITE indicator shall cycle from black to white back to black.
- If result (1) is not as specified, troubleshoot AC System (PARA 9-2-1).
 - If result (2) is not as specified, go to TABLE NO. 15-2-2-3.

NOTE

With ESU 160200-201 installed, test set LOW OIL PRESSURE (L6) lamp will not go on with LOP/NORM/HI OIL TEMP (S3) switch at NORM.

- h. Momentarily press test set LAMP TEST (S4) switch.

RESULT:

- (1) Test set lamps L1 through L5 shall go on, then go off when switch is released.
 - (2) Test set LOW OIL PRESSURE (L6) lamp shall be on when APU is below 20% RPM.
- If results are not as specified, replace test set.

NOTE

- Read steps i. through n. completely before proceeding.
- Steps i. through n. must be accomplished within 29 to 40 seconds. If not accomplished within time specified (with ESU 160200-201 installed), a FAIL TO LIGHT (o ● ● o) indication shall result. With any other ESU installed a FAIL TO START (● o o ●) indication shall result.

- i. Simultaneously place upper console APU CONTR switch to ON and press test set START/STOP RESET (S6) switch.
- j. Place the test set switches listed in Chart No. 15-2-2-1 and observe specified indications.

CHART NO. 15-2-2-1. ESU BITE Indications.			
TEST SET TIMER (SECONDS)	TEST SET SETTING	TEST SET LAMP INDICATION	ESU BITE INDICATION
2	RPM COARSE (S2) 5%	L1 - ON	1 WHITE
3	RPM COARSE (S2) 15%	L2 - ON	2 WHITE
5	LOP/NORM/HI/ OIL TEMP (S3) NORM	L6 - OFF	----
9	RPM COARSE (S2) 70%	L1 - OFF	3 WHITE
11	RPM COARSE (S2) 90% RPM FINE (F1) +5	L3 - ON	4 WHITE THEN ALL SHALL TURN BLACK BEFORE 3.5 SECONDS

RESULT: If results are not as specified, replace ESU (PARA 9-4-39).

- k. On upper console, place AIR SOURCE to APU.
- l. Press and hold No. 1 engine start button.

RESULT: Test set BYPASS VALVE (L4) lamp shall go on.

- If result is not as specified, replace ESU (PARA 9-4-39).

- m. Release No. 1 engine start button.
- n. Press and hold No. 2 engine start button.

RESULT: Test set BYPASS VALVE (L4) lamp shall go on.

- If result is not as specified, replace ESU (PARA 9-4-39).

- o. Release No. 2 engine start button.
- p. If no other check is to be performed, go to Shutdown.

15-2-2.4 Shutdown.

- a. Make sure upper console AIR SOURCE HEAT/START switch is in OFF position.
- b. Make sure upper console APU CONTR switch is in OFF position.
- c. Turn off electrical power (PARA 1-6-16).
- d. If connected, disconnect test set and reconnect helicopter harness connectors to their mating connectors.
- e. Install panel and soundproofing material over ESU on cabin ceiling (PARA 2-4-69).

TABLE NO. 15-2-2-3. ESU BITE Indicators Do Not Cycle When BATT Switch Is Placed ON.

TEST OR INSPECTION	CORRECTIVE ACTION
I	1. Check for 28 VDC between P260-5 and ground. Step 1. If voltage is as specified, go to 2. Step 2. If voltage is not as specified, go to 3. 2. Check continuity between P260-6 and ground. Step 1. If continuity is present, replace ESU (PARA 9-4-39). Go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 10. 3. Install an insulated jumper wire between P242-n and P242-q. Go to 4.
	4. Check for 28 VDC between P260-5 and ground. Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, go to 7. 5. Remove jumper wire from P242. Go to 6. 6. Replace left relay panel (PARA 9-4-2). Go to 10. 7. Remove jumper wire from P242. Go to 8.
	8. Check for 28 VDC between P242-q and ground. Step 1. If voltage is as specified, repair/replace wiring between P242-n and P260-5 (Appendix F). Go to 10. Step 2. If voltage is not as specified, go to 9.
I	9. Check for 28 VDC between APU CONTR INST circuit breaker terminal 1 and ground. Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P242-q (Appendix F). Go to 10. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 10. 10. Procedure completed.

TABLE NO. 15-2-2-4. Fuel Leaks From APU Combustor Drain Line With Prime Boost Pump Running.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect P305 from start fuel valve, P306 from main fuel valve, and P303 from ignition exciter. Go to 2.	
2. Check for fuel leakage.	Step 1. If fuel leaks are present, go to 3. Step 2. If fuel leaks are not present, replace ESU (PARA 9-4-39). Go to 5.
3. Disconnect line to main fuel manifold and purge fuel into container. Go to 4.	
4. Using APU CONTR switch, motor APU and check for fuel flow.	Step 1. If fuel flows, replace main fuel valve (49-20-05). Go to 5. Step 2. If fuel does not flow, replace start fuel valve (49-20-05). Go to 5.
5. Procedure completed.	

TABLE NO. 15-2-2-5. ESU BITE Indicator Shows ● o ● o (Shorted Thermocouple).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect P289 from APU. Go to 2.	
2. Check continuity between P261-1 and P261-2.	Step 1. If continuity is present, repair/replace wiring (Appendix F) between, then go to 6: P261-1 and P289-1 P261-2 and P289-2 Step 2. If continuity is not present, go to 3.
3. Replace APU wiring harness (49-20-05). Go to 4.	
4. Start and operate APU (Appropriate Flight Manual). Go to 5.	
5. Does APU operate normally?	

TABLE NO. 15-2-2-5. ESU BITE Indicator Shows ● o ● o (Shorted Thermocouple). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 6. Step 2. If APU does not operate normally and ESU BITE indicators show o ● ● o (fail to light), shut down APU (Appropriate Flight Manual), then replace ESU for helicopters with ESU 160200-201 (PARA 9-4-39). Go to 6.
6. Procedure completed.	

TABLE NO. 15-2-2-6. ESU BITE Indicator Shows o ● ● o (Fail To Light) For Helicopters With ESU 160200-201.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check for fuel vapor at exhaust duct or overboard drain.	Step 1. If vapor is present, go to 2. Step 2. If vapor is not present, go to 13.
2. Perform Dynamic and Static checks.	Step 1. If speed sequence is good, replace igniter plug (49-20-05). Go to 3.
3. Start and operate APU (Appropriate Flight Manual). Go to 4.	
4. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual), then go to 5.
5. Install an insulated jumper wire between P303-A and P303-B. Go to 6.	
6. Check continuity between J289-17 and ground.	Step 1. If continuity is present, go to 7. Step 2. If continuity is not present, go to 11.

TABLE NO. 15-2-2-6. ESU BITE Indicator Shows o ● ● o (Fail To Light) For Helicopters With ESU 160200-201. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Remove jumper wire. Go to 8.	
8. Replace exciter cable (49-20-05). Go to 9.	
9. Start and operate APU (Appropriate Flight Manual). Go to 10.	
10. Does APU operate normally?	
Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24.	
Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual), then replace ignition exciter (49-20-05). Go to 24.	
11. Remove jumper wire. Go to 12.	
12. Replace APU wiring harness (49-20-05). Go to 13.	
13. Perform Dynamic and Static checks.	
Step 1. If speed sequence is good, replace start fuel nozzle (49-20-05). Go to 14.	
14. Start and operate APU (Appropriate Flight Manual). Go to 15.	
15. Does APU operate normally?	
Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24.	
Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual), then go to 16.	
16. Install an insulated jumper wire between P305-A and P305-B. Go to 17.	
17. Check continuity between J289-17 and ground.	
Step 1. If continuity is present, go to 18.	
Step 2. If continuity is not present, go to 22.	
18. Remove jumper wire. Go to 19.	

TABLE NO. 15-2-2-6. ESU BITE Indicator Shows o ● ● o (Fail To Light) For Helicopters With ESU 160200-201. (Cont)

	TEST OR INSPECTION CORRECTIVE ACTION
	19. Replace fuel start valve (49-20-05). Go to 20. 20. Start and operate APU (Appropriate Flight Manual). Go to 21. 21. Does APU operate normally?
	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally and ESU BITE indicator shows fail to light, shut down APU (Appropriate Flight Manual), then replace start nozzle purge valve (49-20-05). Go to 24.
	22. Remove jumper wire. Go to 23.
	23. Replace APU wiring harness (49-20-05). Go to 24.
	24. Procedure completed.

TABLE NO. 15-2-2-7. ESU BITE Indicator Shows o ● ● o (Shorted Oil Pressure Switch) For Helicopters With ESU 160200-600.

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check continuity between: J289-7 and J289-16 J289-7 and J289-23 Step 1. If continuity is present, go to 2. Step 2. If continuity is not present, replace APU wiring harness (49-20-05). Go to 7. Step 3. If trouble remains, replace low oil pressure switch (49-20-05). Go to 7.
	2. Check for 28 VDC between P304-C and ground.
	Step 1. If voltage is as specified, replace low oil pressure switch (49-20-05). Go to 7. Step 2. If voltage is not as specified, go to 3.

TABLE NO. 15-2-2-7. ESU BITE Indicator Shows o ● ● o (Shorted Oil Pressure Switch) For Helicopters With ESU 160200-600. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
3. Check for 28 VDC between P289-16 and ground.	Step 1. If voltage is as specified, replace APU wiring harness (49-20-05). Go to 7. Step 2. If voltage is not as specified, go to 4.
4. Check for 28 VDC between ESU J1-16 and ground.	Step 1. If voltage is as specified, repair/replace wiring between P261-16 and P289-16 (Appendix F). Go to 7. Step 2. If voltage is not as specified, go to 5.
5. Check for 28 VDC between P260-5 and ground.	Step 1. If voltage is as specified, replace ESU (PARA 9-4-39). Go to 7. Step 2. If voltage is not as specified, go to 6.
6. Check continuity between J242-q and J242-n.	Step 1. If continuity is present, repair/replace wiring between P242-n and P260-5 (Appendix F). Go to 7. Step 2. If continuity is not present, replace left relay panel (PARA 9-4-2). Go to 7.
7. Procedure completed.	

TABLE NO. 15-2-2-8. APU Motor Does Not Motor (No Rotation).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check for any of the following BITE indications:	o ● ● ● ● o ● ● ● o o o o o o o

- Step 1. If any indication listed is present, replace ESU (PARA 9-4-39). Go to 9.

Step 2. If any indication listed is not present, go to 2.

TABLE NO. 15-2-2-8. APU Motor Does Not Motor (No Rotation). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. Make sure APU CONTR switch is placed to OFF. Go to 3.	
3. Press APU start valve manual override lever. Go to 4.	
4. Check that APU start motor operates.	Step 1. If motor operates, go to 5. Step 2. If motor does not operate, go to 6.
5. Perform Dynamic and Static checks.	Step 1. If ESU is good, replace APU start valve (PARA 7-4-53). Go to 9.
6. Disconnect start motor from APU accessory pad. Go to 7.	
7. Using a ½ inch Allen wrench, turn splined shaft on accessory pad. Check that shaft turns.	Step 1. If shaft turns, go to 8. Step 2. If shaft does not turn, replace seized APU (PARA 15-4-1). Go to 9.
8. Attempt to turn splined shaft of APU start motor.	Step 1. If shaft turns, replace APU start valve (PARA 7-4-53). Go to 9. Step 2. If shaft does not turn, replace APU start motor (PARA 15-4-7). Go to 9.
9. Procedure completed.	

TABLE NO. 15-2-2-9. ESU BITE Indicator Shows ● o o ● (Fail To Start).

TEST OR INSPECTION	CORRECTIVE ACTION
1. What was APU speed BITE indication before it changed to fail to start indication?	Step 1. If APU speed was over 14% and less than 70% (o o ● ●), go to 2. Step 2. If APU speed was over 70% and less than 90% (o o o ●), go to 15.

TABLE NO. 15-2-2-9. ESU BITE Indicator Shows ● o o ● (Fail To Start). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. Check that main fuel valve actuates (determined by feeling the valve for a thump) during start sequence.	Step 1. If valve actuates, go to 3. Step 2. If valve does not actuate, go to 13.
3. Loosen nut on inlet fuel filter inlet. Place rag around nut. Go to 4.	
4. On upper console panel, place FUEL PUMP switch to APU BOOST. Go to 5.	
5. Check that fuel is going into rag.	Step 1. If fuel is present, go to 6. Step 2. If fuel is not present, go to 11.
6. Tighten nut. Go to 7.	
7. Loosen nut on inlet fuel filter outlet. Place rag around nut. Go to 8.	
8. Place upper console FUEL PUMP switch to APU BOOST. Go to 9.	
9. Check that fuel is going into rag.	Step 1. If fuel is present, go to 10. Step 2. If fuel is not present, replace filter (49-20-05). Go to 24.
10. Check for restricted outlet filter.	Step 1. If filter is restricted, replace fuel pump (49-20-05). Go to 24. Step 2. If filter is not restricted, replace main fuel nozzles/manifold assembly (49-20-05). Go to 24.
11. Tighten nut. Go to 12.	
12. Troubleshoot Fuel Prime/Boost System (PARA 10-2-3). Go to 24.	
13. Check for 28 VDC between P306-A and ground.	Step 1. If voltage is as specified, go to 14. Step 2. If voltage is not as specified, perform Dynamic and Static checks.

TABLE NO. 15-2-2-9. ESU BITE Indicator Shows ● o o ● (Fail To Start). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
14. Check continuity between P306-B and ground.	Step 1. If continuity is present, replace main fuel valve (49-20-05). Go to 24. Step 2. If continuity is not present, replace APU wiring harness (49-20-05). Go to 24.
15. Make sure all bleed-air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 16.	
16. Restart APU (Appropriate Flight Manual). Go to 17.	
17. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual), then go to 18.
18. Replace fuel manifold (49-20-05). Go to 19.	
19. Restart APU (Appropriate Flight Manual). Go to 20.	
20. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual), then go to 21.
21. Replace fuel pump/acceleration control (49-20-05). Go to 22.	
22. Restart APU (Appropriate Flight Manual). Go to 23.	
23. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 24. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual), then replace APU (PARA 15-4-1). Go to 24.

TABLE NO. 15-2-2-9. ESU BITE Indicator Shows ● o o ● (Fail To Start). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
24. Procedure completed.

TABLE NO. 15-2-2-10. ESU BITE Indicator Shows o ● o o (Loss Of Speed Data) For Helicopters With ESU 160200-600.

TEST OR INSPECTION
CORRECTIVE ACTION
1. Check for 1,190 to 1,450 ohms between P261-12 and P261-13.
Step 1. If resistance is as specified, replace ESU (PARA 9-4-39). Go to 8.
Step 2. If resistance is not as specified, go to 2.
2. Install an insulated jumper wire between P302-A and P302-B. Go to 3.
3. Check continuity between P261-12 and P261-13.
Step 1. If continuity is present, go to 4.
Step 2. If continuity is not present, go to 6.
4. Remove jumper wire from P302. Go to 5.
5. Replace magnetic pickup (speed sensor) (49-20-05). Go to 8.
6. Remove jumper wire from P302. Go to 7.
7. Replace APU wiring harness (49-20-05). Go to 8.
Step 1. If problem remains, repair/replace wiring (Appendix F) between, then go to 8:
P261-12 and P289-12
P261-13 and P289-13
8. Procedure completed.

TABLE NO. 15-2-2-11. ESU BITE Indicator Shows o o ● ● (Overtemperature).

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Perform Dynamic and Static checks. Step 1. If APU overtemperature is good, go to 2 .
	2. Replace combustor drain valve (49-20-05). Go to 3.
	3. Start and operate APU (Appropriate Flight Manual). Go to 4.
	4. Does APU operate normally? Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 8 . Step 2. If APU does not operate normally and ESU BITE indicator shows overtemperature, shut down APU (Appropriate Flight Manual), then go to 5 .
	5. Replace maximum fuel valve (49-20-05). Go to 6.
	6. Start and operate APU (Appropriate Flight Manual). Go to 7.
	7. Does APU operate normally? Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 8 . Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual), then replace fuel pump/acceleration control (49-20-05). Go to 8 .
	8. Procedure completed.

TABLE NO. 15-2-2-12. ESU BITE Indicator Shows o o o ● (Low Oil Pressure).

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check APU oil sump for proper level. Step 1. If oil level is good, go to 2 . Step 2. If oil level is not good, service APU oil sump (PARA 1-3-9), as required. Go to 5 .

TABLE NO. 15-2-2-12. ESU BITE Indicator Shows o o o • (Low Oil Pressure). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. Perform Dynamic and Static checks.	Step 1. If APU oil pressure is good, replace APU low oil pressure switch (49-20-05). Go to 3.
3. Start and operate APU (Appropriate Flight Manual). Go to 4.	
4. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 5. Step 2. If APU does not operate normally and ESU BITE indicator shows low oil pressure, shut down APU (Appropriate Flight Manual), then replace APU wiring harness (49-20-05). Go to 5.
5. Procedure completed.	

TABLE NO. 15-2-2-13. ESU BITE Indicator Shows • • o o (Open Thermocouple).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between P261-1 and P261-2.	Step 1. If continuity is present, replace ESU (PARA 9-4-39). Go to 3. Step 2. If continuity is not present, go to 2.
2. Check continuity between J289-1 and J289-2.	Step 1. If continuity is present, repair/replace wiring (Appendix F) between, then go to 3: P261-1 and P289-1 P261-2 and P289-2 Step 2. If continuity is not present, replace APU thermocouple and harness (49-20-05). Go to 3.
3. Procedure completed.	

TABLE NO. 15-2-2-14. ESU BITE Indicator Shows o o ● o (External Short) After o ● ● ● (Start Fuel Valve And Exciter Signal Out) For Helicopters With ESU 163291-100.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between J303-B and ground.	Step 1. If continuity is present, replace ignition exciter (49-20-05). Go to 5. Step 2. If continuity is not present, go to 2.
2. Check continuity between J305-A and ground.	Step 1. If continuity is present, replace start fuel valve (49-20-05). Go to 5. Step 2. If continuity is not present, go to 3.
3. With P303 and P305 disconnected, check continuity between J289-17 and ground.	Step 1. If continuity is present, replace APU electrical harness (49-20-05). Go to 5. Step 2. If continuity is not present, go to 4.
4. Check continuity between P261-17 and ground.	Step 1. If continuity is present, repair/replace wiring between P261-17 and P289-17 (Appendix F). Go to 5. Step 2. If continuity is not present, replace ESU (PARA 9-4-39). Go to 5.
5. Procedure completed.	

TABLE NO. 15-2-2-15. ESU BITE Indicator Shows o o ● o (External Short) After o o ● ● (Main Fuel Valve Signal Out 14%) For Helicopters With ESU 163291-100.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between J306-A and ground.	Step 1. If continuity is present, replace main fuel valve (49-20-05). Go to 4. Step 2. If continuity is not present, go to 2.
2. With P306 disconnected, check continuity between J289-9 and ground.	

TABLE NO. 15-2-2-15. ESU BITE Indicator Shows o o ● o (External Short) After o o ● ● (Main Fuel Valve Signal Out 14%) For Helicopters With ESU 163291-100. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If continuity is present, replace APU electrical harness (49-20-05). Go to 4. Step 2. If continuity is not present, go to 3.	3. With P306 disconnected and J289 connected, check continuity between P261-9 and ground.
Step 1. If continuity is present, repair/replace wiring between P261-9 and P289-9 (Appendix F). Go to 4. Step 2. If continuity is not present, replace ESU (PARA 9-4-39). Go to 4.	4. Procedure completed.

TABLE NO. 15-2-2-16. ESU BITE Indicator Shows o o ● o (External Short) After o o o o (90% RPM Switch On) Or ● ● ● ● (Ready For Service) For Helicopters With ESU 163291-100.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between J307-A and ground. Step 1. If continuity is present, replace maximum fuel valve (49-20-05). Go to 4. Step 2. If continuity is not present, go to 2.	2. With P307 disconnected, check continuity between J289-15 and ground.
Step 1. If continuity is present, replace APU wiring harness (49-20-05). Go to 4. Step 2. If continuity is not present, go to 3.	3. Check continuity between P261-15 and ground.
Step 1. If continuity is present, repair/replace wiring between P261-15 and P289-15 (Appendix F). Go to 4. Step 2. If continuity is not present, replace ESU (PARA 9-4-39). Go to 4.	4. Procedure completed.

TABLE NO. 15-2-2-17. APU Motors But Shuts Down (APU ON Capsule Goes Off And APU FAIL Capsule Goes On).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check for any of the following BITE indications: o ● ● ● ● o ● ● ● o o o o o o o Step 1. If any indication listed is present, replace ESU (PARA 9-4-39). Go to 10. Step 2. If any indication listed is not present, go to 2.	2. Reset ESU. Go to 3.
3. Check for 1,190 to 1,450 ohms between P261-12 and P261-13. Step 1. If resistance is as specified, replace ESU (PARA 9-4-39). Go to 10. Step 2. If resistance is not as specified, go to 4.	4. Install an insulated jumper wire between P302-A and P302-B. Go to 5.
5. Check continuity between P261-12 and P261-13. Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, go to 8.	6. Remove jumper wire from P302. Go to 7.
7. Remove and reinstall magnetic pickup (speed sensor) to be sure of correct adjustment (49-20-05). Step 1. If trouble remains, replace magnetic pickup (speed sensor) (49-20-05). Go to 10.	8. Remove jumper wire from P302. Go to 9.
9. Repair/replace wiring (Appendix F), as required between, then go to 10: P261-12 and P289-12 P261-13 and P289-13	

TABLE NO. 15-2-2-17. APU Motors But Shuts Down (APU ON Capsule Goes Off And APU FAIL Capsule Goes On). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
10. Procedure completed.

TABLE NO. 15-2-2-18. ESU BITE Indicator Shows o ● o ● (Underspeed).

TEST OR INSPECTION
CORRECTIVE ACTION
1. Check helicopter configuration.
Step 1. ESU 160200-600, go to 2. Step 2. All other ESUs, go to 3.
2. Troubleshoot System for loss of speed data o ● o o (TABLE NO. 15-2-2-10).
Step 1. If trouble remains, go to 3.
3. Make sure all bleed air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 4.
4. Perform Dynamic and Static checks.
Step 1. If speed sequence is good, go to 5.
5. Loosen nut on inlet fuel filter. Place rag around nut. Go to 6.
6. On upper console, place FUEL PUMP switch to APU BOOST. Go to 7.
7. Check that fuel is going into rag.
Step 1. If fuel is present, go to 8. Step 2. If fuel is not present, go to 19.
8. Install an insulated jumper wire between P307-A and P307-D. Go to 9.
9. Check continuity between J289-15 and ground.
Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, go to 17.

TABLE NO. 15-2-2-18. ESU BITE Indicator Shows o ● o ● (Underspeed). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	10. Remove jumper wire. Go to 11.
	11. Replace maximum fuel valve (49-20-05). Go to 12.
	12. Start and operate APU (Appropriate Flight Manual). Go to 13.
	13. Does APU operate normally?
	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 21.
	Step 2. If APU does not operate normally and ESU BITE indications shows underspeed, shut down APU (Appropriate Flight Manual), then go to 14.
	14. Replace APU fuel manifold (49-20-05). Go to 15.
	15. Start and operate APU (Appropriate Flight Manual). Go to 16.
	16. Does APU operate normally?
	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 21.
	Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual), then replace APU fuel pump/acceleration control (49-20-05). Go to 21.
	17. Remove jumper wire. Go to 18.
	18. Replace APU wiring harness (49-20-05). Go to 21.
	19. Tighten nut. Go to 20.
	20. Troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 21.
	21. Procedure completed.

TABLE NO. 15-2-2-19. ESU BITE Indicator Shows ● ● o ● (Overspeed).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Perform Dynamic and Static checks.	Step 1. If speed sequence is good, go to 2.
2. Install an insulated jumper wire between P305-A and P305-B. Go to 3.	
3. Check continuity between J289-17 and ground.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, go to 10.
4. Remove jumper wire. Go to 5.	
5. Start and operate APU (Appropriate Flight Manual). Go to 6.	
6. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 12. Step 2. If APU does not operate normally and ESU BITE indicator shows overspeed, shut down APU (Appropriate Flight Manual), then go to 7.
7. Replace APU start fuel valve (49-20-05). Go to 8.	
8. Start and operate APU (Appropriate Flight Manual). Go to 9.	
9. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual), then go to 12. Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual), then replace APU fuel pump/acceleration control (49-20-05). Go to 12.
10. Remove jumper wire. Go to 11.	
11. Repair/replace wiring (Appendix F), as required between P261-17 and P289-17, then go to 12.	
12. Procedure completed.	

WARNING

Failure to disconnect electrical connectors will permit fuel to enter combustion chamber and create a fire hazard.

TABLE NO. 15-2-2-20. ESU BITE Indicator Shows o o ● ● (Overtemperature).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Make sure all bleed air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 2.	2. Perform Dynamic and Static checks. Step 1. If APU overtemperature and overspeed is good, go to 3.
3. Replace combustor drain valve (49-20-05). Go to 4.	4. Start and operate APU (Appropriate Flight Manual). Go to 5. 5. Does APU operate normally? Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual). Go to 15. Step 2. If APU does not operate normally and ESU BITE indicator shows overtemperature, shut down APU (
6. Disconnect P305, P306, and P307 from fuel valves. Disconnect P303 from ignition exciter. Disconnect fuel line from main fuel valve to the fuel manifold. Go to 7.	7. Place suitable containers under open fuel lines. Go to 8. 8. Motor APU and check for fuel flow from main fuel valve. Step 1. If fuel is present, replace main fuel valve (49-20-05). Go to 15. Step 2. If fuel is not present, go to 9.
9. Leave electrical connectors disconnected. Reconnect fuel line to fuel manifold. Disconnect fuel line at outlet of maximum fuel valve. Go to 10.	

Appropriate Flight
Manual). Go to 6.

TABLE NO. 15-2-2-20. ESU BITE Indicator Shows o o ● ● (Overtemperature). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
10. Place suitable containers under open fuel lines. Go to 11.	
11. Motor APU and check for fuel at maximum fuel valve outlet.	
	Step 1. If fuel is present, replace maximum fuel valve (49-20-05). Go to 15.
	Step 2. If fuel is not present, go to 12.
12. Reconnect all electrical connectors and fuel lines. Go to 13.	
13. Start and operate APU (Appropriate Flight Manual). Go to 14.	
14. Does APU operate normally?	
	Step 1. If APU operates normally, System is operational. Shut down APU (Appropriate Flight Manual). Go to 15.
	Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Replace fuel pump/acceleration control (49-20-05). Go to 15.
15. Procedure completed.	

TABLE NO. 15-2-2-21. APU Operates, But APU ON Capsule Is Off.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.	
2. Check that APU ON capsule goes on.	
	Step 1. If capsule goes on, go to 3.
	Step 2. If capsule does not go on, go to 10.
3. Install an insulated jumper wire between P260-5 and P260-11. Go to 4.	
4. Place BATT switch to ON. Go to 5.	
5. Check that APU ON capsule goes on.	
	Step 1. If capsule goes on, replace ESU (PARA 9-4-39). Go to 11.

TABLE NO. 15-2-2-21. APU Operates, But APU ON Capsule Is Off. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If capsule does not go on, go to 6 .
	6. Check helicopter configuration.
	Step 1. EMEP , go to 7 .
	Step 2. W/O EMEP , go to 9 .
EMEP	7. Remove and check Pin Filter Adapter (PFA) P118 (PARA 9-4-98).
	Step 1. If PFA is good, go to 7 .
	Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 11 .
EMEP	8. Install PFA (PARA 9-4-98). Go to 9.
	9. Check for 28 VDC between P118-B and ground.
	Step 1. If voltage is as specified, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 11 .
	Step 2. If voltage is not as specified, repair/replace wiring between P118-B and P260-11 (Appendix F). Go to 11 .
	10. Check APU ON capsule lamps.
	Step 1. If lamps are good, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 11 .
	Step 2. If lamps are not good, replace lamps (PARA 8-4-20). Go to 11 .
	11. Procedure completed.

TABLE NO. 15-2-2-22. APU OIL TEMP HI Capsule Goes On, ESU BITE Indicator Shows ● ● ● ○ (High Oil Temperature).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check APU oil sump for proper oil level.	Step 1. If oil level is good, go to 2 . Step 2. If oil level is not good, service APU oil sump (PARA 1-3-9). Go to 3 .

TABLE NO. 15-2-2-22. APU OIL TEMP HI Capsule Goes On, ESU BITE Indicator Shows ● ● ● o
 (High Oil Temperature). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. Perform Dynamic and Static checks.	Step 1. If APU high oil temperature is good, replace APU high oil temperature switch (49-20-05). Go to 3.
3. Procedure completed.	

TABLE NO. 15-2-2-23. Fuel Leaks From Combustor Drain Line With APU Running.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect P305 from start fuel valve. Go to 2.	
2. Check for fuel leakage.	Step 1. If fuel leaks are present, go to 3. Step 2. If fuel leaks are not present, replace ESU (PARA 9-4-39). Go to 5.
3. Cycle valve five times and check that leakage stops.	Step 1. If leakage stops, go to 4. Step 2. If leakage does not stop, replace start fuel valve (49-20-05). Go to 5.
4. Check main fuel supply for contamination.	Step 1. If contamination is present, purge Fuel System. Go to 5. Step 2. If contamination is not present, replace start fuel valve (49-20-05). Go to 5.
5. Procedure completed.	

TABLE NO. 15-2-2-24. ESU BITE Indicators Do Not Cycle When APU CONTR Switch Is Placed ON.

	TEST OR INSPECTION CORRECTIVE ACTION
	1. Check for 28 VDC between P260-1 and ground. Step 1. If voltage is as specified, replace ESU (PARA 9-4-39). Go to 12. Step 2. If voltage is not as specified, go to 2.
	2. Check for 28 VDC between J200-R and ground. Step 1. If voltage is as specified, repair/replace wiring between P200-R and P260-1 (Appendix F). Go to 12. Step 2. If voltage is not as specified, go to 3.
	3. Check for 28 VDC between P231-V and ground. Step 1. If voltage is as specified, go to 4. Step 2. If voltage is not as specified, go to 8.
	4. Check for 28 VDC between APU T-handle terminal C and ground. Step 1. If voltage is as specified, go to 5. Step 2. If voltage is not as specified, repair/replace wiring between APU T-handle terminal C and J231-V (Appendix F). Go to 12.
	5. Check for 28 VDC between APU T-handle switch terminal NC and ground. Step 1. If voltage is as specified, go to 6. Step 2. If voltage is not as specified, replace switch (PARA 12-4-9). Go to 12.
	6. Check for 28 VDC between APU CONTR switch terminal 2 and ground. Step 1. If voltage is as specified, go to 7. Step 2. If voltage is not as specified, repair/replace wiring between APU T-handle switch terminal NC and APU CONTR switch (Appendix F). Go to 12.
	7. Check for 28 VDC between APU CONTR switch terminal 3 and ground. Step 1. If voltage is as specified, repair/replace wiring between switch terminal 3 and J200-R (Appendix F). Go to 12. Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 12. 8. Check continuity between P231-V and P242-n.

**TABLE NO. 15-2-2-24. ESU BITE Indicators Do Not Cycle When APU CONTR Switch Is Placed ON.
(Cont)**

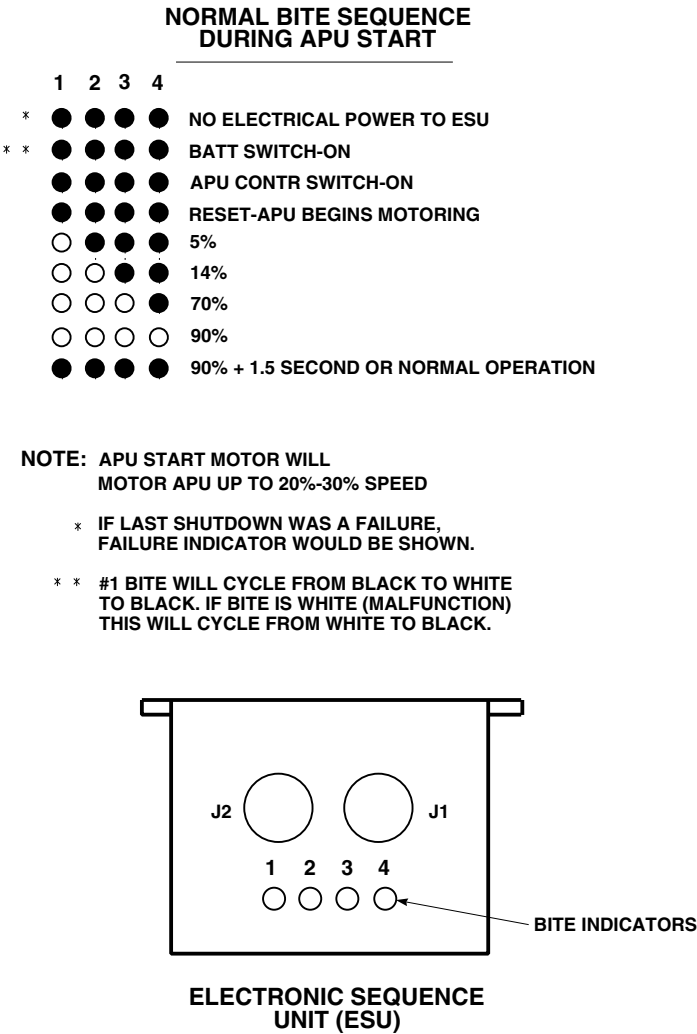
TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, go to 9. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.
9. Check continuity between P242-k and ground.	Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.
10. Check for 28 VDC between APU CONTR INST circuit breaker terminal 1 and ground.	Step 1. If voltage is as specified, go to 11. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 12.
11. Check continuity between APU CONTR INST circuit breaker terminal 1 and P242-p.	Step 1. If continuity is present, replace left relay panel (PARA 9-4-2). Go to 12. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 12.
12. Procedure completed.	

TABLE NO. 15-2-2-25. APU Does Not Shut Down When APU CONTR Switch Is Placed OFF.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 VDC between P260-2 and P260-6.	Step 1. If voltage is as specified, replace ESU (PARA 9-4-39). Go to 5. Step 2. If trouble remains, replace main fuel valve or maximum fuel valve (49-20-05). Go to 5. Step 3. If voltage is not as specified, go to 2.
2. Check for 28 VDC between P260-2 and ground.	

TABLE NO. 15-2-2-25. APU Does Not Shut Down When APU CONTR Switch Is Placed OFF. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If voltage is as specified, repair/replace wiring between P260-2 and P260-6 (Appendix F). Go to 5. Step 2. If voltage is not as specified, go to 3.
3. Check for 28 VDC between J200-P and ground.	Step 1. If voltage is as specified, repair/replace wiring between P200-P and P260-2 (Appendix F). Go to 5. Step 2. If voltage is not as specified, go to 4.
4. Check for 28 VDC between APU CONTR switch terminal 1 and ground.	Step 1. If voltage is as specified, repair/replace wiring between switch terminal 1 and J200-P (Appendix F). Go to 5. Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to 5.
5. Procedure completed.	



NOTES

- 1. HELICOPTERS WITH APU ESU
160200-201.
- 2. HELICOPTERS WITH APU ESU
160200-600.
- 3. HELICOPTERS WITH APU ESU
163290-100.

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FIGURE 15-2-2-1. ELECTRONIC SEQUENCE UNIT (ESU) BITE INDICATOR CONDITIONS
(SHEET 1 OF 2)

FAILURE BITE INDICATIONS ASSOCIATED WITH SHUTDOWN DURING START OF OPERATION

(SEE NOTE 1)

	1	2	3	4	
	○	○	●	●	OVERTEMPERATURE
	●	●	○	●	OVERSPEED
	○	●	○	●	UNDERSPEED
	●	○	○	●	FAIL TO START
	○	○	○	●	LOW OIL PRESSURE
	○	●	●	○	FAIL TO LIGHT
	●	●	○	○	OPEN THERMOCOUPLE
+ *	○	●	●	●	PROCESSOR BOARD FAILURE
*	●	○	●	●	SENSOR / DATA BOARD FAILURE
*	●	○	○	○	PROCESSOR SEQUENCE FAIL
*	○	○	○	○	NO DATA

* ESU RELATED FAILURE ONLY
+ APU FAIL LIGHT WILL NOT BE ON

(SEE NOTE 2)

	1	2	3	4	
	○	○	●	●	OVERTEMPERATURE
	●	●	○	●	OVERSPEED
	○	●	○	●	UNDERSPEED
	●	○	○	●	FAIL TO START
	○	○	○	●	LOW OIL PRESSURE
	○	●	●	○	SHORTED OIL PRESSURE SWITCH
	●	●	○	○	OPEN THERMOCOUPLE
+ *	○	●	●	●	PROCESSOR BOARD FAILURE
*	●	○	●	●	SENSOR / DATA BOARD FAILURE
*	●	○	○	○	PROCESSOR SEQUENCE FAIL
*	○	○	○	○	NO DATA
	○	●	○	○	LOSS OF SPEED DATA

* ESU RELATED FAILURE ONLY
+ APU FAIL LIGHT WILL NOT BE ON

**WARNING BITE
INDICATORS**

	1	2	3	4	
	●	●	●	○	HIGH OIL TEMPERATURE
+ ●	○	●	○	○	SHORTED THERMOCOUPLE

THESE INDICATIONS WILL NOT CAUSE
APU TO SHUT DOWN; ALSO, INDICATIONS
WILL NOT BE SHOWN AFTER A NORMAL
SHUTDOWN

+ THIS MALFUNCTION WILL CAUSE A
SHUTDOWN DURING NEXT ATTEMPTED
START WITH A FAIL LIGHT BITE
INDICATION (SEE NOTE 1)

(SEE NOTE 3)

	1	2	3	4	
	○	○	●	●	OVERTEMPERATURE
	●	●	○	●	OVERSPEED
	○	●	○	●	UNDERSPEED
	●	○	○	●	FAIL TO START
	○	○	○	●	LOW OIL PRESSURE
	○	●	●	○	SHORTED OIL PRESSURE SWITCH
	●	●	○	○	OPEN THERMOCOUPLE
+ *	○	●	●	●	PROCESSOR BOARD FAILURE
*	●	○	●	●	SENSOR / DATA BOARD FAILURE
*	●	○	○	○	PROCESSOR SEQUENCE FAIL
*	○	○	○	○	NO DATA
	○	○	●	○	EXTERNAL SHORT
	○	●	○	○	LOSS OF SPEED DATA

* ESU RELATED FAILURE ONLY
+ APU FAIL LIGHT WILL NOT BE ON

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FIGURE 15-2-2-1. ELECTRONIC SEQUENCE UNIT (ESU) BITE INDICATOR CONDITIONS
(SHEET 2 OF 2)

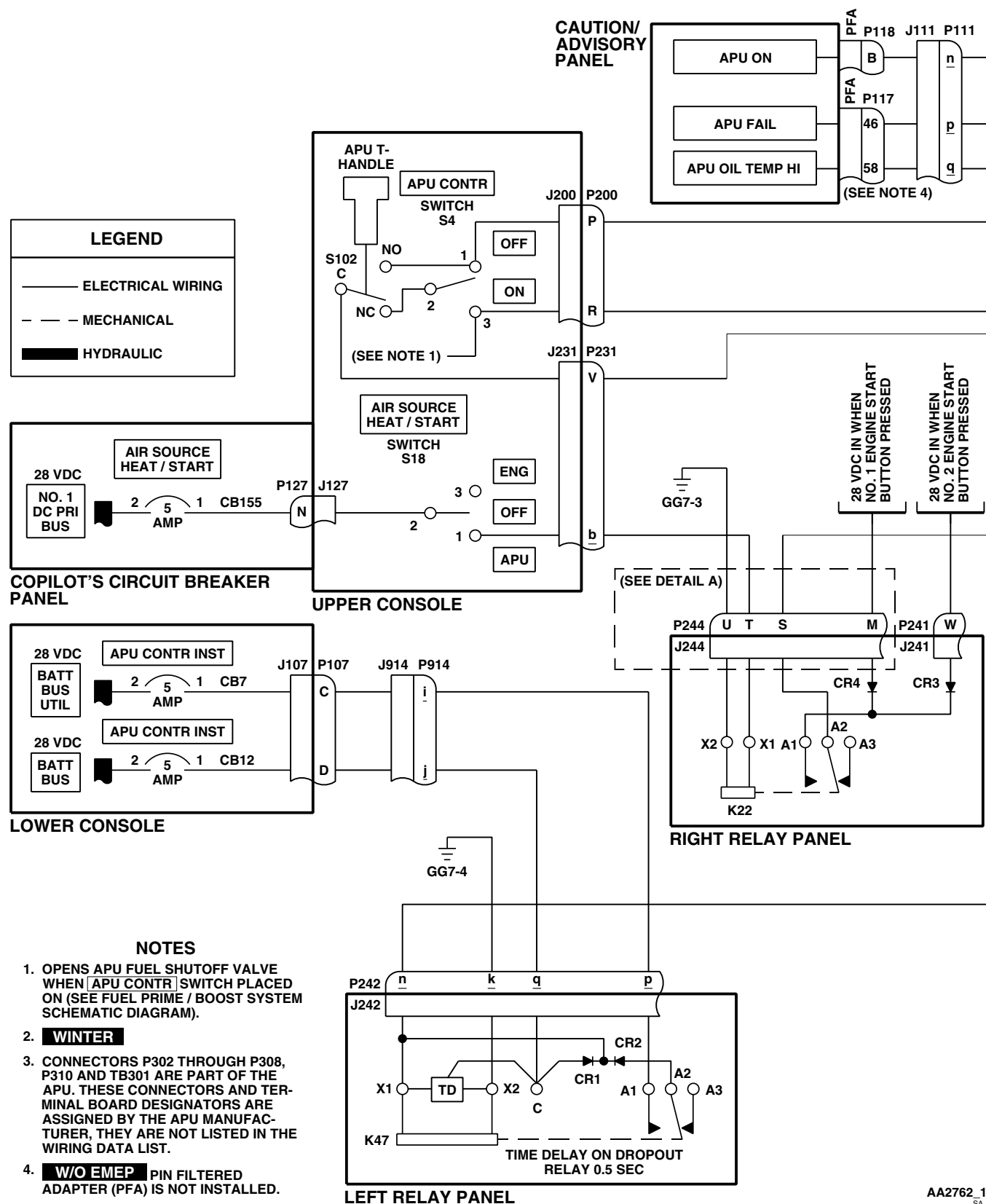


FIGURE 15-2-2-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 4)

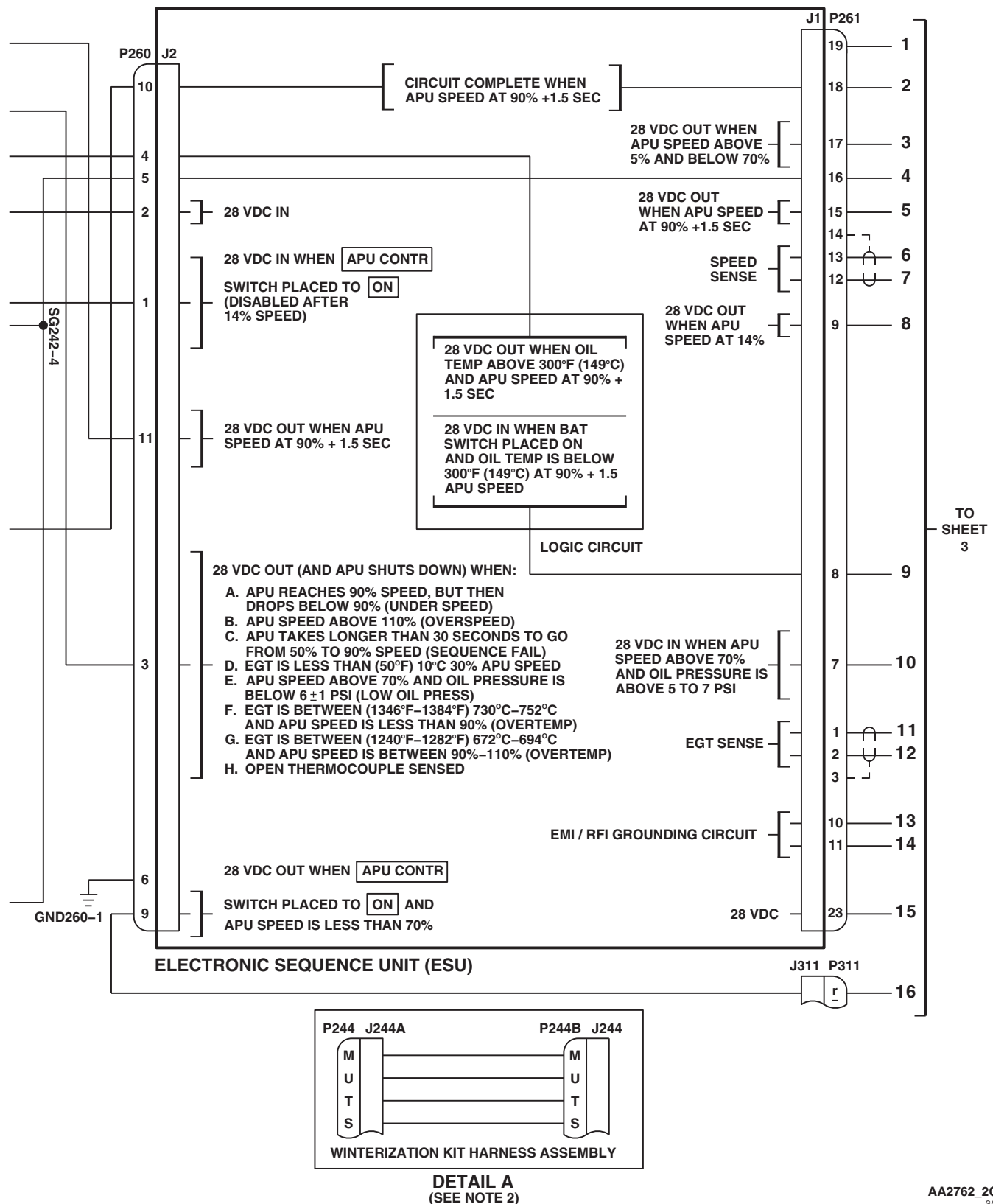


FIGURE 15-2-2-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 4)

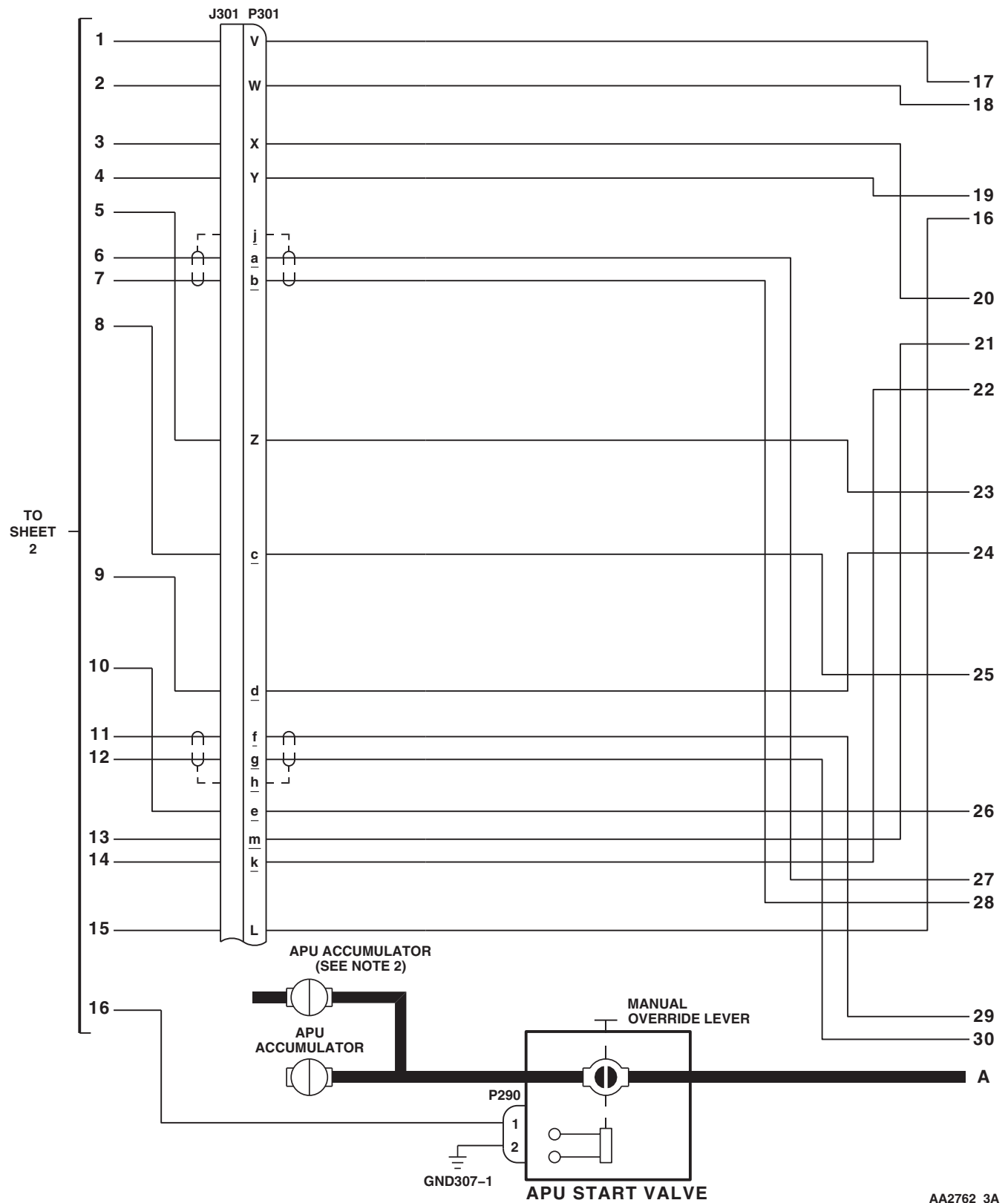


FIGURE 15-2-2-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 4)

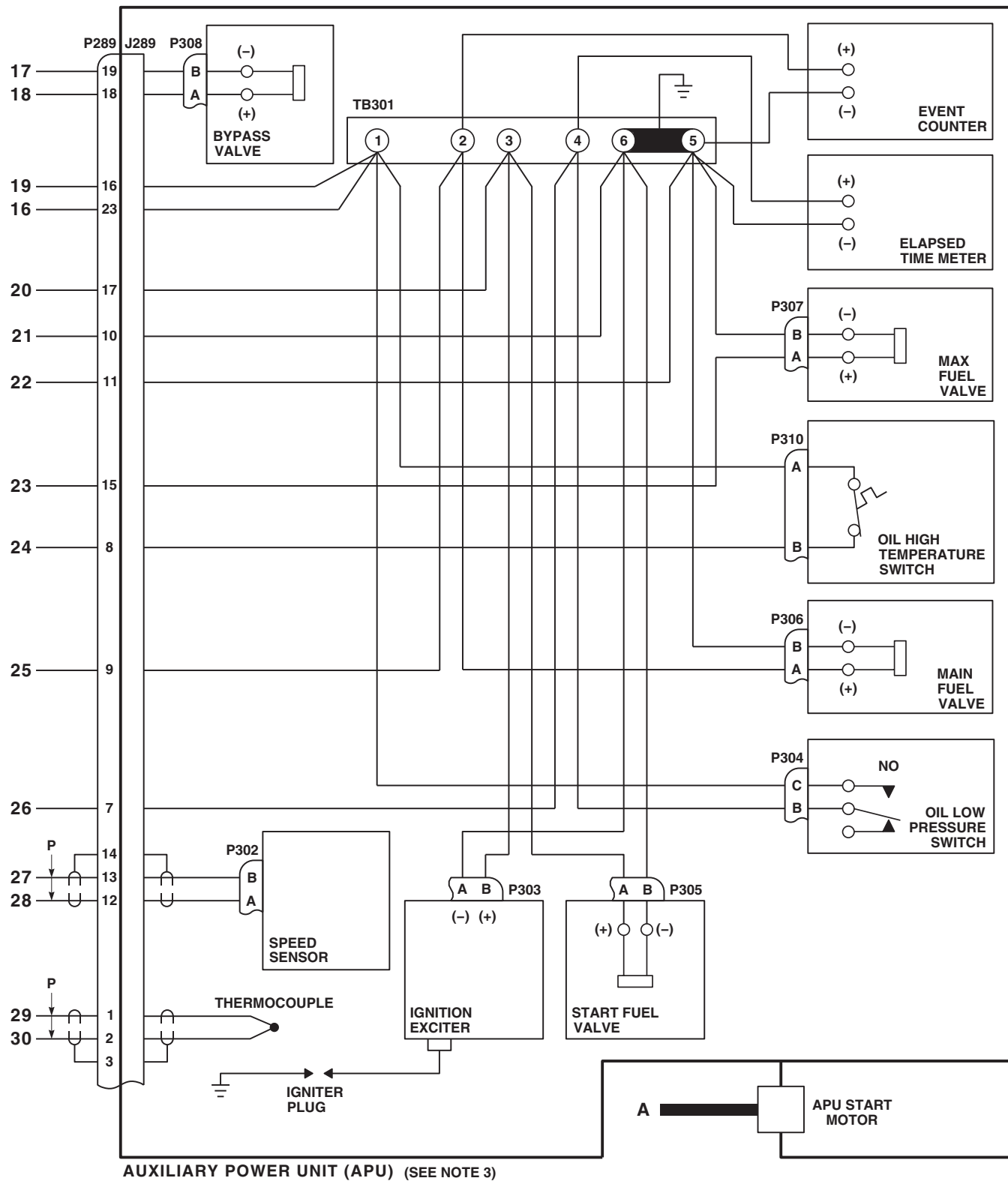
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FIGURE 15-2-2-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 4)

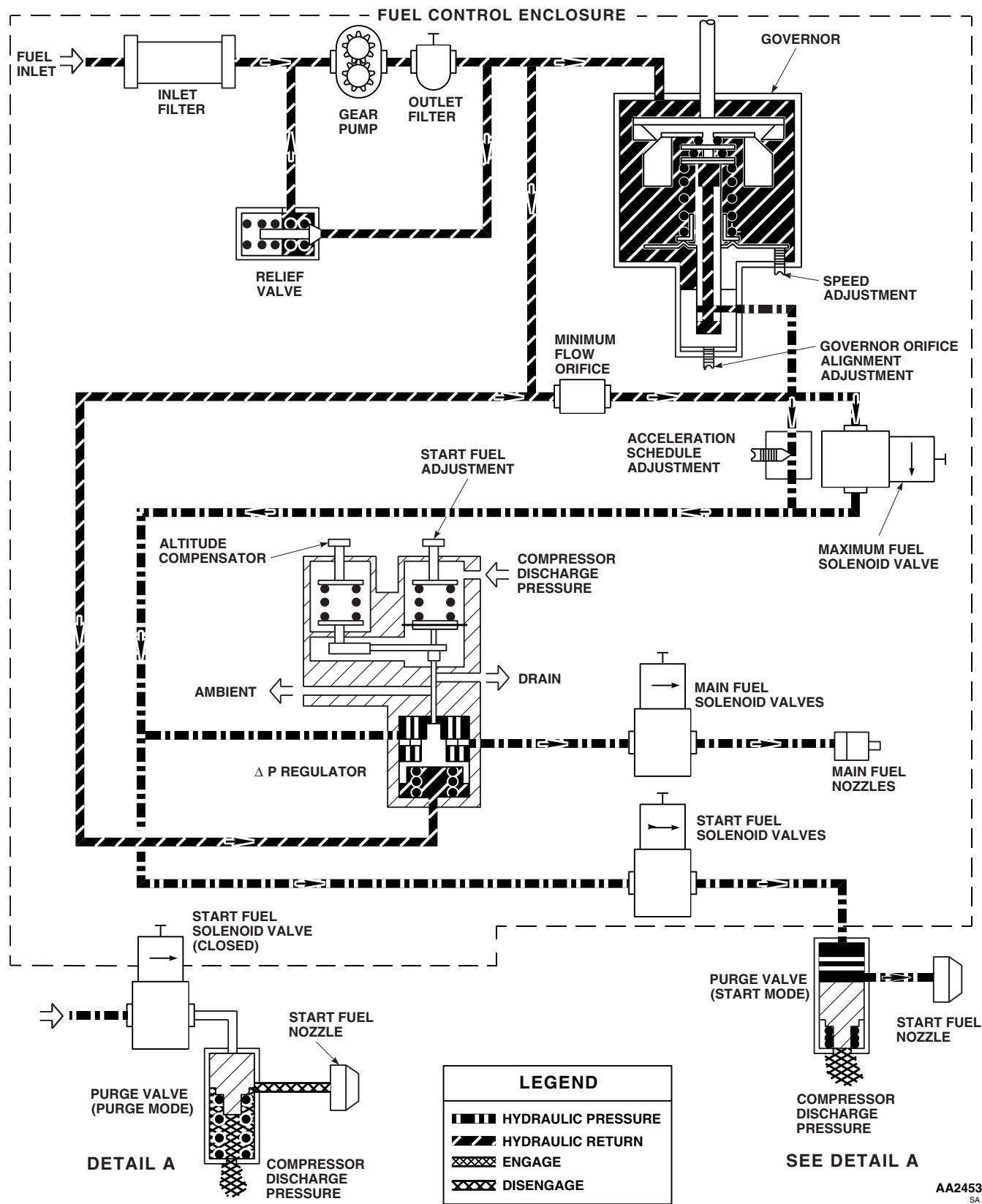


FIGURE 15-2-2-3. AUXILIARY POWER UNIT FUEL SYSTEM SCHEMATIC

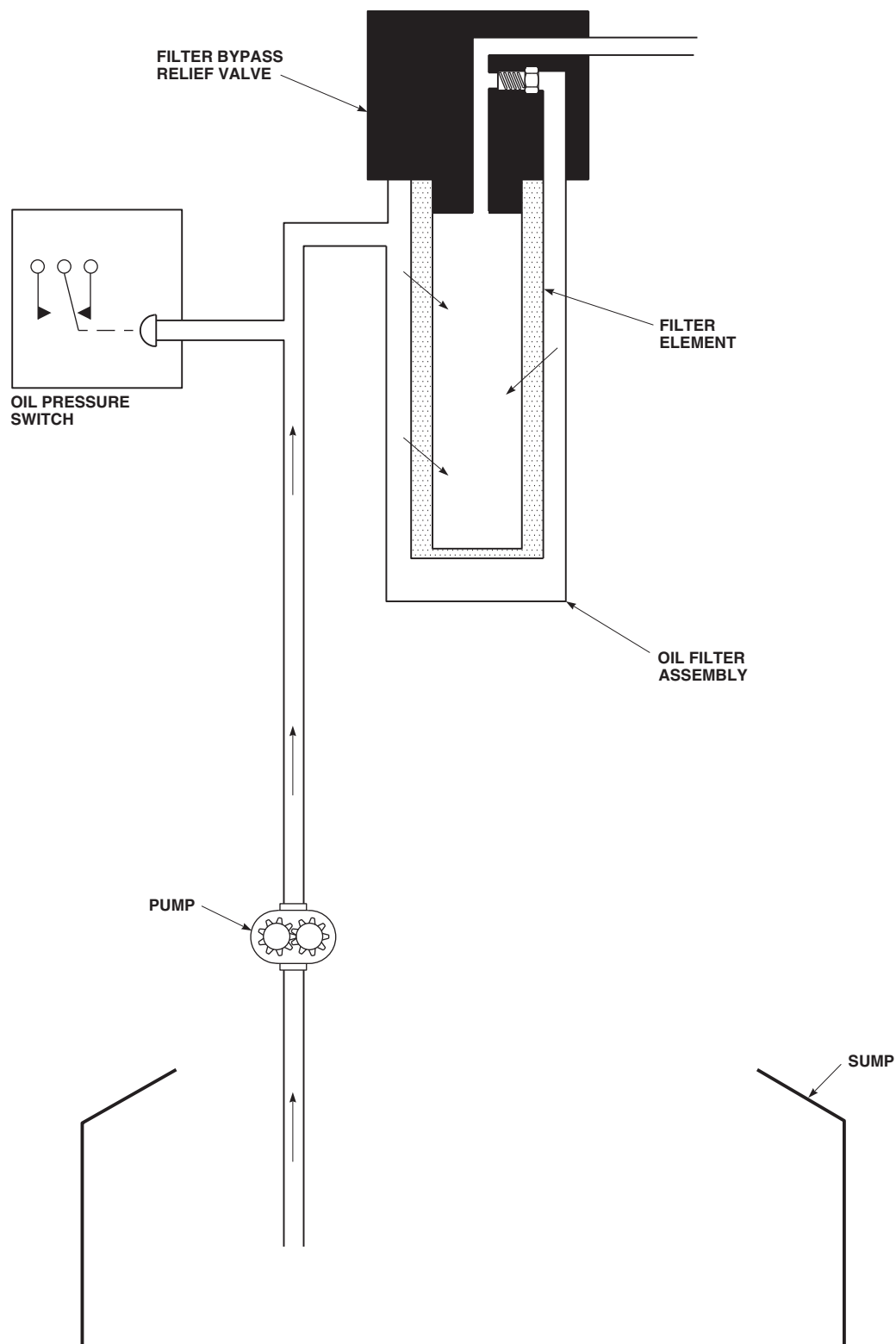
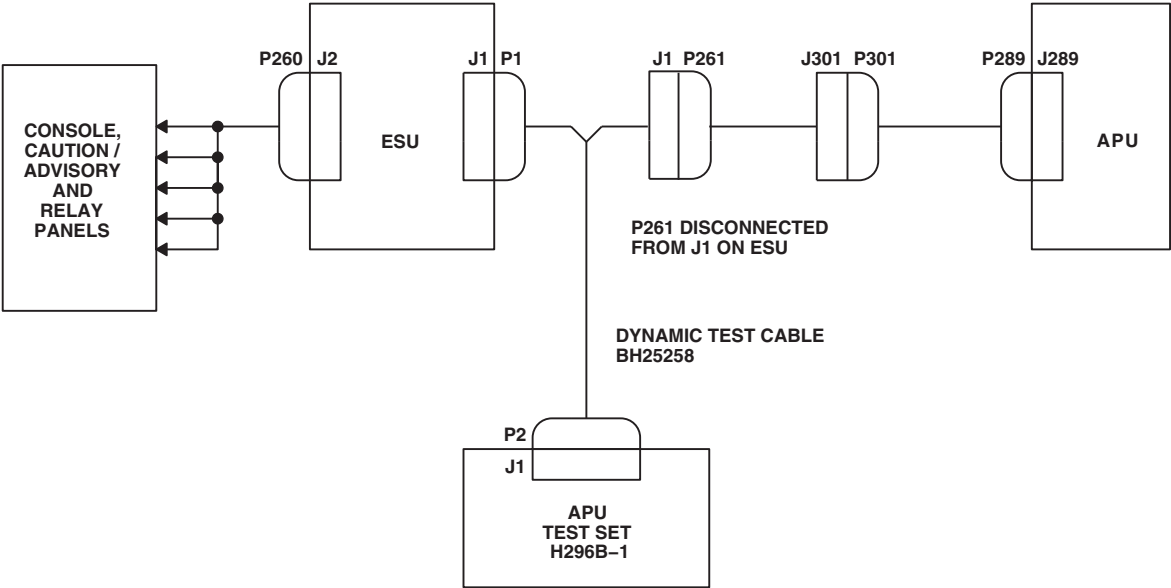
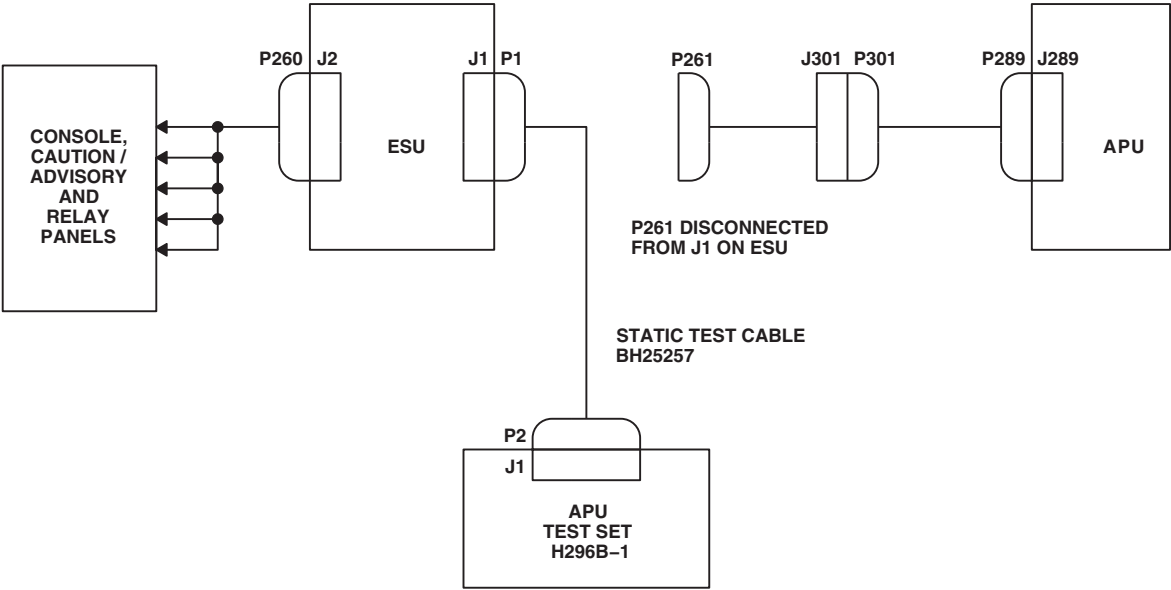
AA2454
SA

FIGURE 15-2-2-4. AUXILIARY POWER UNIT LUBRICATION SYSTEM SCHEMATIC



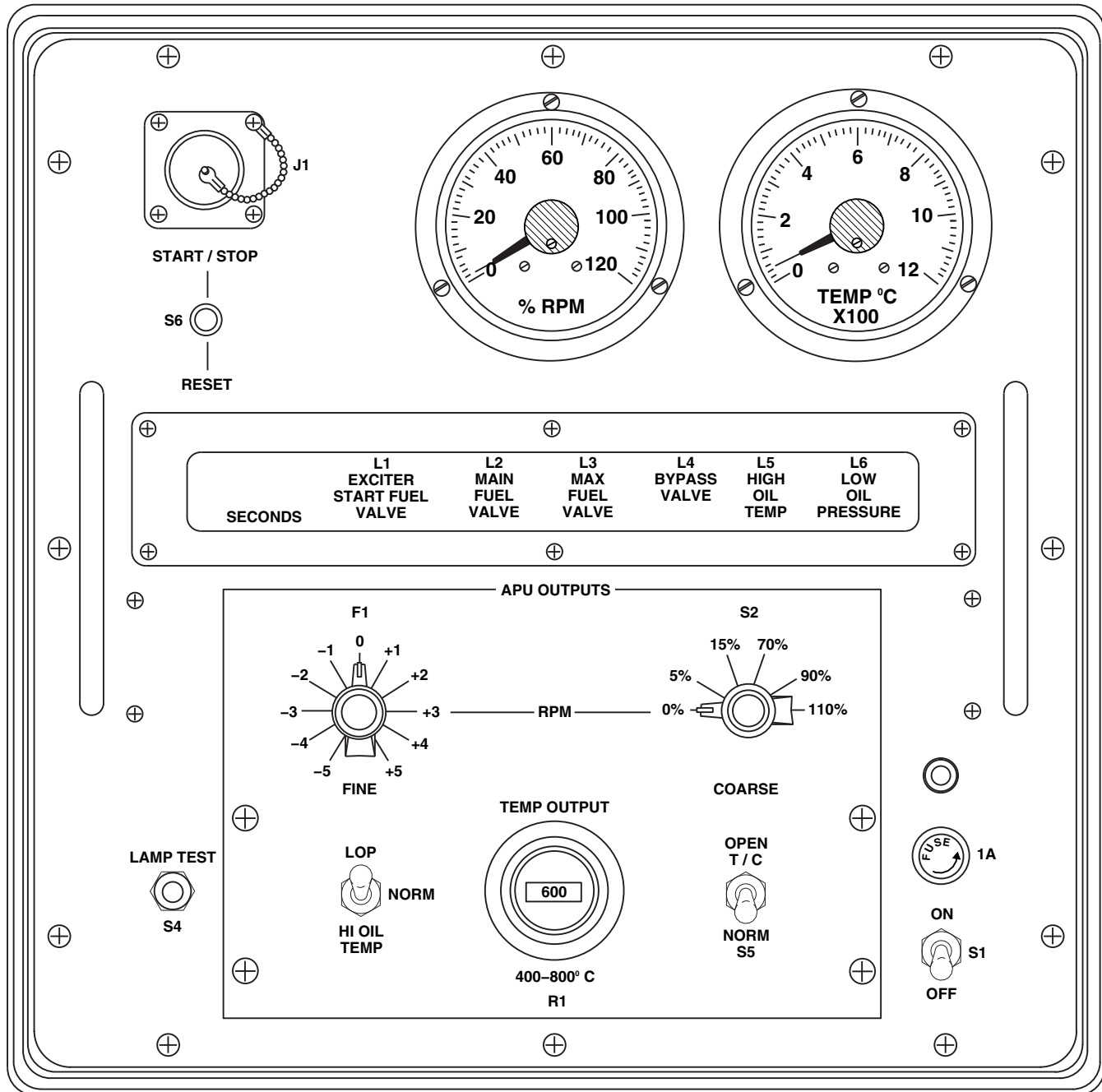
APU DYNAMIC TEST CONDITIONS



APU STATIC TEST CONDITIONS

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FIGURE 15-2-2-5. AUXILIARY POWER UNIT TEST SET-UP (SHEET 1 OF 2)



APU TEST SET, H296B-1

FIGURE 15-2-2-5. AUXILIARY POWER UNIT TEST SET-UP (SHEET 2 OF 2)

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15-2-3 AUXILIARY POWER UNIT SYSTEM WITH 5 BIT DESU.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 27/AN
- Multimeter, Digital, Fluke 45
- APU Test Set (test set), Simulator, Temp./Speed, H296B-1

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16
- PARA 7-4-8
- Appropriate Flight Manual

Equipment Conditions

- APU Accumulator Properly Serviced
- Battery Fully Charged
- External Electrical Power Available
- No. 1 Fuel Tank Serviced

WARNING

To prevent injury to personnel and damage to helicopter, have fire guard check APU fuel drain. Wait at least two minutes after APU shutdown before attempting a restart. **DO NOT RESTART** APU until fuel drainage has stopped.

CAUTION

To prevent possible damage to APU, do not operate APU more than 30 minutes when main engines and rotor are running and outside temperature is greater than 43°C (109°F). Continuous operation of APU is permitted at temperatures up to 51°C (124°F) only if main engines and rotors are not running.

CAUTION

If onboard fire extinguishing is required before a generator becomes operational, RESERVE fire extinguisher must be used to fight fire.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

NOTE

To prevent discharge of APU accumulator, disconnect P290 from APU start valve.

NOTE

For proper results, switch positions and order of operation must be strictly followed.

Special Instructions

The APU operational/troubleshooting procedure is subdivided as follows:

- (1) APU System

- (2) Dynamic
- (3) Static

- (4) Shutdown

15-2-3.1 APU System.

NOTE

Do not connect external electrical power to helicopter before starting or during APU system operational/troubleshooting procedure.

- a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- b. Make sure upper console BATT switch and GENERATORS APU switch are placed to RESET/OFF (center position).
- c. Check bleed air pneumatic tube couplings for loose clamps, cuts, and/or incorrect installation at all connecting points (PARA 7-4-8).

RESULT: All couplings shall be securely installed with clamps and there shall be no cuts in pneumatic tube lines.

- If result is not as specified, repair/replace couplings and lines (PARA 7-4-8).
- d. Remove interior furnishings as required from cabin ceiling to visually check digital electronic sequence unit (DESU) indicators.

NOTE

- An observer must be in the cabin area to watch the DESU indicators during the operational/troubleshooting procedure.
- The observer shall call out the procedural steps to the operator in the cockpit. This will allow the observer to watch for a change in DESU indication as the operator does the next procedural step.

- e. Visually check DESU indicators.

RESULT: All indicators shall be black.

- If result is not as specified, troubleshoot system for indicated malfunction.

NOTE

- Do not attempt to start the APU if BATT LOW capsule on instrument panel is on.
- CHART NO. 15-2-3-1 lists the sequence of events for a normal start.
- The APU will not start if o ● ● ● ● (DESU Failure) is displayed.
- If No. 1 DESU indicator is white, due to malfunction, it shall only cycle from white to black.

- f. Place upper console BATT switch to ON. Watch DESU indicators.

RESULT: No. 1 DESU indicator shall cycle from black to white to black.

- If result is not as specified, go to TABLE NO. 15-2-3-1.

- g. Place upper console FUEL PUMP switch to APU BOOST.

RESULT:

- (1) No fuel leakage shall occur from APU combustor drain line (below helicopter).
- (2) Fuel prime boost motor shall be heard.
- If result (1) is not as specified, go to TABLE NO. 15-2-3-2.

15-2-3-2

- If result (2) is not as specified, troubleshoot fuel prime boost system (PARA 10-2-3).

h. Place APU CONTR switch to ON.

RESULT: ●●●●● (Power Reset) is shown.

- If result is not as specified:

DESU INDICATION	GO TO
○●●●● (DESU Failure)	TABLE NO. 15-2-3-3
○●●○● (Shorted Oil Pressure Switch)	TABLE NO. 15-2-3-4
●○●○● (Thermocouple Shorted)	TABLE NO. 15-2-3-5
●●○○● (Thermocouple Open Circuit Fault)	TABLE NO. 15-2-3-5
○○○○● (Power Interrupt)	TABLE NO. 15-2-3-6
●●●●○ (RTL Indicator Failure)	TABLE NO. 15-2-3-7
○●●●○ (MALF Indicator Failure)	TABLE NO. 15-2-3-7
○○●●○ (Hot Indicator Failure)	TABLE NO. 15-2-3-7
●●○●○ (Main Fuel Valve Failure)	TABLE NO. 15-2-3-8
●○○●○ (Start Circuit Failure)	TABLE NO. 15-2-3-9
○○○●○ (Start Fuel/Exciter Failure)	TABLE NO. 15-2-3-10

DESU INDICATION	GO TO
●●●○○ (Illogical Switch Input)	TABLE NO. 15-2-3-9

NOTE

- Prime boost pump must be running for all APU motoring and start operations.
- A minimal amount of fuel may be evident during and shortly after start.
- If the APU shuts down, the observer shall take note of the last APU speed DESU indication before the malfunction DESU indication appears.
- Make sure upper console APU T-handle is pushed in or APU will not start.
- If a restart is being attempted, the original malfunction DESU indication will be shown.

i. Start APU (Appropriate Flight Manual).

RESULT:

- (1) APU shall start and DESU indicators shall cycle as follows:

DESU INDICATION	
●●●●●	Power On Reset
●●●●○	Start Command Received
○●●●●	5%
○○●●○	14%
○○○●○	70%
○○○○○	90%
●●●●●	90% + 1.5 seconds (ready to load)

- (2) APU ON capsule (caution/advisory warning panel) and capsule on instrument panel shall go on.

- (3) APU FAIL capsule (caution/advisory warning panel) and capsule on instrument panel shall be off.
- (4) APU OIL TEMP HI capsule (caution/advisory warning panel) shall be off.
- (5) No fuel leakage shall occur from APU combustor drain line (below helicopter).

- If APU does not rotate:

DESU INDICATION	GO TO
● ● ● ● ● (Power On Reset)	TABLE NO. 15-2-3-11
● ● ● ● ● (Start Command Received)	TABLE NO. 15-2-3-12
○ ● ● ● ● (DESU Failure)	TABLE NO. 15-2-3-13
● ○ ○ ● ● (Fail To Rotate)	TABLE NO. 15-2-3-14
○ ● ● ○ ● (Shorted Oil Pressure Switch)	TABLE NO. 15-2-3-4
○ ○ ○ ○ ● (Power Interrupt)	TABLE NO. 15-2-3-6
● ○ ○ ● ○ (Start Circuit Failure)	TABLE NO. 15-2-3-9

- If APU shutdowns during start:

DESU INDICATION	GO TO
● ● ● ● ○ (Start Command Received)	TABLE NO. 15-2-3-12
○ ● ● ● ○ (Start Fuel and Exciter On)	TABLE NO. 15-2-3-13

DESU INDICATION

○ ○ ● ● ○
(Main Fuel On)

○ ○ ○ ● ○
(Start Fuel and Exciter Off)

○ ○ ○ ○ ○
(90% RPM)

○ ● ● ● ●
(DESU Failure)

● ● ○ ● ●
(Overspeed)

○ ● ○ ● ●
(Underspeed)

○ ○ ○ ● ●
(Low Oil Pressure)

● ○ ● ○ ●
(Thermocouple Shorted)

● ● ○ ○ ●
(Thermocouple Open Circuit Fault)

○ ● ○ ○ ●
(Loss Of Speed Data)

○ ○ ○ ○ ●
(Power Interrupt)

● ● ○ ○ ○
(Main Fuel Valve Failure)

● ○ ○ ● ○
(Start Circuit Failure)

○ ● ● ○ ○
(Fail To Light)

GO TO

TABLE NO.
15-2-3-15

TABLE NO.
15-2-3-10

TABLE NO.
15-2-3-16

TABLE NO.
15-2-3-13

TABLE NO.
15-2-3-17

TABLE NO.
15-2-3-18

TABLE NO.
15-2-3-19

TABLE NO.
15-2-3-5

TABLE NO.
15-2-3-5

TABLE NO.
15-2-3-20

TABLE NO.
15-2-3-6

TABLE NO.
15-2-3-8

TABLE NO.
15-2-3-9

TABLE NO.
15-2-3-21

15-2-3-4

DESU INDICATION	GO TO
● ○ ● ○ ○ (Fail To Accel (Start Fuel))	TABLE NO. 15-2-3-22
○ ○ ● ○ ○ (Fail To Accel (Start & Main))	TABLE NO. 15-2-3-22
● ● ○ ○ ○ (Fail To Accel (Main Fuel))	TABLE NO. 15-2-3-8
● ● ○ ○ ○ (Slow Accel (Time Out))	TABLE NO. 15-2-3-8
● ○ ○ ○ ○ (Overtemperature (Start))	TABLE NO. 15-2-3-23
○ ○ ○ ○ ○ (APU Start Switch Failure)	TABLE NO. 15-2-3-9

- If APU shuts down while at ready to load:

DESU INDICATION	GO TO
● ● ● ● ● (Normal Operation)	TABLE NO. 15-2-3-11
○ ● ● ● ● (DESU Failure)	TABLE NO. 15-2-3-13
○ ○ ● ● ● (Overtemperature (Run))	TABLE NO. 15-2-3-24
● ● ○ ● ● (Overspeed)	TABLE NO. 15-2-3-17
○ ● ○ ● ● (Underspeed)	TABLE NO. 15-2-3-18
○ ○ ○ ● ● (Low Oil Pressure)	TABLE NO. 15-2-3-19

DESU INDICATION	GO TO
● ○ ● ○ ● (Thermocouple Shorted)	TABLE NO. 15-2-3-5
● ● ○ ○ ● (Thermocouple Open Circuit Fault)	TABLE NO. 15-2-3-5
○ ● ○ ○ ● (Loss Of Speed Data)	TABLE NO. 15-2-3-20
○ ○ ○ ○ ● (Power Interrupt)	TABLE NO. 15-2-3-6
● ● ○ ● ○ (Main Fuel Valve Failure)	TABLE NO. 15-2-3-8

- If APU starts and runs, but ● ● ● ● ○ (RTL Indicator Failure), ○ ● ● ● ○ (MALF Indicator Failure), or ○ ○ ● ● ○ (Hot Indicator Failure) is displayed, go to TABLE NO. 15-2-3-7.

- Recharge APU accumulator by placing GENERATORS APU switch to ON. Visually check accumulator for proper charge (pressure gauge and tape indicator).

RESULT: APU ACCUM LOW capsule (caution/ advisory warning panel) shall go off in 90 seconds and APU accumulator shall be charged.

- If result is not as specified, troubleshoot Hydraulic System (PARA 7-2-1).

- With qualified personnel at the controls, start No. 1 engine (Appropriate Flight Manual).

RESULT: No. 1 engine shall start.

- If result is not as specified, do DYNAMIC and STATIC checks.

- With qualified personnel at the controls, start No. 2 engine (Appropriate Flight Manual).

RESULT: No. 2 engine shall start.

- If result is not as specified, do DYNAMIC and STATIC checks.

m. Shut down No. 1 and No. 2 engines (Appropriate Flight Manual).

n. Shut down APU (Appropriate Flight Manual).

RESULT:

- (1) APU shall shut down.
 - (2) APU ON capsule (caution/advisory warning panel) shall go off.
- If result (1) is not as specified, place FUEL PUMP switch to OFF and go to TABLE NO. 15-2-3-25.
 - If result (2) is not as specified, disconnect P260 on DESU.
 - (a) If APU ON capsule goes off, replace DESU (PARA 9-4-39).
 - (b) If APU ON capsule does not go off, troubleshoot Caution/Advisory System (PARA 8-2-4).

o. Check DESU indicators after APU is shut down.

RESULT: All DESU indicators shall be black.

- If result is not as specified, troubleshoot System for indicated malfunction.

p. Place upper console FUEL PUMP switch to OFF.

RESULT: Fuel pump boost motor shall not be heard.

- If result is not as specified, troubleshoot Fuel Prime Boost System (PARA 10-2-3).
- q. Visually check that there is no fuel or fuel vapors coming from combustor drain line (below helicopter).
- If result is not as specified, check APU combustor drain check valve. Replace if necessary (TM 55-2835-208-23).
- r. Place upper console BATT switch to OFF.
- s. If no other check is to be done, go to SHUTDOWN.

CHART NO. 15-2-3-1. Normal Start Sequence.

APU SPEED SEQUENCE (%RPM)

EVENT

A.5%

1. Start fuel solenoid opens
2. Ignition exciter goes on
3. 20 - 30-second counter (inside DESU) starts
- *4. EGT overtemperature shutdown circuit (inside DESU) limit extended to 740°C (1376°F)
5. Fuel pressure built up due to motoring speed
- *6. Thermocouple checked for open condition (inside DESU)

CHART NO. 15-2-3-1. Normal Start Sequence. (Cont)

APU SPEED SEQUENCE (%RPM)	EVENT
	7. DESU internal checks for sequence failure and data failure
	8. Ignition starts
	*9. Overspeed circuit indicates Normal Operation
	***DESU INDICATION: o ● ● ● o (START FUEL & EXCITER ON)
B.14%	1. Main fuel solenoid opens
	2. Ignition continues
	3. DESU internal self-checks continue
	*4. Thermocouple checked for open condition (inside DESU)
	***DESU INDICATION: o o ● ● o (MAIN FUEL ON)
C.30%	*1. Inside DESU - APU exhaust gas temperature is checked for being above 10°C (50°F)
	2. DESU internal self-checks continue
D.70%	1. APU start valve (hydraulic) - closes
	2. Start fuel valve - closes
	3. Purge valve - opens (due to air pressure)
	4. Ignition - off
	*5. Oil pressure circuit - on
	6. DESU internal self-checks are made
	*7. Thermocouple checked for open condition (inside DESU)
	***DESU INDICATION: o o o ● o (START FUEL & EXCITER OFF)
E.90%	1. This occurs in DESU:

CHART NO. 15-2-3-1. Normal Start Sequence. (Cont)

APU SPEED SEQUENCE (%RPM)

EVENT

	*a. 20 - 30-second counter - disabled
	*b. APU fail circuit - deactivated
	*c. Underspeed circuit - activated
	d. 1.5-second timer - activated
	e. DESU internal self-checks continue
	*f. Thermocouple checked for open condition (inside DESU)
	***DESU INDICATION: o o o o o (90%RPM)
F.90% - 1.5 Seconds	1. Maximum fuel valve - opens
	2. APU ON capsule (caution/advisory panel) - on
	3. DESU internal self-checks continue
	**4. Thermocouple checked for short condition (inside DESU)
	**5. High oil temperature check
	*6. EGT shutdown limit - reduced to 656°C - 679°C (1240°F - 1280°F)
	***DESU INDICATION: ● ● ● ● ● (READY TO LOAD)

G. APU Operating - 102% (Unloaded)

APU Operating - 100% (Loaded)

*DESU checks on system operation. If check is not satisfactory, APU will shut down.

**Warning indication that will appear as a DESU indication, but will not cause APU to shut down.

***DESU indication will appear on DESU when APU has reached corresponding speed during start sequence.

15-2-3.2 Dynamic.

NOTE

An observer must be in the cabin area to watch the DESU BITE indicators,

one operator in the cockpit to work the APU controls, and one operator to operate the test set during the operational/troubleshooting procedure.

a. Do APU check, if not already done.

15-2-3-8

- b. Turn off electrical power, if not already done.
- c. Connect test set (test set cable BH25258), see Figure 15-2-2-6 (Sheet 1), APU Dynamic Test Conditions.
- d. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
- e. Turn on external electrical power (PARA 1-6-16).

RESULT:

- (1) Caution/advisory panel EXT PWR CONNECTED capsule shall go on.
- (2) DESU No. 1 BITE indicator shall cycle from black to white back to black.

- If result (1) is not as specified, troubleshoot AC system (PARA 9-2-1).
- If result (2) is not as specified, go to TABLE NO. 15-2-2-3.

- f. Place APU test set switches as indicated:

SWITCH	POSITION
POWER (S1)	ON
OPEN T/C/NORM (S5)	NORM
TEMP OUTPUT 400 - 800°C (R1)	0°C
LOP/NORM/HI OIL TEMP (S3)	NORM
RPM FINE (F1)	0
RPM COARSE (S2)	0%

NOTE

L6 LOW OIL PRESSURE lamp shall be on when APU is below 20%.

- g. Press test set LAMP TEST switch (S4).

RESULT: Test set display lamps L1 through L5 shall go on.

- If result is not as specified, replace test set.
- h. On upper console place FUEL PUMP switch to APU BOOST.
 - i. Make sure that test % RPM, TEMP °CX100, and SECONDS displays indicate 0.

NOTE

Read steps j. and k. completely before proceeding.

- j. Simultaneously place upper console APU CONTR to ON and press test set START/STOP RESET (S6) switch to start timer.

NOTE

- Test set HIGH OIL TEMP lamp (L5) will not go on unless oil temperature becomes greater than 135° to 149°C (275° to 300°F).
 - Overtemperature is 740° to 776°C (1364° to 1429°F) for less than 90% RPM and 672° to 694°C (1242° to 1281°F) if RPM is between 90% and 100%.
 - It should not take over 10 seconds in normal temperature, or 40 seconds in cold temperatures for APU to reach 90% RPM.
- k. Observe test set % RPM gauge and timer for proper sequence of test set lamps as indicated in TABLE NO. 15-2-2-2.
 - l. Observe DESU BITE indications during APU start.

RESULT: DESU BITE indicates shall cycle to (● ● ● ●) READY FOR SERVICE.

- If result is not as specified, go to APU SYSTEM check step h.

- m. On upper console, place GENERATORS APU switch to ON to recharge APU accumulator.
- n. After APU accumulator has been charged within 90 seconds, place GENERATORS APU switch to RESET (center position).

NOTE

Test set LOW OIL PRESSURE lamp (L6) shall go on when APU decelerates to 20%.

- o. Place upper console APU CONTR switch to OFF.

RESULT: APU shall shut down.

- If result is not as specified, go to TABLE NO. 15-2-2-24.

- p. If no other checks are to be done, go to Shutdown.

CHART NO. 15-2-3-2. Test Set Lamp Sequence Chart.

RPM-PERCENT	5%	14%	70%	90%	90% + 1.5 TO 3.5 SEC
L1 EXCITER/ START FUEL VALVE	ON	ON			
L2 MAIN FUEL VALVE		ON	ON	ON	ON
L3 MAX FUEL VALVE					ON
L4 BYPASS VALVE					
L5 HIGH OIL TEMP					
L6 LOW OIL PRESSURE	ON	ON/OFF 20%			

15-2-3.3 Static.**NOTE**

An observer must be in the cabin area to watch the DESU BITE indicators, one operator in the cockpit to work the APU controls, and one operator to operate the test set during the operational/troubleshooting procedure.

- a. Do APU check, if not already done.
- b. Turn off electrical power, if not already done.
- c. Disconnect P290 from APU start valve.
- d. Connect test set (test set cable BH25257), refer to Figure 15-2-2-6 (Sheet 1), APU Static Test Conditions.
- e. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

15-2-3-10

f. Place APU test set switches as indicated:

SWITCH	POSITION
POWER (S1)	ON
TEMP OUTPUT 400-800°C (R1)	580°C
LOP/NORM/HI OIL TEMP (S3)	LOP
RPM FINE (F1)	0
RPM COARSE (S2)	0%
OPEN T/C/NORM (S5)	NORM

g. Turn on external electrical power (PARA 1-6-16).

RESULT:

- (1) Caution/advisory panel EXT PWR CONNECTED capsule shall go on.
 - (2) DESU No. 1 BITE indicator shall cycle from black to white back to black.
- If result (1) is not as specified, troubleshoot AC system (PARA 9-2-1).
 - If result (2) is not as specified, go to TABLE NO. 15-2-2-3.

h. Momentarily press test set LAMP TEST (S4) switch.

RESULT:

- (1) Test set lamps L1 through L5 shall go on, then go off when switch is released.
- (2) Test set LOW OIL PRESSURE (L6) lamp shall be on when APU is below 20% RPM.

- If results are not as specified, replace test set.

NOTE

- Read steps i. through n. completely before proceeding.
 - Steps i. through n. must be accomplished within 29 to 40 seconds. If not accomplished within time specified, a FAIL TO START (● o o ●) indication shall result.
- i. Simultaneously place upper console APU CONTR switch to ON and press test set START/STOP RESET (S6) switch.
 - j. Place the test set switches to the following settings while observing for the specified indications:

TEST SET TIMER (SECONDS)	TEST SET SETTING	TEST SET LAMP INDICATION	DESU BITE INDICATION
2	RPM COARSE (S2) 5%	L1 - ON	1 WHITE
3	RPM COARSE (S2) 15%	L2 - ON	2 WHITE
5	LOP/NORM/HI/ OIL TEMP (S3) NORM	L6 - OFF	----
9	RPM COARSE (S2) 70%	L1 - OFF	3 WHITE

11	RPM COARSE (S2) 90% RPM FINE (F1) +5	L3 - ON	4 WHITE THEN ALL SHALL TURN BLACK BEFORE 3.5 SECONDS
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RESULT: If results are not as specified, replace DESU (PARA 9-4-39).

k. On upper console, place AIR SOURCE to APU.

l. Press and hold No. 1 engine start button.

RESULT: Test set BYPASS VALVE (L4) lamp shall go on.

- If result is not as specified, replace DESU (PARA 9-4-39).

m. Release No. 1 engine start button.

n. Press and hold No. 2 engine start button.

RESULT: Test set BYPASS VALVE (L4) lamp shall go on.

- If result is not as specified, replace DESU (PARA 9-4-39).

o. Release No. 2 engine start button.

p. If no other check is to be done, go to Shutdown.

15-2-3.4 Shutdown.

a. Make sure upper console AIR SOURCE HEAT/START switch is in OFF position.

b. Make sure upper console APU CONTR switch is in OFF position.

c. Turn off electrical power (PARA 1-6-16).

d. If connected, disconnect test set and reconnect helicopter harness connectors to their mating connectors.

e. Install interior furnishings as required over DESU on cabin ceiling.

TABLE NO. 15-2-3-1. DESU Indicators Do Not Cycle When BATT Switch Is Placed to ON.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P260-5 and P260-6.	Step 1. If voltage is as specified, replace DESU (PARA 9-4-39). Go to 7. Step 2. If voltage is not as specified, go to 2.
2. Check for 28 vdc between P260-5 and ground.	Step 1. If voltage is as specified, repair/replace wiring between P260-6 and ground (Appendix F). Go to 7. Step 2. If voltage is not as specified, go to 3.
3. Check continuity between P242-n and P260-5.	

1. Check for 28 vdc between P260-5 and P260-6.

Step 1. If voltage is as specified, replace DESU (PARA 9-4-39). Go to 7.

Step 2. If voltage is not as specified, go to 2.

2. Check for 28 vdc between P260-5 and ground.

Step 1. If voltage is as specified, repair/replace wiring between P260-6 and ground (Appendix F). Go to 7.

Step 2. If voltage is not as specified, go to 3.

3. Check continuity between P242-n and P260-5.

TABLE NO. 15-2-3-1. DESU Indicators Do Not Cycle When BATT Switch Is Placed to ON. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.	4. Check for 28 vdc between P242-q and P242-k.
Step 1. If voltage is as specified, replace left relay panel (PARA 9-4-2). Go to 7. Step 2. If voltage is not as specified, go to 5.	5. Check continuity between P242-k and ground.
Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 7.	6. Check for 28 vdc between APU CONTR INST circuit breaker terminal 1 and ground.
Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P242-q (Appendix F). Go to 7. Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 7.	7. Procedure completed.

TABLE NO. 15-2-3-2. Fuel Leaks From APU Combustor Drain Line With Prime Boost Pump Running.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect P305 from start fuel valve. Go to 2.	2. Disconnect P306 from main fuel valve. Go to 3.
3. Disconnect P303 from ignition exciter. Go to 4.	4. Check for fuel leakage.
Step 1. If any fuel leakage is present, go to 5. Step 2. If fuel leakage is not present, replace DESU (PARA 9-4-39). Go to 7.	5. Disconnect line to main fuel manifold and purge fuel into container. Go to 6.

**TABLE NO. 15-2-3-2. Fuel Leaks From APU Combustor Drain Line With Prime Boost Pump Running.
(Cont)**

TEST OR INSPECTION	CORRECTIVE ACTION
6. Using APU CONTR switch, motor APU and check for fuel flow.	Step 1. If fuel flows, replace main fuel valve (TM 55-2835-208-23). Go to 7. Step 2. If fuel does not flow, replace start fuel valve (TM 55-2835-208-23). Go to 7.
7. Procedure completed.	

TABLE NO. 15-2-3-3. o ● ● ● ● (DESU Failure) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place APU Switch to OFF then to ON.	Step 1. If ● ● ● ● ● is displayed, continue with APU SYSTEM check. Go to 2. Step 2. If o ● ● ● ● is displayed, replace DESU (PARA 9-4-39). Go to 2.
2. Procedure completed.	

TABLE NO. 15-2-3-4. o ● ● o ● (Shorted Oil Pressure Switch) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Disconnect P304 from oil pressure switch. Go to 2.	
2. Place APU Switch to OFF then to ON.	Step 1. If ● ● ● ● ● is displayed, replace oil pressure switch (TM 55-2835-208-23). Go to 5. Step 2. If o ● ● o ● is displayed, go to 3.
3. Check continuity between:	P261-7 and P289-7 P261-16 and P289-16 P261-23 and P289-23

TABLE NO. 15-2-3-4. o • • o • (Shorted Oil Pressure Switch) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 5. 4. Check continuity between: J289-7 and P304-B J289-16 and P304-C J289-23 and P304-C Step 1. If continuity is present, replace DESU (PARA 9-4-39). Go to 5. Step 2. If continuity is not present, replace APU electrical harness (TM 55-2835-208-23). Go to 5. 5. Procedure completed.

TABLE NO. 15-2-3-5. • o • o • (Thermocouple Shorted) Or • • o o • (Thermocouple Open Circuit Fault) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
	1. Check continuity between: P261-1 and P289-1 P261-2 and P289-2 Step 1. If continuity is present, replace DESU (PARA 9-4-39). Go to 2. Step 2. If trouble remains, replace APU electrical harness (TM 55-2835-208-23). Go to 2. Step 3. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 2. 2. Procedure completed.

TABLE NO. 15-2-3-6. o o o o ● (Power Interrupt) Is Displayed.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for greater than 18 vdc between P260-5 and P260-6.

Step 1. If voltage is as specified, go to 2.

Step 2. If voltage is not as specified, go to **TABLE NO. 15-2-3-1. Go to 3.**

2. Place APU Switch to OFF then to ON.

Step 1. If ● ● ● ● ● is displayed, continue with APU SYSTEM check. Go to 3.

Step 2. If o o o o ● is displayed, replace DESU (PARA 9-4-39). Go to 3.

3. Procedure completed.
-

TABLE NO. 15-2-3-7. ● ● ● ● o (RTL Indicator Failure), o ● ● ● o (MALF Indicator Failure), Or o o ● ● o (Hot Indicator Failure) Is Displayed.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check continuity between:

P117-46 and P260-3

P117-58 and P260-4

P118-B and P260-11

Step 1. If continuity is present, replace DESU (PARA 9-4-39). Go to 2.

Step 2. If continuity is not present, repair/replace wiring as required (Appendix F).
Go to 2.

2. Procedure completed.
-

TABLE NO. 15-2-3-8. ● ● o ● o (Main Fuel Valve Failure) Is Displayed.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect P306 from main fuel valve. Go to 2.
2. Place APU Switch to OFF then to ON.

TABLE NO. 15-2-3-8. ● ● o ● o (Main Fuel Valve Failure) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If ● ● ● ● ● is displayed, replace main fuel valve (TM 55-2835-208-23). Go to 5. Step 2. If ● ● o ● o is displayed, go to 3.
3. Check continuity between:	J289-9 and P306-A J289-11 and P306-B
	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, replace APU electrical harness (TM 55-2835-208-23). Go to 5.
4. Check continuity between:	P261-9 and P289-9 P261-11 and P289-11
	Step 1. If continuity is present, replace DESU (PARA 9-4-39). Go to 5. Step 2. If continuity is not present, repair/replace wiring as required (Appendix F). Go to 5.
5. Procedure completed.	

TABLE NO. 15-2-3-9. ● o o ● o (Start Circuit Failure) Or ● ● ● o o (Illogical Switch Inputs) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 28 vdc between P290-1 and P290-2.	Step 1. If voltage is as specified, replace APU start valve (TM 55-2835-208-23). Go to 4. Step 2. If voltage is not as specified, go to 2.
2. Check for 28 vdc between P290-1 and ground.	Step 1. If voltage is as specified, repair/replace wiring between P290-2 and ground (Appendix F). Go to 4. Step 2. If voltage is not as specified, go to 3.

TABLE NO. 15-2-3-9. ● o o ● o (Start Circuit Failure) Or ● ● ● o o (Illogical Switch Inputs) Is Displayed. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
3. Check continuity between P260-9 and P290-1. Step 1. If continuity is present, replace DESU (PARA 9-4-39). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4. 4. Procedure completed.

TABLE NO. 15-2-3-10. o o o ● o (Start Fuel/Exciter Failure) Is Displayed.

TEST OR INSPECTION CORRECTIVE ACTION
1. Disconnect P303 from ignition exciter. Go to 2. 2. Place APU Switch to OFF then to ON. Step 1. If ● ● ● ● ● is displayed, replace ignition exciter (TM 55-2835-208-23). Go to 7. Step 2. If o o o ● o is displayed, go to 3. 3. Disconnect P305 from start fuel valve. Go to 4. 4. Place APU Switch to OFF then to ON. Step 1. If ● ● ● ● ● is displayed, replace start fuel valve (TM 55-2835-208-23). Go to 7. Step 2. If o o o ● o is displayed, go to 5. 5. Check continuity between: P289-10 and P303-A P289-10 and P305-B P289-17 and P303-B P289-17 and P305-A Step 1. If continuity is present, go to 6. Step 2. If continuity is not present, replace APU electrical harness (TM 55-2835-208-23). Go to 7. 6. Check continuity between:

TABLE NO. 15-2-3-10. o o o ● o (Start Fuel/Exciter Failure) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	P261-10 and P289-10 P261-17 and P289-17 Step 1. If continuity is present, replace DESU (PARA 9-4-39). Go to 4. Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 4. 7. Procedure completed.

TABLE NO. 15-2-3-11. ● ● ● ● ● (Power On Reset) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Go to TABLE NO. 15-2-3-12.	Step 1. If trouble is corrected, go to 7. Step 2. If trouble remains, go to 2.
2. Check for 28 vdc between P260-1 and ground.	Step 1. If voltage is as specified, replace DESU (PARA 9-4-39). Go to 7. Step 2. If voltage is not as specified, go to 3.
3. Check for 28 vdc between APU CONTR switch terminal 3 and ground.	Step 1. If voltage is as specified, repair/replace wiring between terminal 3 and P260-1 (Appendix F). Go to 7. Step 2. If voltage is not as specified, go to 4.
4. Check for 28 vdc between APU CONTR switch terminal 2 and ground.	Step 1. If voltage is as specified, replace switch (PARA 9-4-5). Go to 6. Step 2. If voltage is not as specified, go to 5.
5. Check for 28 vdc between APU T-Handle switch terminal NC and ground.	Step 1. If voltage is as specified, repair/replace wiring between APU T-handle switch terminal NC and APU CONTR switch (Appendix F). Go to 7. Step 2. If voltage is not as specified, go to 6.

TABLE NO. 15-2-3-11. ● ● ● ● ● (Power On Reset) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
6. Check for 28 vdc between APU T-handle terminal C and ground.	Step 1. If voltage is as specified, replace switch (PARA 12-4-9). Go to 7. Step 2. If voltage is not as specified, repair/replace wiring between terminal C and P242-n. Go to 7.
7. Procedure completed.	

TABLE NO. 15-2-3-12. ● ● ● ● o (Start Command Received) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Place APU CONTR switch to OFF. Go to 2.	
2. Wait 2 seconds. Go to 3.	
3. Place APU switch to run.	Step 1. If o ● ● ● ● (DESU Failure), then ● ● ● ● ● (Power On Rest) is displayed, go to 4. Step 2. If o ● ● ● ● (DESU Failure), then ● ● ● ● ● (Power On Rest) is not displayed, replace DESU (PARA 9-4-39). Go to 5.
4. Start APU (Appropriate Flight Manual).	Step 1. If APU rotates, continue with APU SYSTEM check. Go to 5. Step 2. If APU does not rotate, troubleshoot the DESU indication that is displayed. Go to 5.
5. Procedure completed.	

TABLE NO. 15-2-3-13. o ● ● ● ● (DESU Failure) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Reset both APU CONTR INST circuit breakers. Go to 2.	

TABLE NO. 15-2-3-13. o ● ● ● ● (DESU Failure) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
2. Place APU CONTR switch to ON.	
Step 1. If o ● ● ● ● (DESU Failure), then ● ● ● ● ● (Power On Rest) is displayed, go to 3 .	
Step 2. If o ● ● ● ● (DESU Failure), then ● ● ● ● ● (Power On Rest) is not displayed, replace DESU (PARA 9-4-39). Go to 4 .	
3. Start APU (Appropriate Flight Manual).	
Step 1. If APU rotates, continue with APU SYSTEM check. Go to 4 .	
Step 2. If APU does not rotate, troubleshoot the DESU indication that is displayed. Go to 4 .	
4. Procedure completed.	

TABLE NO. 15-2-3-14. ● o o ● ● (Fail To Rotate) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check that main fuel valve actuates (determined by feeling the valve for a thump) during start sequence.	
Step 1. If valve actuates, go to 2 .	
Step 2. If valve does not actuate, go to 14 .	
2. Loosen nut on inlet fuel filter inlet. Go to 3.	
3. Place rag around nut. Go to 4.	
4. Place upper console FUEL PUMP switch to APU BOOST. Go to 5.	
5. Check that fuel is going into rag.	
Step 1. If fuel is present, go to 6 .	
Step 2. If fuel is not present, go to 12 .	
6. Tighten nut. Go to 7.	

TABLE NO. 15-2-3-14. ● o o ● ● (Fail To Rotate) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Loosen nut on inlet fuel filter outlet. Go to 8.	
8. Place rag around nut. Go to 9.	
9. Place upper console FUEL PUMP switch to APU BOOST. Go to 10.	
10. Check that fuel is going into rag.	
	Step 1. If fuel is present, go to 11. Step 2. If fuel is not present, replace filter (TM 55-2835-208-23). Go to 16.
11. Check for restricted outlet filter.	
	Step 1. If filter is restricted, replace fuel pump (TM 55-2835-208-23). Go to 16. Step 2. If filter is not restricted, replace main fuel nozzles/manifold assembly (TM 55-2835-208-23). Go to 16.
12. Tighten nut. Go to 13.	
13. Troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 16.	
14. Check for 28 vdc between P306-A and ground.	
	Step 1. If voltage is as specified, go to 15. Step 2. If voltage is not as specified, do DYNAMIC and STATIC checks. Go to 16.
15. Check continuity between P306-B and ground.	
	Step 1. If continuity is present, replace main fuel valve (TM 55-2835-208-23). Go to 16. Step 2. If continuity is not present, replace APU wiring harness (TM 55-2835-208-23). Go to 16.
16. Procedure completed.	

TABLE NO. 15-2-3-15. o o ●● o (Main Fuel Valve On) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between J306-A and ground.	Step 1. If continuity is present, replace main fuel valve (PARA 10-4-8). Go to 4. Step 2. If continuity is not present, go to 2.
2. Leave P306 disconnected. Go to 3.	
3. Check continuity between J289-9 and ground.	Step 1. If continuity is present, replace APU electrical harness (TM 55-2835-208-23). Go to 8. Step 2. If continuity is not present, go to 4.
4. Leave P306 disconnected. Go to 5.	
5. Connect P289 to J289. Go to 6.	
6. Check continuity between P261-9 and ground.	Step 1. If continuity is present, repair/replace wiring between P261-9 and P289-9 (Appendix F). Go to 7. Step 2. If continuity is not present, replace DESU (PARA 9-4-39). Go to 7.
7. Connect P306 to main fuel valve. Go to 8.	
8. Procedure completed.	

TABLE NO. 15-2-3-16. o o o o o (90% RPM) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check continuity between J307-A and ground.	Step 1. If continuity is present, replace maximum fuel valve (TM 55-2835-208-23). Go to 6. Step 2. If continuity is not present, go to 2.
2. Leave P307 disconnected. Go to 3.	

TABLE NO. 15-2-3-16. o o o o o (90% RPM) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
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3. Check continuity between J289-15 and ground.

Step 1. If continuity is present, replace APU electrical harness (TM 55-2835-208-23).

Go to **6**.

Step 2. If continuity is not present, go to **4**.

4. Check continuity between P261-15 and ground.

Step 1. If continuity is present, repair/replace wiring between P261-15 and P289-15 (Appendix F). Go to **5**.

Step 2. If continuity is not present, replace DESU (PARA 9-4-39). Go to **5**.

5. Connect P307 to maximum fuel valve. Go to 6.**6. Procedure completed.****TABLE NO. 15-2-3-17. ●● o ●● (Overspeed) Is Displayed.**

TEST OR INSPECTION	CORRECTIVE ACTION
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1. Do DYNAMIC and STATIC checks. Go to 2.**2. If speed sequence is good, go to 3.****3. Install an insulated jumper wire between P305-A and P305-B. Go to 4.****4. Check continuity between J289-17 and ground.**

Step 1. If continuity is present, go to **5**.

Step 2. If continuity is not present, go to **11**.

5. Remove jumper wire. Go to 6.**6. Start and operate APU (Appropriate Flight Manual). Go to 7.****7. Does APU operate normally?**

Step 1. If APU operates normally, System is operational. Go to **13**.

TABLE NO. 15-2-3-17. ●● o ● ● (Overspeed) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If APU does not operate normally and DESU indicator shows overspeed, go to 8 .
8. Replace APU start fuel valve (TM 55-2835-208-23). Go to 9.	
9. Start and operate APU (Appropriate Flight Manual). Go to 10.	
10. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 13 . Step 2. If APU does not operate normally, replace APU fuel pump/acceleration control (TM 55-2835-208-23). Go to 13 .
11. Remove jumper wire. Go to 12.	
12. Replace APU electrical harness (TM 55-2835-208-23). Go to 13.	
13. Procedure completed.	

TABLE NO. 15-2-3-18. o ● o ● ● (Underspeed) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Make sure all bleed-air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 2.	
2. Do DYNAMIC and STATIC checks. Go to 3.	
3. If speed sequence is good, go to 4.	
4. Loosen nut on inlet fuel filter inlet. Place rag around nut. Go to 5.	
5. Place upper console FUEL PUMP switch to APU BOOST. Go to 6.	
6. Check that fuel is going into rag.	Step 1. If fuel is present, go to 7 . Step 2. If fuel is not present, go to 19 .

TABLE NO. 15-2-3-18. o ● o ● ● (Underspeed) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Tighten nut. Go to 8.	
8. Install an insulated jumper wire between P307-A and P307-B. Go to 9.	
9. Check continuity between J289-15 and ground.	Step 1. If continuity is present, go to 10. Step 2. If continuity is not present, go to 17.
10. Remove jumper wire. Go to 11.	
11. Replace maximum fuel valve (TM 55-2835-208-23). Go to 12.	
12. Start and operate APU (Appropriate Flight Manual). Go to 13.	
13. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 21. Step 2. If APU does not operate normally and o ● o ● ● is displayed, go to 14.
14. Replace APU fuel manifold (TM 55-2835-208-23). Go to 15.	
15. Start and operate APU (Appropriate Flight Manual). Go to 16.	
16. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 21. Step 2. If APU does not operate normally, replace APU fuel pump/acceleration control (TM 55-2835-208-23). Go to 21.
17. Remove jumper wire. Go to 18.	
18. Replace APU harness (TM 55-2835-208-23). Go to 21.	
19. Tighten nut. Go to 20.	
20. Troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 21.	
21. Procedure completed.	

TABLE NO. 15-2-3-19. o o o ● ● (Low Oil Pressure) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check APU oil sump for proper level.	Step 1. If oil level is good, go to 2. Step 2. If oil level is not good, service APU oil sump as required (Appropriate Flight Manual). Go to 5.
2. Do DYNAMIC and STATIC checks. Go to 3.	
3. If APU oil pressure is good, replace APU low oil pressure switch (TM 55-2835-208-23). Go to 4.	
4. Start and operate APU (Appropriate Flight Manual). Go to 5.	
5. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 6. Step 2. If APU does not operate normally and o o o ● ● is displayed, replace APU wiring harness (TM 55-2835-208-23). Go to 6.
6. Procedure completed.	

TABLE NO. 15-2-3-20. o ● o o ● (Loss Of Speed Data) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Check for 1,190 to 1,450 ohms between P261-12 and P261-13.	Step 1. If resistance is as specified, replace DESU (PARA 9-4-39). Go to 9. Step 2. If resistance is not specified, go to 2.
2. Install an insulated jumper wire between P302-A and P302-B. Go to 3.	
3. Check continuity between P261-12 and P261-13.	Step 1. If continuity is present, go to 4. Step 2. If continuity is not present, go to 6.
4. Remove jumper wire from P302. Go to 5.	

TABLE NO. 15-2-3-20. o ● o o ● (Loss Of Speed Data) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
5. Replace speed sensor (magnetic pickup) (TM 55-2835-208-23). Go to 9.	
6. Remove jumper wire from P302. Go to 7.	
7. Replace APU wiring harness (TM 55-2835-208-23). Go to 8.	
8. If problem remains, repair/replace wiring (Appendix F) between, then go to 9. P261-12 and P289-12 P261-13 and P289-13	
9. Procedure completed.	

TABLE NO. 15-2-3-21. o ● ● o o (Fail To Light) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. What was APU speed when o ● ● o o (Fail To Light) was displayed?	Step 1. If APU speed was over 14% (o o o●o) and less than 70% (o o o●o), go to 2. Step 2. If APU speed was over 70% (o o o●o) and less than 90% (o o o o o), go to 15.
2. Check that main fuel valve actuates (determined by feeling the valve for a thump) during start sequence.	Step 1. If valve actuates, go to 3. Step 2. If valve does not actuate, go to 13.
3. Loosen nut on inlet fuel filter inlet. Place rag around nut. Go to 4.	
4. Place upper console FUEL PUMP switch to APU BOOST. Go to 5.	
5. Check that fuel is going into rag.	Step 1. If fuel is present, go to 6. Step 2. If fuel is not present, go to 11.
6. Tighten nut. Go to 7.	

TABLE NO. 15-2-3-21. o ● ● o o (Fail To Light) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Loosen nut on inlet fuel filter outlet. Place rag around nut. Go to 8.	
8. Place upper console FUEL PUMP switch to APU BOOST. Go to 9.	
9. Check that fuel is going into rag.	
	Step 1. If fuel is present, go to 10. Step 2. If fuel is not present, replace filter (TM 55-2835-208-23). Go to 24.
10. Check for restricted outlet filter.	
	Step 1. If filter is restricted, replace fuel pump (TM 55-2835-208-23). Go to 24. Step 2. If filter is not restricted, replace main fuel nozzles/manifold assembly (TM 55-2835-208-23). Go to 24.
11. Tighten nut. Go to 12.	
12. Troubleshoot Fuel Prime Boost System (PARA 10-2-3). Go to 24.	
13. Check for 28 vdc between P306-A and ground.	
	Step 1. If voltage is as specified, go to 14. Step 2. If voltage is not as specified, do DYNAMIC and STATIC checks.
14. Check continuity between P306-B and ground.	
	Step 1. If continuity is present, replace main fuel valve (TM 55-2835-208-23). Go to 24. Step 2. If continuity is not present, replace APU wiring harness (TM 55-2835-208-23). Go to 24.
15. Make sure all bleed-air ducts are closed and driven equipment is unloaded. Check air inlet for FOD. Go to 16.	
16. Start APU (Appropriate Flight Manual). Go to 17.	
17. Does APU operate normally?	
	Step 1. If APU operates normally, System is operational. Go to 24. Step 2. If APU does not operate normally, go to 18.

TABLE NO. 15-2-3-21. o ● ● o o (Fail To Light) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
18. Replace fuel manifold (TM 55-2835-208-23). Go to 19.	
19. Start APU (Appropriate Flight Manual). Go to 20.	
20. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 24. Step 2. If APU does not operate normally, go to 21.
21. Replace fuel pump/acceleration control (TM 55-2835-208-23). Go to 22.	
22. Start APU (Appropriate Flight Manual). Go to 23.	
23. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 24. Step 2. If APU does not operate normally, replace APU (PARA 15-4-1). Go to 24.
24. Procedure completed.	

TABLE NO. 15-2-3-22. ● o ● o o (Fail To Accel (Start Fuel)) Or ● o ● ● o o (Fail To Accel (Start Main)) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Visually check for any of the following DESU indications: ● ● ● ● ● (5%) ● o o ● o (90%)	Step 1. If any indication listed is present, replace DESU (PARA 9-4-39). Go to 10. Step 2. If any indication listed is not present, go to 2.
2. Make sure APU CONTR switch is placed OFF. Go to 3.	
3. Press APU start valve manual override lever. Go to 4.	
4. Check that APU start motor operates.	

TABLE NO. 15-2-3-22. ● ○ ● ○ ○ (Fail To Accel (Start Fuel)) Or ● ○ ● ● ○ ○ (Fail To Accel (Start Main)) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 1. If motor operates, go to 5 .
	Step 2. If motor does not operate, go to 7 .
5. Do DYNAMIC and STATIC checks. Go to 6.	
6. If DESU hydraulic start valve is good, replace APU hydraulic start valve (PARA 7-4-53). Go to 10.	
7. Disconnect start motor from APU accessory pad. Go to 8.	
8. Using a 1/2 inch allen wrench, turn splined shaft on accessory pad. Check that shaft turns.	
	Step 1. If shaft turns, go to 9 .
	Step 2. If shaft does not turn, replace seized APU (PARA 15-4-1). Go to 10 .
9. Attempt to turn splined shaft of APU start motor.	
	Step 1. If shaft turns, replace APU hydraulic start valve (PARA 7-4-53). Go to 10 .
	Step 2. If shaft does not turn, replace APU start motor (PARA 15-4-7). Go to 10 .
10. Procedure completed.	

TABLE NO. 15-2-3-23. ● ○ ○ ○ ○ (Overtemperature Start) Is Displayed.

TEST OR INSPECTION	CORRECTIVE ACTION
1. Do DYNAMIC and STATIC checks. Go to 2.	
2. If APU overtemperature is good, go to 3.	
3. Replace combustor drain valve (TM 55-2835-208-23). Go to 4.	
4. Start and operate APU (Appropriate Flight Manual). Go to 5.	
5. Does APU operate normally?	
	Step 1. If APU operates normally, System is operational. Go to 9 .

TABLE NO. 15-2-3-23. ● o o o o (Overtemperature Start) Is Displayed. (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
	Step 2. If APU does not operate normally and ● o o o o (Overtemperature Start) is displayed, go to 6 .
6. Replace maximum fuel valve (TM 55-2835-208-23). Go to 7.	
7. Start and operate APU (Appropriate Flight Manual). Go to 8.	
8. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 9. Step 2. If APU does not operate normally, replace fuel pump/acceleration control (TM 55-2835-208-23). Go to 9.
9. Procedure completed.	

WARNING

Failure to disconnect electrical connectors will permit fuel to enter combustion chamber and create a fire hazard.

TABLE NO. 15-2-3-24. o o ● ● ● (Overtemperature (Run)).

TEST OR INSPECTION	CORRECTIVE ACTION
1. Make sure all bleed-air ducts are closed and driven equipment is unloaded. Go to 2.	
2. Check air inlet for FOD. Go to 3.	
3. Do DYNAMIC and STATIC checks. Go to 4.	
4. If APU overtemperature and overspeed is good, go to 5.	
5. Replace combustor drain valve (TM 55-2835-208-23). Go to 6.	
6. Start and operate APU (Appropriate Flight Manual). Go to 7.	

TABLE NO. 15-2-3-24. o o ● ● ● (Overtemperature (Run)). (Cont)

TEST OR INSPECTION	CORRECTIVE ACTION
7. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 21. Step 2. If APU does not operate normally and o o ● ● ●(Overtemperature (Run) is displayed, shut down APU (Appropriate Flight Manual). Go to 8.
8. Disconnect P305, P306, and P307 from fuel valves. Go to 9.	
9. Disconnect P303 from ignition excitor. Go to 10.	
10. Disconnect fuel line from main fuel valve to the fuel manifold. Go to 11.	
11. Place suitable containers under open fuel lines. Go to 12.	
12. Motor APU and check for fuel flow from main fuel valve.	Step 1. If fuel is present, replace main fuel valve (PARA 10-4-8). Go to 23. Step 2. If fuel is not present, go to 13.
13. Leave electrical connectors disconnected. Go to 14.	
14. Reconnect fuel line to fuel manifold. Go to 15.	
15. Disconnect fuel line at outlet of maximum fuel valve. Go to 16.	
16. Place suitable containers under open fuel lines. Go to 17.	
17. Motor APU and check for fuel at maximum fuel valve outlet.	Step 1. If fuel is present, replace maximum fuel valve (TM 55-2835-208-23). Go to 23. Step 2. If fuel is not present, go to 18.
18. Reconnect all electrical connectors and fuel lines. Go to 19.	
19. Start and operate APU (Appropriate Flight Manual). Go to 20.	
20. Does APU operate normally?	Step 1. If APU operates normally, System is operational. Go to 21.

TABLE NO. 15-2-3-24. ○ ○ ● ● ● (Overtemperature (Run)). (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If APU does not operate normally, shut down APU (Appropriate Flight Manual). Go to **22**.

21. Shut down APU (Appropriate Flight Manual). Go to 23..

22. Replace fuel pump/acceleration control (TM 55-2835-208-23). Go to 23..

23. Procedure completed.

TABLE NO. 15-2-3-25. APU Does Not Shut Down When APU CONTR Switch Is Placed OFF.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check for 28 vdc between P260-2 and P260-6.

Step 1. If voltage is as specified, replace DESU (PARA 9-4-39). Go to **5**.

Step 2. If trouble remains, replace main fuel valve or maximum fuel valve (TM 55-2835-208-23). Go to **5**.

Step 3. If voltage is not as specified, go to **2**.

2. Check for 28 vdc between P260-2 and ground.

Step 1. If voltage is as specified, repair/replace wiring between P260-2 and P260-6 (Appendix F). Go to **5**.

Step 2. If voltage is not as specified, go to **3**.

3. Check for 28 vdc between J200-P and ground.

Step 1. If voltage is as specified, repair/replace wiring between P200-P and P260-2 (Appendix F). Go to **5**.

Step 2. If voltage is not as specified, go to **4**.

4. Check for 28 vdc between APU CONTR switch terminal 1 and ground.

Step 1. If voltage is as specified, repair/replace wiring between switch terminal 1 and J200-P (Appendix F). Go to **5**.

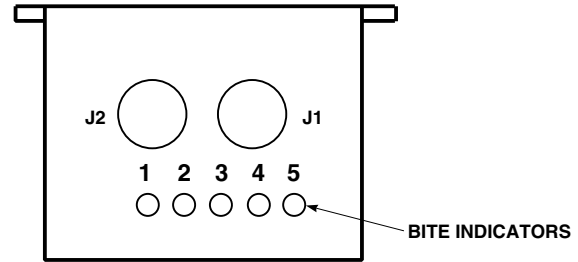
Step 2. If voltage is not as specified, replace switch (PARA 9-4-5). Go to **5**.

TABLE NO. 15-2-3-25. APU Does Not Shut Down When APU CONTR Switch Is Placed OFF. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION
5. Procedure completed.

NORMAL BITE SEQUENCE DURING APU START

	1	2	3	4	5	
* ● ● ● ● ●	●	●	●	●	●	NO ELECTRICAL POWER TO DESU
* * ● ● ● ● ●	●	●	●	●	●	BATT SWITCH-ON
● ● ● ● ●	●	●	●	●	●	APU CONTR SWITCH-ON
● ● ● ● ●	●	●	●	●	●	POWER ON RESET (APU BEGINS MOTORING)
● ● ● ● ○	●	●	●	●	○	START COMMAND RECEIVED
○ ● ● ● ●	○	●	●	●	●	START FUEL AND EXCITER ON (5%)
○ ○ ● ● ○	○	○	●	●	○	MAIN FUEL ON (14%)
○ ○ ○ ● ○	○	○	○	●	○	START FUEL AND EXCITER OFF (70%)
○ ○ ○ ○ ○	○	○	○	○	○	90%
● ● ● ● ●	●	●	●	●	●	READY TO LOAD (90% + 1.5 SECOND OR NORMAL OPERATION)



**DIGITAL ELECTRONIC SEQUENCE
UNIT (DESU)**

NOTE: APU START MOTOR WILL
MOTOR APU UP TO 20%-30% SPEED

- * IF LAST SHUTDOWN WAS A FAILURE,
FAILURE INDICATOR WOULD BE SHOWN.
- * * #1 BITE WILL CYCLE FROM BLACK TO WHITE
TO BLACK. IF BITE IS WHITE (MALFUNCTION)
THIS WILL CYCLE FROM WHITE TO BLACK.

FAILURE BITE INDICATIONS ASSOCIATED WITH SHUTDOWN DURING START OF OPERATION

	1	2	3	4	5	
○ ○ ● ● ●	○	○	●	●	●	OVERTEMPERATURE (RUN)
● ● ○ ● ●	●	●	○	●	●	OVERSPEED
○ ● ○ ● ●	○	●	○	●	●	UNDERSPEED
● ○ ○ ● ●	●	○	○	●	●	FAIL TO ROTATE
○ ○ ○ ● ●	○	○	○	●	●	LOW OIL PRESSURE
○ ● ● ○ ○	○	●	●	○	○	SHORTED OIL PRESSURE SWITCH
● ● ○ ○ ●	●	●	○	○	●	OPEN CIRCUIT FAULT THERMOCOUPLE
○ ○ ○ ○ ○	○	○	○	○	○	APU START SWITCH FAILURE
○ ● ○ ○ ○	○	●	○	○	○	LOSS OF SPEED DATA
● ● ○ ● ●	●	●	○	●	●	OVERSPEED
○ ○ ○ ○ ●	○	○	○	○	●	POWER INTERRUPT
● ● ● ● ○	●	●	●	●	○	RTL INDICATOR FAILURE
○ ● ● ● ○	○	●	●	●	○	MALFUNCTION INDICATOR FAILURE
● ○ ● ● ○	●	○	●	●	○	MAXIMUM FUEL VALVE FAILURE
○ ○ ● ● ○	○	○	●	●	○	HOT INDICATOR FAILURE
● ● ○ ● ○	●	●	○	●	○	MAIN FUEL VALVE FAILURE
○ ● ○ ● ○	○	●	○	●	○	ANTI - SURGE VALVE FAILURE
● ○ ○ ● ○	●	○	○	●	○	START CIRCUIT FAILURE
○ ○ ○ ● ○	○	○	○	●	○	START FUEL EXCITER FAILURE
● ● ● ○ ○	●	●	●	○	○	ILLOGICAL SWITCH INPUTS
○ ● ● ○ ○	○	●	●	○	○	FAIL TO LIGHT
● ○ ● ○ ○	●	○	●	○	○	FAIL TO ACCELERATE (START FUEL)
○ ○ ● ○ ○	○	○	●	○	○	FAIL TO ACCELERATE (START & MAIN)
● ● ○ ○ ○	●	●	○	○	○	FAIL TO ACCELERATE (MAIN FUEL)
○ ● ○ ○ ○	○	●	○	○	○	SLOW ACCELERATE (TIME OUT)
● ○ ○ ○ ○	●	○	○	○	○	OVERTEMPERATURE (START)
○ ○ ○ ○ ○	○	○	○	○	○	APU START SWITCH FAILURE

WARNING BITE INDICATORS

	1	2	3	4	5	
● ● ● ○ ●	●	●	●	○	●	HIGH OIL TEMPERATURE
+ ● ○ ● ○ ●	+	●	○	●	○	SHORTED THERMOCOUPLE

THESE INDICATIONS WILL NOT CAUSE
APU TO SHUT DOWN; ALSO, INDICATIONS
WILL NOT BE SHOWN AFTER A NORMAL
SHUTDOWN

- + THIS MALFUNCTION WILL CAUSE A
SHUTDOWN DURING NEXT ATTEMPTED
START WITH A FAIL LIGHT BITE
INDICATION (SEE NOTE)

FIGURE 15-2-3-1. DIGITAL ELECTRONIC SEQUENCE UNIT (DESU)
BITE INDICATOR CONDITIONS

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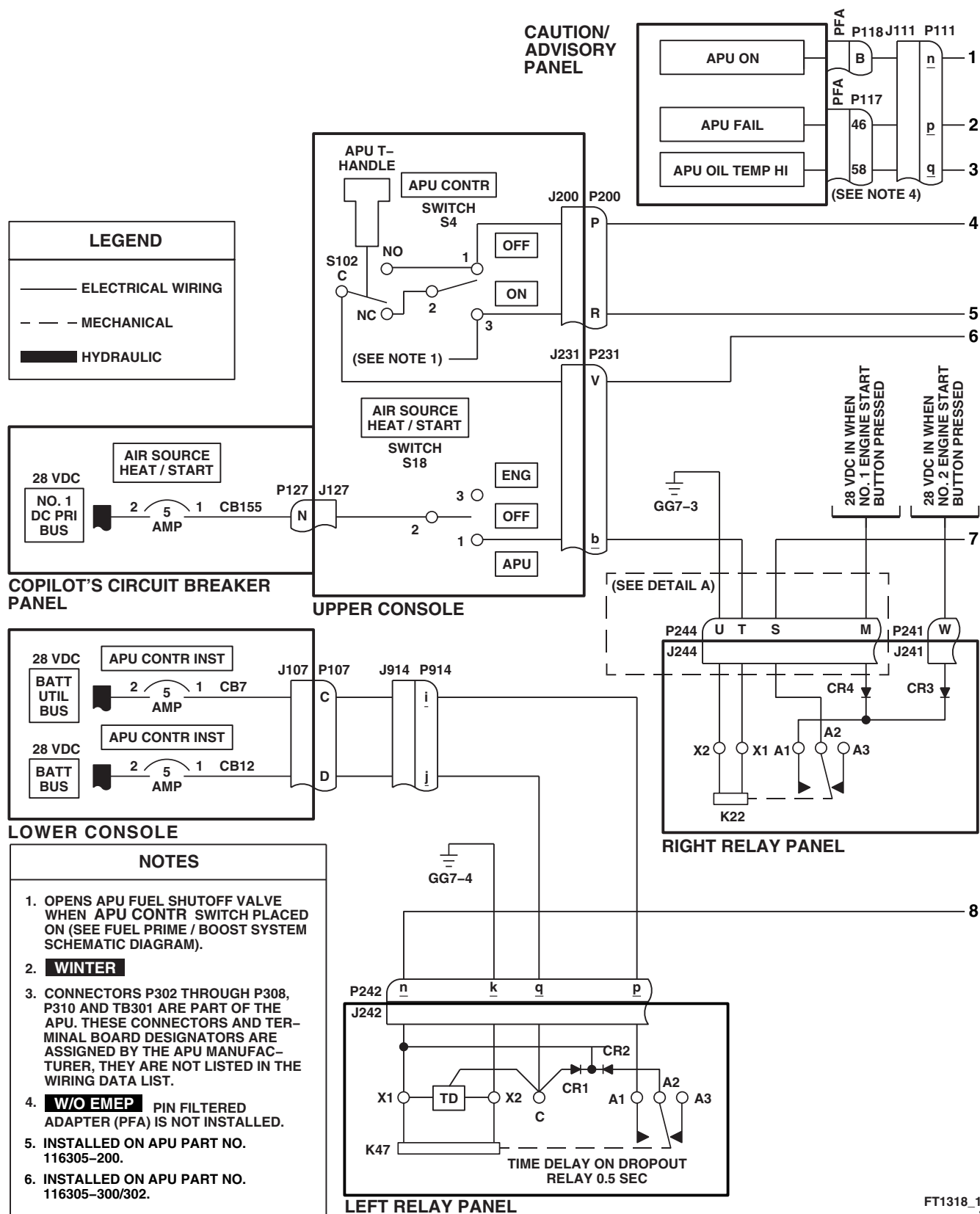


FIGURE 15-2-3-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 5)

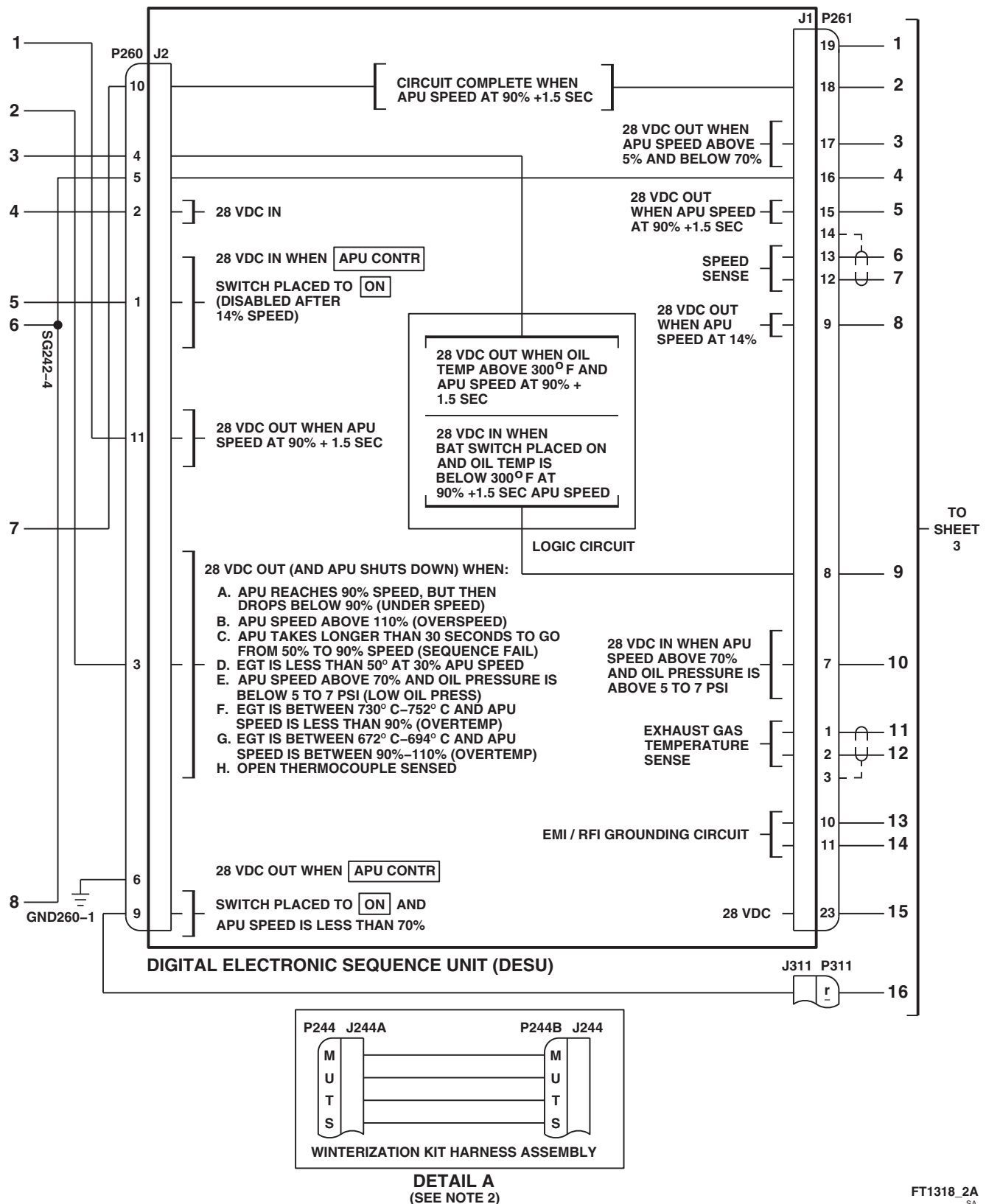


FIGURE 15-2-3-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 5)

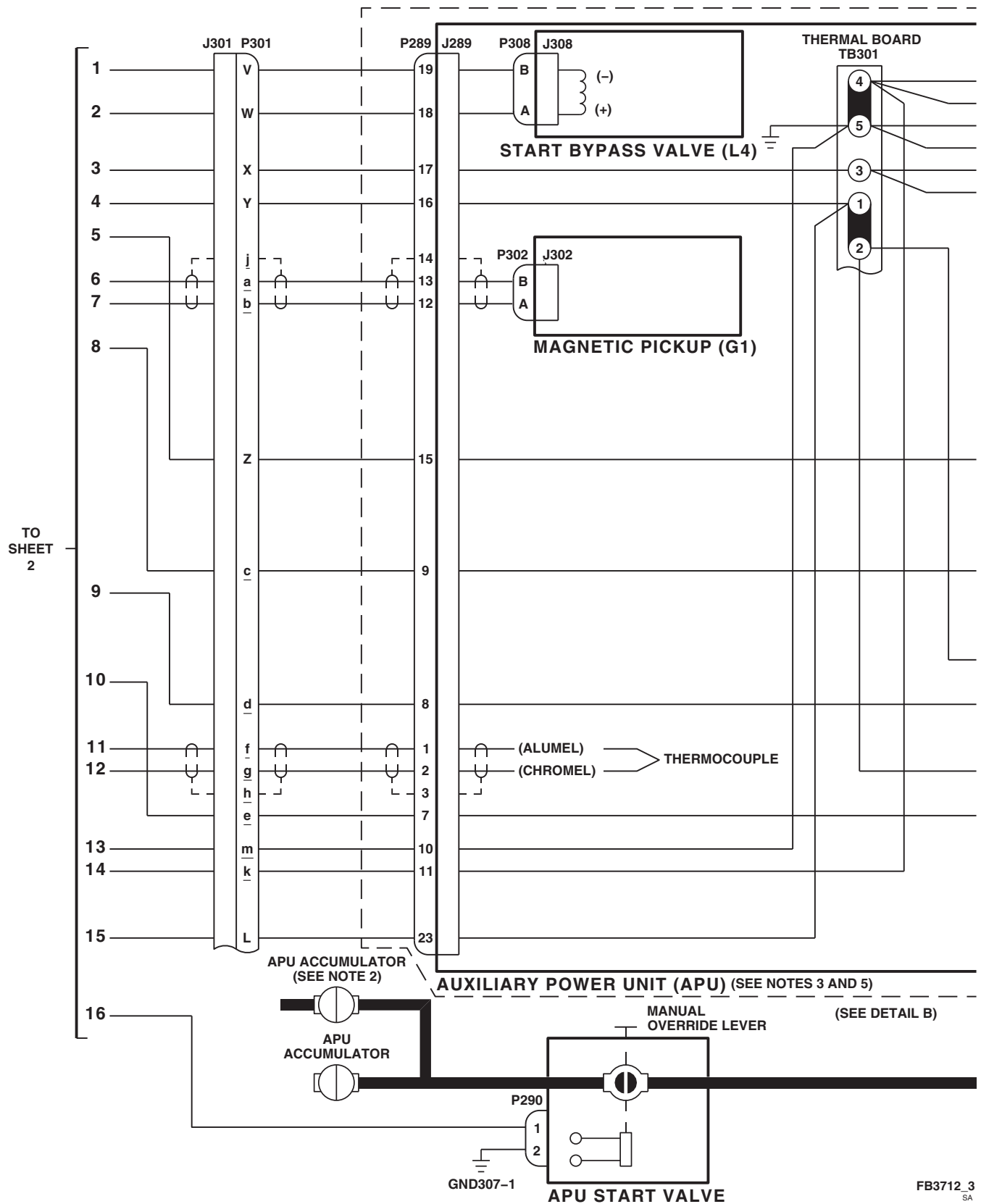


FIGURE 15-2-3-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 3 OF 5)

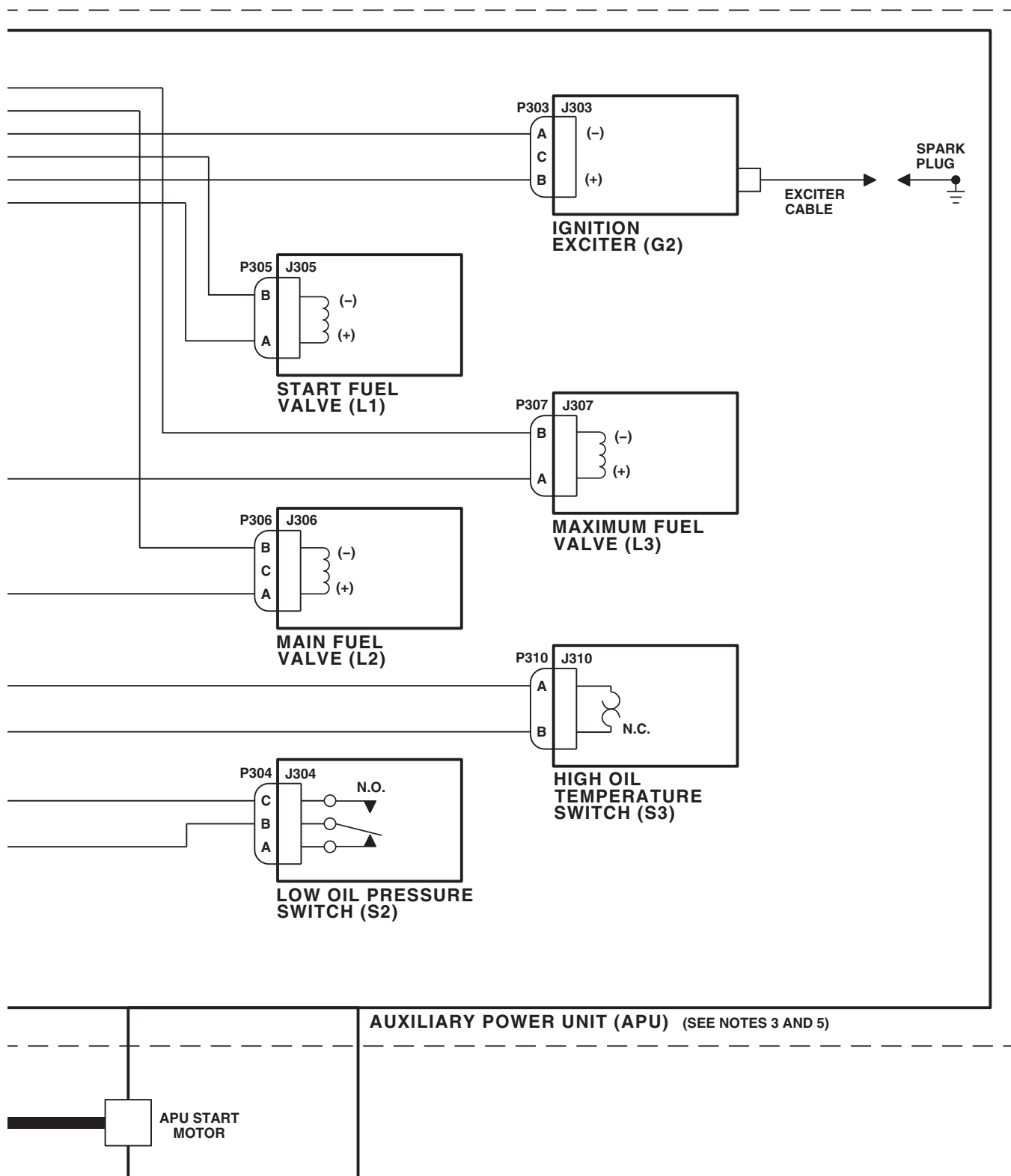
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FIGURE 15-2-3-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 4 OF 5)

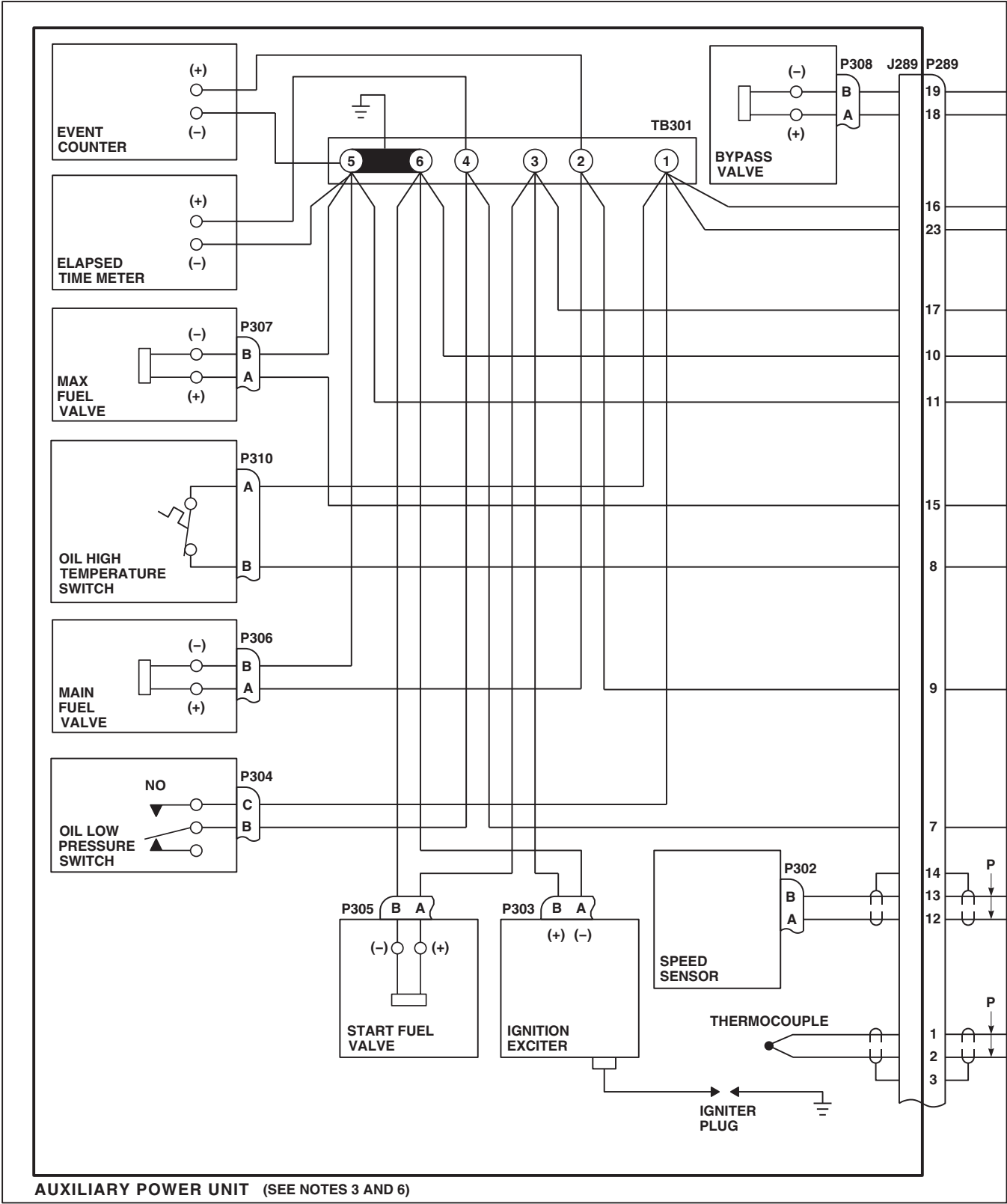


FIGURE 15-2-3-2. AUXILIARY POWER UNIT SYSTEM SCHEMATIC DIAGRAM (SHEET 5 OF 5)

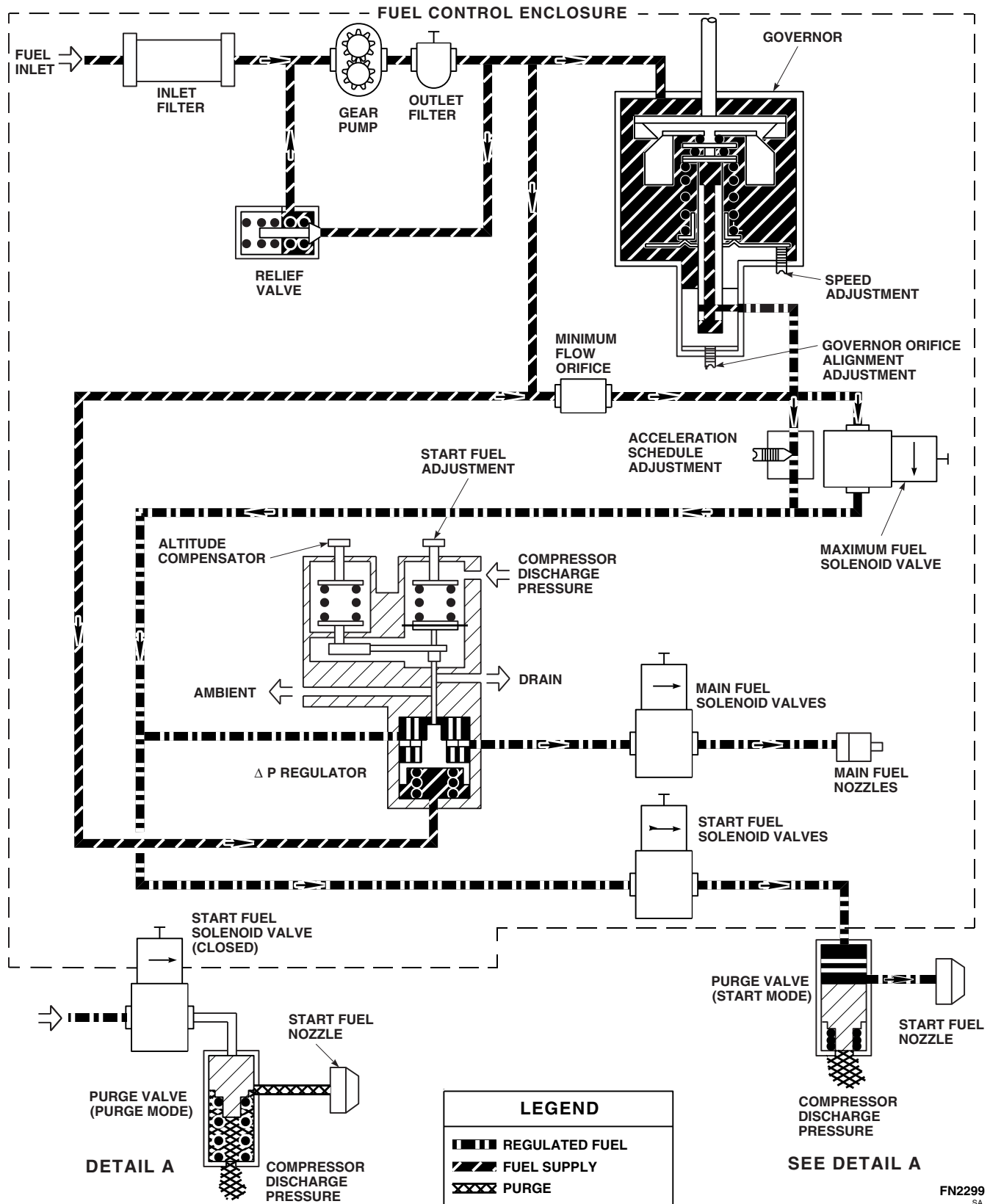
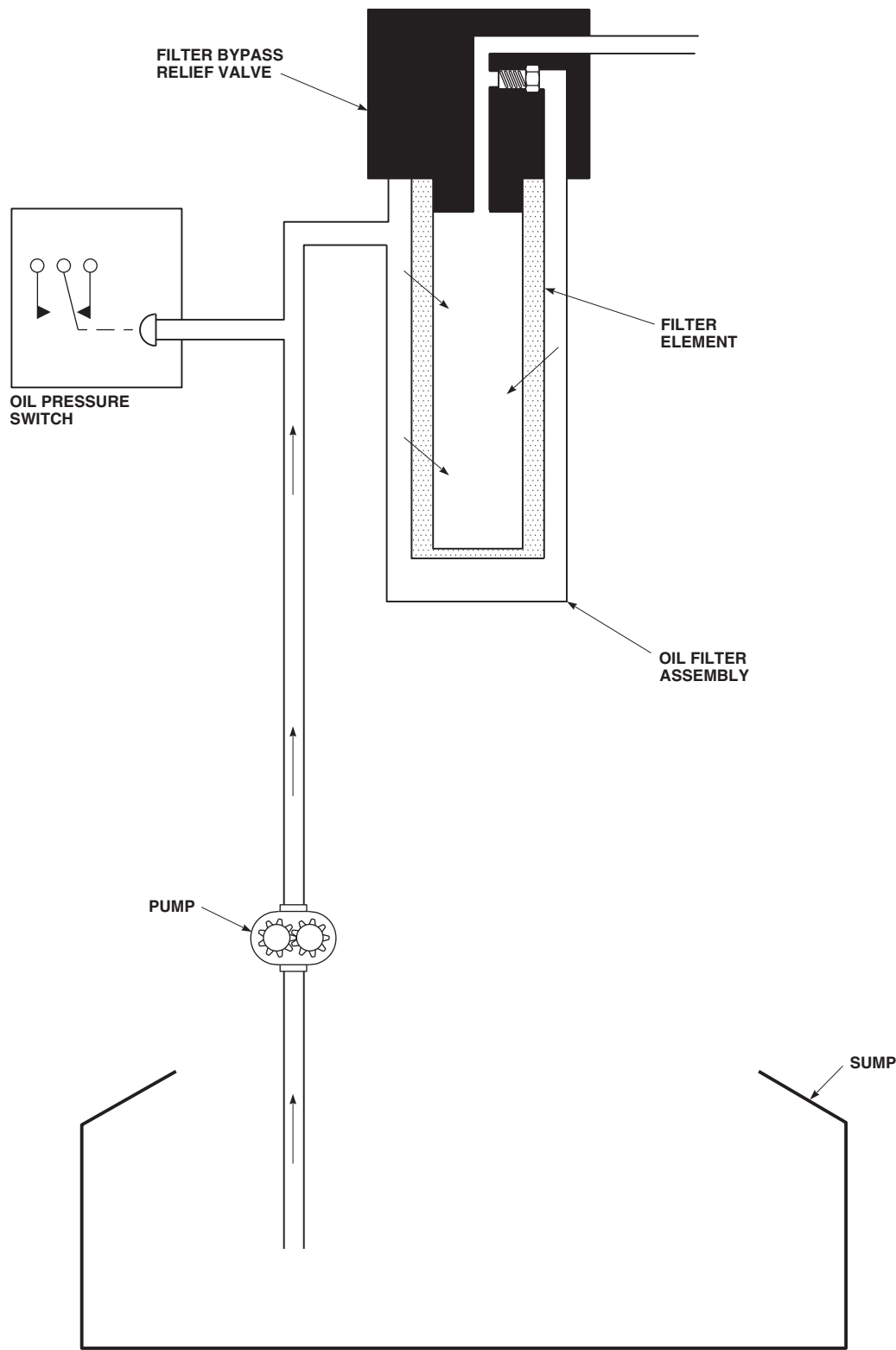
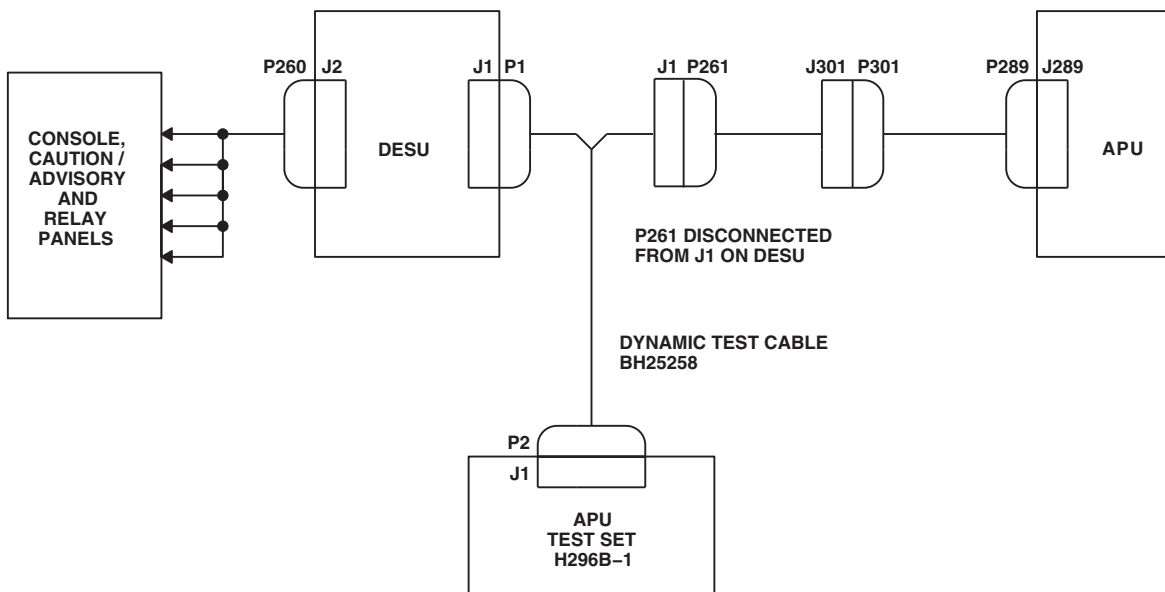


FIGURE 15-2-3-3. AUXILIARY POWER UNIT FUEL SYSTEM SCHEMATIC

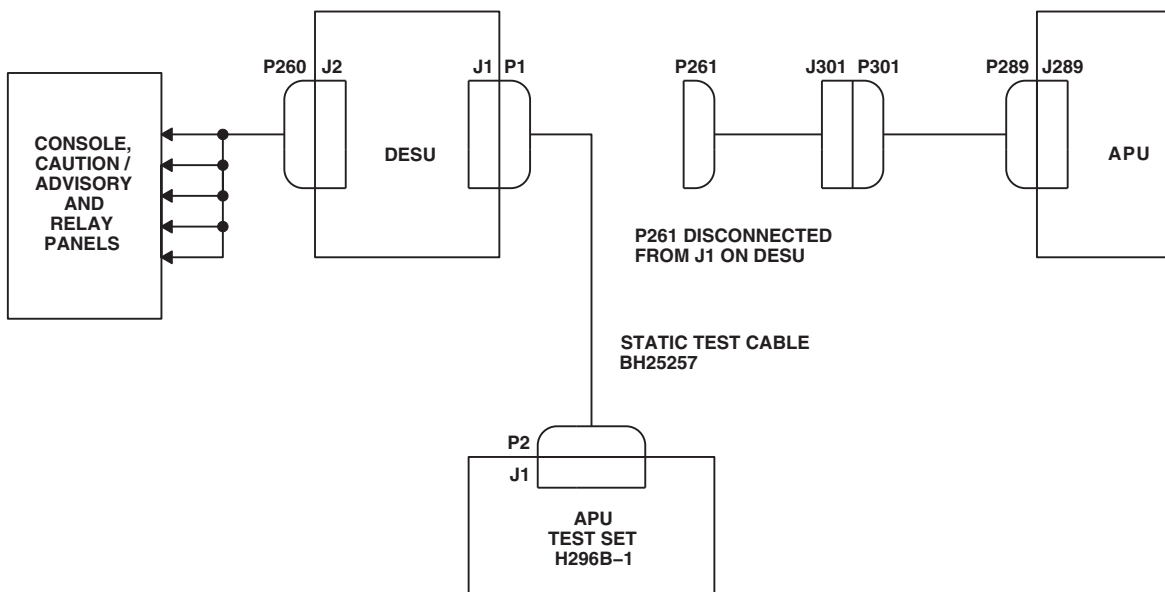


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FIGURE 15-2-3-4. AUXILIARY POWER UNIT LUBRICATION SYSTEM SCHEMATIC



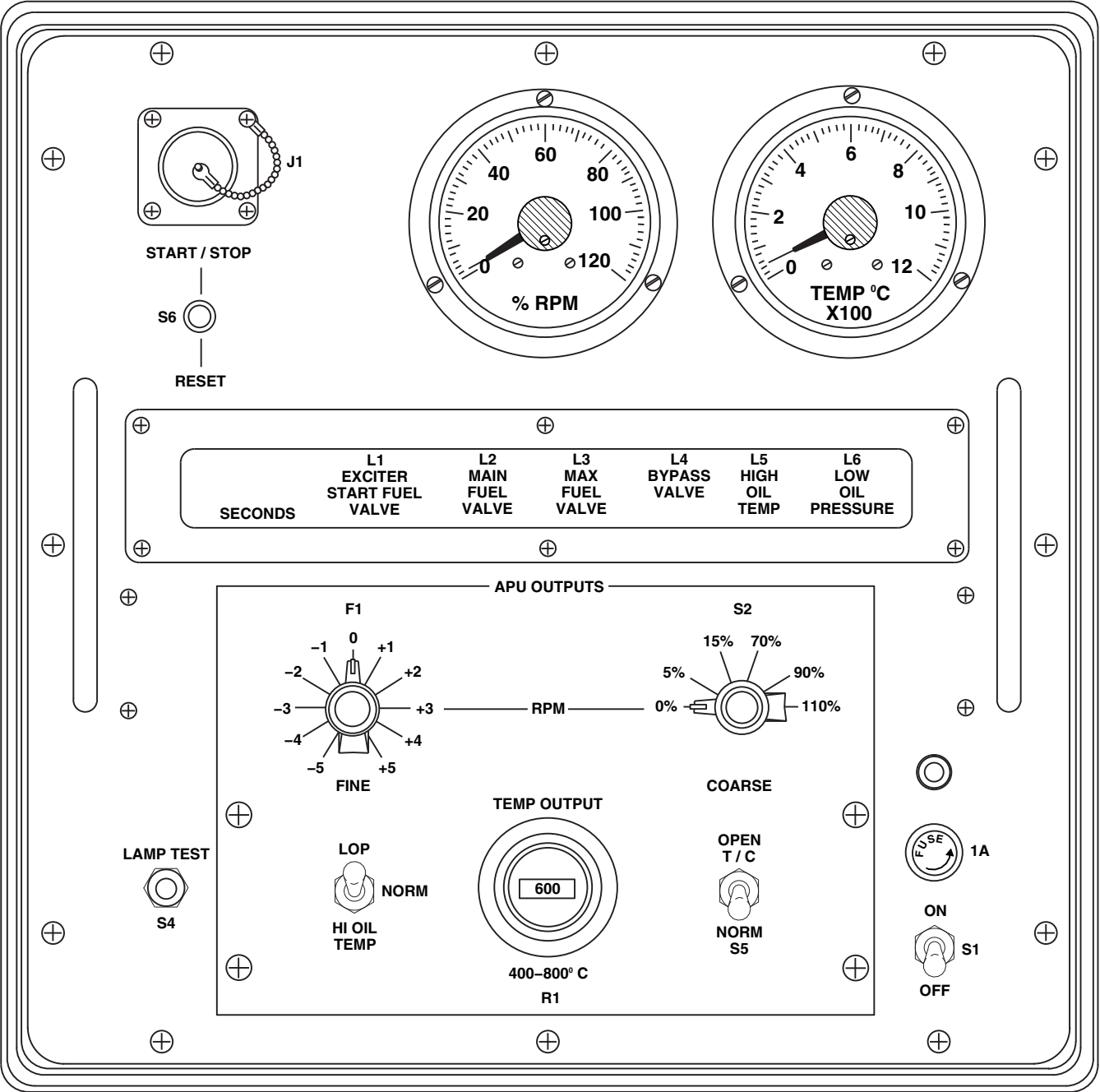
APU DYNAMIC TEST CONDITIONS



APU STATIC TEST CONDITIONS

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FIGURE 15-2-3-5. AUXILIARY POWER UNIT TEST SETUP (SHEET 1 OF 2)



APU TEST SET, H296B-1

FIGURE 15-2-3-5. AUXILIARY POWER UNIT TEST SETUP (SHEET 2 OF 2)

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15-2-3-45/(15-2-3-46 Blank)

SECTION 15-3
ON AIRCRAFT INSPECTIONS
NOT APPLICABLE

SECTION 15-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
15-4-1	Auxiliary Power Unit (APU)
15-4-2	APU Exhaust Duct
15-4-3	APU Exhaust Pipe
15-4-4	APU Shroud Assembly
15-4-5	APU Front Mounting Lugs
15-4-6	APU Rear Mounting Lug
15-4-7	APU Starter Motor
15-4-8	APU Spline Adapter
15-4-9	Prime APU Fuel Inlet Hose
15-4-10	APU Fuel Shutoff Valve
15-4-11	APU Fuel Inlet Hose
15-4-12	APU Drain Lines
15-4-13	APU Front Inboard Support Mount
15-4-14	APU Front Outboard Support Mount
15-4-15	APU Front Inboard and Outboard Support Mount Repair
15-4-16	APU Rear Support Mount

15-4-1 **AUXILIARY POWER UNIT (APU).**

PARAGRAPH BREAKDOWN

	PAGE
15-4-1.1 Replace APU	15-4-1-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- APU Adapter, 70700-20423-041
- Cabin Maintenance Crane, 70700-20300-042 (Alternate for Hoist)
- Hoist, Capacity to Support APU (150 lbs)
- Hoisting Sling, 70700-20323-042
- Maintenance Trailer
- Nonmetallic Mallet
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wrench

Materials/Parts

- Adhesive, Item 53, Appendix D
- Antiseize Compound, Item 74, Appendix D
- Grease, Item 144, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- Appendix E
- PARA 1-3-9
- PARA 1-3-10
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-108
- PARA 7-3-2
- PARA 7-4-4
- PARA 9-4-19
- PARA 15-2-1
- PARA 15-2-2
- PARA 15-4-2
- PARA 15-4-7
- PARA 15-4-8
- PARA 15-4-9

- PARA 18-4-1

15-4-1.1 Replace APU.

15-4-1.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If removing APU for storage or BENCH level maintenance, preserve APU (Appendix E).
- If cabin maintenance crane will be used, erect cabin maintenance crane (PARA 18-4-1).
- Remove APU access doors (PARA 2-4-108).
- Remove clamp securing electrical leads to airframe.
- Disconnect electrical connector, P414, from fire detector on rear panel (Figure 15-4-1-1, Sheet 1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Loosen stud fasteners securing fuel control upper cover. Remove fuel control upper cover.

NOTE

Removal of fuel inlet hose bonding jumper is not required unless fuel inlet hose is being removed from helicopter.

- If necessary, remove screw, washers, clamp, and nut securing bonding jumper to fuel inlet hose.

CAUTION

Damage to fuel filter will result if union in fuel control lower cover turns when removing fuel inlet hose. Hold union with wrench while removing fuel inlet hose to avoid damage to fuel filter.

- Hold base of union in fuel control lower cover with wrench, disconnect fuel inlet hose from union. Cap union in lower cover and plug hose.

- Disconnect fuel control box drain hose, gear box vent hose, compartment drain hose, combustor drain hose, and fire extinguisher hose. Plug hoses and cap ports.
- Depressurize APU accumulator (PARA 1-3-10).

WARNING

Injury to personnel and damage to equipment will result if APU start valve is operated with personnel working in area of APU. Do not operate APU start valve when starter motor is being removed.

- Tag APU start valve warning personnel not to operate.
- Remove nuts, washers, APU starter motor, and gasket from mounting pad on APU. Discard gasket.
- Tie APU starter motor to beam. Make sure hoses are not kinked.
- Disconnect electrical connectors, P252 and P279, from APU generator.
- Disconnect electrical connector, P289, from bracket on APU.
- Install protective caps and plugs on electrical connectors.
- Remove APU exhaust duct (PARA 15-4-2).
- Remove clamp securing check valve to pneumatic tube at lower front panel half (Figure 15-4-1-1, Sheet 2, Detail B).
- Loosen stud fasteners securing front panel assembly and rear panel of shroud assembly to exhaust firewall (Figure 15-4-1-1, Sheet 2,

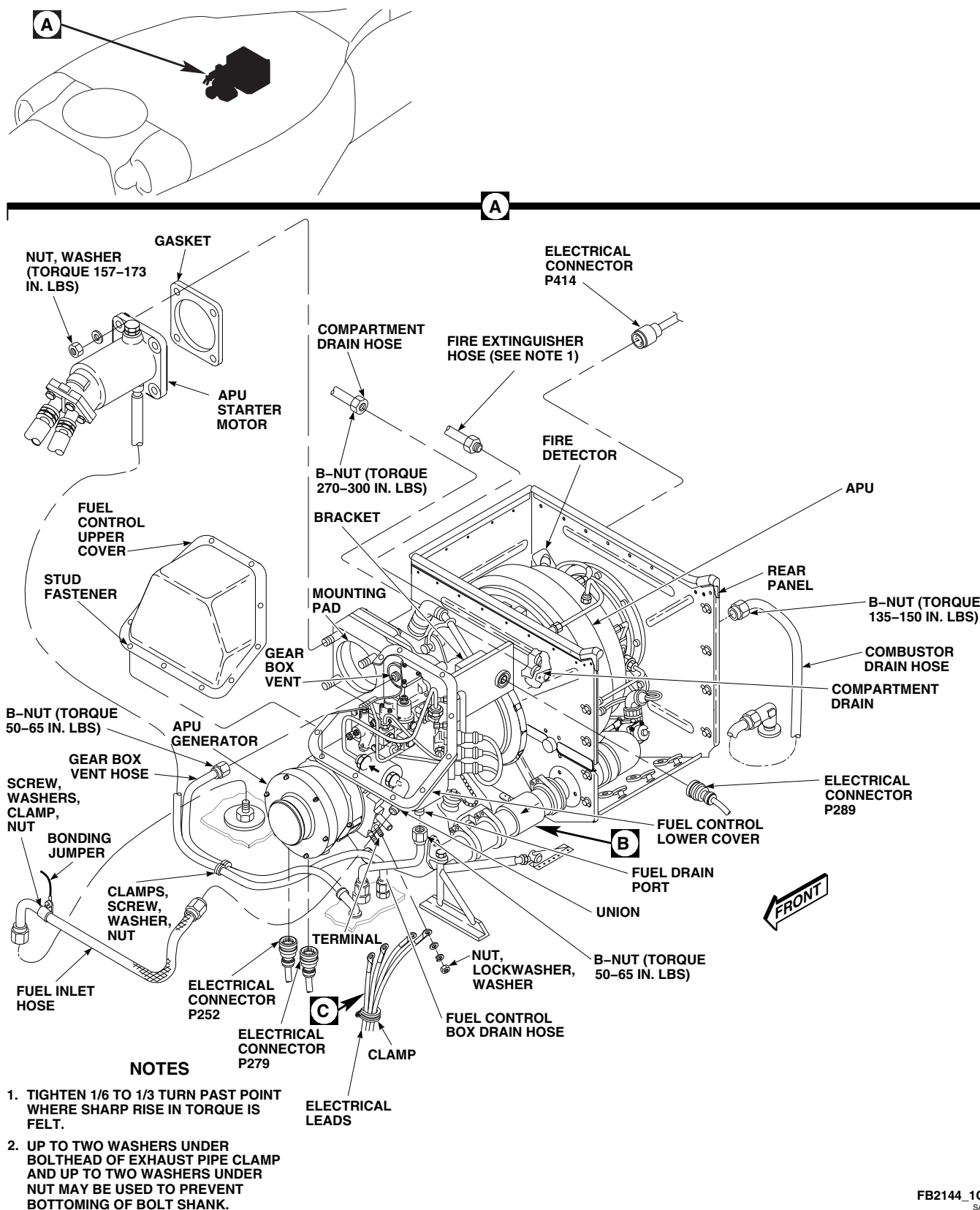


FIGURE 15-4-1-1. REPLACE APU (SHEET 1 OF 4)

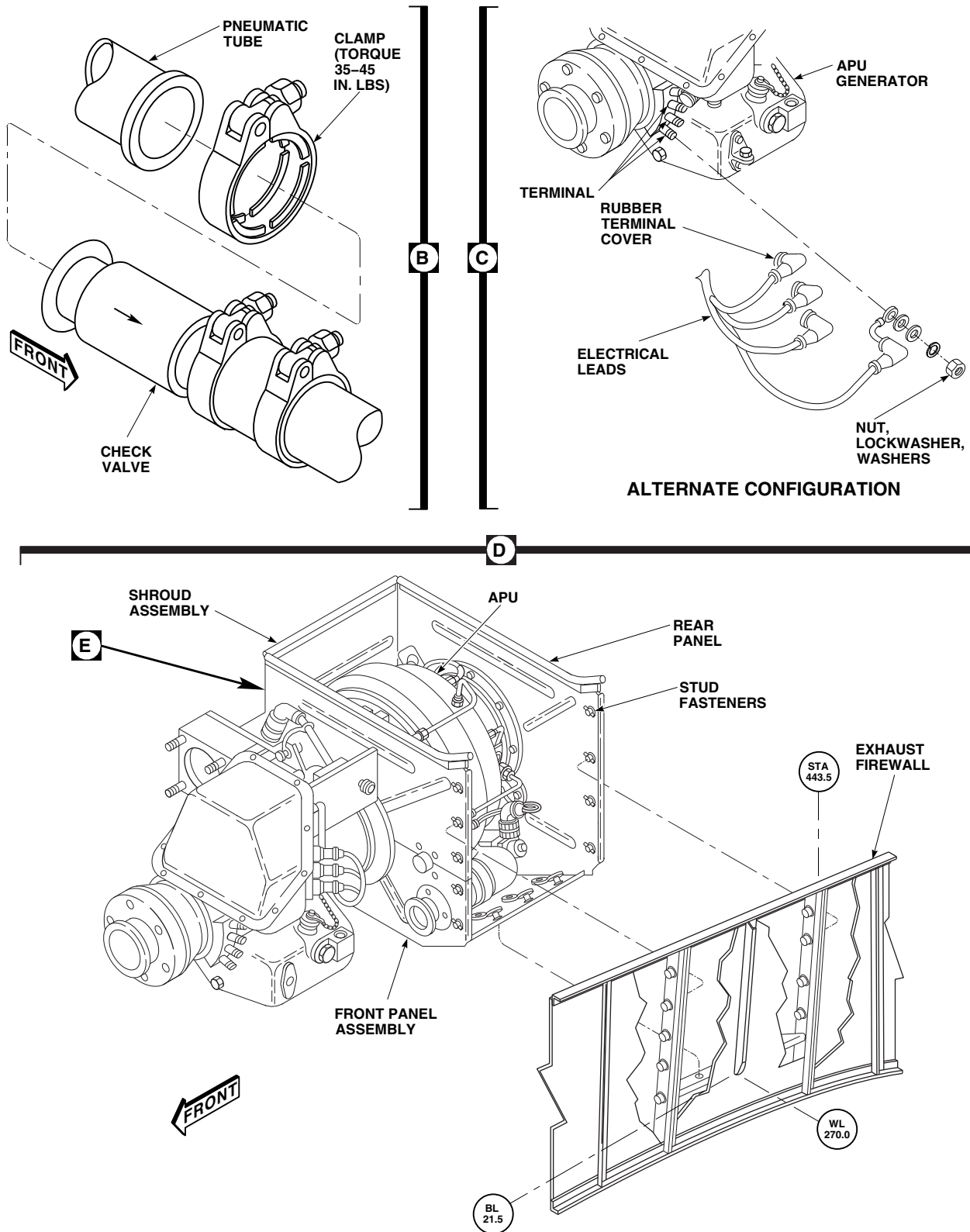


FIGURE 15-4-1-1. REPLACE APU (SHEET 2 OF 4)

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Detail D). Make sure APU (with shroud assembly) is separated from exhaust firewall.

- v. Install hoisting sling on APU (Figure 15-4-1-1, Sheet 3, Detail E). Attach sling to cabin maintenance crane or hoist. Take up slack.
- w. Remove bolts and washers securing front mounting lugs to front support mounts (Figure 15-4-1-1, Sheet 3, Detail F). Operate hoist/cabin maintenance crane and support full weight of APU.
- x. Slide APU to rear, disengaging rear mounting lug from rear support mount. Raise APU up enough to disconnect generator wiring.

NOTE

Metal terminal cover will not be reused.

- y. If metal terminal cover is installed, remove screws and washers securing metal terminal cover, or slide rubber terminal covers back on electrical leads (Figure 15-4-1-1, Sheet 1, Detail A or Sheet 2, Detail C).
- z. Tag and disconnect electrical leads from terminals on APU generator by removing nuts, lockwashers, and washers.

WARNING

- Injury to personnel or damage to equipment will result if cabin maintenance crane is used with excessive force to free APU from helicopter. Do not allow cabin maintenance crane hook to retract fully against pulley when lifting APU from helicopter.
 - Injury to personnel or damage to equipment will result if caution is not taken when hoisting APU. Keep area clear when hoisting APU.
- aa. With assistant operating hoist/cabin maintenance crane, remove APU from helicopter (Figure 15-4-1-1, Sheet 3, Detail E).

- ab. Remove bolts and washers securing front mounting lugs to APU gear box sump, and remove mounting lugs (Figure 15-4-1-1, Sheet 3, Detail F).
- ac. Install APU on maintenance trailer/APU adapter assembly with bolts and washers (Figure 15-4-1-1, Sheet 3, Detail G). If maintenance trailer/APU adapter assembly is not available, support APU to prevent damage to components on bottom of APU.
- ad. Remove hoisting sling from APU.
- ae. On upper and lower front panel halves, remove bolts, washers, sleeve halves, grommet, and plate (Figure 15-4-1-1, Sheet 4, Detail H).
- af. Remove band clamp securing upper and lower front panel halves together.
- ag. On lower front panel half, remove screws and washers securing cover halves around bleed-air outlet flange. Remove cover halves from lower front panel half.
- ah. Remove screws and washers securing upper and lower front panel halves to shroud assembly. Unlock stud fasteners and remove front panel halves.
- ai. Remove lockwire from combustor drain in rear panel.
- aj. Loosen and remove rear mounting lug from compressor (Figure 15-4-1-1, Sheet 3, Detail F).
- ak. Remove shroud assembly surrounding APU, carefully sliding off combustor drain (Figure 15-4-1-1, Sheet 4, Detail H).
- al. Remove combustor drain from combustor fuel drain check valve.
- am. Remove packing from APU combustor fuel drain check valve. Cap port. Discard packing.
- an. Loosen bolt, washer(s), and nut securing exhaust pipe. Remove exhaust pipe from APU.
- ao. Remove APU generator (PARA 9-4-19).

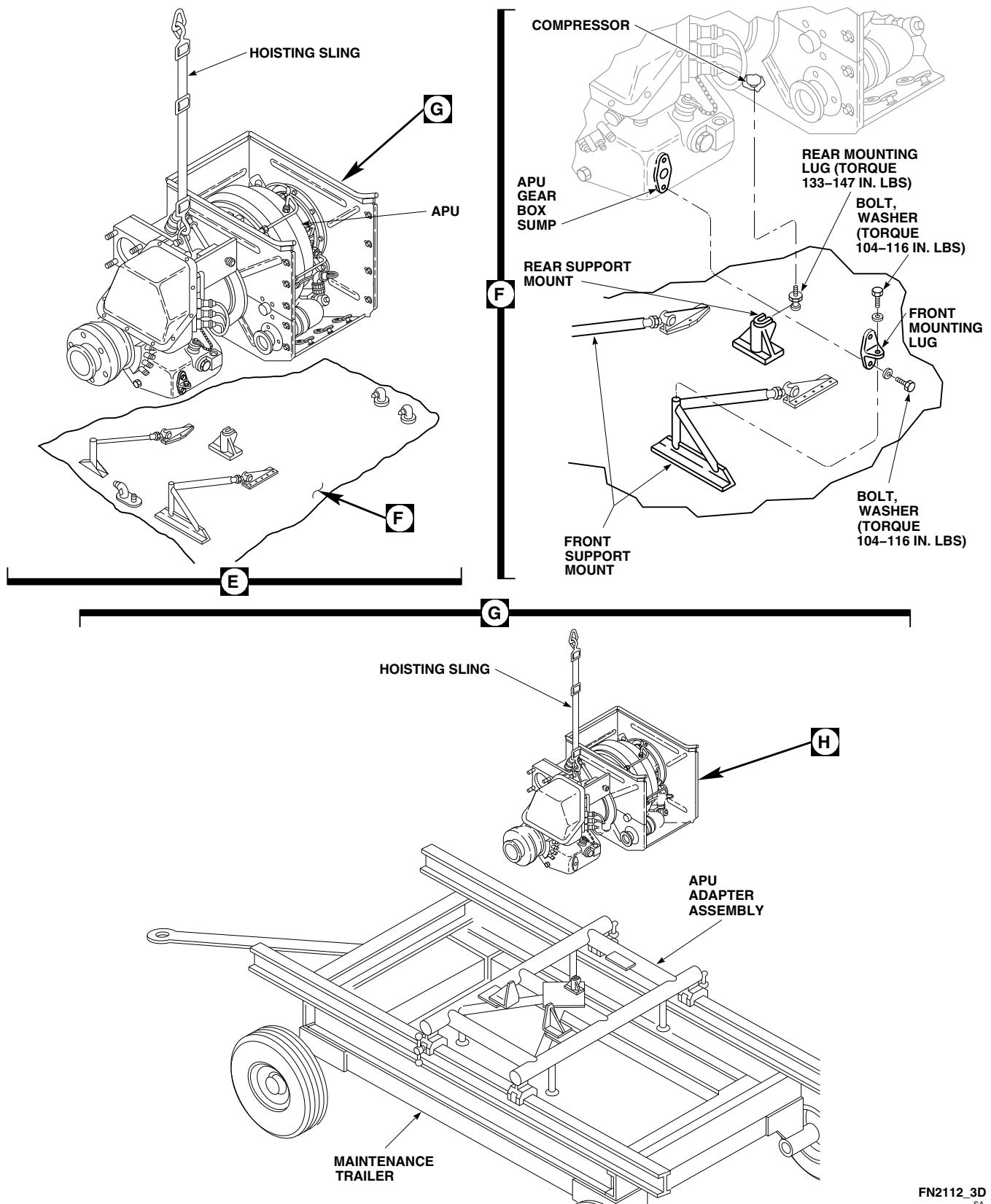
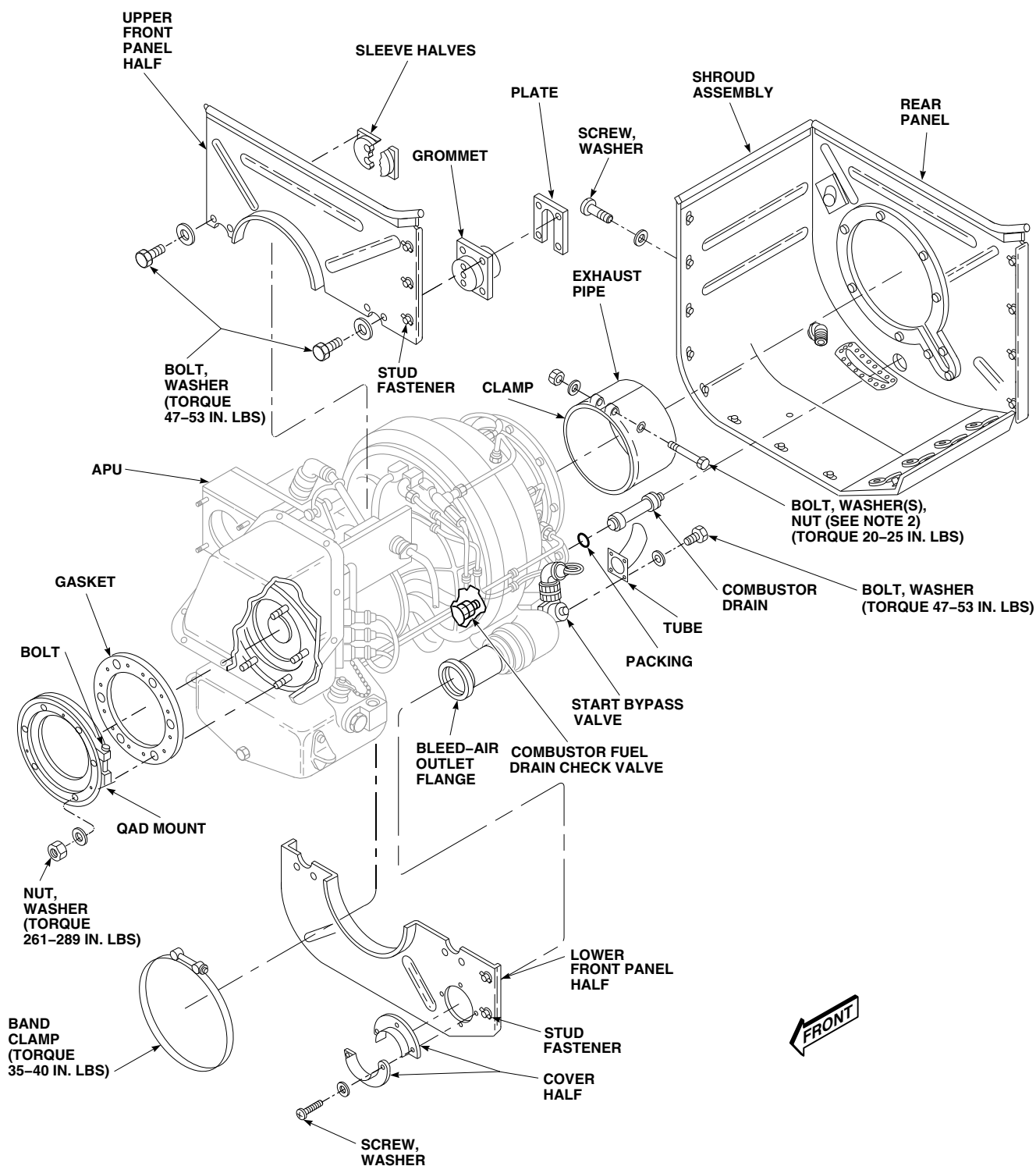


FIGURE 15-4-1-1. REPLACE APU (SHEET 3 OF 4)

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FIGURE 15-4-1-1. REPLACE APU (SHEET 4 OF 4)

- ap. Loosen bolt on QAD mount securing QAD mount to APU. Remove nuts, washers, QAD mount, and gasket from APU.
- aq. Remove bolts and washers securing tube to start bypass valve. Cover port.

15-4-1.1.2 Inspect.

- a. Inspect APU check valve (PARA 7-4-4).
- b. If APU is being replaced because of internal failure, check pneumatic system for possible metal contamination (PARA 7-3-2).
- c. If installed, inspect rubber terminal covers on electrical leads for cracks. **NONE ALLOWED.** If cracked, replace terminal covers.
- d. Inspect APU generator spline adapter (PARA 15-4-8).
- e. Inspect APU starter motor coupling shaft (PARA 15-4-7).

15-4-1.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Depreserve replacement APU (Appendix E).
- c. Install QAD mount as follows (Figure 15-4-1-1, Sheet 4, Detail H):
 - (1) Place gasket and QAD mount on APU. Secure with QAD mount bolt installed on left side of APU.
 - (2) Install washers and nuts.
 - (3) Torque nuts in crisscross pattern in increments as follows:
 - (a) **TORQUE NUTS TO 50 INCH-POUNDS.**
 - (b) **TORQUE NUTS TO 150 INCH-POUNDS.**
 - (c) **TORQUE NUTS TO 250 INCH-POUNDS.**

- (d) **TORQUE NUTS TO A FINAL TORQUE OF 261 - 289 INCH-POUNDS.**

- d. Install APU generator (PARA 9-4-19).
- e. Install tube as follows:
 - (1) Remove port caps and secure tube to start bypass valve using bolts and washers.
 - (2) **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
 - (3) Using lockwire, Item 197, Appendix D, lockwire bolts together.
- f. Install exhaust pipe as follows:
 - (1) Using antiseize compound, Item 74, Appendix D, coat internal beaded surface of exhaust pipe.

NOTE

Up to two washers under bolthead and up to two washers under nut on clamp may be used to prevent bottoming of bolt shank.

- (2) Position exhaust pipe on flange of APU. Make sure exhaust pipe does not overlap APU exhaust case doubler. **TORQUE CLAMP BOLT TO 20 - 25 INCH-POUNDS.**
- (3) Using nonmetallic mallet, tap clamping surface lightly. **CHECK CLAMP BOLT TORQUE FOR 20 - 25 INCH-POUNDS.**
- (4) Continue tapping clamping surface until two successive trials show no loss in torque.
- g. Install combustor drain as follows:
 - (1) Remove port cover, and install packing on APU combustor fuel drain check valve.
 - (2) Install combustor drain on combustor fuel drain check valve.

CAUTION

Damage to APU combustor housing will occur if shroud assembly is installed incorrectly. Make sure clearance exists between shroud assembly and housing of combustor fuel drain check valve.

- (3) Insert combustor drain through hole in rear panel.
- (4) Install shroud assembly around APU.
- (5) Using lockwire, Item 197, Appendix D, lockwire combustor drain to cover on rear panel.
- h. Install upper and lower front panel halves as follows:

CAUTION

Damage will occur if ignition lead, thermocouple lead, and wiring for valves and switches are installed incorrectly. To prevent chafing and pinching when panel halves are installed, take extreme care when installing panel halves.

- (1) Position upper and lower front panel halves on shroud assembly and secure with screws and washers.
- (2) Install band clamp around front panel halves.
- (3) **TORQUE BAND CLAMP TO 35 - 40 INCH-POUNDS.**
- i. Install ignitor leads and electrical wiring as follows:
 - (1) Install ignitor leads and electrical wiring inside sleeve halves.
 - (2) Secure sleeve halves to upper and lower front panel halves with bolts and washers.

- (3) **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

NOTE

Lockwire bolts only on right side of upper front panel half (looking toward front).

- (4) Using lockwire, Item 197, Appendix D, lockwire bolts together.
- j. On upper and lower front panel halves, install grommet and plate as follows:
 - (1) Position grommet and plate over tube bundle.
 - (2) Secure grommet and plate to upper and lower front panel halves with bolts and washers.
 - (3) **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
- k. On lower front panel half, position cover halves around bleed-air outlet flange and secure with screws and washers.
- l. Install rear mounting lug as follows:
 - (1) Install rear mounting lug in compressor (Figure 15-4-1-1, Sheet 3, Detail F).
 - (2) **TORQUE LUG TO 133 - 147 INCH-POUNDS.**
 - (3) Using lockwire, Item 196, Appendix D, lockwire lug to shroud assembly.
- m. Install hoisting sling on APU (Figure 15-4-1-1, Sheet 3, Detail E). Attach sling to hoist/cabin maintenance crane. Take up slack.
- n. Check to make sure clearance exists between shroud assembly and housing of combustor fuel drain check valve.

WARNING

Injury to personnel or damage to equipment will result if caution is not taken when hoisting APU. Keep area clear when hoisting APU.

- o. With assistant operating hoist/cabin maintenance crane, support full weight of APU. Remove bolts and washers and disengage APU from maintenance trailer/APU adapter assembly (Figure 15-4-1-1, Sheet 3, Detail G). Remove front mounting lugs from APU adapter assembly.
- p. Install front mounting lug (TOP side up) to APU gear box sump as follows:
 - (1) Coat mating surface of mounting lug with primer, Item 257, Appendix D.
 - (2) Secure front mounting lug (TOP side up) to APU gear box sump, while primer is still wet, with bolts and washers (Figure 15-4-1-1, Sheet 3, Detail F).
 - (3) **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**
 - (4) Clean excess primer from mounting lug and attaching hardware using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- q. Raise APU and lower into compartment enough to connect wiring to generator.
- r. Connect electrical leads to terminals on generator using nuts, lockwashers, and washers. Remove tags (Figure 15-4-1-1, Sheet 1, Detail A or Sheet 2, Detail C).

NOTE

Do not install metal terminal cover. Metal terminal cover will be replaced with rubber terminal covers.

- s. Install rubber terminal covers over terminals (Figure 15-4-1-1, Sheet 2, Detail C).

- t. Remove caps and plugs from electrical connectors.
- u. Inspect and treat electrical connectors (PARA 1-7-20).
- v. Connect electrical connector, P289, to side of APU (Figure 15-4-1-1, Sheet 1, Detail A).
- w. Connect electrical connectors, P252 and P279, to APU generator.
- x. Install APU starter motor as follows:
 - (1) Position gasket on mounting pad of APU.
 - (2) Using grease, Item 144, Appendix D, lubricate APU starter motor coupling shaft.
 - (3) Secure starter motor on mounting pad with washers and nuts.
 - (4) **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**
- y. Engage rear mounting lug in rear support mount, and slide APU toward front along engine exhaust firewall (Figure 15-4-1-1, Sheet 3, Detail F).
- z. Attach APU shroud assembly panels to exhaust firewall, and lock stud fasteners (Figure 15-4-1-1, Sheet 2, Detail D).
- aa. Install front mounting lugs as follows:
 - (1) Secure front mounting lugs to front support mounts with bolts and washers (Figure 15-4-1-1, Sheet 3, Detail F).
 - (2) **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**
 - (3) Using lockwire, Item 197, Appendix D, lockwire bolts together.
- ab. Remove hoisting sling from APU and hoist or cabin maintenance crane.
- ac. Install APU exhaust duct (PARA 15-4-2).

- ad. Seal all seams and joined areas of enclosure formed by shroud assembly, upper and lower front panel halves, and exhaust firewall with adhesive, Item 53, Appendix D.
- ae. Remove caps and plugs from electrical connectors.
- af. Inspect and treat electrical connectors (PARA 1-7-20).
- ag. Connect electrical connector, P414, to fire detector on rear panel (Figure 15-4-1-1, Sheet 1, Detail A).
- ah. Remove caps and plugs and connect fire extinguisher hose. **TIGHTEN NUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- ai. Remove caps and plugs and connect compartment drain hose to compartment drain. **TORQUE B-NUT TO 270 - 300 INCH-POUNDS.**
- aj. Remove caps and plugs and connect combustor drain hose to combustor drain. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS.**
- ak. Remove caps and plugs and connect gear box vent hose to gear box vent. **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**
- al. Remove caps and plugs and connect fuel control box drain hose to fuel drain port. **TORQUE B-NUT TO 50 - 65 INCH-POUNDS.**
- am. Install check valve to pneumatic tube at lower front panel assembly and secure with clamp (Figure 15-4-1-1, Sheet 2, Detail B). **TORQUE CLAMP TO 35 - 45 INCH-POUNDS.**

CAUTION

Damage to fuel filter will result if union in fuel control lower cover turns when installing fuel inlet hose. Hold union with wrench while installing fuel inlet hose to avoid damage to fuel filter.

- an. Hold fuel inlet port union in fuel control base with wrench. Remove plug and loosely connect fuel inlet hose to union (Figure 15-4-1-1, Sheet 1, Detail A).
- ao. If bonding jumper was removed from fuel inlet hose, install clamp on hose and connect bonding jumper to clamp with screw, washers, and nut.
- ap. Prime APU fuel inlet hose (PARA 15-4-9) and check for leaks.
- aq. Install fuel control upper cover on fuel control base and lock stud fasteners.
- ar. Secure generator electrical leads to airframe with clamp.
- as. Check electrical leads, fuel lines, and drain lines for contact. If contact occurs, reposition electrical leads, fuel lines, and drain lines. Install extra clamps as required.
- at. Service APU (PARA 1-3-9).
- au. Service APU accumulator (PARA 1-3-10).
- av. Make sure area is clean and free of foreign material.
- aw. Install APU access doors (PARA 2-4-108).
- ax. If installed, disassemble and remove cabin maintenance crane (PARA 18-4-1).
- ay. Perform operational check of APU (PARA 15-2-1 or PARA 15-2-2).

15-4-2 APU EXHAUST DUCT.

PARAGRAPH BREAKDOWN

		PAGE
15-4-2.1	Replace APU Exhaust Duct	15-4-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 53, Appendix D

References

- Appendix D
- PARA 1-6-16

15-4-2.1 Replace APU Exhaust Duct.

15-4-2.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open APU compartment access doors.
- Remove screws and washers securing mounting flange of exhaust duct to rear shroud of APU (Figure 15-4-2-1, Detail A).
- Remove bolts and saddle washers securing exhaust duct to APU panel.
- Pull duct through APU panel until mounting flange reaches APU panel. Rotate duct until flange finger points toward front. Remove duct.

15-4-2.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- With finger on mounting flange pointing toward front, install exhaust duct through APU panel (Figure 15-4-2-1, Detail A).
- Apply bead of adhesive, Item 53, Appendix D, to mounting flange and secure exhaust duct to rear shroud of APU with screws and washers.
- Secure exhaust duct to APU panel with bolts and saddle washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
- Make sure area is clean and free of foreign material.
- Close APU compartment access doors.

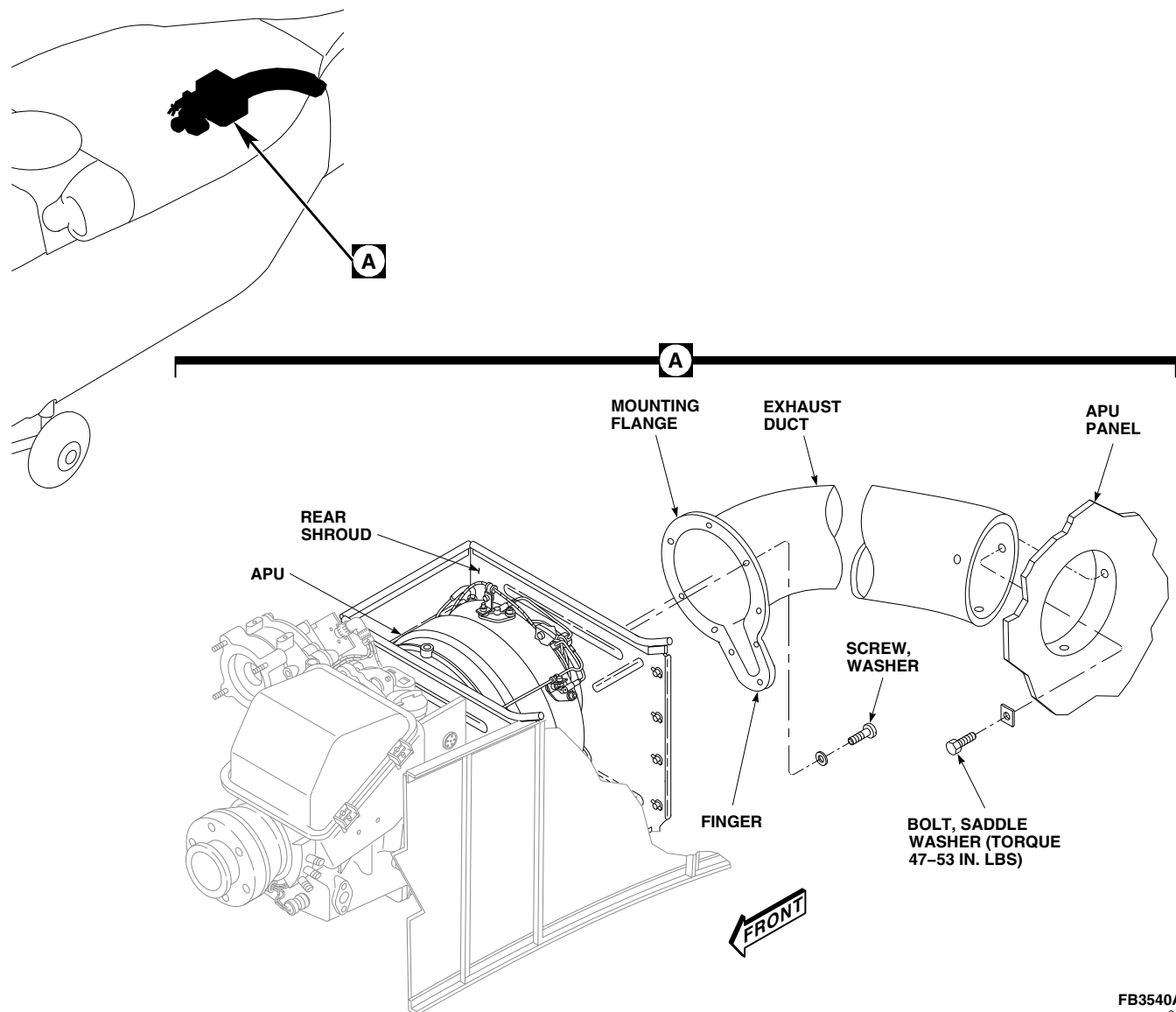
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FIGURE 15-4-2-1. REPLACE APU EXHAUST DUCT

15-4-3 APU EXHAUST PIPE.

PARAGRAPH BREAKDOWN

	PAGE
15-4-3.1 Replace APU Exhaust Pipe	15-4-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Mallet
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D

References

- Appendix D
- PARA 1-6-16

15-4-3.1 Replace APU Exhaust Pipe.

15-4-3.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open APU compartment access doors.
- Loosen bolt, washer(s), and nut securing exhaust pipe (Figure 15-4-3-1, Detail A).
- Slide exhaust pipe out through exhaust duct.

15-4-3.1.2 Install.

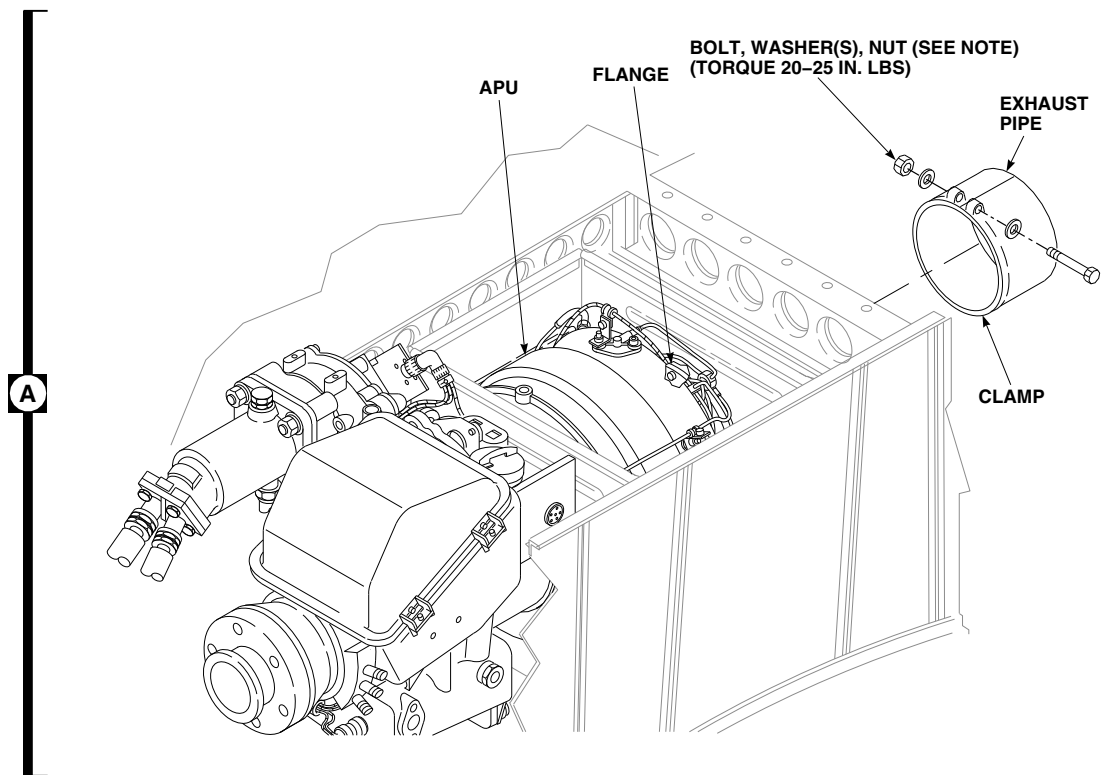
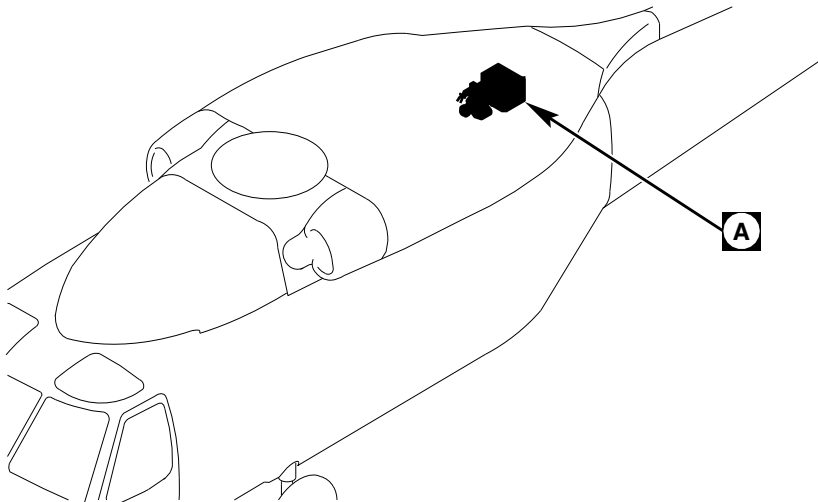
- Turn off all electrical power (PARA 1-6-16).
- Using antiseize compound, Item 74, Appendix D, coat internal beaded surface of exhaust pipe.

NOTE

Up to two washers under bolthead and up to two washers under nut on clamp

may be used to prevent bottoming of bolt shank.

- Pass exhaust pipe up through exhaust duct. Install pipe on flange of APU. Make sure exhaust pipe does not overlap APU exhaust case doubler. **TORQUE CLAMP BOLT TO 20 - 25 INCH-POUNDS** (Figure 15-4-3-1, Detail A).
- Using nonmetallic mallet, tap clamping surface lightly. **CHECK CLAMP BOLT TORQUE FOR 20 - 25 INCH-POUNDS.**
- Continue tapping clamping surface until two successive trials show no loss of torque.
- Make sure area is clean and free of foreign material.
- Close APU compartment access doors.

**NOTE**

UP TO TWO WASHERS UNDER
BOLTHEAD AND UP TO TWO
WASHERS UNDER NUT ON CLAMP
MAY BE USED TO PREVENT
BOTTOMING OF BOLT SHANK.

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FIGURE 15-4-3-1. REPLACE APU EXHAUST PIPE

15-4-4 APU SHROUD ASSEMBLY.

PARAGRAPH BREAKDOWN

		PAGE
15-4-4.1	Replace APU Shroud Assembly	15-4-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Adhesive, Item 53, Appendix D
- Lockwire, Item 196, Appendix D

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 15-4-1

15-4-4.1 Replace APU Shroud Assembly.

15-4-4.1.1 Remove.

- a. Remove APU (PARA 15-4-1).
- b. On upper and lower front panel halves, remove bolts, washers, sleeve halves, grommet, and plate (Figure 15-4-4-1, Detail A).
- c. Remove band clamp securing upper and lower front panel halves together.
- d. On lower front panel half, remove screws and washers securing cover halves around bleed-air outlet flange. Remove cover halves from lower front panel half.
- e. Remove screws and washers securing upper and lower front panel halves to shroud assembly. Unlock stud fasteners and remove front panel halves.

- f. Remove lockwire from combustor drain in rear panel.
- g. Loosen and remove rear mounting lug from compressor.
- h. Remove shroud assembly surrounding APU, carefully sliding off combustor drain.
- i. Remove elbow, washer, and nut from shroud assembly (Figure 15-4-4-1, Detail B).

15-4-4.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install elbow in shroud assembly, facing front, and secure with washer and nut (Figure 15-4-4-1, Detail B). **TORQUE NUT TO 240 - 289 INCH-POUNDS.**

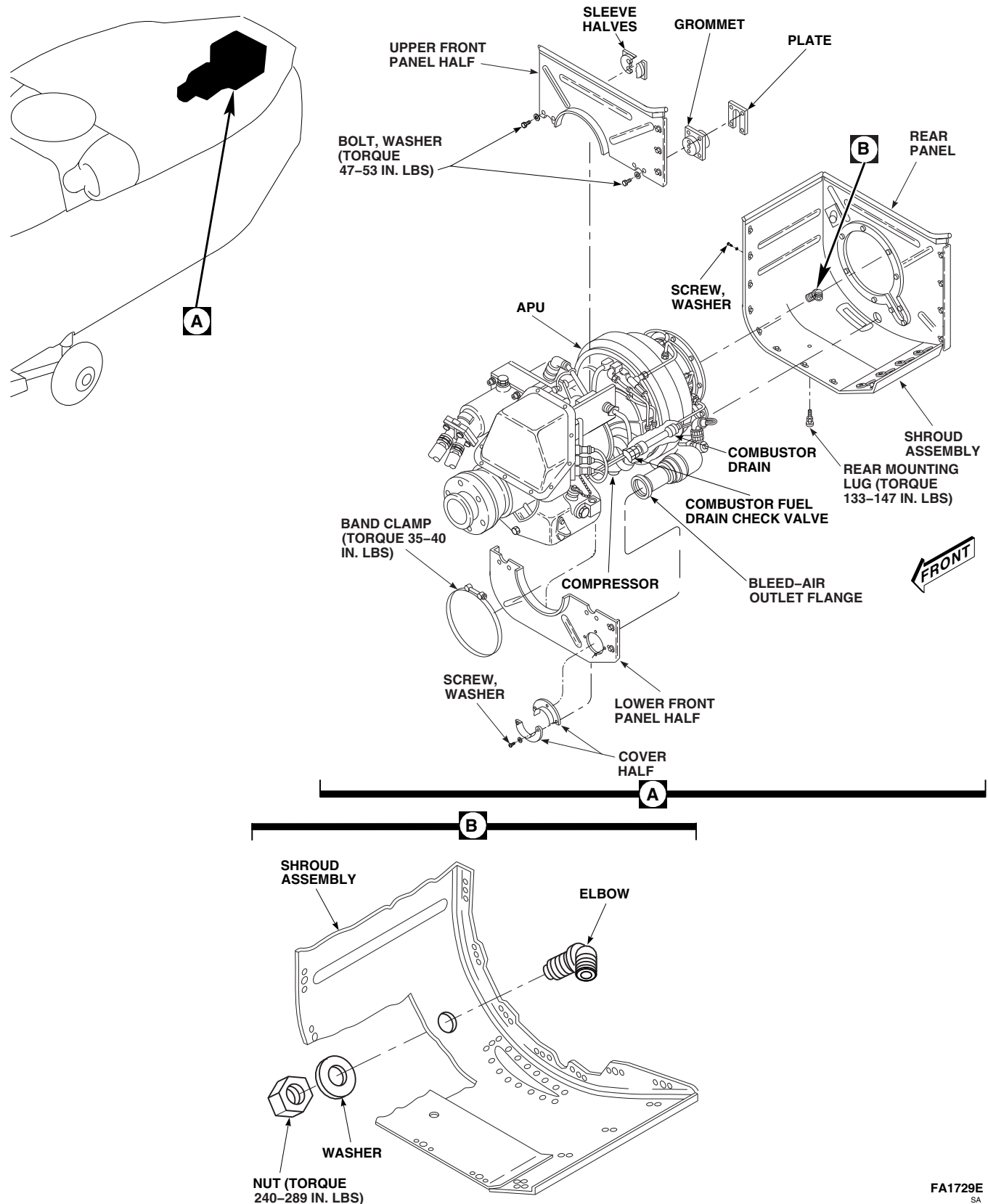


FIGURE 15-4-4-1. REPLACE APU SHROUD ASSEMBLY

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CAUTION

Damage to APU combustor housing will occur if shroud assembly is installed incorrectly. Make sure clearance exists between shroud assembly and housing of combustor fuel drain check valve.

- c. Install shroud assembly around APU (Figure 15-4-4-1, Detail A).
- d. Install rear mounting lug as follows:
 - (1) Install rear mounting lug in compressor.
 - (2) **TORQUE LUG TO 133 - 147 INCH-POUNDS.**
 - (3) Using lockwire, Item 196, Appendix D, lockwire lug to shroud assembly.
- e. Check to make sure clearance exists between shroud assembly and housing of combustor fuel drain check valve.
- f. Using lockwire, Item 197, Appendix D, lockwire combustor drain to cover on rear panel.
- g. Install upper and lower front panel halves as follows:

CAUTION

Damage will occur if ignition lead, thermocouple lead, and wiring for valves and switches are installed incorrectly. To prevent chafing and pinching when panel halves are installed, take extreme care when installing panel halves.

- (1) Position upper and lower front panel halves on shroud assembly and secure with screws and washers.
- (2) Install band clamp around front panel halves.

- (3) **TORQUE BAND CLAMP TO 35 - 40 INCH-POUNDS.**

- h. Install ignitor leads and electrical wiring as follows:

- (1) Install ignitor leads and electrical wiring inside sleeve halves.
- (2) Secure sleeve halves to upper and lower front panel halves with bolts and washers.

- (3) **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

NOTE

Lockwire bolts only on right side of upper front panel half (looking toward front).

- (4) Using lockwire, Item 197, Appendix D, lockwire bolts together.

- i. On upper and lower front panel halves, install grommet and plate as follows:

- (1) Position grommet and plate over tube bundle.
- (2) Secure grommet and plate on upper and lower front panel halves with bolts and washers.

- (3) **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

- j. On lower front panel half, position cover halves around bleed-air outlet flange and secure with screws and washers.

- k. Seal all seams and joined areas of enclosure formed by shroud assembly, upper and lower front panel halves, and exhaust firewall with adhesive, Item 53, Appendix D.

- 1. Install APU (PARA 15-4-1).

15-4-5 APU FRONT MOUNTING LUGS.

PARAGRAPH BREAKDOWN

	PAGE
15-4-5.1 Replace APU Front Mounting Lugs	15-4-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Hoisting Sling, 70700-20323-042
- Torque Wrench, 30 - 150 in. lbs
- Wood Block

Materials/Parts

- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 15-4-1

15-4-5.1 Replace APU Front Mounting Lugs.

NOTE

Replacement of left and right front mounting lugs is the same.

15-4-5.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open APU access doors.
- Place wood block under APU gear box sump or use hoisting sling to support APU (PARA 15-4-1).
- Remove bolt and washer securing front mounting lug to front support mount (Figure 15-4-5-1, Detail A).

- Remove bolts and washers securing front mounting lug to APU gear box sump, and remove mounting lug.

15-4-5.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install front mounting lug (TOP side up) on APU gear box sump as follows:
 - Coat mating surface of mounting lug with primer, Item 257, Appendix D.
 - Secure front mounting lug (TOP side up) on APU gear box sump, while primer is still wet, with bolts and washers (Figure 15-4-5-1, Detail A).
 - TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**

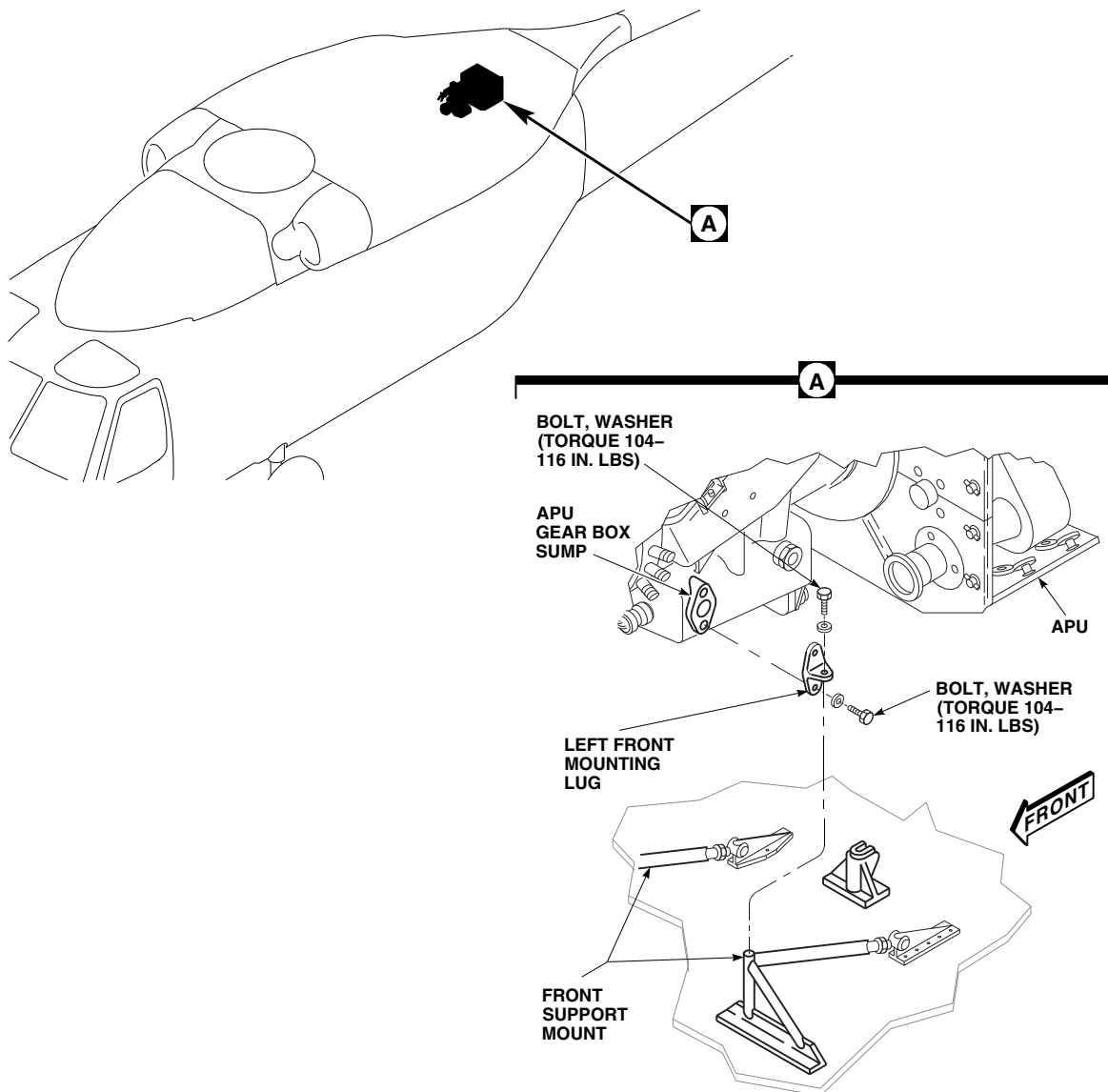
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FIGURE 15-4-5-1. REPLACE APU FRONT MOUNTING LUGS

- (4) Clean excess primer from mounting lug and attaching hardware using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- c. Remove wood block or hoisting sling supporting APU (PARA 15-4-1).
- d. Secure front mounting lug on front support mount with bolt and washer. **TORQUE BOLT TO 104 - 116 INCH-POUNDS.** Lockwire, Item 197, Appendix D, bolts together.
- e. Make sure area is clean and free of foreign material.
- f. Close APU access doors.

15-4-6 APU REAR MOUNTING LUG.

PARAGRAPH BREAKDOWN

	PAGE
15-4-6.1 Replace APU Rear Mounting Lug	15-4-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 15-4-1

15-4-6.1 Replace APU Rear Mounting Lug.

15-4-6.1.1 Remove.

- Remove APU (PARA 15-4-1).
- Loosen and remove rear mounting lug from compressor (Figure 15-4-6-1, Detail A).

15-4-6.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install rear mounting lug in compressor (Figure 15-4-6-1, Detail A).

c. TORQUE LUG TO 133 - 147 INCH-POUNDS.

- Lockwire, Item 196, Appendix D, lug to shroud assembly.
- Make sure area is clean and free of foreign material.

f. Install APU (PARA 15-4-1).

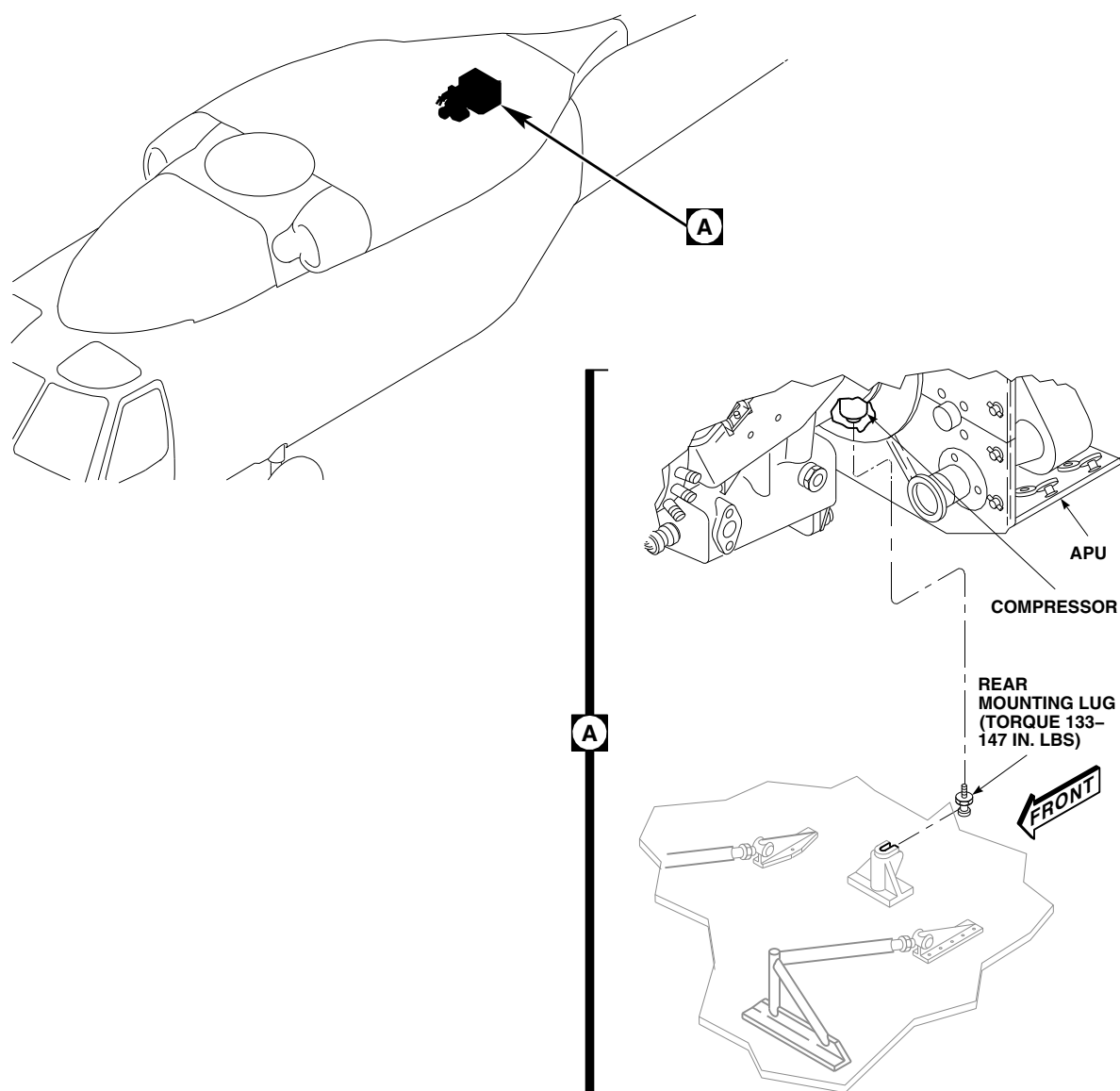
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FIGURE 15-4-6-1. REPLACE APU REAR MOUNTING LUG

15-4-7 APU STARTER MOTOR.

PARAGRAPH BREAKDOWN

	PAGE
15-4-7.1 Replace APU Starter Motor	15-4-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Micrometer
- Pin, 0.096 inch-diameter
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Grease, Item 144, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D

- Lockwire, Item 194, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-3-10
- PARA 1-6-16
- PARA 7-3-1
- PARA 15-2-1
- PARA 15-2-2

15-4-7.1 Replace APU Starter Motor.

15-4-7.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Depressurize APU accumulator (PARA 1-3-10).

- Open APU access doors.

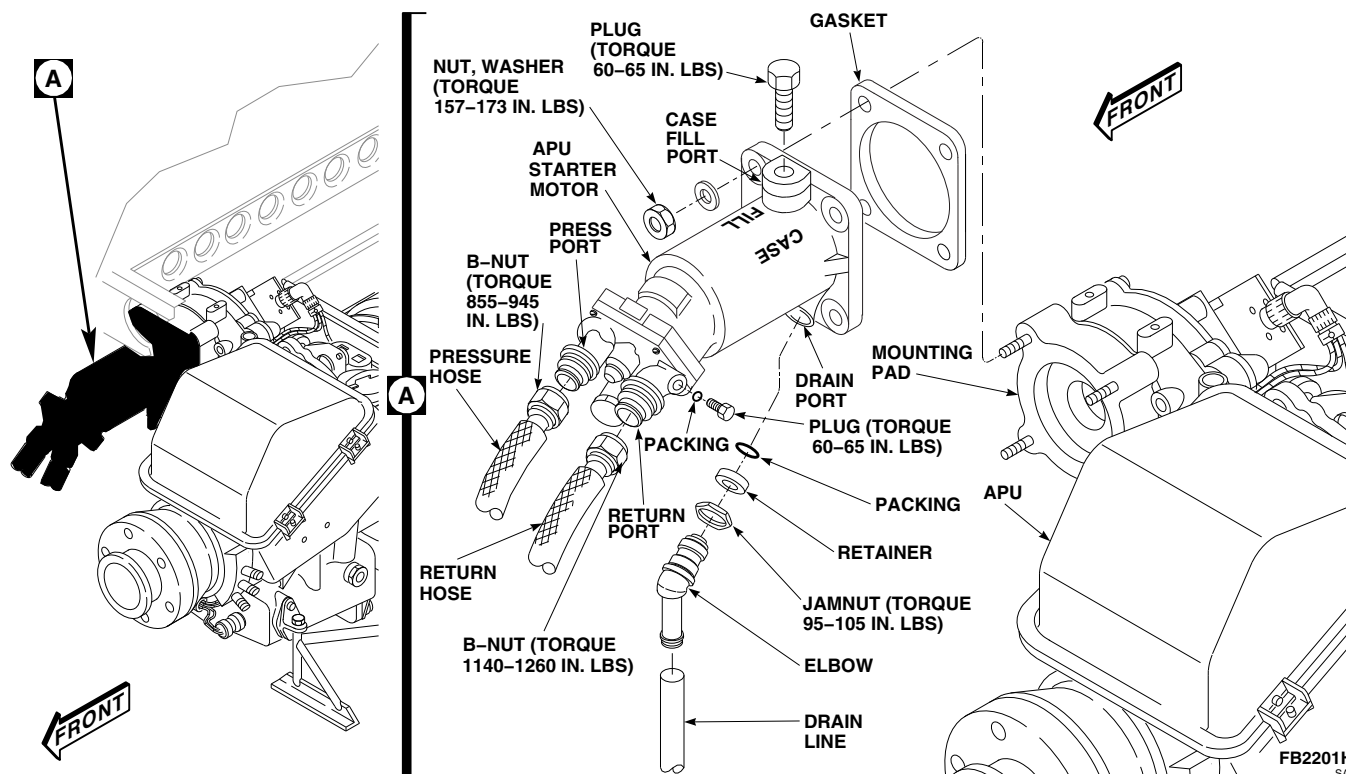


FIGURE 15-4-7-1. REPLACE APU STARTER MOTOR

WARNING

Injury to personnel and damage to equipment will result if APU start valve is operated with personnel working in area of APU. Do not operate APU start valve when starter motor is being replaced.

- d. Tag APU start valve warning personnel not to operate.
- e. Disconnect pressure and return hoses from PRESS and RETURN ports on starter motor. Cap ports on starter motor and plug hoses (Figure 15-4-7-1, Detail A).
- f. Disconnect drain line from elbow on starter motor. Cap elbow and plug drain line.
- g. Remove nuts, washers, APU starter motor, and gasket from mounting pad on APU. Discard gasket.

- h. Remove jamnut, elbow, retainer, and packing from drain port. Discard packing. Cap elbow and plug drain port. If plug interferes with fuel control unit cover, replace with low profile plug.

15-4-7.1.2 Inspect.

- a. Inspect APU starter motor coupling shaft for spline wear as follows:
 - (1) Place 0.096-inch-diameter pin between two splines.
 - (2) Place second 0.096-inch-diameter pin between two splines directly opposite first pin. Secure pins.
 - (3) Using micrometer, measure outside diameter of shaft splines, over pins. **OUTSIDE DIAMETER MUST NOT BE LESS THAN 0.7343-INCH.** If coupling shaft is worn beyond limits, replace APU starter motor.

- (4) Repeat step a. (1) through a. (3) with pins located in four different positions.
- b. Refer to PARA 7-3-1 for shaft seal leakage limits.

15-4-7.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove cap from threaded end of elbow. Using petrolatum, Item 234, Appendix D, lubricate packing, retainer, and elbow threads.
- c. Install jamnut, retainer, and packing onto elbow. Turn jamnut and push retainer and packing down until firmly against lower threaded section of elbow (Figure 15-4-7-1, Detail A).
- d. Remove protective plug from drain port in APU starter motor.
- e. While keeping jamnut from turning, tighten elbow in drain port until sharp rise in torque is felt. **TIGHTEN ELBOW AND JAMNUT 1 ½ TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN ELBOW NO MORE THAN 1 ADDITIONAL TURN TO POSITION ELBOW TOWARD FRONT.**
- f. **TORQUE JAMNUT TO 95 - 105 INCH-POUNDS WHILE KEEPING ELBOW FROM TURNING.**
- g. Fill starter motor with hydraulic fluid as follows:
 - (1) Remove plug from CASE FILL port on starter motor.

NOTE

For arctic operation, -29°F (-34°C), use hydraulic fluid, Item 154, Appendix D.

- (2) Fill starter motor completely with hydraulic fluid, Item 155, Appendix D.

- (3) Install plug in CASE FILL port.
- (4) **TORQUE PLUG TO 60 - 65 INCH-POUNDS.**
- (5) Using lockwire, Item 194, Appendix D, lockwire plug to body of starter motor.
- h. Position gasket on mounting pad of APU.
- i. Using grease, Item 144, Appendix D, lubricate APU starter motor coupling shaft.
- j. Secure starter motor on mounting pad with washers and nuts.
- k. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**
- l. Remove cap and plug and connect drain line to elbow on starter motor.
- m. Remove caps from PRESS and RETURN ports on starter motor.
- n. Remove plug and connect pressure hose to PRESS port.
- o. **TORQUE B-NUT TO 855 - 945 INCH-POUNDS.**
- p. Remove plug and connect return hose to RETURN port.
- q. **TORQUE B-NUT TO 1140 - 1260 INCH-POUNDS.**
- r. Remove tag from APU starter valve.
- s. Service APU accumulator (PARA 1-3-10).
- t. Make sure area is clean and free of foreign material.
- u. Close APU access doors.
- v. Perform operational check of APU (PARA 15-2-2 and PARA 15-2-1).

15-4-8 APU SPLINE ADAPTER.

PARAGRAPH BREAKDOWN

		PAGE
15-4-8.1	Replace APU Spline Adapter	15-4-8-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Insertion Tool, 1106841-1
- Machinist’s Scribe
- Powertrain Repairer’s Toolkit
- Puller Kit, Mechanical

Materials/Parts

- Acetone, Item 25, Appendix D
- Lubricating Oil, Item 202, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 9-4-19
- PARA 15-4-1

15-4-8.1 Replace APU Spline Adapter.

15-4-8.1.1 Remove.

- If required, remove APU (PARA 15-4-1).
- Remove APU generator (PARA 9-4-19).

CAUTION

- Spline adapter is a press fit and is difficult to remove. Use of hammer or excessive force to remove spline adapter could cause ring gear separation and damage APU. Do not use hammer or excessive force to remove APU spline adapter.
- Damage will occur if wrong spline adapter removal tool is used. Make sure

removal tool is matched with appropriate APU.

- Using mechanical puller kit, remove spline adapter as follows (Figure 15-4-8-1, Detail A):
 - Mount puller to accessory drive shaft face.
 - Insert removal key through inside diameter of adapter and rotate key half the angle between teeth and lock in place.
 - Back jackscrew until spline adapter is free from drive.
 - Remove puller and discard spline adapter.
- Clean female involute drive spline with acetone, Item 25, Appendix D.

15-4-8.1.2 Inspect.

- a. Inspect output drive gear for any broken pieces of spline adapter. If necessary, using machinist's scribe, remove any broken pieces of spline adapter.
- b. Inspect spline in output drive gear for damage. **NONE ALLOWED.** If splines in output drive gear are damaged, replace APU (PARA 15-4-1).

15-4-8.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).

CAUTION

- Do not open bag containing spline adapter until ready for use. Torlon (green) spline adapters will absorb moisture and increase in size.
- Do not install a polyimide spline adapter. Replace spline adapter, even if partially inserted and removed.
- Spline adapter is a press fit and is difficult to install. Use of hammer or excessive force to install spline adapter could cause ring gear separation and damage

APU. Do not use hammer or excessive force to install APU spline adapter.

- b. Mount insertion tool on generator drive mounting surface.
- c. Lightly coat outside surface of spline adapter with light grade lubricating oil, Item 202, Appendix D.
- d. Insert tapered end of spline adapter into APU female spline drive (Figure 15-4-8-1, Detail A).

NOTE

Make sure spline adapter is inserted perpendicular with mounting surface.

- e. Using hand pressure, insert spline adapter approximately $\frac{1}{4}$ -inch into APU female spline drive.
- f. Rotate handle on insertion tool in clockwise direction causing spline adapter to advance into APU female spline until spline adapter is no more than $\frac{1}{8}$ -inch from APU output drive spline front face. Remove insertion tool.
- g. Install APU generator (PARA 9-4-19).
- h. If removed, install APU (PARA 15-4-1).
- i. Make sure area is clean and free of foreign material.

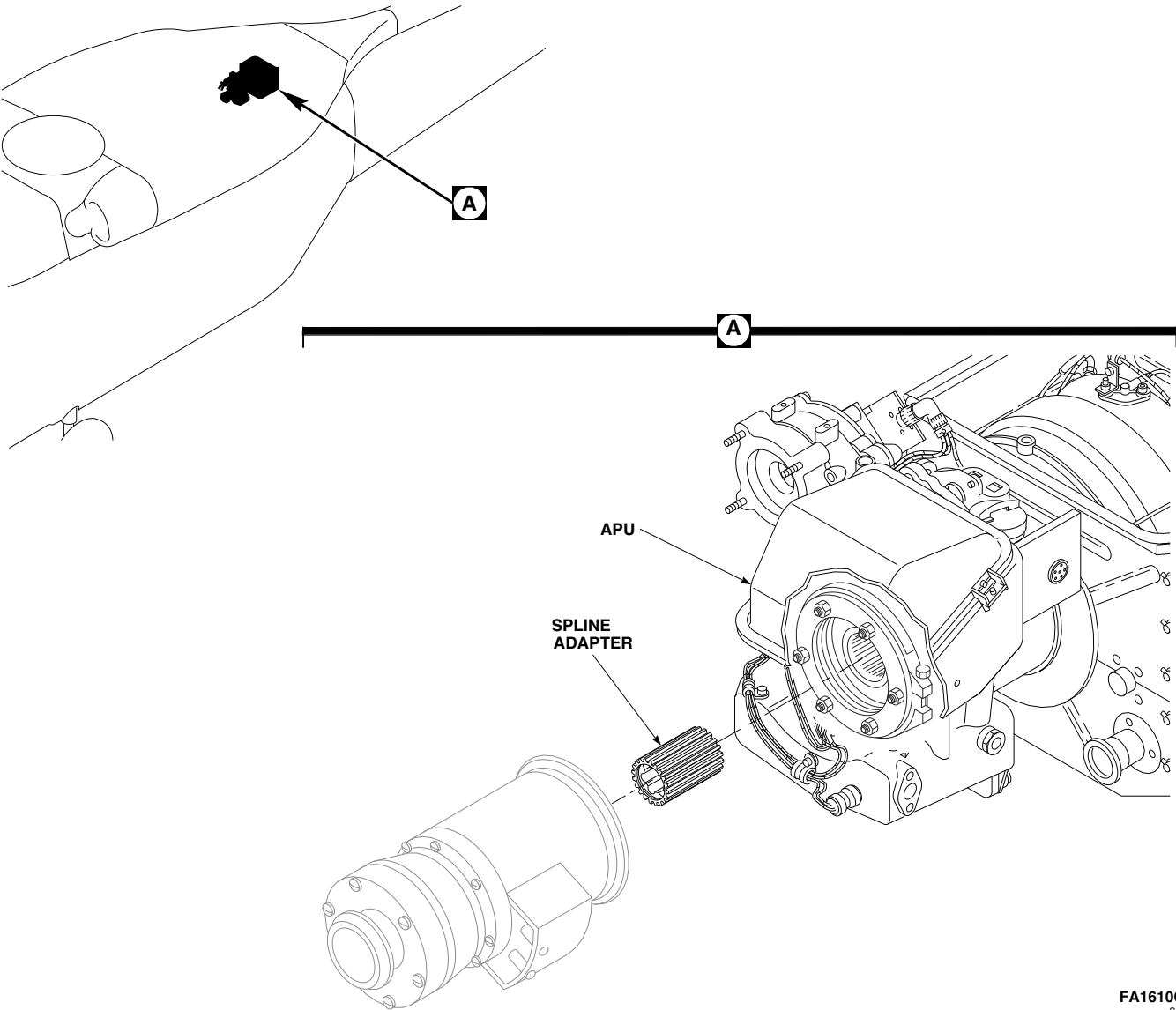


FIGURE 15-4-8-1. REPLACE APU SPLINE ADAPTER

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15-4-9 PRIME APU FUEL INLET HOSE.

PARAGRAPH BREAKDOWN

	PAGE
15-4-9.1 Prime APU Fuel Inlet Hose	15-4-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Wrench

Materials/Parts

- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-6-16

15-4-9.1 Prime APU Fuel Inlet Hose.

- Turn off all electrical power (PARA 1-6-16).
- Open APU access doors.
- Loosen stud fasteners securing fuel control upper cover to fuel control lower cover (Figure 15-4-9-1, Detail A). Remove fuel control upper cover.

CAUTION

Damage to fuel filter will result if union in fuel control lower cover turns while loosening fuel inlet hose. Hold union with wrench while loosening fuel inlet hose to avoid damage to fuel filter.

- Securing union inside fuel control lower cover with wrench, loosen B-nut on fuel inlet hose to allow fuel to leak when priming fuel inlet hose.
- On upper console, place BATT switch ON, or connect external electrical power to helicopter (PARA 1-6-16).

WARNING

To prevent injury to personnel or damage to equipment, make sure APU T-handle on upper console is not pulled out.

- Hold machinery towel, Item 211, Appendix D, around fuel inlet hose B-nut.
- On upper console, have assistant place FUEL PUMP switch in cockpit to APU BOOST. **WHEN FUEL LEAKS FROM FUEL INLET HOSE B-NUT, TIGHTEN B-NUT AND PLACE FUEL PUMP SWITCH OFF. TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- On upper console, place BATT switch OFF, or disconnect external electrical power from helicopter (PARA 1-6-16).
- Install fuel control upper cover and secure with stud fasteners.

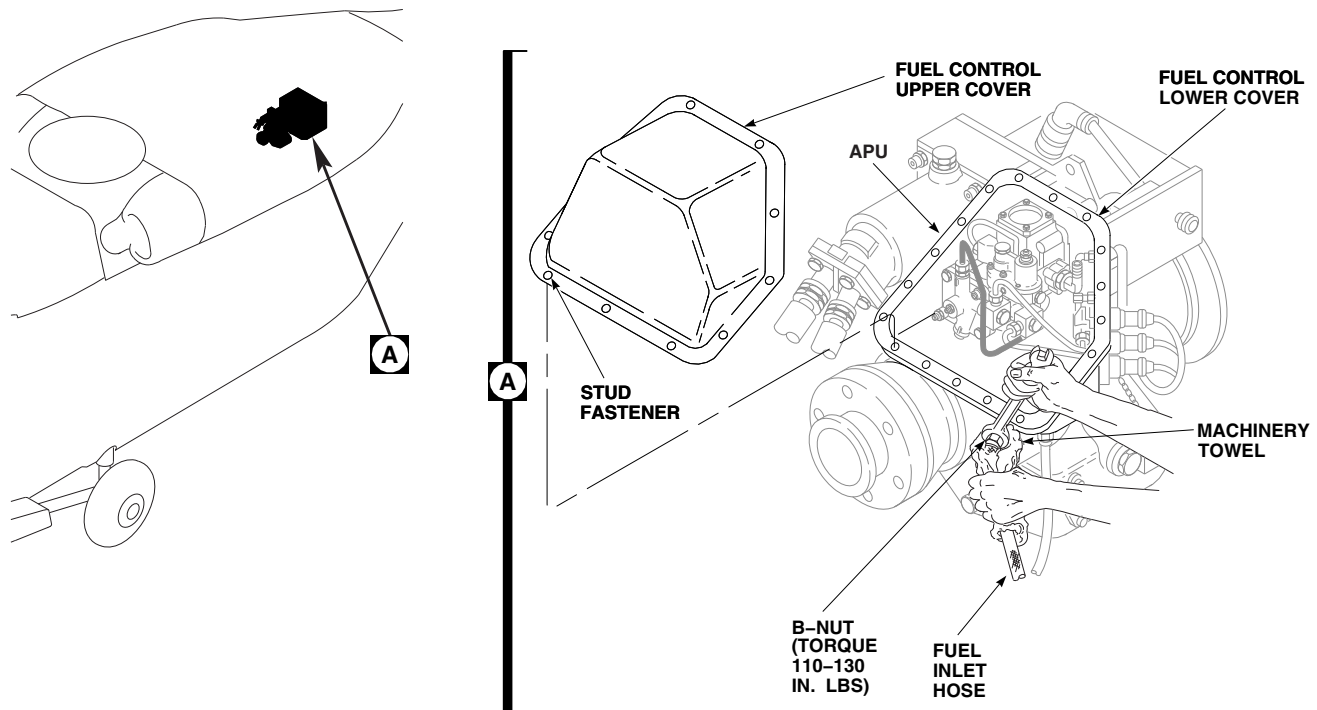
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FIGURE 15-4-9-1. PRIME APU FUEL INLET HOSE

- j. Make sure area is clean and free of foreign material.
- k. Close APU access doors.

15-4-10 APU FUEL SHUTOFF VALVE.

PARAGRAPH BREAKDOWN

	PAGE
15-4-10.1 Replace APU Fuel Shut-off Valve	15-4-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 10-2-3
- PARA 13-5-3
- PARA 13-5-6
- PARA 15-4-9

15-4-10.1 Replace APU Fuel Shutoff Valve.

15-4-10.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Gain access to fuel cell compartment as follows:
 - Remove left side rear passenger’s seats (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left stowage compartment (PARA 2-4-69).
 - Remove cargo netting from stowage compartment.

- If installed, remove condenser pallet (PARA 13-5-3) and evaporator pallet (PARA 13-5-6).

- Unlock fasteners securing right panel to enclosure on top of fuel cells. Remove panel.
- Disconnect electrical connector, P306, from APU fuel shutoff valve (Figure 15-4-10-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Disconnect fuel hose from union on fuel shut-off valve. Cap union and plug hose.
- Open APU access doors.

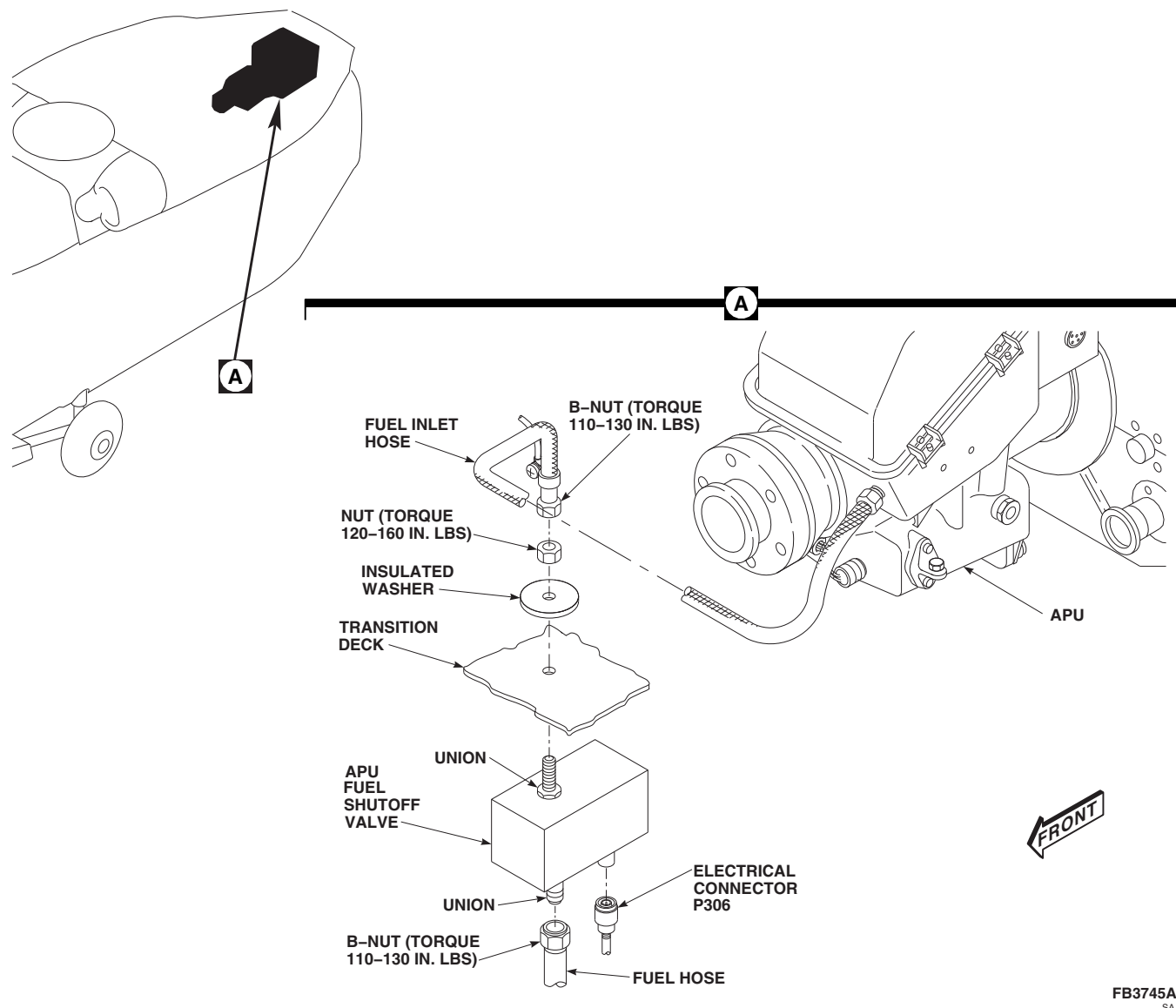
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FIGURE 15-4-10-1. REPLACE APU FUEL SHUTOFF VALVE

- h. Place machinery towels, Item 211, Appendix D, on transition deck around fuel inlet hose.
- i. Disconnect fuel inlet hose from union on fuel shutoff valve (protruding through transition deck) and plug fuel inlet hose.
- j. With assistant inside helicopter holding fuel shutoff valve, remove nut and insulated washer securing fuel shutoff valve to transition deck. Remove valve.
- k. Cap fuel shutoff valve union.

15-4-10.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove caps from APU fuel shutoff valve unions.
- c. With assistant inside helicopter holding fuel shutoff valve in place, secure fuel shutoff valve to transition deck with insulated washer and nut. **TORQUE NUT TO 120 - 160 INCH-POUNDS** (Figure 15-4-10-1, Detail A).

- d. Coat threads on fuel shutoff valve unions with antiseize compound, Item 74, Appendix D.
- e. Remove plug from fuel inlet hose and connect to union on fuel shutoff valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- f. Remove plug from fuel hose and connect to union on fuel shutoff valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS.**
- g. Remove protective caps and plugs from electrical connectors.
- h. Inspect and treat electrical connectors (PARA 1-7-20).
- i. Connect electrical connector, P306, to fuel shutoff valve.
- j. Prime fuel inlet and fuel hoses (PARA 15-4-9) and check for leaks.
- k. Perform operational check of fuel prime boost system (PARA 10-2-3).
- l. Remove machinery towels from APU compartment.
- m. Make sure APU compartment area is clean and free of foreign material.
- n. Close APU access doors.
- o. Make sure fuel cell area is clean and free of foreign material.
- p. Install right panel on enclosure on top of fuel cells. Lock fasteners.
- q. Close fuel cell compartment as follows:
 - (1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

15-4-11 APU FUEL INLET HOSE.

PARAGRAPH BREAKDOWN

	PAGE
15-4-11.1 Replace APU Fuel Inlet Hose	15-4-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Wrench

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Machinery Towel, Item 211, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 15-4-9

15-4-11.1 Replace APU Fuel Inlet Hose.

15-4-11.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open APU access doors.
- Loosen stud fasteners securing fuel control upper cover to fuel control lower cover. Remove fuel control upper cover (Figure 15-4-11-1, Detail A).
- Remove screw, washers, and nut securing clamp and bonding jumper to fuel inlet hose. Remove clamp and bonding jumper from fuel inlet hose.

CAUTION

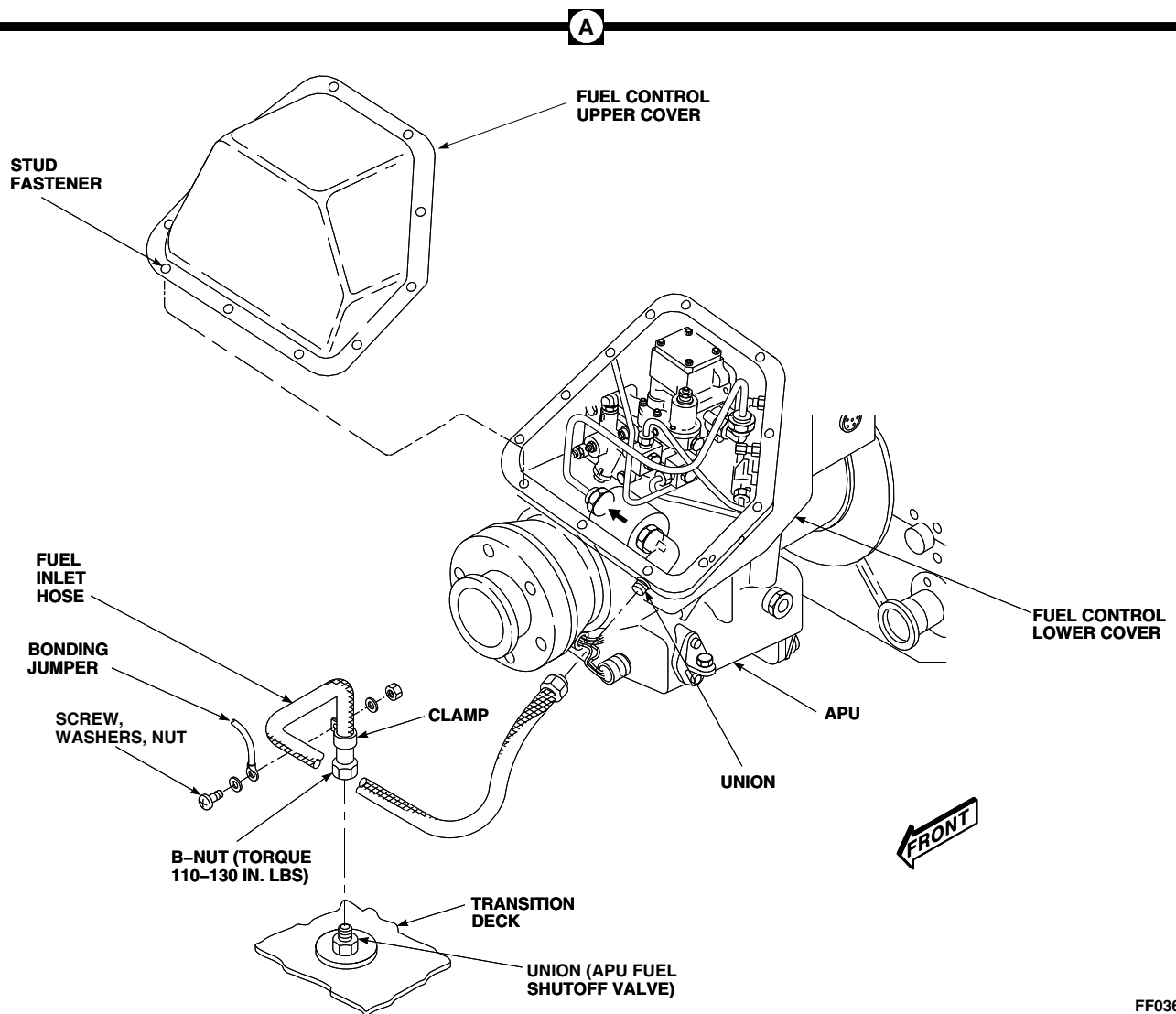
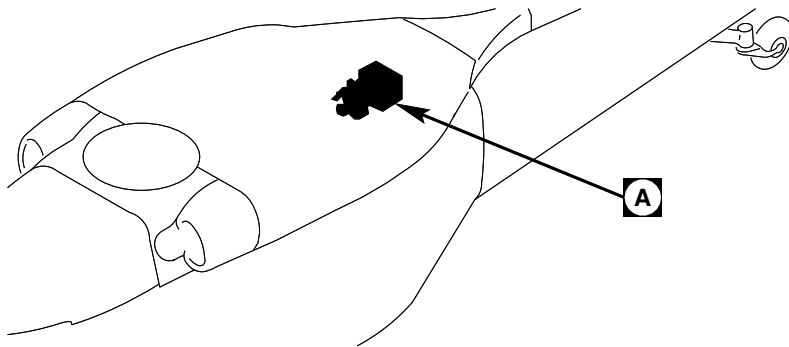
Damage to fuel filter will result if union in fuel control lower cover turns when

removing fuel inlet hose. Hold union with wrench while removing fuel inlet hose to avoid damage to fuel filter.

- While holding union inside fuel control cover with wrench, disconnect fuel inlet hose from union in fuel control lower cover. Cap union in lower cover and plug hose.
- Place machinery towels, Item 211, Appendix D, on transition deck around fuel inlet hose.
- Disconnect fuel inlet hose from union on APU fuel shutoff valve (protruding through transition deck). Cap fuel shutoff valve union and plug hose.

15-4-11.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).



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FIGURE 15-4-11-1. REPLACE APU FUEL INLET HOSE

- b. Remove cap and coat threads on APU fuel shutoff valve union with antiseize compound, Item 74, Appendix D.
- c. Remove plug and connect fuel inlet hose to union on fuel shutoff valve. **TORQUE B-NUT TO 110 - 130 INCH-POUNDS** (Figure 15-4-11-1, Detail A).
- d. Remove cap and plug and loosely secure fuel inlet hose to union in fuel control lower cover.
- e. Secure bonding jumper on fuel inlet hose with clamp, screw, washers, and nut.
- f. Prime APU fuel inlet hose (PARA 15-4-9) and check for leaks.

15-4-12 APU DRAIN LINES.

PARAGRAPH BREAKDOWN

	PAGE
15-4-12.1 Replace APU Drain Lines	15-4-12-1
INITIAL SETUP	• PARA 2-4-45
Tools	• PARA 2-4-69
• Aircraft Mechanic’s Toolkit • Torque Wrench, 30 - 150 in. lbs	• PARA 13-5-3
Materials/Parts	• PARA 13-5-6
• Acetone, Item 25, Appendix D	
References	
• Appendix D • PARA 1-6-16	

15-4-12.1 Replace APU Drain Lines.

15-4-12.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).

b. Open APU access doors.

c. If installed, open transition section avionics door.

d. If helicopter is not equipped with transition section avionics door, gain access to fuel cell compartment as follows:

(1) Remove left side rear passenger’s seats (PARA 2-4-45).

(2) Remove soundproofing panel from rear bulkhead covering left stowage compartment (PARA 2-4-69).

(3) Remove cargo netting from stowage compartment.
- (4) If installed, remove ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).

e. In cabin, remove clamps, screws, washers, and nuts securing drain line to rigid drain line in cabin ceiling. Leave clamps installed on rigid drain line (Figure 15-4-12-1, Sheet 1, Detail A).

f. In transition section, remove clamps, screws, washers, and nuts securing drain line to rigid drain line on rear of left side bulkhead. Leave clamps installed on rigid drain line.

g. In cabin, disconnect drain line from elbow in cabin ceiling and from front of rear bulkhead and remove from helicopter (Figure 15-4-12-1, Sheet 2, Detail B and Detail C).

h. In transition section, disconnect drain line from elbow in rear of rear bulkhead and from grommet in floor. Remove drain line from helicopter

(Figure 15-4-12-1, Sheet 2, Detail C and Detail D).

- i. In APU compartment, disconnect APU starter motor drain line from elbow.
- j. While holding elbow in cabin ceiling, have assistant remove jamnut and washer from elbow in APU compartment. Remove elbow and packing and remove retainer from cabin ceiling (Figure 15-4-12-1, Sheet 2, Detail B).
- k. In transition section, remove jamnut, washer, and elbow from rear bulkhead (Figure 15-4-12-1, Sheet 2, Detail C).

15-4-12.1.2 Inspect.

- a. Check drain line identification tapes for security and condition. If missing or unreadable tapes are evident, replace as required.
- b. Check grommet for cracks and wear. If damaged, replace grommet (Figure 15-4-12-1, Sheet 2, Detail D).

15-4-12.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Clean fittings and area around fittings with acetone, Item 25, Appendix D.
- c. In APU compartment, install packing and retainer on elbow and install elbow in cabin ceiling. Install washer and jamnut on elbow in cabin ceiling. Position elbow facing rear. **TORQUE JAMNUT TO 85 - 105 INCH-POUNDS** (Figure 15-4-12-1, Sheet 2, Detail B).
- d. In APU compartment, connect APU starter motor drain line on elbow.

- e. In transition section, install elbow, washer, and jamnut in rear bulkhead. Position elbow facing down and outboard. Do not torque jamnut at this time (Figure 15-4-12-1, Sheet 2, Detail C).

- f. In transition section, connect drain line to elbow and through grommet in floor (Figure 15-4-12-1, Sheet 2, Detail C and Detail D).

- g. In cabin, secure drain line to existing clamps on rigid drain line with clamps, screws, washers, and nuts down rear side of left rear bulkhead (Figure 15-4-12-1, Sheet 1, Detail A).

h. **TORQUE JAMNUT ON REAR BULKHEAD ELBOW TO 85 - 105 INCH-POUNDS.**

- i. In cabin, install drain line through ceiling beam and connect to elbows. Secure to existing clamps on rigid drain line with clamps, screws, washers, and nuts (Figure 15-4-12-1, Sheet 1, Detail A and Sheet 2, Detail B and Detail C).

- j. Close fuel cell compartment as follows:

- (1) If removed, install ECS condenser pallet (PARA 13-5-3) and ECS evaporator pallet (PARA 13-5-6).
- (2) Install cargo netting.
- (3) Install soundproofing panel (PARA 2-4-69).
- (4) Install rear passenger's seats (PARA 2-4-45).

- k. If opened, close transition section avionics door.

- l. Close APU access doors.

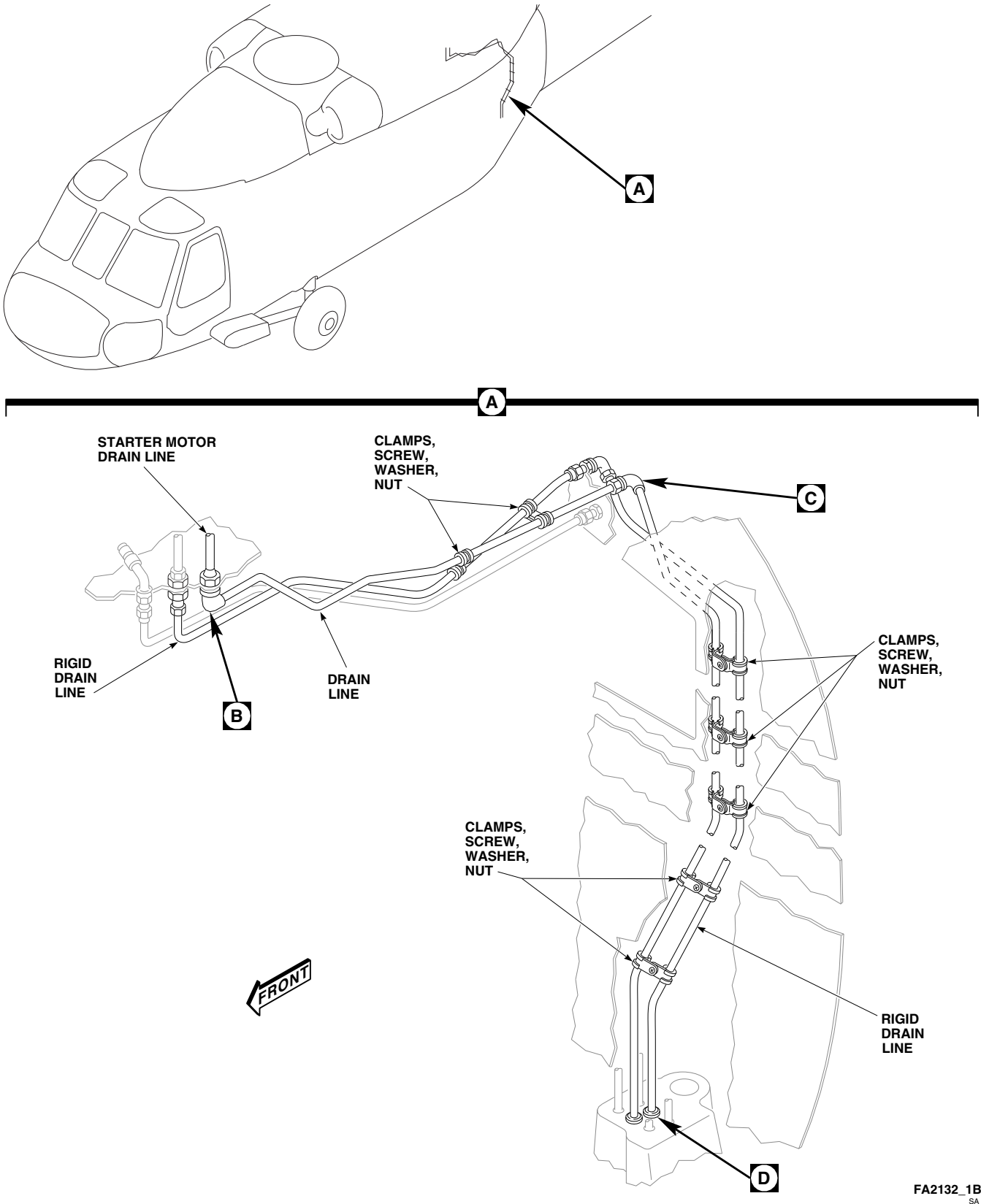
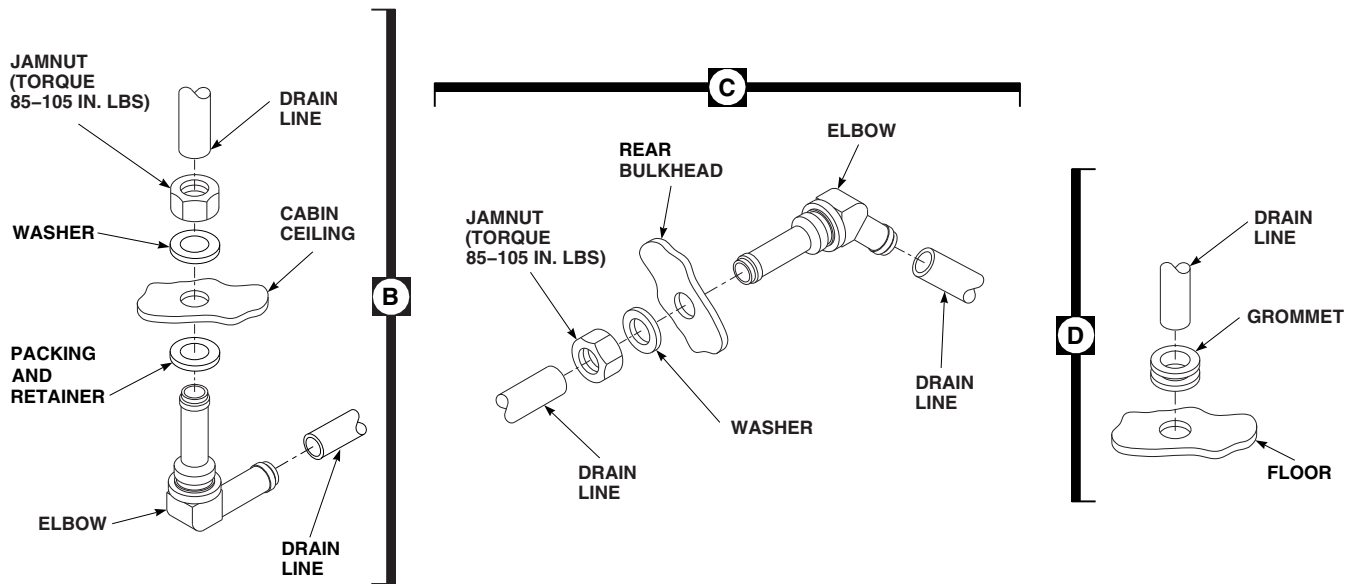


FIGURE 15-4-12-1. REPLACE APU DRAIN LINES (SHEET 1 OF 2)



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FIGURE 15-4-12-1. REPLACE APU DRAIN LINES (SHEET 2 OF 2)

15-4-13 APU FRONT INBOARD SUPPORT MOUNT.

PARAGRAPH BREAKDOWN

	PAGE
15-4-13.1 Replace APU Front Inboard Support Mount	15-4-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Blind Rivet Gun Kit
- Hoisting Sling, 70700-20323-042
- Torque Wrench, 30 - 150 in. lbs
- Wood Block

Materials/Parts

- Sealing Compound, Item 281, Appendix D

References

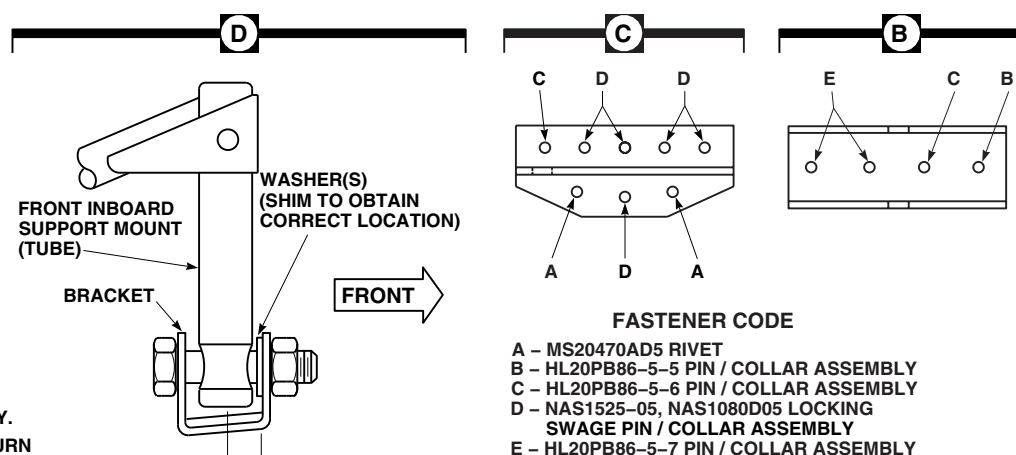
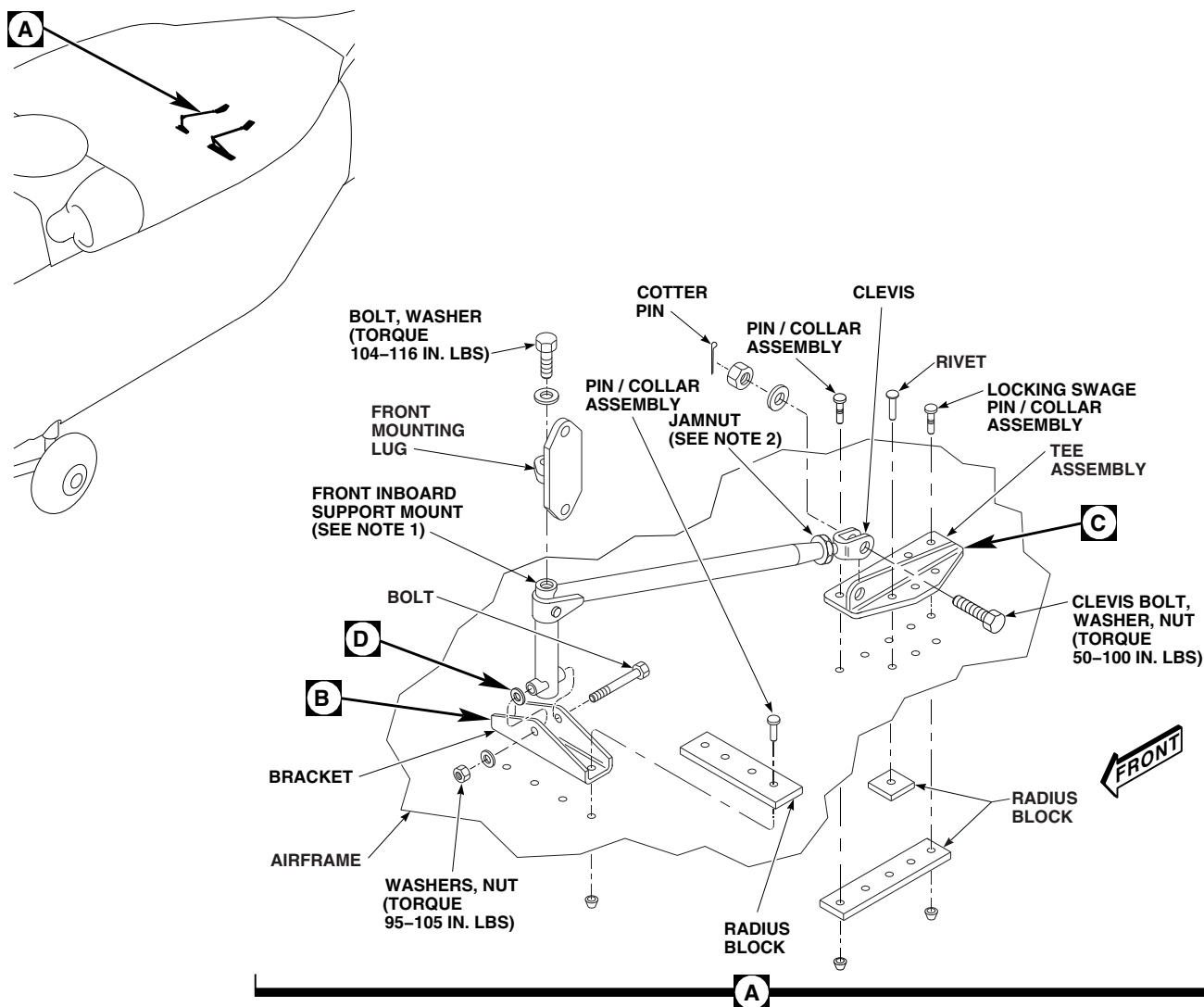
- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-1
- PARA 2-6-5
- PARA 13-5-3
- PARA 15-4-1

15-4-13.1 Replace APU Front Inboard Support Mount.

15-4-13.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open APU access doors.
- Remove bolt and washer securing front mounting lug to front inboard support mount (Figure 15-4-13-1, Detail A). Remove mounting lug.
- Place wood block under APU sump, or use hoisting sling to support APU (PARA 15-4-1).
- Remove bolt, washer(s), and nut securing support mount to bracket.

- Remove cotter pin, nut, washer, and clevis bolt securing clevis of support mount to tee assembly.
- Gain access to support mount area as follows:
 - Remove left side rear passenger’s seats (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left stowage compartment (PARA 2-4-69).
 - Remove cargo netting from stowage compartment.
 - If installed, remove ECS condenser pallet (PARA 13-5-3).



NOTES

1. APU NOT SHOWN FOR CLARITY.
2. TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 15-4-13-1. REPLACE APU FRONT INBOARD SUPPORT MOUNT

- h. Position assistant inside rear compartment to help remove fasteners.
- i. Remove pin/collar assemblies securing radius block and bracket to airframe. Remove bracket and radius block.
- j. Remove rivets, locking swage pin/collar assemblies, and pin/collar assemblies securing tee assembly and radius blocks to airframe. Remove tee assembly and radius blocks.
- k. Touch up damaged protective finishes (PARA 2-6-5).
- i. Secure support mount to bracket with bolt, washer(s), and nut. **TORQUE NUT TO 95 - 105 INCH-POUNDS** (Figure 15-4-13-1, Detail A).
- j. Loosen jamnut and adjust clevis end of support mount to line up with hole in tee assembly. Secure clevis to tee assembly with clevis bolt, washer, and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS**. Install cotter pin.
- k. **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

15-4-13.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using sealing compound, Item 281, Appendix D, coat all tee assembly and bracket fasteners (PARA 2-6-1).
- c. Position assistant inside rear compartment to help install fasteners.
- d. Position replacement bracket and radius block on airframe (Figure 15-4-13-1, Detail A).
- e. Install required pin and collar assemblies securing bracket and radius block (Figure 15-4-13-1, Detail A and Detail B).
- f. Position tee assembly and radius blocks on airframe (Figure 15-4-13-1, Detail A).
- g. Using blind rivet gun, install required rivets. Install locking swage pin/collar assemblies, and pin/collar assemblies securing tee assembly and radius blocks to airframe (Figure 15-4-13-1, Detail A and Detail C).
- h. Install front inboard support mount in bracket with bolt. Shim support mount using washer to obtain 0.340-inch between centerline of tube and inside, front of bracket (Figure 15-4-13-1, Detail D).
- l. Line up support mount with front mounting lug on APU. Remove wood block or hoisting sling supporting APU (PARA 15-4-1).
- m. Install bolt and washer securing front mounting lug to support mount. **TORQUE BOLT TO 104 - 116 INCH-POUNDS**.
- n. Clean and seal support mount parts to airframe for corrosion protection and waterproofing (PARA 2-6-1).
- o. Touch up paint as required (PARA 2-6-5).
- p. Make sure area is clean and free of foreign material.
- q. Close access to front inboard support mount area as follows:
 - (1) If removed, install ECS condenser pallet (PARA 13-5-3).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).
- r. Close APU access doors.

15-4-14 APU FRONT OUTBOARD SUPPORT MOUNT.

PARAGRAPH BREAKDOWN

	PAGE
15-4-14.1 Replace APU Front Outboard Support Mount	15-4-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Blind Rivet Gun Kit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 281, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-1
- PARA 2-6-5
- PARA 13-5-3
- PARA 15-4-1

15-4-14.1 Replace APU Front Outboard Support Mount.

15-4-14.1.1 Remove.

- Remove APU (PARA 15-4-1).
- Remove cotter pin, nut, washer, and clevis bolt securing clevis to tee assembly (Figure 15-4-14-1, Detail A).
- Gain access to front outboard support mount as follows:
 - Remove left side rear passenger’s seats (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left stowage compartment (PARA 2-4-69).

(3) Remove cargo netting from stowage compartment.

(4) If installed, remove ECS condenser pallet (PARA 13-5-3).

- Position assistant inside rear compartment to help remove and install fasteners.
- Remove rivets and pin/collar assemblies securing front outboard support mount and radius blocks to airframe. Remove front outboard support mount and radius blocks.
- Remove locking swage pin/collar assemblies securing tee assembly and radius blocks to airframe. Remove tee assembly and radius blocks.

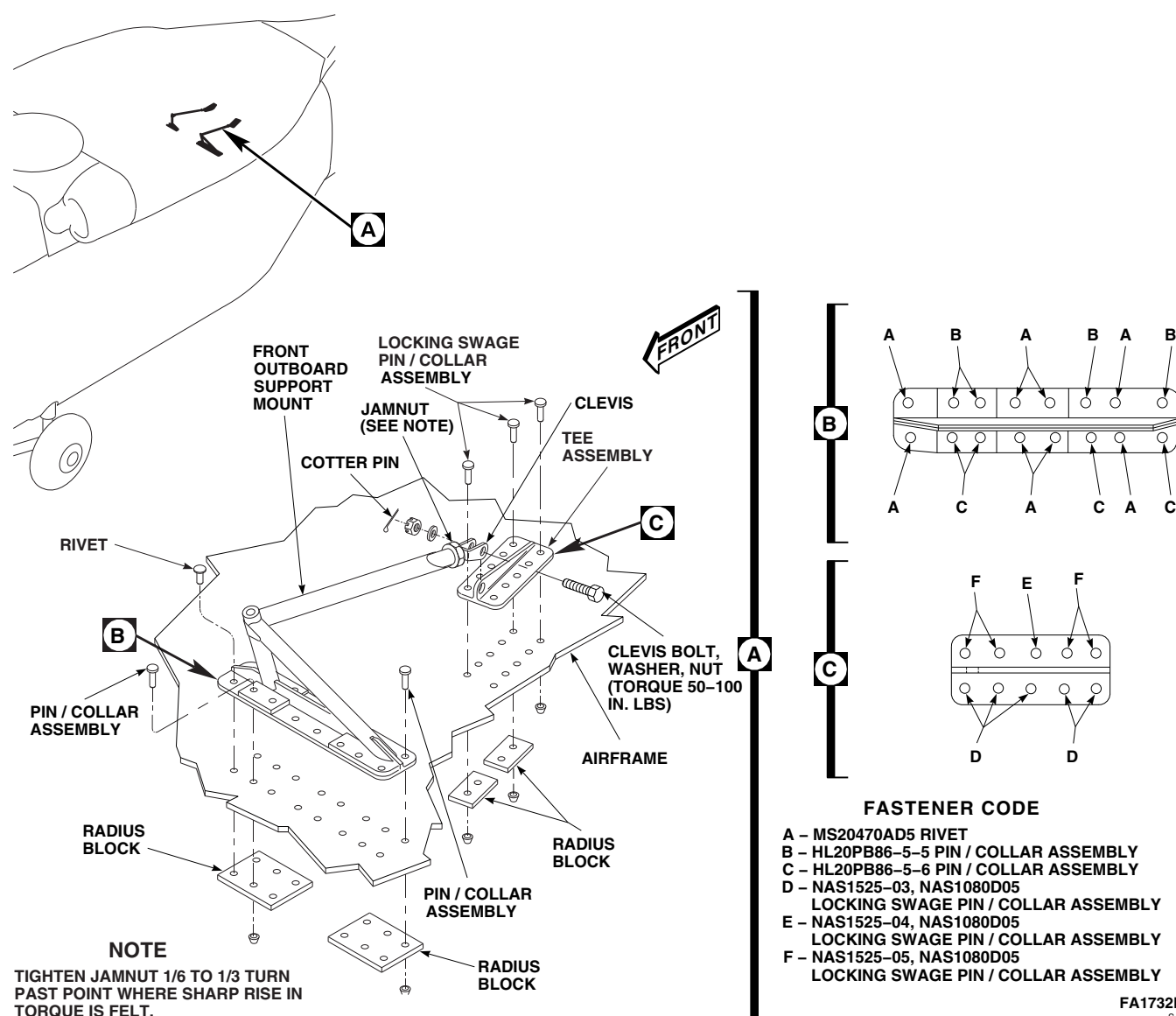


FIGURE 15-4-14-1. REPLACE APU FRONT OUTBOARD SUPPORT MOUNT

- g. Touch up damaged protective finishes (PARA 2-6-5).

15-4-14.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Using sealing compound, Item 281, Appendix D, coat all tee assembly and front outboard support mount fasteners (PARA 2-6-1).
- Position tee assembly and radius blocks on airframe (Figure 15-4-14-1, Detail A).
- Install required locking swage pin/collar assemblies securing tee assembly and radius

blocks to airframe (Figure 15-4-14-1, Detail A and Detail C).

- Position front outboard support mount and radius blocks on airframe (Figure 15-4-14-1, Detail A).
- Using blind rivet gun, install required rivets. Install pin/collar assemblies securing front outboard support mount and radius blocks to airframe (Figure 15-4-14-1, Detail A and Detail B).
- Loosen jamnut and adjust clevis of support mount to line up with hole in tee assembly.

Secure clevis to tee assembly with clevis bolt, washer, and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.** Install cotter pin (Figure 15-4-14-1, Detail A).

- h. **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- i. Install APU (PARA 15-4-1).
- j. Clean and seal support mount parts to airframe for corrosion protection and waterproofing (PARA 2-6-1).
- k. Touch up paint as required (PARA 2-6-5).

- l. Make sure area is clean and free of foreign material.
- m. Close access to front outboard support mount area as follows:
 - (1) If removed, install ECS condenser pallet (PARA 13-5-3).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).

15-4-15 APU FRONT INBOARD AND OUTBOARD SUPPORT MOUNT REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
15-4-15.1 Repair APU Front Inboard and Outboard Support Mounts	15-4-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Primer, Item 258, Appendix D

References

- AMS-QQ-P-416
- Appendix D
- MIL-S-18729

- MIL-S-6736
- MIL-S-6758
- PARA 2-6-1
- PARA 2-6-5
- PARA 15-4-1
- PARA 15-4-13
- PARA 15-4-14
- QQ-A-250
- QQ-A-367

15-4-15.1 Repair APU Front Inboard and Outboard Support Mounts.

NOTE

Front APU support mounts (inboard and outboard) include cadmium-plated 4130 steel tubular welded assemblies, 4130 steel attaching angles and bracket, and two aluminum alloy tee extrusion brackets.

- Evaluate damage to aluminum parts (Figure 15-4-15-1, Detail A and Detail B) (PARA 2-6-

1). Damage greater than negligible limits to extrusion requires replacement of part.

- Replace worn bushing in tee brackets. Press-fit wet with primer, Item 258, Appendix D.
- Evaluate damage and make repairs to steel welded assemblies (PARA 2-6-1).
- Refer to PARA 15-4-1 for hoisting, removing, and installing APU.
- Refer to PARA 15-4-13 and PARA 15-4-14 for removal of front inboard and outboard support mounts.

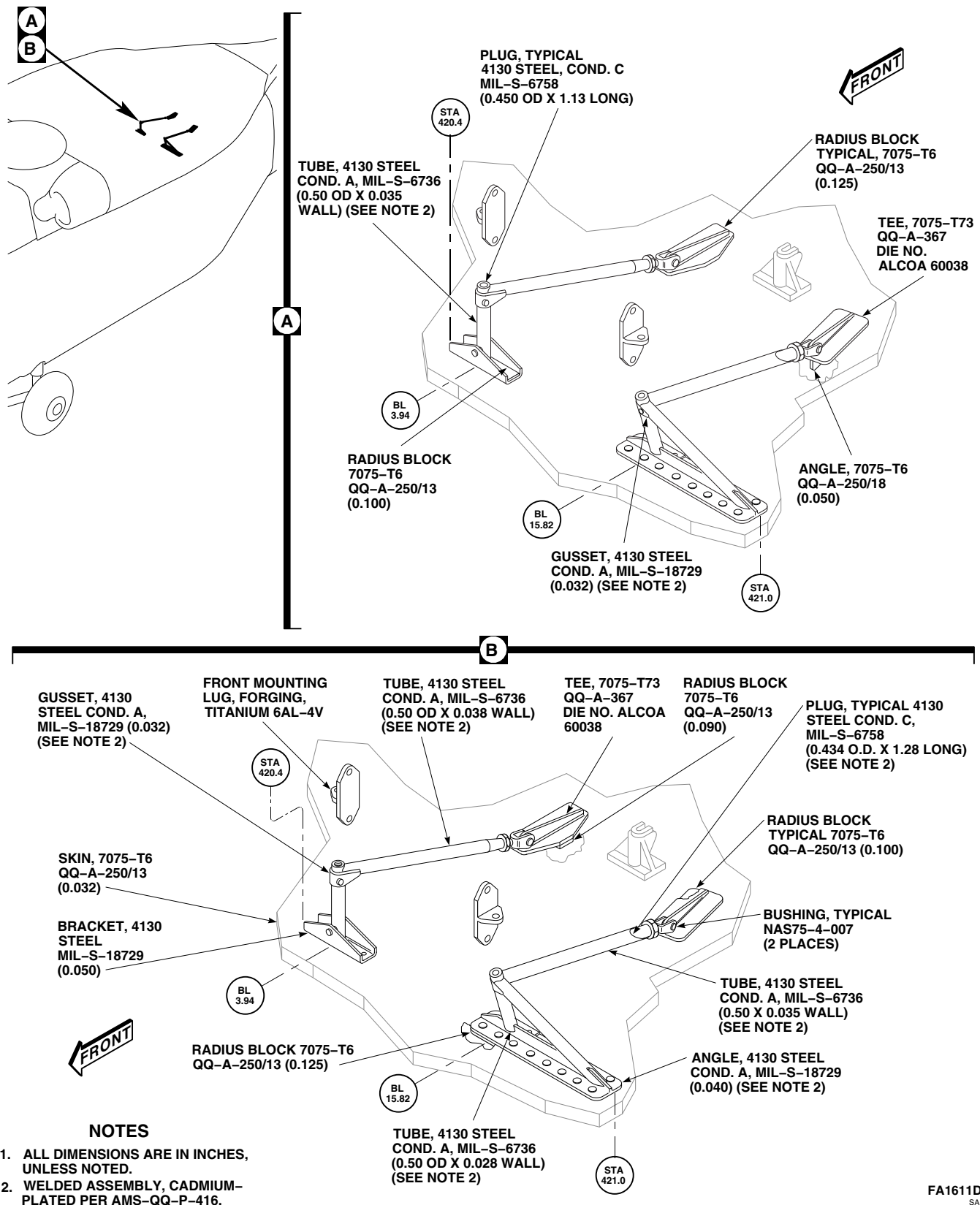
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FIGURE 15-4-15-1. REPAIR APU FRONT INBOARD AND OUTBOARD SUPPORT MOUNTS

- f. Refer to PARA 2-6-1 for corrosion protection, watertightness, and sealing support parts to structure.
- g. Touch up paint as required (PARA 2-6-5).

15-4-16 APU REAR SUPPORT MOUNT.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
15-4-16.1	Replace APU Rear Support Mount	15-4-16-1	15-4-16.2	Repair APU Rear Support Mount	15-4-16-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Hi-Lok Fastener, HL20PB86-5-3
- Sealing Compound, Item 281, Appendix D

References

- AMS 5355
- Appendix D

- MIL-M-20693
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-5
- PARA 13-5-3
- PARA 15-4-1
- QQ-A-250

15-4-16.1 Replace APU Rear Support Mount.

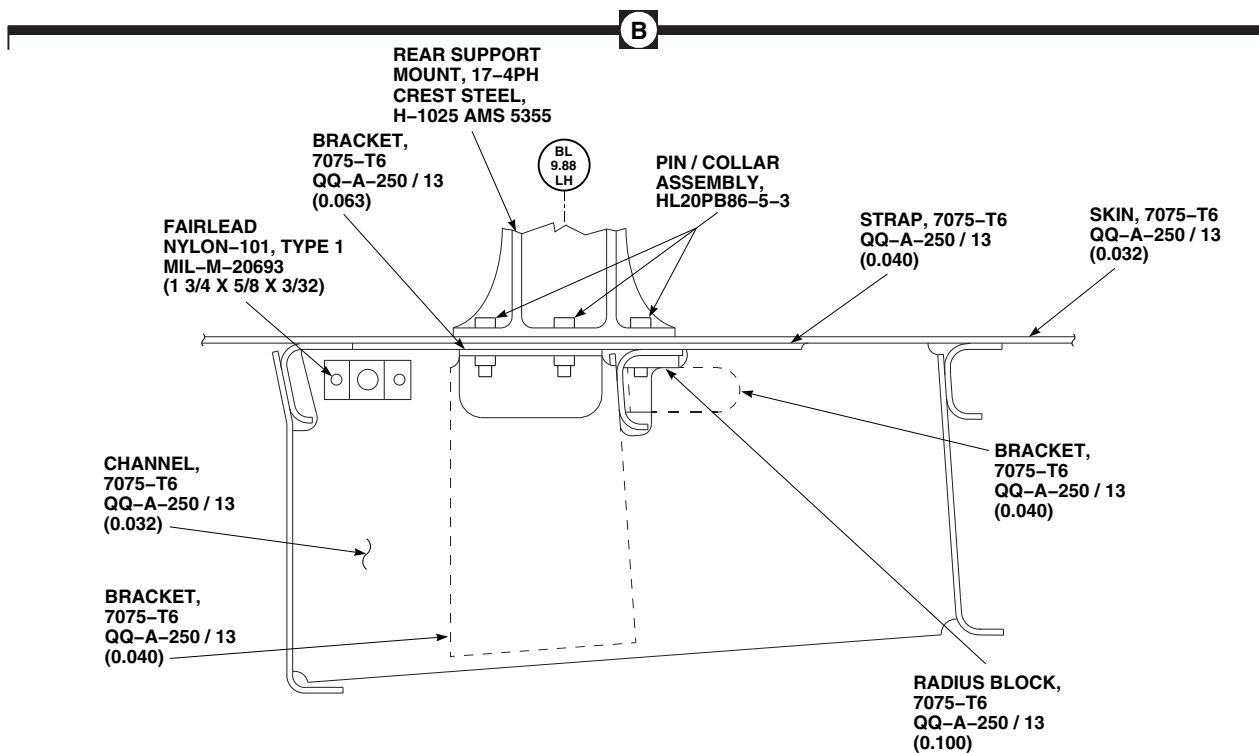
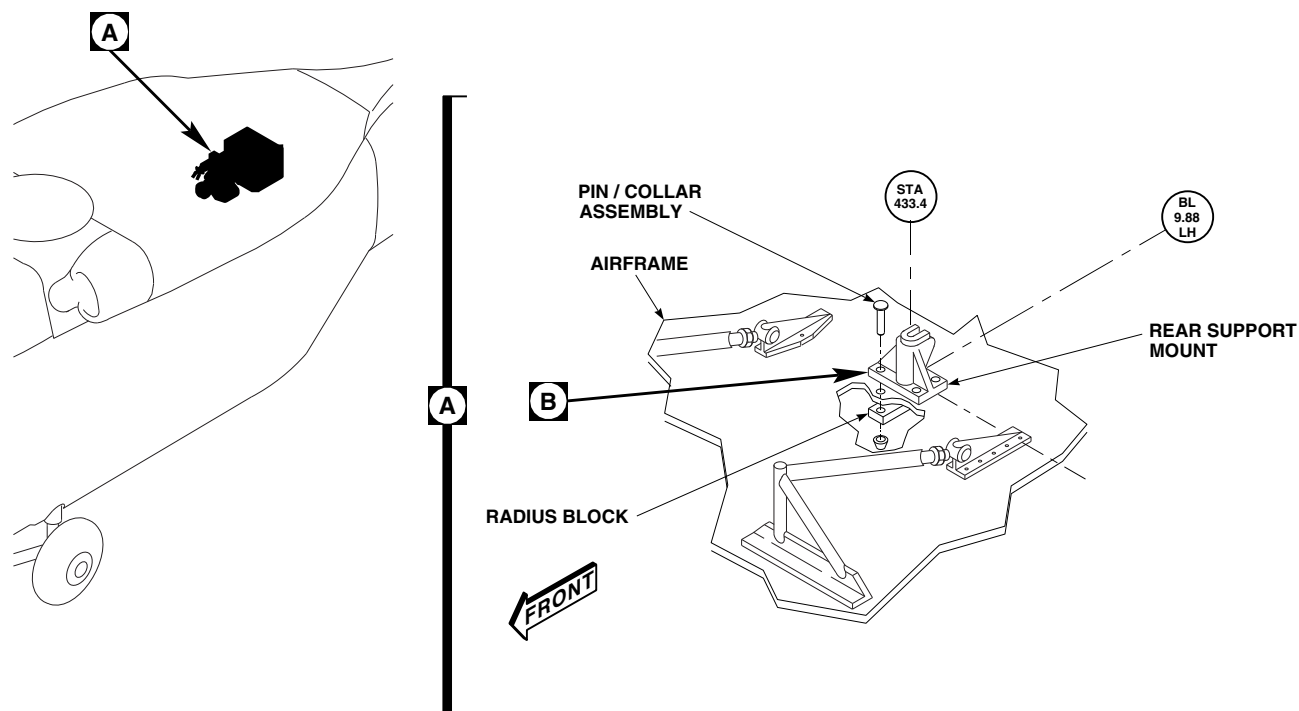
15-4-16.1.1 Remove.

- Remove APU (PARA 15-4-1).
- Gain access to rear support mount as follows:
 - Remove left side rear passenger’s seats (PARA 2-4-45).
 - Remove soundproofing panel from rear bulkhead covering left stowage compartment (PARA 2-4-69).

- Remove cargo netting from stowage compartment.

- If installed, remove ECS condenser pallet (PARA 13-5-3).

- Position assistant inside rear compartment to help remove and install fasteners.
- Remove pin/collar assemblies securing rear support mount and radius block to airframe (Figure 15-4-16-1, Detail A and Detail B).
- Touch up damaged protective finishes (PARA 2-6-5).



LOOKING FORWARD - STA 433.44 (LEFT)

NOTE

ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 15-4-16-1. REPLACE/REPAIR APU REAR SUPPORT MOUNT

15-4-16.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Coat all rear support mount pin/collar assemblies with sealing compound, Item 281, Appendix D (PARA 2-6-1).
- c. Position rear support mount and radius block on airframe (Figure 15-4-16-1, Detail A).
- d. Install required pin/collar assemblies securing rear support mount and radius block to airframe (Figure 15-4-16-1, Detail A and Detail B).
- e. For corrosion protection and waterproofing, seal support mount parts to structure (PARA 2-6-1).
- f. Touch up paint as required (PARA 2-6-5).
- g. Install APU (PARA 15-4-1).
- h. Close access to rear support mount area as follows:
 - (1) If removed, install ECS condenser pallet (PARA 13-5-3).
 - (2) Install cargo netting.
 - (3) Install soundproofing panel (PARA 2-4-69).
 - (4) Install rear passenger's seats (PARA 2-4-45).
- i. Make sure area is clean and free of foreign material.

15-4-16.2 Repair APU Rear Support Mount.

NOTE

APU rear support mount consists of mounting lug attached to APU and support mount installed on transition sec-

tion upper plating at STA 433.44. Mounting lug, which engages support mount, is made from 17-4PH corrosion-resistant steel, and threaded into APU base. Support mount, made from 17-4PH steel casting, is installed on upper plating support structure with Hi-Lok fasteners, HL20PB86-5-3. Support structure, made from formed aluminum alloy sheet stock, reinforces area between stringers for installing support mount (Figure 15-4-16-1, Detail B).

- a. Evaluate damage to support structure (PARA 2-6-1).

NOTE

Negligible damage to mounting lug and support fitting is limited to nicks, scratches, and abrasions or wear, no deeper than 5% of material thickness at point of damage.

- b. Inspect mounting lug and support fittings for damage. Evaluate as follows:
 - (1) Damage within negligible limits may be repaired by blending damage smoothly to surrounding area and polishing with abrasive cloth, Item 6, Appendix D.
 - (2) Damage greater than negligible limits must be reviewed by engineering authority for damage evaluation and either repair or replacement.
- c. Refer to PARA 2-6-2 for typical support structure repair.
- d. Refer to PARA 15-4-1 for hoisting, removing, or installing APU.
- e. Refer to PARA 2-6-1 for corrosion protection, watertightness, and sealing support parts to structure.
- f. Touch up paint as required (PARA 2-6-5).

SECTION 15-5

MAINTENANCE PROCEDURES (BENCH)

NOT APPLICABLE

